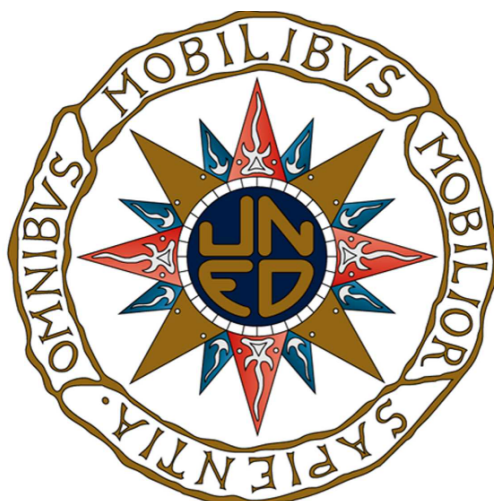


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TRABAJO FIN DE MÁSTER

Classical and Rough Volatility Models

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Aus dem Paradies, das Cantor uns geschaffen hat, soll uns niemand vertreiben können.

We shall never be expelled from the paradise Cantor created for us.

David Hilbert

Abstract

The aim of this work is to present a natural progression from the classical foundations of stochastic calculus to the study and application of modern tools such as Rough Path Theory, with the ultimate goal of employing these tools in the field of financial modelling.

Part I develops the machinery: Chapter 1 covers the basics of stochastic calculus, constructs the Itô integral from the ground up, and pays some attention to the regularity of Brownian paths. Chapter 2 builds the financial theory on that foundation: contingent claims and arbitrage, the fundamental theorems of asset pricing, the Black–Scholes framework, and the local and stochastic volatility models devised to repair it.

Part II describes modern approaches and tools. Chapter 3 presents the theory of rough paths: the empirical evidence that volatility is rough, the precise sense in which the semimartingale theory of Part I becomes unavailable, and Lyons’ resolution through the extension of a path by its iterated integrals. Chapter 4 brings the theory to data, where the models are calibrated and implemented.

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Keywords: stochastic calculus; Itô integral; Brownian motion; quadratic variation; volatility; Black–Scholes; local and stochastic volatility; fractional Brownian motion; rough paths; rough volatility; calibration.

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Notation

A recurring difficulty in a text of this kind is that probabilists and practitioners do not agree on the notations they use. The table below fixes the necessary conventions. Very standard symbols or those introduced for local use only are not included here.

Probability space and information

$(\Omega, \mathcal{F}, \mathbb{P})$	Complete probability space; $\mathbb{P}$ is always the <i>physical</i> measure.
$\mathbb{Q}$	Risk-neutral measure.
$(\mathcal{F}_t)_{t \in [0, T]}$	Filtration, always satisfying the usual conditions (Definition 1.6).
$(\mathcal{F}_t^W)_{t \in [0, T]}$	Augmented filtration generated by the Brownian motion W .
$\mathbb{E}[\cdot], \mathbb{E}^{\mathbb{Q}}[\cdot], \mathbb{V}[\cdot]$	Expectation under $\mathbb{P}$, and under a named measure. Variance.
$\mathcal{B}(E)$	Borel σ -algebra of the topological space E .
$\mathbb{1}_A$	Indicator function of the event A .
$\varphi_X(u)$	Characteristic function $\mathbb{E}[e^{iuX}]$.

Processes

W	Brownian motion.
$\mathbf{W}$	Multidimensional Brownian motion with independent components.
W^H	Fractional Brownian motion with Hurst parameter H .
H	Hurst parameter, always.
μ	The <i>drift</i> of a process, or the mean of a distribution.
σ	A <i>diffusion coefficient</i> ; $\Sigma = \sigma\sigma^\top$ is the diffusion matrix.
ρ	A <i>correlation</i> , always; $P = (\rho_{ij})$ is a correlation matrix (Definition 1.71).
$p_{s,x}(t, y), p_T$	Transition density of a Markov process, and the terminal density of S_T .
$V_1(f)_t, V_p(f)_t$	Total variation and p -variation of f on $[0, t]$.
$[X], \langle X \rangle$	Realised (pathwise) and predictable quadratic variation of X , respectively.
$L_{\text{ad}}^2([0, T] \times \Omega)$	Adapted integrands square-integrable in $dt \otimes d\mathbb{P}$.
$\mathcal{L}(\Omega, L^p[0, T])$	The larger, pathwise class: adapted, with $\int_0^T f ^p dt < \infty$ almost surely.

Finance

S_t	Spot price of the underlying.
F_t or $F(t, T)$	Forward price for delivery at T .
$B(t, T), D(t, T)$	Price at t of a zero-coupon bond maturing at T , and the discount factor from T back to t .
S^0	Numerator asset.
G_T, Φ	Terminal payoff (a random variable) and payoff function.
K	Strike price; $k = \log(K/F)$ is log-moneyness.
r, q	Short rate and continuous dividend (or repo) yield.
$\sigma_{\text{imp}}, \sigma_{\text{loc}}, \sigma_{\text{real}}$	Implied, local and realised volatility; $w = \sigma_{\text{imp}}^2 T$ is total implied variance.
V	The value of a derivative.
π, Π_t	A trading strategy, and the value of a hedging portfolio.
$N(\cdot)$	Standard normal distribution function.

Rough paths

$\mathbf{X} = (X, \mathbb{X})$	A rough path: a trajectory <i>plus</i> a second level it does not determine.
$\mathbb{X}_{s,t}, \mathbb{W}_{s,t}$	That second level, a function of <i>two</i> times and never a path.
$A_{s,t}$	Lévy area: the antisymmetric part of $\mathbb{X}_{s,t}$, its only free datum for a pair.
α	Hölder exponent of a rough path, always in $(\frac{1}{3}, \frac{1}{2}]$; distinct from H .
$S(\mathbf{X}), S^N$	Signature of a rough path, and its truncation at level N .

Introduction

1 Objective and Scope

The primary objective of this dissertation is to provide a solid review of *volatility*, aiming to bridge the gap between readers who come from mathematics — and who often lack expertise in finance — and those who come from finance, and who may not always see the depth of the machinery they rely on. To that end, the text keeps a clear and rigorous mathematical mindset while maintaining continuous references to practical and financial interpretations.

Part I aims to provide a thorough introduction to the subject. Chapter 1 develops the probabilistic machinery: measure-theoretic preliminaries, Brownian motion, the Itô integral and stochastic differential equations. Particular attention is paid there to the regularity of Brownian trajectories, since it is precisely the delicate balance between roughness and tractability — non-vanishing quadratic variation on the one hand, unbounded total variation on the other — that both makes Itô’s theory necessary and marks the limits of its reach. Chapter 2 then builds the financial theory on top of it: contingent claims and arbitrage, the fundamental theorems of asset pricing, the Black–Scholes framework, and some local and stochastic volatility models. A reader already fluent in stochastic calculus may begin at Chapter 2.

Part II is devoted to more recent developments in volatility modelling, including Rough Paths and Rough Volatility theories (Chapter 3). The latter is introduced from a general and mathematical perspective: although it finds direct applications in finance, the theory of rough paths has far-reaching implications in other areas of mathematics as well. These chapters aim to highlight the necessity of more sophisticated tools to describe and operate in the market.

Finally, these theoretical constructs are brought to practice in Chapter 4, where two calibrations are performed against the same market smiles: an implied volatility surface fitted by a physics-informed neural network, on which the static no-arbitrage conditions are imposed as penalties and, on the extrapolated wings, proved outright; and the rough Heston model of Chapter 3. We also include an example from in pure rough paths framework. Here, the models are numerically implemented, providing an applied perspective and illustrating the strengths and challenges associated with each approach.

Some final thoughts are left for the Conclusions, where a general overview of the work done will be given, along with reflections on possible directions for further research.

2 A worthy note on context

The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve.

Eugene Wigner expressed his admiration in *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*, his 1960 article in *Communications on Pure and Applied Mathematics* [72]. It is a sentiment many would agree with: the effectiveness of mathematics in describing Nature is both remarkable and fascinating. Moreover, over the last century its branches have reached into areas where, not so long ago, mathematicians were considered too abstract to be of much use.

In 1827, Robert Brown, observing under a microscope the particles released by pollen grains suspended in water, noticed something unexpected: the particles wandered along irregular, hard-to-describe paths. Physics was not yet equipped to tackle such a problem. What was needed belonged to what we now call *statistical mechanics*, the study of collective behaviour of particles and molecules, intimately tied to *thermodynamics*. The decisive step came later, in 1905, when Albert Einstein provided a theory of this motion — the same year in which he also explained the photoelectric effect, introduced special relativity and gave the first statement of the mass–energy equivalence formula, $E = mc^2$. What a year.

Far from pollen grains, a parallel story was unfolding in finance. In 1900, Louis Bachelier, a doctoral student of Henri Poincaré, defended a thesis on the modelling of stock prices. His model is what we would now call an arithmetic Brownian motion, and it rested on shaky probabilistic foundations: Kolmogorov and his peers were not yet born, and even Lebesgue had not yet published his work on measure theory. Worse, it allowed negative prices — a point critics did not miss.

The common thread in both stories is *Brownian motion*. This stochastic process is a central object in probability: geometers find its fractal dimension fascinating, while probabilists value the rich family of processes built upon it. The idea of treating “random noise” as a genuine mathematical object took shape gradually, but it was Kiyoshi Itô’s work on *stochastic integration* that truly changed the picture.

Finance came to the subject later. Today mathematics is everywhere in banks, consultancy and audit firms, underpinning the valuation of derivatives, that is, assigning a *fair price* (according to market data) to contracts involving random future cash flows. Pricing is the orientation of this dissertation. It is only one of several uses of the same machinery — risk management and portfolio optimisation are others — but it is the one in which the mathematics is most visible.



Part I

Foundations and Classical Models

Chapter 1

Stochastic Foundations

We assume the reader is familiar with the foundations of probability theory. Standard references include, for example, Billingsley's classic text [8], as well as the treatments in [27] and [46]. Nevertheless, we briefly recall the basic probabilistic framework in order to fix notation and ensure clarity.

We will work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with Ω the sample space, $\mathcal{F}$ a σ -algebra of events, and $\mathbb{P}$ a probability measure. A **random variable** is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$, the target usually being $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, and particularly $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ in most of our applications.

Convention. Every probability space in this dissertation is assumed *complete*: if $A \in \mathcal{F}$ has $\mathbb{P}(A) = 0$, then every $B \subseteq A$ lies in $\mathcal{F}$ and has $\mathbb{P}(B) = 0$. Completeness is not a formality, and we shall see in Definition 1.6 exactly what it buys.

Convention (The symbols $\square$ and $\blacksquare$). A result whose *statement* ends with a hollow $\square$ is quoted **without proof**; a reference is given whenever the result is not entirely standard. A filled $\blacksquare$ closes an *argument*: it marks the end of a proof, whether that proof is given in full, reduced to a sketch, or deferred to Appendix B.

1.1 Probabilistic Preliminaries

1.1.1 Processes, filtrations and stopping times

The notion of *stochastic process* generalises that of a single random variable by allowing evolution in an external parameter, usually interpreted as time. To accommodate both discrete-time and continuous-time settings in a unified framework, we let $\mathbb{T}$ denote the **index set**, representing time. Typically, $\mathbb{T}$ is a subset of $\mathbb{R}$, such as:

$$\mathbb{T} \in \{\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, [0, T], [0, \infty), \mathbb{R}\},$$

depending on the application.

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a (complete) probability space, and let $\mathbb{T}$ be a time index set. A **stochastic process** is a collection of random variables $\{X_t : t \in \mathbb{T}\}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in a measurable space $(E, \mathcal{E})$, such that the map

$$X : \mathbb{T} \times \Omega \rightarrow E, \quad (t, \omega) \mapsto X_t(\omega)$$

is jointly measurable with respect to $\mathcal{B}(\mathbb{T}) \otimes \mathcal{F}$. In particular:

- For each fixed $t \in \mathbb{T}$, the function $\omega \mapsto X_t(\omega)$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
- For each fixed $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called the *sample path* or *trajectory* of the process.

We may denote the process by $X = \{X_t\}_{t \in \mathbb{T}}$, and abbreviate $X_t(\omega)$ as $X(t)$ or simply X_t when context allows. We say X is a *discrete-time* process whenever $\mathbb{T}$ is $\mathbb{N}$ or $\mathbb{Z}$, and a *continuous-time* process if $\mathbb{T}$ is $[0, T]$ for some $T > 0$ or $[0, \infty)$. This is a statement about the index set only, and is independent of whether the sample paths are continuous: the Poisson process of Example 1.22 is a continuous-time process with discontinuous paths.

This formulation includes many important examples such as Markov chains, Poisson processes, and Brownian motion. Note that the nature of the index set $\mathbb{T}$ often dictates the regularity assumptions or path properties we may require (e.g., right-continuity, càdlàg, continuity in probability, etc.). Analysing the dynamic evolution of these processes requires us to quantify their expected future behaviour given currently available information. This idea is contained in the **conditional expectation**. Its existence relies on the following profound result:

Theorem 1.2 (Radon–Nikodym). *Let ν and λ be two σ -finite measures on the same measurable space $(\Omega, \mathcal{F})$. Suppose that ν is absolutely continuous with respect to λ , denoted $\nu \ll \lambda$; that is,*

$$\lambda(A) = 0 \quad \Rightarrow \quad \nu(A) = 0, \quad \text{for all } A \in \mathcal{F}. \quad (1.1)$$

Then there exists a measurable function $f : \Omega \rightarrow [0, \infty)$, unique up to λ -null sets, such that for all $A \in \mathcal{F}$,

$$\nu(A) = \int_A f \, d\lambda. \quad (1.2)$$

This function f is called the Radon–Nikodym derivative of ν with respect to λ , and is denoted by $f = d\nu/d\lambda$. $\square$

We shall need the statement in slightly greater generality than above, since the measure we are about to construct might not be positive.

Corollary 1.3 (Signed version). *Let ν be a finite signed measure on $(\Omega, \mathcal{F})$ and λ a σ -finite positive measure with $\nu \ll \lambda$. Then there exists $f \in L^1(\Omega, \mathcal{F}, \lambda)$, real-valued and unique up to λ -null sets, satisfying (1.2).*

Proof: Apply the Jordan decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. Both $\nu^\pm \ll \lambda$, so Theorem 1.2 yields non-negative densities $f^\pm$, and $f := f^+ - f^-$ works; integrability follows from $\int (f^+ + f^-) \, d\lambda = |\nu|(\Omega) < \infty$. $\blacksquare$

As a consequence, if $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ (that is, $\int |X| \, d\mathbb{P}$ is finite) and $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra, the *conditional expectation* of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to $\mathbb{P}$ -a.e. equality) $\mathcal{G}$ -measurable function satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_G X \, d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \quad (1.3)$$

Conditional expectation inherits the properties one would expect of an averaging operator: it preserves expectations and order, it is linear, it acts as the identity on $\mathcal{G}$ -measurable variables and as the plain expectation on independent ones, known factors may be taken out of it, and it satisfies the conditional Jensen inequality $\phi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\phi(X) \mid \mathcal{G}]$ for convex ϕ . These, together with the conditional forms of the Fatou, monotone and dominated convergence theorems, are recorded as Propositions A.1–A.4 in Appendix A; we shall use them throughout without comment. Two further properties, however, are stated here.



Proposition 1.4 (The tower property and the L^2 projection). *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. Then:*

i) (**Tower property**) *If $\mathcal{H}$ is a sub- σ -algebra of $\mathcal{G}$ and $X \in L^1(\Omega)$, then*

$$\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] \quad \text{almost surely.}$$

ii) (**L^2 projection**) *If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Equivalently, it is the unique $\mathcal{G}$ -measurable Y minimising $\mathbb{E}[(X - Y)^2]$, and*

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}]) Z] = 0 \quad \text{for every } Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}).$$

□

In the study of stochastic processes, it is essential to formalise how information accumulates over time. This is done through the concept of a **filtration**, which models the *evolution of knowledge* or *information available* up to each time t .

Definition 1.5 (Filtration). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be a time index set. A **filtration** is a family $\{\mathcal{F}_t\}_{t \in \mathbb{T}}$ of sub- σ -algebras of $\mathcal{F}$ satisfying:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all } s \leq t. \quad (1.4)$$

That is, the filtration is *increasing* in t . We interpret $\mathcal{F}_t$ as the collection of events whose outcomes are known or observable up to t . A stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$ is said to be **adapted** to the filtration $\{\mathcal{F}_t\}$ if, for each $t \in \mathbb{T}$, the random variable X_t is measurable with respect to $\mathcal{F}_t$. We say that $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ is a **filtered probability space**.

Definition 1.6 (The usual conditions). A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ is said to satisfy the **usual conditions** if

i) $\mathcal{F}_0$ contains every $\mathbb{P}$ -null set of $\mathcal{F}$ (*completeness*); and

ii) the filtration is *right-continuous*, that is,

$$\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{n \geq 1} \mathcal{F}_{t+1/n}, \quad t \in [0, T]. \quad (1.5)$$

Convention 1.7. From this point onwards, every filtered probability space is assumed to satisfy the usual conditions.

Completeness is what allows us to modify a process on a null set without leaving the filtration, as we shall do whenever we choose a continuous version of an object defined only up to indistinguishability. Right-continuity is what makes first hitting times of open sets into stopping times; it does explicit work in Proposition 1.10 below.

Recall that if χ is a real-valued random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the σ -algebra generated by χ is denoted $\sigma(\chi)$ and defined as:

$$\sigma(\chi) := \{A \in \mathcal{F} : A = \chi^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R})\}. \quad (1.6)$$

Example 1.8 (Natural Filtration). Given a stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$, the **natural filtration** generated by X is defined by:

$$\mathcal{F}_t^X := \sigma(X_s : s \leq t), \quad (1.7)$$

the smallest σ -algebra with respect to which all variables X_s for $s \leq t$ are measurable. This filtration encodes exactly the information revealed by observing the process up to time t .

Definition 1.9 (Stopping time). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a filtration $\{\mathcal{F}_t\}_{t \in \mathbb{T}}$. A mapping $\tau : \Omega \rightarrow \mathbb{T}$ is a *stopping time* if

$$\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t \quad \text{for each } t \in \mathbb{T}.$$

Proposition 1.10. Let $(\mathcal{F}_t)_{t \in [0, T]}$ be a right-continuous filtration. Then, a random variable $\tau : \Omega \rightarrow [0, T]$ is a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$ if and only if

$$\{\omega \in \Omega : \tau(\omega) < t\} \in \mathcal{F}_t, \quad \text{for all } t \in [0, T].$$

Proof: Suppose first that τ is a stopping time. Since $\{\tau < t\} = \bigcup_{n \geq 1} \{\tau \leq t - 1/n\}$ and each $\{\tau \leq t - 1/n\}$ lies in $\mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, the union lies in $\mathcal{F}_t$. This implication uses nothing beyond monotonicity of the filtration.

Now, suppose $\{\tau < t\} \in \mathcal{F}_t$ for every t . Then for each $n \geq 1$ we have $\{\tau < t + 1/n\} \in \mathcal{F}_{t+1/n}$, and since $\{\tau \leq t\} = \bigcap_{n \geq 1} \{\tau < t + 1/n\}$,

$$\{\tau \leq t\} \in \bigcap_{n \geq 1} \mathcal{F}_{t+1/n} = \mathcal{F}_{t^+} = \mathcal{F}_t,$$

the last equality being exactly the right-continuity (1.5). ■

Note that the converse uses right-continuity essentially, and fails without it.

Example 1.11. Let X be a measurable adapted process on $[0, T]$ with $\int_0^T |X_s|^2 ds < \infty$ almost surely. For $n \in \mathbb{N}$ set

$$A_n(\omega) = \left\{ t \in [0, T] : \int_0^t |X_s(\omega)|^2 ds > n \right\},$$

and let $\tau_n(\omega) = \inf A_n(\omega)$ if $A_n(\omega) \neq \emptyset$, and $\tau_n(\omega) = T$ otherwise. Since $t \mapsto \int_0^t |X_s|^2 ds$ is continuous and non-decreasing, the threshold n is exceeded strictly before t if and only if it is exceeded at some rational time strictly before t , so that for $t \in (0, T]$

$$\{\tau_n < t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}}} \left\{ \int_0^r |X_s|^2 ds > n \right\} \in \mathcal{F}_t, \quad (1.8)$$

whereas $\{\tau_n < 0\} = \emptyset \in \mathcal{F}_0$. By Proposition 1.10, τ_n is a stopping time.

The sequence $(\tau_n)_{n \in \mathbb{N}}$ just constructed is a *localising sequence*, and it is the prototype of every localisation argument in this chapter: one stops the process at the first time it leaves a bounded region, works in the well-behaved class available there, and lets the region exhaust the space.

Example 1.12 (First hitting time). Let X be a right-continuous process adapted to $\{\mathcal{F}_t\}$. If $U \subseteq \mathbb{R}$ is an *open set*, then the *first hitting time*

$$\tau_U(\omega) = \inf\{t > 0 : X_t(\omega) \in U\} \quad (1.9)$$

is a stopping time. Indeed, since U is open and X is right-continuous, the path enters U strictly before t if and only if it lies in U at some *rational* time strictly before t : if $X_r \in U$ for some $r < t$, right-continuity places X in the open set U throughout a nonempty interval $[r, r + \varepsilon)$, which contains a rational, so $\{\tau_U < t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}}} \{X_r \in U\} \in \mathcal{F}_t$ and Proposition 1.10 applies.



1.1.2 Fundamental Distributions

Before presenting some cornerstone distributions, recall that the **characteristic function** of a (real) random variable X is defined by

$$\varphi_X(u) := \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}. \quad (1.10)$$

It is the Fourier transform of the probability distribution of X .

Its elementary properties (normalisation, uniform continuity, Hermitian symmetry, behaviour under affine maps and under sums of independent variables, and the correspondence between differentiability at the origin and existence of moments) are collected in Proposition A.5 of Appendix A, and are used below without further comment.

Theorem 1.13 (Fourier inversion). *Let X be a random variable with distribution function F_X . If φ_X is integrable, then F_X admits a continuous density f_X given by*

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iux} \varphi_X(u) du. \quad (1.11)$$

□

In the general case, without assuming a density, the distribution function F_X can still be recovered via the *Lévy inversion formula*: for any $a < b$ that are *continuity points* of F_X ,

$$F_X(b) - F_X(a) = \lim_{R \rightarrow \infty} \frac{1}{2\pi} \int_{-R}^R \frac{e^{-iau} - e^{-ibu}}{iu} \varphi_X(u) du. \quad (1.12)$$

The restriction to continuity points is necessary and not a technicality: at a jump of F_X the right-hand side converges to the average of the left and right limits rather than to F_X itself. Since F_X is monotone it has at most countably many discontinuities, so continuity points are dense and the formula determines F_X everywhere by right-continuity.

We deduce that if two random variables X and Y have the same characteristic function, i.e. $\varphi_X(u) = \varphi_Y(u)$ for all $u \in \mathbb{R}$, then X and Y have the same distribution.

Theorem 1.14 (Lévy's Continuity Theorem). *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables with characteristic functions φ_{X_n} .*

- If $X_n \xrightarrow{d} X$ for some random variable X , then $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$ for every $u \in \mathbb{R}$.
- Conversely, suppose $\varphi_{X_n}(u) \rightarrow \varphi(u)$ for every $u \in \mathbb{R}$, where $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ is some function continuous at $u = 0$. Then φ is the characteristic function of a random variable X , and $X_n \xrightarrow{d} X$. □

We now describe certain distributions of widespread importance and usefulness. The first place must be given to the normal distribution.

Example 1.15 (Normal Distribution). Let $\mu \in \mathbb{R}$ and $\sigma > 0$, and define the real-valued random variable $X : \Omega \rightarrow \mathbb{R}$ with distribution $X \sim \mathcal{N}(\mu, \sigma^2)$, that is, X has the probability density function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (1.13)$$

Then X is said to be normally distributed with mean μ and variance σ^2 . Thus, a normal distribution is characterised by its first two moments.

The normal distribution is fundamental in probability theory and statistics. It appears as the limiting distribution in the Central Limit Theorem and models many naturally occurring phenomena. Its characteristic function takes the form

$$\varphi_X(u) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right). \quad (1.14)$$

The resemblance of its density to the *heat kernel* (see [30]) is not a coincidence, and will be explained in Subsection 1.2.6.

Example 1.16 (Multivariate Normal Distribution). An $\mathbb{R}^d$ -valued random vector X follows a *multivariate normal distribution* with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$, written similarly as $X \sim \mathcal{N}_d(\mu, \Sigma)$, if $a^\top X \sim \mathcal{N}(a^\top \mu, a^\top \Sigma a)$ for every $a \in \mathbb{R}^d$.

Here Σ is necessarily symmetric positive semi-definite. If it is invertible the law is absolutely continuous on $\mathbb{R}^d$ with density $f_X(x) = (2\pi)^{-d/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$; if it is singular, X is supported on a lower-dimensional affine subspace. In either case the characteristic function is

$$\varphi_X(u) = \mathbb{E}[e^{iu^\top X}] = \exp\left(iu^\top \mu - \frac{1}{2}u^\top \Sigma u\right), \quad u \in \mathbb{R}^d. \quad (1.15)$$

The characteristic function φ_X of (1.15) extends to an entire analytic function on $\mathbb{C}^d$. Moreover, a sequence of random vectors $X_n \sim \mathcal{N}_d(\mu_n, \Sigma_n)$ converges in distribution to $\mathcal{N}_d(\mu, \Sigma)$ if and only if $\mu_n \rightarrow \mu$ in $\mathbb{R}^d$ and $\Sigma_n \rightarrow \Sigma$ entrywise.

Any symmetric positive semi-definite matrix Σ admits a matrix factor $\Sigma = LL^\top$ (e.g., via Cholesky or spectral decomposition), yielding the representation $X \stackrel{d}{=} \mu + LZ$ for $Z \sim \mathcal{N}_d(0, I_d)$. What sets the multivariate case apart from the one-dimensional one is geometry: L acts as an anisotropic scaling followed by a rotation, deforming the spherical contours of Z into oriented ellipsoids and thereby introducing directional cross-correlations.

The following are easily verified through Theorem 1.14 (or by evaluating characteristic functions directly via (1.15)):

- Let $X \sim \mathcal{N}_d(\mu_X, \Sigma_X)$ and $Y \sim \mathcal{N}_d(\mu_Y, \Sigma_Y)$ be two *independent* multivariate normal random vectors in $\mathbb{R}^d$. For matrices $A, B \in \mathbb{R}^{m \times d}$ and a vector $c \in \mathbb{R}^m$,

$$AX + BY + c \sim \mathcal{N}_m(A\mu_X + B\mu_Y + c, A\Sigma_X A^\top + B\Sigma_Y B^\top).$$

If X and Y are jointly normal with cross-covariance matrix $\Sigma_{XY} = \text{Cov}(X, Y) \in \mathbb{R}^{d \times d}$, the resulting covariance matrix generalises to $A\Sigma_X A^\top + B\Sigma_Y B^\top + A\Sigma_{XY} B^\top + B\Sigma_{XY}^\top A^\top$. Degenerate distributions with singular covariance matrices (including constant vectors where $\Sigma = 0$) are fully accommodated within this framework.

- While independence always implies zero correlation ($\Sigma_{XY} = 0$), the converse holds *if* the combined vector (X, Y) is jointly normal.

Remark 1.17 (Normal marginals do not make a normal vector). Let $X \sim \mathcal{N}(0, 1)$ and let ξ be a random sign independent of X , so that $\mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = \frac{1}{2}$. Put $Y := \xi X$. Conditioning on ξ shows at once that $Y \sim \mathcal{N}(0, 1)$, and

$$\mathbb{E}[XY] = \mathbb{E}[\xi] \mathbb{E}[X^2] = 0,$$

so X and Y are uncorrelated standard normal variables. They are very far from independent: $|X| = |Y|$ almost surely. Moreover, the sum $X + Y = (1 + \xi)X$ vanishes with probability $\frac{1}{2}$, hence is not normal at all.



The moral is that “normal” is a property of a *joint law on* $\mathbb{R}^d$, not a property of a list of marginals. The pair (X, Y) above has normal marginals on $\mathbb{R}$, but it is not multivariate normal in $\mathbb{R}^2$ and the standard linear-combination and independence properties fail for it.

The normal distribution owes its central place to the following result (in a weak form):

Theorem 1.18 (Central Limit Theorem). *Let $\{X_n\}_{n=1}^\infty$ be independent, identically distributed (i.i.d.) random variables with finite mean μ and variance $\sigma^2 > 0$. Then*

$$S_n := \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (1.16)$$

Proof: Put $Y_k := (X_k - \mu)/\sigma$, so that the Y_k are i.i.d. with $\mathbb{E}[Y_1] = 0$ and $\mathbb{E}[Y_1^2] = 1$, and $S_n = n^{-1/2} \sum_{k=1}^n Y_k$. Write φ for the common characteristic function of the Y_k .

Since $\mathbb{E}[Y_1^2] < \infty$, Proposition A.5 vi) makes φ twice differentiable with $\varphi(0) = 1$, $\varphi'(0) = i\mathbb{E}[Y_1] = 0$ and $\varphi''(0) = i^2\mathbb{E}[Y_1^2] = -1$. Taylor’s theorem with Peano remainder therefore gives

$$\varphi(u) = 1 - \frac{1}{2}u^2 + o(u^2), \quad u \rightarrow 0. \quad (1.17)$$

By independence and items iv) and v) of the same proposition, $\varphi_{S_n}(u) = \varphi(u/\sqrt{n})^n$.

Fix $u \in \mathbb{R}$ and set $z_n := \varphi(u/\sqrt{n})$ and $w_n := 1 - u^2/(2n)$. Both have modulus at most 1 once $n \geq u^2$, the first because characteristic functions are bounded by 1 and the second because $0 \leq w_n \leq 1$ for such n . The elementary estimate $|z^n - w^n| \leq n|z - w|$, valid for $|z|, |w| \leq 1$ and obtained by telescoping $z^n - w^n = \sum_{j=0}^{n-1} z^j(z - w)w^{n-1-j}$, then yields

$$|\varphi_{S_n}(u) - w_n^n| \leq n|\varphi(u/\sqrt{n}) - w_n| = n \cdot o\left(\frac{u^2}{n}\right) \rightarrow 0,$$

the middle equality being (1.17) applied at $u/\sqrt{n} \rightarrow 0$. Since $w_n^n = (1 - u^2/2n)^n \rightarrow e^{-u^2/2}$, we conclude that $\varphi_{S_n}(u) \rightarrow e^{-u^2/2}$ for every $u \in \mathbb{R}$.

The limit is continuous at the origin, so Theorem 1.14 applies and S_n converges in distribution to the law with characteristic function $e^{-u^2/2}$, which is $\mathcal{N}(0, 1)$. ■

Theorem 1.19 (Cramér–Wold device). *Let X_n, X be $\mathbb{R}^d$ -valued random vectors. Then, the limit $X_n \xrightarrow{d} X$ holds if and only if $a^\top X_n \xrightarrow{d} a^\top X$ for every $a \in \mathbb{R}^d$.*

The proof is immediate from Theorem 1.14: the characteristic function of $a^\top X_n$ evaluated at 1 is $\varphi_{X_n}(a)$, so convergence of all one-dimensional projections is precisely pointwise convergence of φ_{X_n} , and the limit is continuous at the origin because it is a characteristic function. ■

Corollary 1.20 (Multivariate Central Limit Theorem). *Let $\{X_n\}_{n \geq 1}$ be i.i.d. $\mathbb{R}^d$ -valued random vectors with mean μ and finite covariance Σ . Then*

$$\frac{1}{\sqrt{n}} \sum_{k=1}^n (X_k - \mu) \xrightarrow{d} \mathcal{N}_d(0, \Sigma). \quad (1.18)$$

Proof: Fix $a \in \mathbb{R}^d$. The variables $a^\top X_k$ are i.i.d. real, with mean $a^\top \mu$ and variance $a^\top \Sigma a$, so the one-dimensional Central Limit Theorem 1.18 gives $n^{-1/2} \sum_k a^\top (X_k - \mu) \xrightarrow{d} \mathcal{N}(0, a^\top \Sigma a)$. By Example 1.16 that limit is the law of $a^\top Z$ for $Z \sim \mathcal{N}_d(0, \Sigma)$. Since a was arbitrary, Theorem 1.19 yields (1.18). The degenerate case $a^\top \Sigma a = 0$ is included, the limit then being 0. ■

For a multivariate normal vector — and, as Remark 1.17 showed, it is *joint* normality that is needed, normality of each marginal separately being insufficient — uncorrelatedness does imply independence. Indeed, if Σ is block-diagonal then (1.15) factorises as a product of the characteristic functions of the corresponding sub-vectors, which is exactly independence.

Example 1.21 (Log-Normal Distribution). A random variable X is said to follow a **log-normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted

$$X \sim \text{Log}\mathcal{N}(\mu, \sigma^2), \quad (1.19)$$

if the logarithm of X is normally distributed, i.e.,

$$\log X \sim \mathcal{N}(\mu, \sigma^2). \quad (1.20)$$

The probability density function of $X \sim \text{Log}\mathcal{N}(\mu, \sigma^2)$, after a straightforward change-of-variable calculation, is

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \quad x > 0. \quad (1.21)$$

The formula $\mathbb{E}[X^k] = \exp(k\mu + \frac{1}{2}k^2\sigma^2)$ holds for every $k \in \mathbb{R}$, not merely for $k \in \mathbb{N}$: it is obtained by writing $X^k = e^{kZ}$ with $Z \sim \mathcal{N}(\mu, \sigma^2)$ and evaluating the Gaussian moment generating function, which is finite for every real argument. Moments of *all* real orders, negative ones included, therefore exist and are finite. In particular $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathbb{V}[X] = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$. And yet the log-normal distribution has two unpleasant analytical features.

The first is that its characteristic function admits no closed-form expression, and that the *moment generating function* $\mathbb{E}[e^{uX}]$ is infinite for every $u > 0$. Despite having moments of all orders, **the log-normal distribution is not determined by them**: there is an explicit one-parameter family of mutually distinct densities on $(0, \infty)$ sharing every moment with it, exhibited by Heyde [44] and reproduced in Appendix A.3.

We present two more elementary processes.

Example 1.22 (Random walk and Poisson process). i) Let $\{\xi_n\}$ be i.i.d. with $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2}$. The **simple symmetric random walk** is $X_0 := 0$, $X_n := \sum_{k=1}^n \xi_k$: a discrete-time, integer-valued process with stationary and independent increments. Since $\varphi_\xi(u) = \cos(u)$, independence gives $\varphi_{X_n}(u) = \cos^n(u)$.

ii) Fix $\lambda > 0$. A process $N = \{N_t\}_{t \geq 0}$ with values in $\mathbb{N}_0$ is a **Poisson process with rate λ** if $N_0 = 0$ almost surely, N has independent increments, and

$$\mathbb{P}(N_{t+s} - N_s = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k \in \mathbb{N}_0, \quad s, t \geq 0. \quad (1.22)$$

Its paths are piecewise constant and non-decreasing, with unit jumps at random times; $\mathbb{E}[N_t] = \mathbb{V}[N_t] = \lambda t$ and $\varphi_{N_t}(u) = e^{\lambda t(e^{iu} - 1)}$.

1.1.3 Markov Processes

Markov processes are characterised by a specific form of *memorylessness*. They are particularly suited to describe the evolution of systems where the future depends only on the present state and not on the past history. Some of these processes arise as *diffusions* and in that case, a lot can be said about the distributions.



Definition 1.23 (Markov Process). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X = \{X_t\}_{t \in \mathbb{T}}$ be a stochastic process taking values in a measurable space $(E, \mathcal{E})$. We say that X is a **Markov process** if for every $s < t$ and every measurable set $A \in \mathcal{E}$,

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s) \quad \text{almost surely,} \quad (1.23)$$

where $\mathcal{F}_s = \sigma(X_r : r \leq s)$ is the σ -algebra generated by the history of the process up to time s . More generally, given a filtration $(\mathcal{F}_t)_{t \in [0, T]}$ to which X is adapted, X is said to be Markov *with respect to* $(\mathcal{F}_t)_{t \in [0, T]}$ if the same identity holds with $\mathcal{F}_s$ that filtration; this is the form asserted for Brownian motion in Proposition 1.36(d), and it is strictly stronger, since $(\mathcal{F}_t)_{t \in [0, T]}$ may carry more information than X alone.

The **transition probability** of a Markov process is $\mathbb{P}(X_t \in A \mid X_s = x)$; in order to integrate against the measure it defines we write $P_{s,x}(t, dy) = \mathbb{P}(X_t \in dy \mid X_s = x)$.

There is a rather sharp distinction between finite-state Markov processes and those with infinitely many states. In the finite case, the theory is built around transition matrices: in discrete time the probability of moving from one state to another in a single step is recorded in a matrix P_i , so that the evolution over n steps is the product $\prod_{i=1}^n P_i$. This is simplest in the *time-homogeneous* case, that is, when the transition probability does not depend on t . In continuous time the corresponding object is a matrix exponential rather than such a product.

In the infinite-state setting the dynamics are instead described by operators acting on spaces of functions or measures, so that the study of these processes relies on functional analysis and differential equations as much as on probability. That apparatus, the transition semigroup $(P_t)_{t \geq 0}$, its infinitesimal generator, and the Kolmogorov equations they satisfy is developed in Subsection 1.2.6, once diffusions are available to make it concrete.

1.1.4 Martingales, semimartingales and local martingales

A martingale is a process whose expected future value, conditional on everything observed so far, is its present value: a fair game. Martingales are central to the study of stochastic processes, and two weakenings of the notion are needed alongside it.

Definition 1.24 (Martingale). Let X_t be a stochastic process, adapted to a given filtration $\mathcal{F}_t$. We say it is a **martingale** if:

- $\mathbb{E}[|X_t|] < \infty$ for all t .
- $X_s = \mathbb{E}[X_t \mid \mathcal{F}_s]$ for $s < t$.

We say that X_t is a **supermartingale** if the second condition is replaced by $X_s \geq \mathbb{E}[X_t \mid \mathcal{F}_s]$ and a **submartingale** if $X_s \leq \mathbb{E}[X_t \mid \mathcal{F}_s]$.¹

The following result will be useful when constructing stochastic processes from Itô integrals. Notice that if X_t is a *square-integrable* martingale, then by Jensen's inequality (Proposition A.1, item 5) X_t^2 is a submartingale.

¹The Poisson process of Example 1.22 is not a martingale; the *compensated* process $N_t - \lambda t$ is.

Proposition 1.25 (Doob's submartingale inequality). *Let $Z = \{Z_t\}_{t \in [0, T]}$ be a right-continuous submartingale. Then, for any $\varepsilon > 0$,*

$$\mathbb{P}\left(\sup_{0 \leq t \leq T} Z_t \geq \varepsilon\right) \leq \frac{\mathbb{E}[\max\{Z_T, 0\}]}{\varepsilon}.$$

In particular, for a right-continuous martingale X , $\mathbb{P}(\sup_{0 \leq t \leq T} |X_t| \geq \varepsilon) \leq \mathbb{E}[|X_T|]/\varepsilon$.

Proof: The argument is given in Appendix B.1; see also [50, Thm. 4.5.1]. ■

The bound is in terms of the *terminal* value X_T , not of a running X_t ; this is the whole content of the inequality, since it converts information at a single time into control of the entire path. For the L^2 theory of the Itô integral we shall need the following stronger, L^p form, which is proved by integrating the tail estimate above.

Proposition 1.26 (Doob's L^p maximal inequality). *Let X be a right-continuous martingale and let $p > 1$. Then*

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} |X_t|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_T|^p].$$

In particular, for $p = 2$, $\mathbb{E}[\sup_{t \leq T} |X_t|^2] \leq 4 \mathbb{E}[|X_T|^2]$. □

The case $p = 2$ is the one we shall use repeatedly. It says that controlling the terminal second moment controls the second moment of the whole trajectory — which is exactly what is needed to upgrade convergence of the Itô integral in $L^2(\Omega)$, at each fixed time, to uniform convergence along a subsequence, and hence to continuity of sample paths (Theorem 1.54).

The martingale property is stated at deterministic times, but localisation in Section 1.2 involves evaluating processes at *random* times. For this to make sense we must first say what information is available at a stopping time.

Definition 1.27 (The σ -algebra $\mathcal{F}_\tau$). Let τ be a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$. The σ -algebra of events observable up to τ is

$$\mathcal{F}_\tau := \left\{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t \in [0, T]\right\}. \quad (1.24)$$

The definition says something simple: an event belongs to $\mathcal{F}_\tau$ if, whenever τ has occurred by time t , an observer at time t can already tell whether the event happened. One checks readily that $\mathcal{F}_\tau$ is indeed a σ -algebra, that τ is $\mathcal{F}_\tau$ -measurable, that $\mathcal{F}_\tau = \mathcal{F}_t$ when $\tau \equiv t$ is constant, and that $\varsigma \leq \tau$ implies $\mathcal{F}_\varsigma \subseteq \mathcal{F}_\tau$ for a second stopping time ς . If X is right-continuous and adapted, then X_τ is $\mathcal{F}_\tau$ -measurable on $\{\tau < \infty\}$.

Theorem 1.28 (Doob's optional stopping theorem). *Let $X = \{X_t\}_{t \geq 0}$ be a right-continuous martingale with respect to a filtration $(\mathcal{F}_t)_{t \geq 0}$, and let $\tau_1 \leq \tau_2$ be stopping times with values in $[0, \infty]$. If any one of the following holds*

- i) τ_2 is bounded, i.e. there is $c < \infty$ with $\tau_2 \leq c$ almost surely;*
- ii) X is uniformly integrable (Definition A.9); or*
- iii) there exists $Y \in L^1(\Omega)$ with $|X_t| \leq Y$ for all t ,*



then X_{τ_1} and X_{τ_2} are integrable and

$$\mathbb{E}[X_{\tau_2} \mid \mathcal{F}_{\tau_1}] = X_{\tau_1} \quad \text{almost surely.} \quad (1.25)$$

In particular, taking $\tau_1 \equiv 0$, $\mathbb{E}[X_{\tau_2}] = \mathbb{E}[X_0]$. The corresponding statements with $=$ replaced by $\geq$ or $\leq$ hold for sub- and supermartingales respectively. $\square$

Equation (1.25) is the assertion that a martingale remains a martingale when observed along a random clock. One word first on how it is to be read, since the stopping times are allowed the value $+\infty$ and X_∞ has not been defined. Under i) the question does not arise, both times being bounded by c . Under ii) or iii) the martingale convergence theorem supplies a random variable $X_\infty \in L^1(\Omega)$ with $X_t \rightarrow X_\infty$ almost surely and in L^1 , and X_{τ_i} is to be read as X_∞ on $\{\tau_i = \infty\}$; with that reading the statement is exact, and it is the reading intended throughout.

Two remarks on the hypotheses before we use the theorem. They are not independent: iii) implies ii), since a family dominated by a single integrable variable is uniformly integrable, so iii) is recorded only because it is the form in which the hypothesis is usually easiest to check. More importantly, under the finite-horizon convention of Definition 1.9 — which is in force everywhere else in this dissertation, and under which every stopping time satisfies $\tau_2 \leq T$ — hypothesis i) holds automatically, and the theorem applies unconditionally. The hypotheses therefore do work only on an infinite horizon.

The notion of a martingale is often too restrictive, particularly when considering processes such as stochastic exponentials or certain diffusion models. A relaxation is therefore introduced.

Definition 1.29 (Local Martingale). A process $X = \{X_t\}_{t \in [0, T]}$ is a **local martingale** if there exists an increasing sequence of stopping times $(\tau_n)_{n \geq 1}$ with $\tau_n \uparrow T$ almost surely and $\mathbb{P}(\tau_n = T) \rightarrow 1$, such that each stopped process $X_t^{\tau_n} := X_{t \wedge \tau_n}$ is a martingale. The second requirement is what pins down X_T : without it the τ_n could all stay strictly below T and the terminal value would be unconstrained. On an infinite horizon the condition reads $\tau_n \uparrow \infty$; we shall use the finite-horizon form, which is the one that appears in the proof of Theorem 1.65.

Before introducing the last class of processes, we record two ways of measuring how much a function moves. Both are needed to state the definition that follows, and both will be with us for the rest of the chapter.

Definition 1.30 (Total variation). The **total variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_1(f)_t := \sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|, \quad (1.26)$$

the supremum being over all partitions $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$ of $[0, t]$. If $V_1(f)_t < \infty$ we say f has **bounded variation** on $[0, t]$.

Bounded variation is exactly the condition under which the Riemann–Stieltjes integral $\int_0^t g df$ is defined for every continuous g ; equivalently, it is the condition under which f induces a signed measure df . It is therefore the natural first thing to ask of a prospective integrator.

Definition 1.31 (Quadratic Variation). Let $f : [0, T] \rightarrow \mathbb{R}$ and let $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = t\}$ be a partition of $[0, t]$. Write

$$\|\mathcal{P}\| = \max_{1 \leq k \leq n} (t_k - t_{k-1}). \quad (1.27)$$

If the limit

$$[f]_t := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{k=1}^n \left(f(t_k) - f(t_{k-1}) \right)^2 \quad (1.28)$$

exists, it is called the **quadratic variation** of f on $[0, t]$. Similarly, for $g : [0, T] \rightarrow \mathbb{R}$ we define the **covariation** of f and g on $[0, t]$ as

$$[f, g]_t := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{k=1}^n \left(f(t_k) - f(t_{k-1}) \right) \left(g(t_k) - g(t_{k-1}) \right) \quad (1.29)$$

provided this limit exists, so that $[f]_t = [f, f]_t$.

Two cautions are in order. First, the limit in (1.28) need not exist for an arbitrary f . Second, for a *stochastic* process the sums in (1.28) are random variables, and the limit must be specified as a limit of random variables. We shall always understand it in $L^2(\Omega)$, which for our purposes is both the easiest to obtain and strong enough to imply convergence in probability, and hence almost sure convergence along a subsequence.

If f is continuously differentiable, then $|f(t_k) - f(t_{k-1})| \leq C(t_k - t_{k-1})$ and the sum is bounded by $C^2 \|\mathcal{P}\| t \rightarrow 0$: smooth functions have zero quadratic variation. The conclusion holds more generally for any continuous f of bounded variation: bounding one factor of each square by the largest increment gives $\sum_k (f(t_k) - f(t_{k-1}))^2 \leq \left(\max_k |f(t_k) - f(t_{k-1})| \right) V_1(f)_t$, and uniform continuity on $[0, t]$ sends the maximum to 0 while $V_1(f)_t$ stays finite. A non-zero quadratic variation is therefore a certificate of irregularity, and Proposition 1.36(e) below will assert that Brownian motion carries exactly such a certificate. Quadratic variation acts as a measure of the “roughness” of the path.

There is a second notion of quadratic variation, obtained not by refining partitions but by subtracting from M^2 its martingale part, provided the following classical decomposition. Writing $\mathbb{L}$ for the adapted processes with almost all sample paths left-continuous, the **predictable** σ -algebra is the smallest σ -algebra on $[0, T] \times \Omega$ making every element of $\mathbb{L}$ measurable, and a process is predictable if it is measurable with respect to it.

Theorem 1.32 (Doob–Meyer decomposition). *Let $M = \{M_t\}_{t \in [0, T]}$ be a càdlàg (continue à droite, limite à gauche), square-integrable martingale. Then M^2 admits a unique decomposition*

$$M_t^2 = R_t + \langle M \rangle_t, \quad (1.30)$$

in which R is a càdlàg martingale and $\langle M \rangle$ is predictable, right-continuous and increasing, with $\langle M \rangle_0 = 0$ and $\mathbb{E}[\langle M \rangle_t] < \infty$ for every $t \in [0, T]$. $\square$

The process $\langle M \rangle$ is called the **compensator** of M^2 , or the *predictable quadratic variation* of M . For a proof see [50] or [70].

Predictability is what makes the decomposition *unique*. When the integrator has continuous sample paths, predictable and merely adapted integrands give rise to the same stochastic integrals, which is why the construction of Subsection 1.2.1 will be able to proceed without the notion. The distinction becomes essential the moment the integrator is allowed to jump: if the integrand may “see” a jump as it happens, one can bet on the jump at the instant it occurs, and the resulting integral fails to be a martingale. Predictability forbids exactly this, and its financial reading is that one may not trade on information one does not yet have.

Two notions of quadratic variation are now in play: the pathwise $[M]$ of Definition 1.31, defined by refining partitions, and the predictable $\langle M \rangle$ just obtained, defined by a decomposition. For the processes of this dissertation they agree.



Proposition 1.33. *Let M be a continuous square-integrable martingale. Then the pathwise quadratic variation $[M]_t$ of Definition 1.31 exists as a limit in probability, uniformly on compact time intervals, and coincides, up to indistinguishability, with the compensator $\langle M \rangle_t$ of Theorem 1.32. The limit is not in general an L^2 one: taking $M_t = ZW_t$ with Z independent of W , $\mathcal{F}_0$ -measurable and in $L^2 \setminus L^4$ gives a continuous square-integrable martingale whose $[M]_t = Z^2 t$ is not square-integrable. Furthermore, the polarised quantity*

$$\langle M, N \rangle_t := \frac{1}{2} (\langle M + N \rangle_t - \langle M \rangle_t - \langle N \rangle_t) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_k (M_{t_k} - M_{t_{k-1}}) (N_{t_k} - N_{t_{k-1}}) \quad (1.31)$$

*defines the **quadratic covariation** of two continuous square-integrable martingales, is bilinear and symmetric, and satisfies the Kunita–Watanabe inequality*

$$\int_0^t |d\langle M, N \rangle_s| \leq \sqrt{\langle M \rangle_t} \sqrt{\langle N \rangle_t} \quad \text{almost surely.} \quad (1.32)$$

□

The distinction between $[M]$ and $\langle M \rangle$ is genuine for processes with jumps, where the two differ by the accumulated squared jumps.² Quadratic covariation is what gives meaning to the off-diagonal entries $\rho_{ij} dt$ of the multiplication table in Subsection 1.2.4, and it is the object that the correlation parameter of a stochastic volatility model actually controls.

Finally, there is another related concept, broad enough to serve as a class of *good integrators*. In particular, it extends the role played by Brownian motion as a canonical integrator, while retaining enough structure to allow a general theory of stochastic calculus.

Definition 1.34 (Semimartingale). A process X is called a **semimartingale** if it can be decomposed as

$$X_t = M_t + A_t,$$

where M_t is a càdlàg local martingale and A_t is adapted, càdlàg, of locally bounded variation, with $M_0 = X_0$ and $A_0 = 0$.

This last definition is the correct level of generality for stochastic integration. The two terms are exactly the two mechanisms by which an integrator can be usable: A has bounded variation, so $\int g dA$ exists pathwise as a Lebesgue–Stieltjes integral; and M is a local martingale, so $\int g dM$ exists in the Itô sense.

Granting it, a semimartingale is precisely a process that can be integrated against by splitting the work between deterministic analysis and probability, and the Bichteler–Dellacherie theorem makes the converse precise: semimartingales are, in a suitable topological sense, the *only* processes admitting a reasonable theory of stochastic integration; see [65].

1.1.5 Brownian Motion

Brownian motion is a fundamental stochastic process whose importance has grown throughout many areas of mathematics. It is rich enough to generate a new class of *stochastic differential equations* and to address numerous applied problems involving randomness, yet simple enough to allow explicit calculations. Its properties also enable efficient simulation, making it a cornerstone of Monte Carlo methods across disciplines.

²For the compensated Poisson process $M_t = N_t - \lambda t$ one has $[M]_t = N_t$, which is random, while $\langle M \rangle_t = \lambda t$ is deterministic: every jump of N has size 1, so each contributes 1 to $[M]$.

Definition 1.35 (Brownian Motion). A stochastic process $W : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}$ is called a **Brownian motion** (or **Wiener process**) if the following conditions hold:

i)

$$\mathbb{P}(W_0 = 0) = 1. \quad (1.33)$$

ii)

$$\mathbb{P}(\{\omega : t \mapsto W_t(\omega) \text{ is continuous}\}) = 1. \quad (1.34)$$

iii) For $0 \leq s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t - s) \quad (1.35)$$

so that $W_t \sim \mathcal{N}(0, t)$.

iv) The process has independent increments: for $0 \leq t_1 < t_2 < \dots < t_n$, the random variables

$$\{W_{t_1}, W_{t_2} - W_{t_1}, W_{t_3} - W_{t_2}, \dots, W_{t_n} - W_{t_{n-1}}\} \quad (1.36)$$

are independent.

Proposition 1.36. *Let $(W_t)_{t \geq 0}$ be a Brownian motion. Then:*

a) For all $s, t \geq 0$,

$$\mathbb{E}[W_s W_t] = \min\{s, t\}. \quad (1.37)$$

b) For $t_0 \geq 0$, the process

$$Y(t) = W(t + t_0) - W(t_0) \quad (1.38)$$

is also a Brownian motion.

c) For any $c > 0$, the process

$$Z(t) = \frac{1}{\sqrt{c}} W(ct) \quad (1.39)$$

is also a Brownian motion.

d) With respect to its own natural filtration $(\mathcal{F}_t^W)_{t \in [0, T]}$, Brownian motion is both a Markov process and a martingale.

e) The quadratic variation of W over $[0, t]$ exists and equals

$$[W]_t = t \quad \text{the limit being taken in } L^2(\Omega). \quad (1.40)$$

f) The process $W_t^2 - t$ is a martingale.

Properties b) and c) are called translation invariance and scaling invariance, respectively.

Proof: a) By i) we have $W_t = W_t - W_0$ almost surely, so $W_t \sim \mathcal{N}(0, t)$ by iii). For $s < t$, write $W_t = (W_t - W_s) + W_s$ and use that $W_t - W_s$ is independent of W_s with zero mean:

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s (W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s = \min\{s, t\}.$$

b) $Y_0 = 0$ and the paths of Y are continuous, since those of W are. For $s < t$ we have $Y_t - Y_s = W_{t+t_0} - W_{s+t_0} \sim \mathcal{N}(0, t-s)$, and increments of Y over disjoint intervals are increments of W over disjoint intervals, hence independent.



c) Again $Z_0 = 0$ and the paths are continuous. For $s < t$, $Z_t - Z_s = c^{-1/2}(W_{ct} - W_{cs})$ is normal with mean 0 and variance $c^{-1}(ct - cs) = t - s$; independence of increments is inherited from W because $s \mapsto cs$ preserves disjointness of intervals.

d) Write $\mathcal{F}_s := \mathcal{F}_s^W = \sigma(W_u : u \leq s)$. One step has to be taken first, since condition iv) of Definition 1.35 asserts independence of the increments *from one another*, and what is needed here is independence of $W_t - W_s$ from the whole of $\mathcal{F}_s^W$. The two are the same statement: the sets $\{W_{u_1} \in A_1, \dots, W_{u_k} \in A_k\}$ with $u_1 < \dots < u_k \leq s$ form a π -system generating $\mathcal{F}_s^W$, each such set is described by the increments up to time s , and independence verified on a generating π -system extends to the σ -algebra it generates. It is exactly this step that makes every Brownian motion a Brownian motion for its own natural filtration, as recorded after Definition 1.41 below.

Granting it, integrability is clear from a). For $s < t$, using independence of $W_t - W_s$ from $\mathcal{F}_s$ and $\mathcal{F}_s$ -measurability of W_s ,

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] + \mathbb{E}[W_s \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s] + W_s = W_s,$$

so W is a martingale. The same independence, applied this time to $\mathbb{P}(W_t \in A \mid \mathcal{F}_s) = \mathbb{P}(W_s + (W_t - W_s) \in A \mid \mathcal{F}_s)$, yields $\mathbb{P}(W_t \in A \mid \mathcal{F}_s) = g(W_s)$ with $g(x) = \mathbb{P}(x + \mathcal{N}(0, t-s) \in A)$: the conditional law of W_t given $\mathcal{F}_s$ depends on ω only through W_s , which is the Markov property.

e) Let $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$ and set $X_k = (W_{t_k} - W_{t_{k-1}})^2 - (t_k - t_{k-1})$. The X_k are independent with $\mathbb{E}[X_k] = 0$, and since $\mathbb{E}[Z^4] = 3\sigma^4$ for $Z \sim \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}[X_k^2] = 3(t_k - t_{k-1})^2 - 2(t_k - t_{k-1})^2 + (t_k - t_{k-1})^2 = 2(t_k - t_{k-1})^2.$$

Hence, by independence and orthogonality,

$$\mathbb{E}\left[\left(\sum_k (W_{t_k} - W_{t_{k-1}})^2 - t\right)^2\right] = \sum_k \mathbb{E}[X_k^2] = 2 \sum_k (t_k - t_{k-1})^2 \leq 2\|\mathcal{P}\| t \longrightarrow 0$$

as $\|\mathcal{P}\| \rightarrow 0$, which is (1.40). See also [50, Thm. 4.1.2].

f) Integrability follows from a). For $s < t$, expanding $W_t^2 = (W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2$ and conditioning,

$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = (t - s) + 2W_s \cdot 0 + W_s^2,$$

so $\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$. ■

One might wonder whether such a process actually exists, given the strong set of requirements in Definition 1.35. The construction proceeds in two stages. The first one produces a process with the right finite-dimensional distributions and says nothing whatever about its paths; the second repairs the paths without disturbing the distributions.

The first step is the Daniell–Kolmogorov extension theorem. Conditions i), iii) and iv) of Definition 1.35 prescribe, for every finite set of times $t_1 < \dots < t_n$, the joint law of $(W_{t_1}, \dots, W_{t_n})$: it is the centred Gaussian law on $\mathbb{R}^n$ with covariance $\min\{t_i, t_j\}$, which is positive semi-definite and therefore legitimate. These laws are *consistent*, in the sense that the law prescribed for a larger set of times marginalises to the law prescribed for a smaller one. The extension theorem then asserts that any consistent family of finite-dimensional distributions is realised by some process on the product space $\mathbb{R}^{[0, \infty)}$; see [48, Thm. 2.2.2] or [27, Ch. 8].

What this stage cannot deliver is condition ii). The set of continuous paths is not measurable in the product σ -algebra since membership in it depends on uncountably many coordinates at once, so continuity is invisible to the finite-dimensional data and must be obtained separately. That is the second stage, and it does so from a moment condition alone.

Theorem 1.37 (Kolmogorov's continuity criterion). *Let $X = \{X_t\}_{t \in [0, T]}$ be a stochastic process for which there exist constants $\alpha, \beta, C > 0$ with*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C |t - s|^{1+\beta}, \quad s, t \in [0, T]. \quad (1.41)$$

Then X admits a continuous modification $\tilde{X}$. Moreover, the paths of $\tilde{X}$ are almost surely locally γ -Hölder continuous for every $\gamma \in (0, \beta/\alpha)$; see [48, Thm. 2.2.8]. $\square$

Applied to Brownian motion the criterion is immediate. Since $W_t - W_s \sim \mathcal{N}(0, t - s)$, the Gaussian moments give $\mathbb{E}[|W_t - W_s|^{2m}] = c_m |t - s|^m$ for every $m \in \mathbb{N}$, so (1.41) holds with $\alpha = 2m$ and $\beta = m - 1$. A continuous modification therefore exists, which is what was needed; and the Hölder conclusion yields exponents $\gamma < (m - 1)/2m$, which as $m \rightarrow \infty$ approach $1/2$ from below but never reach it. That the supremum is exactly $1/2$, and the fact that it is not attained, is the subject of the next subsection.

Remark 1.38 (Modification versus indistinguishability). What Theorem 1.37 produces is a process agreeing with X at each fixed time almost surely, not one agreeing with X along whole paths. Two processes X and Y on the same space are **modifications** of each other if $\mathbb{P}(X_t = Y_t) = 1$ for every fixed t , and **indistinguishable** if $\mathbb{P}(X_t = Y_t \text{ for all } t) = 1$. The second is strictly stronger: in the first the exceptional null set is allowed to depend on t , and an uncountable union of null sets need not be null. The two notions do coincide when both processes have right-continuous paths, which is the situation we shall almost always be in.

Now, Daniell–Kolmogorov supplies a process whose finite-dimensional distributions satisfy conditions i), iii) and iv). Kolmogorov's criterion replaces it by a modification with continuous paths, which by Remark 1.38 carries the same finite-dimensional distributions and therefore retains i), iii) and iv) while acquiring ii). Since the criterion is stated on a compact interval, one applies it on each $[0, n]$ and patches, the modifications so obtained agreeing almost surely on overlaps.

Turning back to Example 1.22 on random walk, the following result shows how a piecewise linear interpolation of a rescaled random walk converges to Brownian motion.

Theorem 1.39 (Donsker's invariance principle). *Let $\{\xi_k\}_{k \geq 1}$ be i.i.d. with $\mathbb{E}[\xi_1] = 0$ and $\mathbb{V}[\xi_1] = 1$, and let $X_n = \sum_{k=1}^n \xi_k$ be the associated random walk. Define the rescaled, linearly interpolated process*

$$W_t^{(n)} := \frac{1}{\sqrt{n}} \left(X_{[nt]} + (nt - [nt]) \xi_{[nt]+1} \right), \quad t \in [0, T]. \quad (1.42)$$

Then $W^{(n)} \xrightarrow{d} W$ as $n \rightarrow \infty$, where the convergence is in distribution on the space $C[0, T]$ of continuous functions with the uniform topology, and W is a Brownian motion. $\square$

This gives a constructive route to Brownian motion. The scaling in (1.42) is the whole content of the theorem: space is scaled down by $\sqrt{n}$ while time is sped up by n . That is exactly the relation $\text{space} \sim \sqrt{\text{time}}$ which Definition 1.35 imposes through $W_t - W_s \sim \mathcal{N}(0, t - s)$, and which reappears as the scaling invariance of Proposition 1.36(c). The theorem is the path-space analogue of the Central Limit Theorem, and it explains why Brownian motion is ubiquitous in the same sense that the Gaussian distribution is.

Proposition 1.40. *Let $S(t)$, $t \geq 0$, be a process adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$ such that for any $0 \leq s < t$, the increment $S(t) - S(s)$ is independent of $\mathcal{F}_s$. Then $S(t)$ has independent increments.*



Proof: Let $0 \leq t_1 < t_2 < \dots < t_n$ and write $\Delta_k = S(t_k) - S(t_{k-1})$. We show by induction on n that the vector $(\Delta_1, \dots, \Delta_n)$ has independent components. For $n = 1$ there is nothing to prove. Assume the claim for $n - 1$. Each of $\Delta_1, \dots, \Delta_{n-1}$ is $\mathcal{F}_{t_{n-1}}$ -measurable, since S is adapted and $t_k \leq t_{n-1}$ for $k \leq n - 1$; hence so is any Borel function of them. By hypothesis Δ_n is independent of $\mathcal{F}_{t_{n-1}}$, and therefore independent of $(\Delta_1, \dots, \Delta_{n-1})$. For bounded Borel $g_1, \dots, g_n$ this gives

$$\mathbb{E}\left[\prod_{k=1}^n g_k(\Delta_k)\right] = \mathbb{E}[g_n(\Delta_n)] \mathbb{E}\left[\prod_{k=1}^{n-1} g_k(\Delta_k)\right] = \prod_{k=1}^n \mathbb{E}[g_k(\Delta_k)],$$

the last equality by the induction hypothesis. Since this holds for all such g_k , the increments are independent. $\blacksquare$

Definition 1.41 (Brownian motion with respect to a filtration). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space. A process W is a $(\mathcal{F}_t)_{t \in [0, T]}$ -**Brownian motion** if

- i) W is adapted to $(\mathcal{F}_t)_{t \in [0, T]}$, with $W_0 = 0$ almost surely and almost all paths continuous; and
- ii) for all $s \leq t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and has law $\mathcal{N}(0, t - s)$.

By Proposition 1.40, condition ii) already forces the independence of increments demanded in Definition 1.35 iv), so a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion is in particular a Brownian motion; the content of the definition is the *relative* independence, which iv) alone does not give. Conversely, any Brownian motion is a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion for its own natural filtration $(\mathcal{F}_t^W)_{t \in [0, T]}$, augmented as in Convention 1.7, and this is the filtration we use unless another is named. The distinction is not idle: Girsanov's theorem (Theorem 1.81) changes the measure while keeping the filtration fixed, and its conclusion is exactly that a certain process is a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion under the new measure.

Theorem 1.42 (Strong Markov property). *Let W be a Brownian motion and τ an almost surely finite stopping time. Then the shifted process $\widetilde{W}_s := W_{\tau+s} - W_\tau$, $s \geq 0$, is a Brownian motion independent of $\mathcal{F}_\tau$ (Definition 1.27).*

Proof: See [48, Thm. 2.6.16]. $\blacksquare$

The statement genuinely strengthens Proposition 1.36(d), where the Markov property was asserted at deterministic times only. Fix $a > 0$ and let $\tau_a = \inf\{t \geq 0 : W_t = a\}$. The *reflection principle* is the observation that, on the event $\{\tau_a \leq t\}$, the post- τ_a increments form a Brownian motion independent of $\mathcal{F}_{\tau_a}$.

Theorem 1.43 (Reflection principle). *Let W be a Brownian motion. Let $\overline{W}_t := \max_{0 \leq s \leq t} W_s$ be its **running maximum**. Then for every $a \geq 0$,*

$$\mathbb{P}(\overline{W}_t \geq a) = \mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(W_t > a) = \mathbb{P}(|W_t| > a), \quad (1.43)$$

so that $\overline{W}_t$ and $|W_t|$ have the same law. Moreover the pair $(W_t, \overline{W}_t)$ has joint density

$$f_{(W_t, \overline{W}_t)}(x, m) = \frac{2(2m - x)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2m - x)^2}{2t}\right), \quad m \geq \max\{x, 0\}. \quad (1.44)$$

Proof: (Sketch.) The first chain of equalities is the argument just given: $\{\bar{W}_t \geq a\} = \{\tau_a \leq t\}$ up to a null set, since a continuous path attaining the level a exceeds it immediately afterwards except on the event $\{\bar{W}_t = a\}$, which has probability zero; and conditionally on $\tau_a \leq t$ the reflected path is as likely to finish above a as below, so $\mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(W_t > a)$, again $W_t = a$ having probability zero. For (1.44), the same reflection applies so $\mathbb{P}(W_t \leq x, \bar{W}_t \geq m) = \mathbb{P}(W_t \geq 2m - x)$ for $m \geq \max\{x, 0\}$, and differentiates twice; see [48, Sec. 2.6] or [70]. ■

The construction is shown in Figure 1.1: up to τ_a the two paths coincide, and after it the reflected one is the mirror image of the original in the level a , turning the count of paths finishing above a into the count of those that ever reached it. Note that $\mathbb{E}[\bar{W}_t] = \mathbb{E}[W_t] = \sqrt{2t/\pi}$.

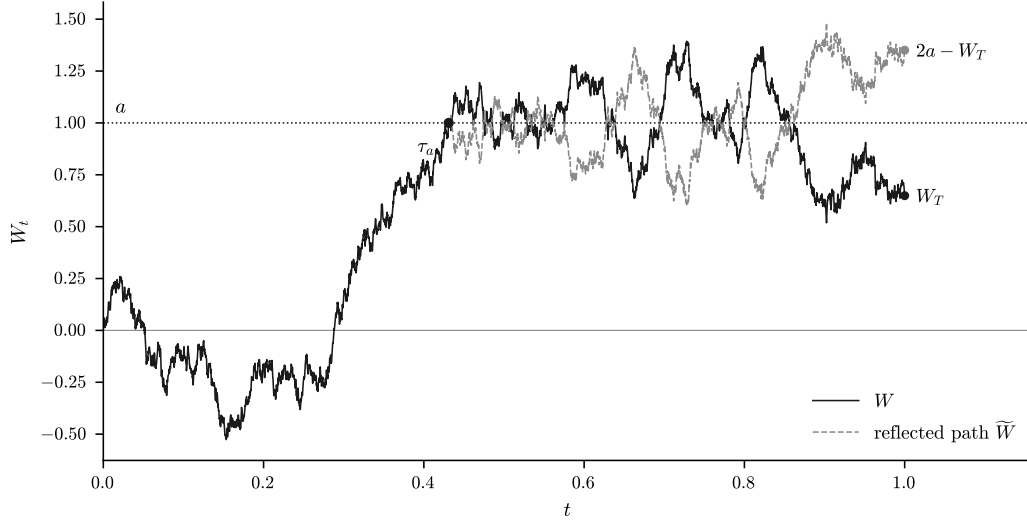


Figure 1.1: The reflection principle. A Brownian path (solid) reaching the level a for the first time at τ_a , together with the path obtained by reflecting it in that level after τ_a (dashed). The two have the same law, they agree on $[0, \tau_a]$, and their terminal values W_T and $2a - W_T$ lie symmetrically about a . Theorem 1.43 is the statement that this pairing is measure-preserving.

1.1.6 Path regularity

We now examine how irregular the trajectories of Brownian motion actually are. This subsection is short, and every statement in it is classical, but it is the most consequential material in the chapter: the construction of the Itô integral in Section 1.2 is a *response* to what follows, and the theory developed in Part II is a response to what happens when these estimates are pushed slightly further.

We begin with the observation that the two notions of variation introduced in Subsection 1.1.4, the quadratic variation of Definition 1.31 and the total variation of Definition 1.30, are in direct competition.

Proposition 1.44. *Let W be a Brownian motion. Then, almost surely, $V_1(W)_t = \infty$ for every $t > 0$.*

Proof: Fix $t > 0$ and a sequence of partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| \rightarrow 0$. On any partition,

$$\sum_k (W_{t_k} - W_{t_{k-1}})^2 \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) \cdot \sum_k |W_{t_k} - W_{t_{k-1}}| \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) V_1(W)_t. \quad (1.45)$$



Suppose, for contradiction, that $V_1(W)_t < \infty$ on an event of positive probability. Sample paths of W are continuous, hence uniformly continuous on the compact $[0, t]$, so the maximal increment on the right of (1.45) tends to 0 as $\|\mathcal{P}_n\| \rightarrow 0$. The whole right-hand side therefore tends to 0, forcing $[W]_t = 0$ on that event. This contradicts (1.40), which gives $[W]_t = t > 0$ in $L^2(\Omega)$ and hence almost surely along a subsequence. ■

This result is key because it shows that **the integral $\int_0^t f dW$ cannot be defined pathwise as a Riemann–Stieltjes integral** for integrands as irregular as W itself — which is the case that matters, the integrand being typically a function of the driving path — because W does not induce a measure on $[0, t]$ along any path. Figure 1.2 shows the two sums of (1.45) pulling apart as the mesh is refined, which is the whole of Proposition 1.44 in one picture.

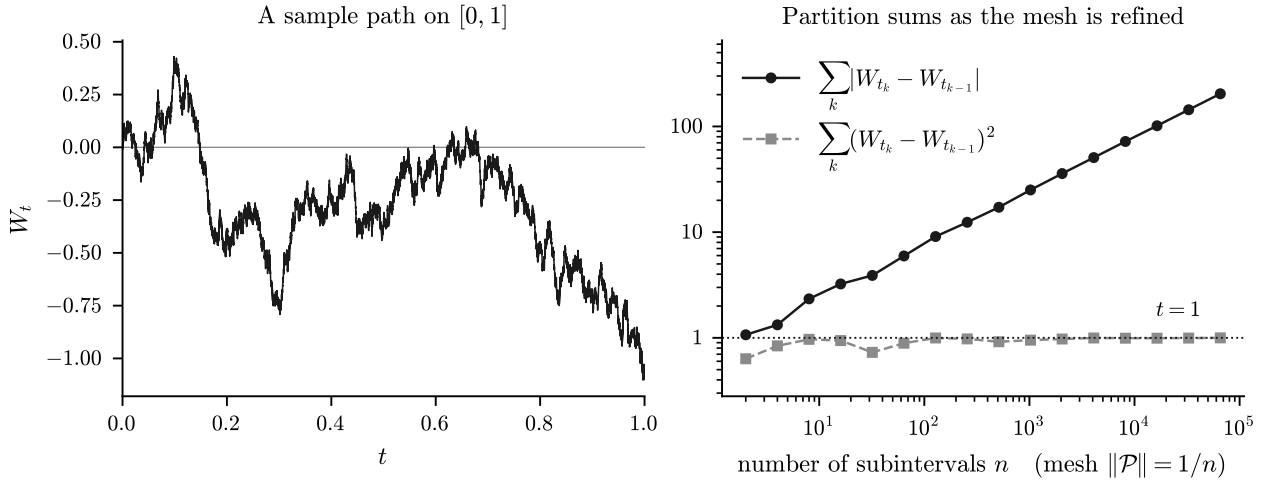


Figure 1.2: A Brownian path on $[0, 1]$ (left) and its partition sums (right, logarithmic on both axes) over the uniform partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| = 1/n$. The sum of absolute increments grows without bound, with slope $\frac{1}{2}$ on these axes, so that it diverges like $\sqrt{n}$ — while the sum of squared increments settles at $t = 1$. This is Proposition 1.44 and (1.40) side by side: the same path has infinite total variation and quadratic variation exactly t , and it is this pair of facts that makes the Riemann–Stieltjes integral unavailable and the Itô integral necessary.

The same competition can be expressed more finely by asking, not whether the sum of increments converges, but for which exponent p the sum of p -th powers does.

Definition 1.45 (p -variation). Let $p \geq 1$. The p -**variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_p(f)_t := \left(\sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|^p \right)^{1/p}, \quad (1.46)$$

and f is said to have *finite p -variation* on $[0, t]$ if $V_p(f)_t < \infty$. We write $f \in \mathcal{C}^{p\text{-var}}([0, t])$.

For $p = 1$ this is Definition 1.30. For $p = 2$ it is *not* the quadratic variation of Definition 1.31, and the difference is not cosmetic: the latter is a limit along partitions of vanishing mesh, the former a supremum over all partitions, and for Brownian motion the first equals t while the second is infinite (Theorem 1.46 ii)). Refining a partition need not increase the sum of squared increments — splitting an increment into two of the same sign strictly decreases it, since $(a + b)^2 > a^2 + b^2$ there — and the supremum is attained instead along coarse partitions chosen to follow the path. If f is continuous and $V_p(f)_t < \infty$, then $V_q(f)_t < \infty$ for every $q > p$, since $|f(t_k) - f(t_{k-1})|^q \leq (\max_k |f(t_k) - f(t_{k-1})|)^{q-p} |f(t_k) - f(t_{k-1})|^p$. Each function thus has a critical exponent below which its p -variation is infinite and above which it is finite, and this exponent is a precise measure of roughness.

Theorem 1.46 (Regularity of Brownian paths). *Let W be a Brownian motion. Then, almost surely:*

- i) *the sample path $t \mapsto W_t$ is locally γ -Hölder continuous for every $\gamma < \frac{1}{2}$, and is locally γ -Hölder for no $\gamma \geq \frac{1}{2}$;*
- ii) *$V_p(W)_t < \infty$ for every $p > 2$ and every t , whereas $V_p(W)_t = \infty$ for every $p \leq 2$;*
- iii) *the sample path is nowhere differentiable: for almost every ω , there is no t at which $t \mapsto W_t(\omega)$ possesses a finite derivative.* $\square$

Part i) was half-proved above via Kolmogorov's criterion; the negative half, like iii), is due to Paley, Wiener and Zygmund, and proofs of all three may be found in [31] or [48]. Part iii) is worth a moment's reflection. A continuous function which is differentiable nowhere is already a celebrated pathology in real analysis, exhibited with some effort by Weierstrass; the theorem says that if one picks a Brownian path at random one obtains such a function with probability one. The pathological case is the meaningful case for integration.

1.2 Stochastic Calculus

Stochastic calculus is a broad area of mathematics which allows, among other things, randomness and uncertainty to be incorporated into differential equations. One of our main aims is to give meaning to expressions of the kind

$$dX_t = a(t, X_t) dt + b(t, X_t) dW_t, \quad X_0 \in \mathbb{R}. \quad (1.47)$$

Here dW_t will be specified in the next subsection. Notice that this generalises the deterministic case: if $b \equiv 0$, then X_t reduces to the solution of the ODE $dX_t = a(t, X_t) dt$. What is being modelled is still a trajectory — a *position* as a function of time — but a random one, drawn from a whole family of possible paths: the evolution of the underlying probability distribution.

We shall first construct a rigorous integral interpretation of such expressions, providing concrete meaning to the stochastic term. Afterwards, we will introduce the stochastic chain rule (Itô's Lemma), which differs from the classical one in a somewhat surprising yet elegant way, and then proceed to study the existence and uniqueness of solutions and some of their properties.

The exposition could be carried out in the full generality anticipated by the semimartingales of Definition 1.34. We keep instead to the classical presentation, integrating against Brownian motion alone; what the general case would require was recorded after Definition 1.34 itself. The aim is to reach the results we actually need without losing sight of them along the way.

1.2.1 The Itô Integral

The stochastic differential equation (1.47) can be reformulated in integral form as

$$X_t - X_0 = \int_0^t a(s, X_s) ds + \int_0^t b(s, X_s) dW_s. \quad (1.48)$$

For each $\omega \in \Omega$, the first integral on the right-hand side is nothing new. Regarding the second one, we may ask whether there is a sensible way of defining

$$M_t = \int_0^t f(s, \omega) dW_s, \quad 0 \leq t \leq T,$$



so that $\{M_t\}_{t \in [0, T]}$ is a martingale. Another important consideration is that the integrand must be *adapted* to the filtration associated with W_t . These requirements naturally lead to the following definitions.

Definition 1.47 (Adapted square-integrable processes). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space.

- i) The space of stochastic processes on $[0, T]$, adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ and such that

$$\int_0^T \mathbb{E}[|f(t)|^2] dt < \infty,$$

will be denoted by $L_{\text{ad}}^2([0, T] \times \Omega)$. This is a Hilbert space with the inner product

$$(f, g)_{\text{ad}} = \int_0^T \mathbb{E}[f(t)g(t)] dt,$$

written with round brackets to keep it apart from the quadratic covariation $\langle M, N \rangle$ of Proposition 1.33.

- ii) We say $f \in L_{\text{ad}}^2([0, T] \times \Omega)$ is a **step stochastic process** if

$$f(t, \omega) = \sum_{i=1}^n \psi(t_{i-1}, \omega) \mathbb{1}_{[t_{i-1}, t_i)}(t),$$

where $\psi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and has finite variance ($\mathbb{E}[\psi(t_{i-1})^2] < \infty$).

The construction of the stochastic integral begins by defining it on step processes and extending by density. For such f , take $t_0 = 0$, $t_n = T$, and define³

$$I(f) = \int_0^T f(t) dW_t := \sum_{i=1}^n \psi(t_{i-1}) (W(t_i) - W(t_{i-1})). \quad (1.49)$$

This mapping $f \mapsto I(f)$ is linear. The following lemmas are key.

Lemma 1.48. *If f is a step stochastic process, then*

$$\mathbb{E}[I(f)] = 0, \quad \mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

Proof: Write $\Delta_i W = W(t_i) - W(t_{i-1})$, so that $I(f) = \sum_{i=1}^n \psi(t_{i-1}) \Delta_i W$. Since $\psi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and $\Delta_i W$ is independent of $\mathcal{F}_{t_{i-1}}$ with mean zero, conditioning gives

$$\mathbb{E}[\psi(t_{i-1}) \Delta_i W] = \mathbb{E}[\psi(t_{i-1}) \mathbb{E}[\Delta_i W \mid \mathcal{F}_{t_{i-1}}]] = 0,$$

and summing yields $\mathbb{E}[I(f)] = 0$. For the second moment, expand the square. If $i < j$ then $\psi(t_{i-1})\psi(t_{j-1})\Delta_i W$ is $\mathcal{F}_{t_{j-1}}$ -measurable while $\Delta_j W$ is independent of it with mean zero, so the cross terms vanish by the same argument. The diagonal terms give, using independence of $\Delta_i W$ from $\mathcal{F}_{t_{i-1}}$ once more,

$$\mathbb{E}[\psi(t_{i-1})^2 (\Delta_i W)^2] = \mathbb{E}[\psi(t_{i-1})^2] (t_i - t_{i-1}) = \int_{t_{i-1}}^{t_i} \mathbb{E}[f(t)^2] dt,$$

³The choice of the left endpoint t_{i-1} ensures that the stochastic integral preserves the martingale property. Variants with different evaluation points lead to different calculi, e.g. Stratonovich calculus, where the classical chain rule holds (see [50]).

since $f \equiv \psi(t_{i-1})$ on $[t_{i-1}, t_i]$. Summing over i gives $\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt$. ■

Note that both computations rest on exactly one structural fact: the integrand is evaluated at the *left* endpoint, which is what makes $\psi(t_{i-1})$ independent of the increment it multiplies.

Lemma 1.49 (Density of step processes). *For every $f \in L^2_{\text{ad}}([0, T] \times \Omega)$, there exists a sequence $\{f_n\}_{n=1}^\infty$ of step stochastic processes such that*

$$\lim_{n \rightarrow \infty} \int_0^T \mathbb{E}[(f(t) - f_n(t))^2] dt = 0.$$

That is, step processes are dense in $L^2_{\text{ad}}([0, T] \times \Omega)$.

Proof: The argument is given in Appendix B.2; see also [50, Lemma 4.3.3]. ■

We are now ready to extend the definition.

Definition 1.50 (Stochastic Integral). Let $f \in L^2_{\text{ad}}([0, T] \times \Omega)$. We define its *stochastic integral* over $[0, T]$ as the limit in $L^2(\Omega)$ of $I(f_n)$ for any approximating sequence of step processes $\{f_n\}$ with $f_n \rightarrow f$ in $L^2_{\text{ad}}([0, T] \times \Omega)$.

The following fundamental result follows.

Theorem 1.51 (Itô's isometry). *Let $f \in L^2_{\text{ad}}([0, T] \times \Omega)$.*

i) The stochastic integral $I(f)$ is a centred random variable and

$$\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

ii) For another $g \in L^2_{\text{ad}}([0, T] \times \Omega)$,

$$\mathbb{E} \left[\left(\int_0^T f(t) dW_t \right) \left(\int_0^T g(t) dW_t \right) \right] = \int_0^T \mathbb{E}[f(t)g(t)] dt.$$

Proof: i) Let $\{f_n\}$ be step processes with $f_n \rightarrow f$ in $L^2_{\text{ad}}([0, T] \times \Omega)$, as furnished by Lemma 1.49. By Lemma 1.48 applied to $f_n - f_m$,

$$\mathbb{E}[(I(f_n) - I(f_m))^2] = \int_0^T \mathbb{E}[(f_n(t) - f_m(t))^2] dt \xrightarrow{n, m \rightarrow \infty} 0,$$

so $\{I(f_n)\}$ is Cauchy in $L^2(\Omega)$ and its limit $I(f)$ exists and is independent of the approximating sequence. Both $f \mapsto \mathbb{E}[I(f)]$ and $f \mapsto \mathbb{E}[I(f)^2]$ are continuous for the L^2 norms involved, so the identities $\mathbb{E}[I(f_n)] = 0$ and $\mathbb{E}[I(f_n)^2] = \int_0^T \mathbb{E}[f_n^2] dt$ pass to the limit.

ii) Apply i) to $f + g$ and expand: $\mathbb{E}[I(f + g)^2] = \mathbb{E}[I(f)^2] + 2\mathbb{E}[I(f)I(g)] + \mathbb{E}[I(g)^2]$ by linearity of I , while the right-hand side expands identically. ■

This is precisely the statement that I is an isometry from $L^2_{\text{ad}}([0, T] \times \Omega)$ into $L^2(\Omega)$, extended from a dense subspace by continuity. It is worth noting that if f is deterministic (i.e. does not depend on ω), then

$$I(f) \sim \mathcal{N} \left(0, \int_0^T f(t)^2 dt \right).$$

For $0 \leq t \leq T$, we have $\int_0^t \mathbb{E}[f(s)^2] ds \leq \int_0^T \mathbb{E}[f(s)^2] ds$, so $f \in L^2_{\text{ad}}([0, t] \times \Omega)$ and the stochastic integral is well defined on $[0, t]$. Consider then the process $X_t = \int_0^t f(s) dW(s)$. We know this process has finite first two moments, and so we can take the conditional expectation of X_t with respect to the given filtration.



Theorem 1.52 (Martingale). *Take $f \in L^2_{\text{ad}}([0, T] \times \Omega)$. Then the process given by $X_t = I(f\mathbb{1}_{[0,t]})$ is a martingale with respect to the filtration $\mathcal{F}_t$.*

Proof: Integrability follows from Itô's isometry, since $\mathbb{E}[X_t^2] = \int_0^t \mathbb{E}[f(s)^2] ds < \infty$ and, for probability spaces, $L^2(\Omega) \subset L^1(\Omega)$. Adaptedness follows from the approximating step processes.

For the martingale identity, take first f a step process and $s < t$ lying in the partition defining it (refining the partition if necessary, which changes nothing). Then $X_t - X_s = \sum_i \psi(t_{i-1})(W(t_i) - W(t_{i-1}))$, the sum running over the subintervals of $[s, t]$. For each such i we have $t_{i-1} \geq s$, so conditioning first on $\mathcal{F}_{t_{i-1}}$ and then on $\mathcal{F}_s$ gives

$$\mathbb{E}[\psi(t_{i-1})(W(t_i) - W(t_{i-1})) \mid \mathcal{F}_s] = \mathbb{E}[\psi(t_{i-1}) \mathbb{E}[W(t_i) - W(t_{i-1}) \mid \mathcal{F}_{t_{i-1}}] \mid \mathcal{F}_s] = 0,$$

by the tower property (Proposition 1.4, item i)) and independence of the increment from $\mathcal{F}_{t_{i-1}}$. Hence $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$.

For general $f \in L^2_{\text{ad}}$, take step processes $f_n \rightarrow f$ and write X^n for the corresponding integrals. Itô's isometry gives $X^n_t \rightarrow X_t$ in $L^2(\Omega)$, hence in $L^1(\Omega)$, and conditional expectation is a contraction on L^1 (Proposition A.1, item 7), so $\mathbb{E}[X^n_t \mid \mathcal{F}_s] \rightarrow \mathbb{E}[X_t \mid \mathcal{F}_s]$ in $L^1(\Omega)$. The left-hand side equals $X^n_s \rightarrow X_s$, and the martingale identity passes to the limit. ■

Being a square-integrable martingale, X has a compensator in the sense of Theorem 1.32, and the isometry identifies it at once. This is the form in which the isometry is used from Chapter 2 onwards, so we record it.

Corollary 1.53 (Quadratic variation of an Itô integral). *Let $f \in L^2_{\text{ad}}([0, T] \times \Omega)$ and let $X_t = \int_0^t f(s) dW_s$. Then*

$$\langle X \rangle_t = \int_0^t f(s)^2 ds, \quad t \in [0, T]. \quad (1.50)$$

Proof: Run the computation of Lemma 1.48 conditionally on $\mathcal{F}_s$ instead of under $\mathbb{E}$ — the cross terms vanish for exactly the same reason, the integrand being evaluated before the increment it multiplies — and pass to the limit as in Theorem 1.51. This gives $\mathbb{E}[(X_t - X_s)^2 \mid \mathcal{F}_s] = \mathbb{E}[\int_s^t f(u)^2 du \mid \mathcal{F}_s]$ for $s < t$. Expanding $X_t^2 - X_s^2 = (X_t - X_s)^2 + 2X_s(X_t - X_s)$ and using $\mathbb{E}[X_t - X_s \mid \mathcal{F}_s] = 0$ then yields

$$\mathbb{E}\left[X_t^2 - \int_0^t f^2 du \mid \mathcal{F}_s\right] = X_s^2 - \int_0^s f^2 du,$$

so $X^2 - \int_0^\cdot f^2 du$ is a martingale. The process $\int_0^\cdot f^2 du$ is continuous, adapted, increasing and null at 0, hence predictable, and the uniqueness in Theorem 1.32 identifies it with $\langle X \rangle$. ■

We must point out that when the integrand is of the form $f(s) = \sigma(s, X_s)$ the Markov property also holds, but we will return to that when we see these processes as diffusions in Subsection 1.2.6.

Theorem 1.54 (Continuity property). *Suppose $f \in L^2_{\text{ad}}([0, T] \times \Omega)$. Then the process given by $X_t = \int_0^t f(s) dW(s)$ admits a continuous modification, with which we shall always work.*

Proof: The argument is given in Appendix B.3; see also [50, Thm. 4.6.2]. ■

Finally, a result that actually links Riemann sums with stochastic integrals will be helpful in the next subsection.

Theorem 1.55. Suppose $f \in L^2_{\text{ad}}([0, T] \times \Omega)$ and assume that $\mathbb{E}[f(t)f(s)]$ is a continuous function of t and s . Then, for partitions of $[0, T]$ given by $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = T\}$,

$$\int_0^T f(t) dW(t) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^n f(t_{i-1}) (W(t_i) - W(t_{i-1})) \quad (1.51)$$

in $L^2(\Omega)$, where the limit is taken as in Definition 1.31.

Proof: The argument is given in Appendix B.4; see also [50, Thm. 4.7.1]. ■

The continuity hypothesis on $\mathbb{E}[f(t)f(s)]$ is what makes this work and cannot simply be dropped: without it the left-endpoint approximation need not converge, and the Riemann-sum interpretation of the Itô integral fails. Theorem 1.55 also isolates the one genuinely arbitrary choice in the whole construction, and it is worth naming the alternative, because Chapter 3 will show that the two are on an equal footing.

Definition 1.56 (Stratonovich integral). Let $F \in \mathcal{C}^1(\mathbb{R})$ and put $f = F(W)$. The **Stratonovich integral** of f against W is the limit of the same Riemann sums as in Theorem 1.55, with the integrand evaluated at the *midpoint* of each subinterval rather than at the left endpoint:

$$\int_0^T f(t) \circ dW(t) := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^n f\left(\frac{t_{i-1} + t_i}{2}\right) (W(t_i) - W(t_{i-1})), \quad (1.52)$$

the limit again being taken in $L^2(\Omega)$.

The midpoint value $f(\frac{t_{i-1} + t_i}{2})$ is not $\mathcal{F}_{t_{i-1}}$ -measurable, so it is correlated with the increment it multiplies, and for a general adapted f the sums in (1.52) need not converge at all. For $f = F(W)$ they do, and the two integrals differ by a computable amount,

$$\int_0^T F(W_t) \circ dW_t = \int_0^T F(W_t) dW_t + \frac{1}{2} \int_0^T F'(W_t) dt, \quad (1.53)$$

the correction being one half of the quadratic covariation $\langle F(W), W \rangle_T$. The Itô integral is the one we shall use throughout Part I: it alone preserves the martingale property, because it evaluates the integrand before the increment it multiplies. The Stratonovich integral preserves the classical chain rule with no second-order term, which makes it the natural choice in differential geometry and in the Wong–Zakai approximation results of Chapter 3.

More generally, we can consider a larger family of integrands. Let $\mathcal{L}(\Omega, L^2[0, T])$ be the family of processes f such that

- $f(t)$ is $\mathcal{F}_t$ -adapted.
- $\int_0^T |f(t)|^2 dt < \infty$ almost surely.

The class $\mathcal{L}(\Omega, L^1[0, T])$, which we shall also need, is defined in exactly the same way with $\int_0^T |f(t)| dt < \infty$ almost surely in place of the second condition.

Notice how we dropped the condition of having finite expectation, so that we have indeed a larger class of processes. Consider for example the process $f(t) = e^{W(t)^2}$. It follows that

$$\mathbb{E}[f(t)^2] = \begin{cases} \frac{1}{\sqrt{1-4t}} & \text{if } 0 \leq t < 1/4 \\ \infty & \text{if } t \geq 1/4. \end{cases} \quad (1.54)$$



Hence $\int_0^1 e^{W(t)^2} dW_t$ does not make sense with the previous construction. These processes are nonetheless worth retaining, and the following holds; its proof lies beyond our scope.

Theorem 1.57 (Local Martingale Property). *Let $f \in \mathcal{L}(\Omega, L^2[0, T])$ and $t \in [0, T]$. Then*

$$X_t = \int_0^t f(s) dW_s$$

is a local martingale. Furthermore, it has a continuous realisation. $\square$

1.2.2 Itô's Chain Rule

One of the most striking features of this new integration is a modified chain rule, generalising the familiar expression $d(f \circ g)(x) = f'(g(x)) dg(x)$. The fact underlying the modification is the non-zero quadratic variation of Brownian motion. The resulting calculus lets us evaluate integrals and solve equations that generalise the already rich world of differential equations.

Definition 1.58 (Itô process). Given an initial random variable X_0 , an Itô process is a stochastic process on $[0, T]$ of the form

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW_s \iff dX_t = \mu(t) dt + \sigma(t) dW_t, \quad X_0 \text{ given,}$$

where we emphasise the equivalent formulations, since the latter is just a way of writing the former. We impose X_0 being $\mathcal{F}_0$ -measurable, $\mu \in \mathcal{L}(\Omega, L^1[0, T])$ and $\sigma \in \mathcal{L}(\Omega, L^2[0, T])$. The letters are those of the notation table: μ is a drift and σ a diffusion coefficient, here allowed to be arbitrary adapted processes rather than functions of (t, X_t) — that specialisation comes in Subsection 1.2.3. Note how the first integral is just a Lebesgue integral for each ω , that is, for each path of W . The second is a proper Itô integral.

We state Itô's rule directly in its general form, using the convenient *differential formulation*. A complete proof is beyond the scope of this dissertation, but a result so well known has been treated by many authors; see for instance [50].

Theorem 1.59 (Itô's Rule). *Let $dX_t = \mu(t) dt + \sigma(t) dW_t$ be an Itô process, and let $V(t, x)$ be a continuous function on $[0, T] \times \mathbb{R}$ with continuous partial derivatives $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial x}$ and $\frac{\partial^2 V}{\partial x^2}$. Then $V(t, X_t)$ is also an Itô process, and*

$$dV(t, X_t) = \left(\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma^2(t) \frac{\partial^2 V}{\partial x^2}(t, X_t) \right) dt + \sigma(t) \frac{\partial V}{\partial x}(t, X_t) dW_t. \quad (1.55)$$

$\square$

A convenient way to derive Equation (1.55) is through a symbolic derivation using the Taylor expansion to first order in dt and second order in dX_t , together with the *Itô table* of Table 1.1.

$\times$	dW_t	dt
dW_t	dt	0
dt	0	0

Table 1.1: Itô multiplication table for a scalar Brownian motion.

Applying the Taylor expansion to $V(t, X_t)$ gives

$$dV(t, X_t) = \frac{\partial V}{\partial t}(t, X_t) dt + \frac{\partial V}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2}(t, X_t) (dX_t)^2.$$

Using the Itô table, we have $(dX_t)^2 = \sigma(t)^2 dt$, so using $dX_t = \mu(t) dt + \sigma(t) dW_t$ yields

$$dV(t, X_t) = \frac{\partial V}{\partial x}(t, X_t) \sigma(t) dW_t + \left[\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma(t)^2 \frac{\partial^2 V}{\partial x^2}(t, X_t) \right] dt,$$

which is precisely Equation (1.55).

We stress that the computation just performed is a *mnemonic*, not a proof: the symbols dW_t and $(dW_t)^2$ have been manipulated as though they were infinitesimals, which they are not. What makes the manipulation legitimate is the content of (1.40) — the second-order term survives precisely because W has non-vanishing quadratic variation — and a genuine proof proceeds by Taylor-expanding along a partition and controlling the remainder in $L^2(\Omega)$. We refer to [50] for the details. The value of the Itô table is that it makes the bookkeeping automatic once one accepts that $\langle W \rangle_t = t$, and we shall use it in that spirit throughout.

As with the ordinary chain rule, this opens the way to evaluating stochastic integrals in terms of others.

Example 1.60. We can use this result to obtain a closed form for the integral $\int_0^t W_s dW_s$. Take $\phi(x) = x^2$ and define $Y_t = \phi(W_t)$, so that

$$dY_t = 2W_t dW_t + \frac{1}{2} \phi''(W_t) (dW_t)^2 = 2W_t dW_t + dt \quad (1.56)$$

so that $\int_0^t W_s dW_s = \frac{1}{2} (W_t^2 - t)$, which is a much simpler expression, and clearly a martingale.

Example 1.61. In finance, the Black–Scholes model relies on the *log-normal* equation, given by $dS_t = \mu S_t dt + \sigma S_t dW_t$, where $\mu, \sigma \in \mathbb{R}$ and $\sigma > 0$. The equation is solved by taking $Y_t = \log(S_t)$, so that

$$dY_t = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 = \mu dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt \quad (1.57)$$

obtaining $Y_t = Y_0 + (\mu - \sigma^2/2)t + \sigma W_t \implies S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$.

Example 1.62 (Lamperti's Transform). Consider a process $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ and assume that σ is *strictly positive* on the range of X and differentiable once with respect to time and once with respect to x . Fix any reference point x_0 in the range of X and define the *deterministic* function

$$\Gamma(t, x) = \int_{x_0}^x \frac{1}{\sigma(t, s)} ds \quad (1.58)$$

Itô's rule requires a deterministic $\mathcal{C}^{1,2}$ function, and changing x_0 shifts Γ by a function of t alone, which alters the drift below but not the conclusion. Consider now the process $Y_t = \Gamma(t, X_t)$:

$$\begin{aligned} dY_t &= \left(\frac{\partial \Gamma}{\partial t} + \mu(t, X_t) \frac{\partial \Gamma}{\partial x} + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 \Gamma}{\partial x^2} \right) dt + \underbrace{\sigma(t, X_t) \frac{\partial \Gamma}{\partial x}}_{=1} dW_t \\ &= \left(\frac{\partial \Gamma}{\partial t} + \frac{\mu(t, X_t)}{\sigma(t, X_t)} - \frac{1}{2} \frac{\partial \sigma}{\partial x}(t, X_t) \right) dt + dW_t \end{aligned} \quad (1.59)$$



so that all the complexity of the original process is contained in the drift. The diffusion coefficient of Y is identically 1, so that Y is driven by the Brownian increments themselves and an increment of Y over a step of length Δt has variance Δt whatever the current state. Two consequences follow: a lattice built on Y moves by the same amount at every node, which is exactly the condition under which a binomial or trinomial tree *recombines*, so that the number of nodes grows quadratically rather than exponentially in the number of steps, allowing for fast numerical approaches. Also, a discretisation of Y no longer has to approximate a varying diffusion coefficient, which is the term limiting the accuracy of the Euler scheme applied to X directly.

1.2.3 Stochastic Differential Equations

The main aim of this subsection is to give conditions under which an Itô process exists and is unique. We follow a direct approach and state the existence and uniqueness theorem. We need a precise definition for *stochastic differential equation* or SDE, and thus some preliminary results. Given an $\mathcal{F}_0$ -measurable random variable X_0 of finite variance, consider the following formal equation on $[0, T]$:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ given.} \quad (1.60)$$

Definition 1.63 (Solution of an SDE). A stochastic process X_t , $0 \leq t \leq T$, is called a solution of (1.60) if it satisfies:

- $\sigma(t, X_t) \in \mathcal{L}(\Omega, L^2[0, T])$ so its Itô integral makes sense for each $t \in [0, T]$.
- Almost all sample paths of $\mu(t, X_t)$ belong to $L^1[0, T]$.
- Equation (1.60) holds almost surely.

We need to establish some growth conditions on μ and σ for the equation to admit a solution on the whole of $[0, T]$, just as in ordinary differential equations. Let us define the following:

Definition 1.64. Let $g(t, x)$ be a measurable function on $[0, T] \times \mathbb{R}$.

- It is said to satisfy a Lipschitz condition (or simply called Lipschitz) in x if there exists $C > 0$ such that $|g(t, x) - g(t, y)| \leq C|x - y|$ for all $0 \leq t \leq T$, $x, y \in \mathbb{R}$.
- Similarly, it is said to satisfy a linear growth condition in x if there exists $C > 0$ such that $|g(t, x)| \leq C(1 + |x|)$ for all $0 \leq t \leq T$, $x \in \mathbb{R}$.

Notice that for $x \geq 0$ we have $1 + x^2 \leq (1 + x)^2 \leq 2(1 + x^2)$, so that the linear growth condition can be restated, for a new constant C' , as $|g(t, x)| \leq C'\sqrt{1 + x^2}$.

The proof of existence and uniqueness rests on a Picard iteration, exactly as for ordinary differential equations, and the comparison step is furnished by Grönwall's inequality (Lemma A.7 in Appendix A); since the argument is by now standard — though, unlike its deterministic counterpart, it invokes the Itô isometry and Doob's inequality at every step — we do not reproduce it here.

Theorem 1.65. *Let μ and σ be measurable functions on $[0, T] \times \mathbb{R}$ satisfying both the Lipschitz and the linear growth conditions in x of Definition 1.64, and let the initial condition X_0 be $\mathcal{F}_0$ -measurable with $\mathbb{E}[X_0^2] < \infty$. Then the stochastic differential equation (1.60) admits a solution, which is unique in the following sense: any two solutions are indistinguishable. Moreover the solution has continuous sample paths.*

Proof: We shall only prove the uniqueness part, the existence being a Picard iteration as indicated above. Let X and Y be two continuous solutions and put $Z_t = X_t - Y_t$, so that

$$Z_t = \int_0^t [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s + \int_0^t [\mu(s, X_s) - \mu(s, Y_s)] ds.$$

We cannot yet take expectations in this identity. By Definition 1.63 the integrand of the stochastic term lies only in the pathwise class $\mathcal{L}(\Omega, L^2[0, T])$, on which the Itô integral is constructed by localisation and is a local martingale rather than a martingale; Itô's isometry, a statement about $L^2_{\text{ad}}([0, T] \times \Omega)$, is therefore not available. Nor do we know that $\mathbb{E}[Z_t^2]$ is finite, and an inequality between infinite quantities would tell us nothing. Both obstructions are removed by the same device. Define, for $n \in \mathbb{N}$,

$$\tau_n := \inf \{t \in [0, T] : |X_t| \geq n \text{ or } |Y_t| \geq n\} \wedge T, \quad (1.61)$$

which is a stopping time. Here the target set $\{|x| \geq n\}$ is *closed*, so Example 1.12 does not apply and continuity of the paths has to be used instead: $\{\tau_n \leq t\} = \{\sup_{s \leq t} (|X_s| \vee |Y_s|) \geq n\}$, and by continuity the supremum may be taken over the rationals of $[0, t]$ together with t itself, which exhibits the event as a countable operation on members of $\mathcal{F}_t$. As the paths are continuous they are bounded on $[0, T]$ for almost every ω , whence $\tau_n \uparrow T$ almost surely.

Fix n and stop the identity above at τ_n . On $\{s \leq \tau_n\}$ the Lipschitz condition then gives $|\sigma(s, X_s) - \sigma(s, Y_s)| \leq C|Z_s| \leq 2Cn$, so the stopped integrand $\mathbb{1}_{[0, \tau_n]}(s) [\sigma(s, X_s) - \sigma(s, Y_s)]$ is bounded, adapted, and hence genuinely an element of $L^2_{\text{ad}}([0, T] \times \Omega)$. Itô's isometry now applies to it, and together with the Lipschitz condition yields

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^{t \wedge \tau_n} [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s \right)^2 \right] &= \int_0^t \mathbb{E} [(\sigma(s, X_s) - \sigma(s, Y_s))^2 \mathbb{1}_{\{s \leq \tau_n\}}] ds \\ &\leq C^2 \int_0^t \mathbb{E} [Z_s^2 \mathbb{1}_{\{s \leq \tau_n\}}] ds. \end{aligned}$$

For the Lebesgue term, the Cauchy–Schwarz inequality followed by the Lipschitz condition gives, pathwise,

$$\left(\int_0^{t \wedge \tau_n} [\mu(s, X_s) - \mu(s, Y_s)] ds \right)^2 \leq T \int_0^t (\mu(s, X_s) - \mu(s, Y_s))^2 \mathbb{1}_{\{s \leq \tau_n\}} ds \leq TC^2 \int_0^t Z_s^2 \mathbb{1}_{\{s \leq \tau_n\}} ds.$$

Write $u_n(t) := \mathbb{E}[Z_{t \wedge \tau_n}^2]$. Since $|Z_{t \wedge \tau_n}| \leq 2n$ we have $u_n(t) \leq 4n^2 < \infty$ for every t , which is the finiteness the argument was missing. Moreover $Z_s^2 \mathbb{1}_{\{s \leq \tau_n\}} \leq Z_{s \wedge \tau_n}^2$ pointwise, so combining the two estimates through $(a + b)^2 \leq 2(a^2 + b^2)$ and setting $\lambda := 2C^2(1 + T)$ gives

$$u_n(t) \leq \lambda \int_0^t u_n(s) ds, \quad 0 \leq t \leq T.$$

This is a Grönwall inequality with $g \equiv 0$ applied to a finite, non-negative function, so Lemma A.7 of Appendix A forces $u_n \equiv 0$. Hence $Z_{t \wedge \tau_n} = 0$ almost surely for each t and each n ; letting $n \rightarrow \infty$ and using $\tau_n \uparrow T$ together with the continuity of Z gives $Z_t = 0$ almost surely for each fixed $t \in [0, T]$.



It remains to upgrade this to indistinguishability, and the step is not vacuous: the null set so far depends on t (Remark 1.38). Let $\{r_n\}$ enumerate the rationals of $[0, T]$ and let Ω_0 be the intersection of the full-measure events $\{Z_{r_n} = 0\}$ with the full-measure event on which Z has continuous paths. Then $\mathbb{P}(\Omega_0) = 1$, and for $\omega \in \Omega_0$ the function $t \mapsto Z_t(\omega)$ is continuous and vanishes on a dense set, hence vanishes identically. Thus X and Y are indistinguishable. ■

Two comments on the hypotheses. First, both conditions are needed, and they do different jobs: the Lipschitz condition delivers uniqueness, whereas the linear growth condition prevents the solution from exploding in finite time.

Second, the word “unique” conceals a distinction that this formulation is too coarse to see. We have asserted *pathwise* uniqueness: given the Brownian motion, the solution is determined. One may instead ask only that the *law* of the solution be determined, which is a weaker requirement and corresponds to the notion of a *weak* solution.

Let us first make the distinction precise, since it is systematically blurred in the applied literature and the blurring causes real confusion.

Definition 1.66 (Strong and weak solutions). Consider $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ with a given $\mathcal{F}_0$ -measurable X_0 .

- i) A **strong solution** is a process X , adapted to the augmented filtration generated by a *given* Brownian motion W and the initial condition, satisfying the equation almost surely. *Pathwise uniqueness* holds if any two strong solutions on the same space, driven by the same W , are indistinguishable.
- ii) A **weak solution** is a pair (X, W) together with a filtered probability space, such that W is a Brownian motion for that filtration and the equation holds. *Uniqueness in law* holds if any two weak solutions induce the same law on $C[0, T]$.

The distinction is one of information. For a strong solution the noise is handed to us and the trajectory is a measurable functional of it — one may, at least in principle, simulate the solution from a given stream of random numbers. For a weak solution we are permitted to construct the noise ourselves, and all we control is the distribution of the result. For pricing this is often enough, since prices are expectations and expectations depend only on the law; for hedging, and for any pathwise statement, it is not.

The relation between the four notions is settled by the Yamada–Watanabe theorem: pathwise uniqueness together with existence of a weak solution implies existence of a strong solution and uniqueness in law. What we require is the companion criterion, which relaxes the Lipschitz condition in exactly the direction needed.

Theorem 1.67 (Yamada–Watanabe). *Consider the one-dimensional stochastic differential equation $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$. Suppose there exist a strictly increasing function $\eta : [0, \infty) \rightarrow [0, \infty)$ with $\eta(0) = 0$ and*

$$\int_{0+} \frac{du}{\eta(u)^2} = \infty, \quad (1.62)$$

and a strictly increasing concave ζ with $\zeta(0) = 0$ and $\int_{0+} du/\zeta(u) = \infty$, such that for all t and all x, y

$$|\sigma(t, x) - \sigma(t, y)| \leq \eta(|x - y|), \quad |\mu(t, x) - \mu(t, y)| \leq \zeta(|x - y|).$$

Then pathwise uniqueness holds for the equation [73]. □

The point of (1.62) is that it admits $\eta(u) = u^\gamma$ for every $\gamma \geq \frac{1}{2}$, since $\int_0^+ u^{-2\gamma} du$ diverges precisely when $2\gamma \geq 1$. Hölder continuity of order $\frac{1}{2}$ is thus permitted for the diffusion coefficient — and only just, the exponent $\frac{1}{2}$ being critical. The drift is treated separately and more leniently, needing only an Osgood-type modulus.

Example 1.68 (The square-root process). Consider the process which will serve as the variance in the Heston model of Chapter 2 (Subsection 2.3.2),

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t, \quad v_0 > 0, \quad (1.63)$$

with $\kappa, \theta, \xi > 0$. The drift is affine, hence Lipschitz. The diffusion coefficient $\sigma(v) = \xi\sqrt{v}$ is not Lipschitz at the origin, but it satisfies $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$, so Theorem 1.67 applies with $\eta(u) = \xi\sqrt{u}$, and pathwise uniqueness holds. A non-negative weak solution exists, so the implication recorded above applies and (1.63) admits a unique *strong* solution, in the sense of Definition 1.66(i), with uniqueness in law. The equation is therefore well posed.

We also need the solution to stay strictly positive: it is to be a variance, and $\sqrt{v_t}$ must make sense. Note that the diffusion coefficient vanishes at $v = 0$ while the drift there equals $\kappa\theta > 0$, so the origin is a point where the noise switches off and the drift pushes upward.

Proposition 1.69 (Feller condition). *Let v solve (1.63) with $v_0 > 0$. If*

$$2\kappa\theta \geq \xi^2, \quad (1.64)$$

then $\mathbb{P}(v_t > 0 \text{ for all } t \in [0, T]) = 1$. If $2\kappa\theta < \xi^2$, the process reaches 0 with positive probability, though it is instantaneously reflecting and does not become negative [32]. $\square$

1.2.4 The multidimensional case

By a multidimensional Brownian motion on $\mathbb{R}^n$ we mean a stochastic process of the form $\mathbf{W}(t) = (W_1(t), W_2(t), \dots, W_n(t))$ formed by n independent Brownian motions on $\mathbb{R}$. Similarly, we can set them to start at $x_0 \in \mathbb{R}^n$ just by a simple translation. It is a martingale and a Markov process that, starting at the origin, has a transition density function $p_t(x) = (2\pi t)^{-n/2} e^{-\|x\|^2/2t}$.

There are certain properties of these processes that dramatically change with dimension. For example, for $n = 1$, it returns to 0 infinitely often, and for $n = 2$ enters arbitrarily small neighbourhoods of 0 infinitely often with probability 1, however the probability that it reaches $\mathbf{0}$ is 0. For $n \geq 3$, the process is *transient*, satisfying $\lim_{t \rightarrow \infty} \|\mathbf{W}(t)\| = \infty$ almost surely.

In terms of fractal dimensions, the *range* $\mathbf{W}([0, t])$ has Hausdorff dimension 2 for $n \geq 2$, and dimension 1 for $n = 1$ (where it is simply an interval). The *graph* $\{(s, \mathbf{W}(s)) : s \in [0, t]\} \subset \mathbb{R}^{1+n}$ behaves differently: it has dimension $3/2$ for $n = 1$, a classical result of Taylor [71], and dimension 2 for $n \geq 2$. These fascinating properties of Brownian motion can be found in [31].

We can therefore define a multidimensional stochastic integral as

$$dX_t = \mu dt + \sum_{j=1}^m \sigma_j dW_t^j \quad (1.65)$$

for an m -dimensional driver, each integral being considered separately. Let $W_1(t), \dots, W_m(t)$ be independent Brownian motions. Consider n Itô processes $X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(n)}$ given by

$$X_t^{(i)} = X_0^{(i)} + \int_0^t \mu_i(s) ds + \sum_{j=1}^m \int_0^t \sigma_{ij}(s) dW_j(s), \quad 1 \leq i \leq n, \quad (1.66)$$



where $\mu_i \in \mathcal{L}(\Omega, L^1[0, T])$ and $\sigma_{ij} \in \mathcal{L}(\Omega, L^2[0, T])$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. Collecting $W = (W_1, \dots, W_m)^\top$ and $X_t = (X_t^{(1)}, \dots, X_t^{(n)})^\top$ into column vectors, $\mu = (\mu_1, \dots, \mu_n)^\top$ likewise, and $\sigma = [\sigma_{ij}]$ into an $n \times m$ matrix, equation (1.66) becomes

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW(s), \quad 0 \leq t \leq T,$$

the integrals being taken componentwise. Growth conditions on μ_i and σ_{ij} must again be imposed for the solution to make sense.

So far, a multidimensional Brownian motion has *independent* components. In modelling one almost always wants the components to be correlated — a falling equity price accompanied by rising volatility, two rates moving together, an exchange rate and the foreign interest rate — and we shall shortly make that precise.

Constructing such an object requires a way of *recognising* a Brownian motion, and the following classical theorem supplies one.

Theorem 1.70 (Lévy's Characterisation Theorem). *Let X be a continuous local martingale on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ with $X_0 = 0$ and $\langle X \rangle_t = t$ almost surely for every $t \in [0, T]$. Then X is an $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion under $\mathbb{P}$.*

More generally, let $X = (X^1, \dots, X^n)$ be a vector of continuous local martingales with $X_0 = 0$ and

$$\langle X^i, X^j \rangle_t = \delta_{ij} t \quad \text{almost surely, for all } i, j \text{ and all } t,$$

where δ_{ij} is the Kronecker delta. Then X is an n -dimensional $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion with independent components.

Proof: The argument is given in Appendix B.5; see also [50, Thm. 8.4.2]. ■

The statement is considerably stronger than it looks. Being a Brownian motion is, by Definition 1.35, a statement about the *joint law* of the whole family $\{X_t\}$: independent, stationary, Gaussian increments. Lévy's theorem says that all of this is already implied by two facts — that X is a local martingale and that its quadratic variation is exactly t . Gaussianity is not assumed anywhere; it is a conclusion.

With the criterion in hand we may define the object we want.

Definition 1.71 (Correlated Brownian motion). Let $P = (\rho_{ij}) \in \mathbb{R}^{n \times n}$ be symmetric and positive semi-definite with $\rho_{ii} = 1$ for all i . An n -dimensional process $\widetilde{\mathbf{W}} = (\widetilde{W}^1, \dots, \widetilde{W}^n)$ is a **Brownian motion with correlation matrix P** if each $\widetilde{W}^i$ is a one-dimensional Brownian motion and

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \rho_{ij} t, \quad 1 \leq i, j \leq n, \quad t \in [0, T]. \quad (1.67)$$

Such a process always exists, and is obtained from an independent one by a linear map. Since P is symmetric positive semi-definite it admits a factorisation $P = LL^\top$ — the Cholesky factorisation when P is positive definite, and a square root more generally. Setting $\widetilde{\mathbf{W}} := L \mathbf{W}$ for an independent $\mathbf{W}$, bilinearity of the quadratic covariation (1.31) gives

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \sum_{k, l} L_{ik} L_{jl} \langle W^k, W^l \rangle_t = \sum_k L_{ik} L_{jk} t = (LL^\top)_{ij} t = \rho_{ij} t,$$

and each component is a continuous local martingale with $\langle \widetilde{W}^i \rangle_t = t$, hence a Brownian motion by Theorem 1.70. With this in hand we may state the formula.

Theorem 1.72 (Multidimensional Itô formula). *Let X be the n -dimensional Itô process of (1.66), driven by an m -dimensional Brownian motion with independent components, and let $V \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R}^n, \mathbb{R})$. Then, almost surely and for every $t \in [0, T]$,*

$$dV(t, X_t) = \frac{\partial V}{\partial t} dt + \sum_{l=1}^n \frac{\partial V}{\partial x_l} dX_t^{(l)} + \frac{1}{2} \sum_{l,h=1}^n \frac{\partial^2 V}{\partial x_l \partial x_h} d\langle X^{(l)}, X^{(h)} \rangle_t, \quad (1.68)$$

where the cross-variations are

$$d\langle X^{(l)}, X^{(h)} \rangle_t = \sum_{j=1}^m \sigma_{lj}(t) \sigma_{hj}(t) dt = (\sigma \sigma^\top)_{lh} dt. \quad (1.69)$$

If instead the driving Brownian motion has correlation matrix $P = (\rho_{jk})$ in the sense of Definition 1.71, (1.68) is unchanged and (1.69) becomes

$$d\langle X^{(l)}, X^{(h)} \rangle_t = \sum_{j,k=1}^m \sigma_{lj}(t) \rho_{jk} \sigma_{hk}(t) dt = (\sigma P \sigma^\top)_{lh} dt. \quad (1.70)$$

□

The proof is the one-dimensional argument of Theorem 1.59 applied coordinatewise, with Table 1.2 supplying the second-order terms; see [50, Ch. 7] or [63, Ch. 4]. Taking $V(x_1, x_2) = x_1 x_2$ and $n = 2$ recovers the Itô product rule.

The Itô multiplication table extends accordingly.

$\times$	dt	dW_t^1	dW_t^2	$\dots$	dW_t^n
dt	0	0	0	$\dots$	0
dW_t^1	0	dt	$\rho_{12} dt$	$\dots$	$\rho_{1n} dt$
dW_t^2	0	$\rho_{21} dt$	dt	$\dots$	$\rho_{2n} dt$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
dW_t^n	0	$\rho_{n1} dt$	$\rho_{n2} dt$	$\dots$	dt

Table 1.2: Itô multiplication table for correlated n -dimensional Brownian motion.

The entries are not a new convention but a restatement of (1.67): the symbol $dW_t^i dW_t^j$ abbreviates $d\langle W^i, W^j \rangle_t = \rho_{ij} dt$. The independent case is recovered by taking $P = \mathbb{I}_{n \times n}$.

Example 1.73. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ function and let $\nabla V = (\frac{\partial V}{\partial x_1}, \frac{\partial V}{\partial x_2}, \dots, \frac{\partial V}{\partial x_n})$ and $\Delta V = \sum_{i=1}^n \frac{\partial^2 V}{\partial x_i^2}$ to obtain

$$V(\mathbf{W}(t)) = V(\mathbf{W}(0)) + \int_0^t \nabla V(\mathbf{W}(s)) \cdot d\mathbf{W}_s + \frac{1}{2} \int_0^t \Delta V(\mathbf{W}(s)) ds.$$

This equation exhibits the deep connection between Brownian motion and diffusions, showing how *one half of the Laplace operator* is the generator of Brownian motion; see Subsection 1.2.6.

Example 1.74. An important example is given by $V(x, y) = xy$. Clearly, $\partial V / \partial x = y$ and also $\partial V / \partial y = x$. Therefore, if we have processes X_t and Z_t , we obtain

$$d(X_t Z_t) = Z_t dX_t + X_t dZ_t + \frac{1}{2} dX_t dZ_t + \frac{1}{2} dX_t dZ_t = Z_t dX_t + X_t dZ_t + dX_t dZ_t \quad (1.71)$$

which is usually known as the Itô product formula, understanding $dX_t dZ_t$ as in Table 1.2.



The following generalisation holds.

Theorem 1.75. *Let $\sigma(t, x)$ be an $n \times m$ matrix-valued function and $\mu(t, x)$ an $\mathbb{R}^n$ -valued function. Assume there exists $C > 0$ such that for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,*

$$\|\sigma(t, x) - \sigma(t, y)\| \leq C\|x - y\|, \quad \|\mu(t, x) - \mu(t, y)\| \leq C\|x - y\|$$

and

$$\|\sigma(t, x)\|^2 \leq C(1 + \|x\|^2), \quad \|\mu(t, x)\|^2 \leq C(1 + \|x\|^2)$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^n$ and, for matrices, the Frobenius norm $\|\sigma\|^2 = \sum_i \sum_j \sigma_{ij}^2$. For any $\mathcal{F}_0$ -measurable random vector X_0 of finite variance, the SDE $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$ has a unique continuous solution. $\square$

1.2.5 Probability measure changes

Another important topic we must cover has to do with probability changes. Brownian motion depends crucially on the underlying probability $\mathbb{P}$. Given a process X_t , it is natural to ask whether there is a probability $\mathbb{Q}$ so that X_t is a Brownian motion with respect to $\mathbb{Q}$. We assume the σ -algebra and the filtration remain unchanged. The change of measure itself is engineered through a single family of processes, which we now introduce.

Definition 1.76 (Exponential Process). Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. We define its exponential process by

$$\mathcal{E}_\alpha(t) = e^{\int_0^t \alpha(s) dW(s) - \frac{1}{2} \int_0^t \alpha^2(s) ds} \quad (1.72)$$

which is precisely the solution of $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW(t)$, $\mathcal{E}_\alpha(0) = 1$ so that $\mathcal{E}_\alpha(t)$ is a local martingale.

This exponential process is important because, when it is a genuine $\mathbb{P}$ -martingale, it serves as the density of a new probability $\mathbb{Q}$ with respect to $\mathbb{P}$.

Theorem 1.77. *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then $\mathcal{E}_\alpha(t)$ is a martingale if, and only if, $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ for all $t \in [0, T]$.*

Proof: For necessity, if $\mathcal{E}_\alpha$ is a martingale then $\mathbb{E}[\mathcal{E}_\alpha(t)] = \mathbb{E}[\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_0]] = \mathbb{E}[\mathcal{E}_\alpha(0)] = 1$ for every t . Conversely, recall that $\mathcal{E}_\alpha(t) = 1 + \int_0^t \alpha(s)\mathcal{E}_\alpha(s)dW_s$, so $\mathcal{E}_\alpha$ is a non-negative local martingale, and let (τ_n) be a localising sequence. A non-negative local martingale is always a supermartingale: apply the conditional Fatou lemma (Proposition A.2) to $\mathcal{E}_\alpha(t \wedge \tau_n) \rightarrow \mathcal{E}_\alpha(t)$. So for $s < t$,

$$\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_s] \leq \mathcal{E}_\alpha(s) \quad \text{almost surely.}$$

Both sides are non-negative and, by hypothesis, have the same expectation, namely 1. A non-negative random variable dominated by another of equal expectation is almost surely equal to it, so the inequality is an equality and $\mathcal{E}_\alpha$ is a martingale. See [50, Thm. 8.7.3]. $\blacksquare$

The criterion just proved is exact, but it is not one that can be checked directly: verifying $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ is very nearly as hard as verifying the martingale property itself. What is wanted is a condition on α alone, and the following supplies one.

Theorem 1.78 (Novikov's condition). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. If $\mathbb{E}\left[e^{\frac{1}{2} \int_0^T \alpha^2(s) ds}\right] < \infty$, then $\mathcal{E}_\alpha(t)$ is a martingale. A proof may be found in [50, Sec. 8.7].* $\square$

This is a much more practical way to check whether $\mathcal{E}_\alpha(t)$ is a martingale, and therefore whether we can construct a new probability $\mathbb{Q}$ such that $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$. For the following Theorem, we need to make use of two auxiliary lemmas.

Lemma 1.79 (Conditional expectation under a change of measure). *Let Λ be a nonnegative integrable random variable with $\mathbb{E}[\Lambda] = 1$, so that $d\mathbb{Q} = \Lambda d\mathbb{P}$ defines a new probability measure (see (1.2)). Then, for any σ -algebra $\mathcal{G} \subset \mathcal{F}$ and Y such that $\mathbb{E}^\mathbb{Q}[|Y|] < \infty$, it follows that*

$$\mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] = \frac{\mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}]}{\mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}]}, \quad \mathbb{Q}\text{-almost surely.} \quad (1.73)$$

Proof: Integrability holds because $\mathbb{E}^\mathbb{P}[|Y|\Lambda] = \mathbb{E}^\mathbb{Q}[|Y|] < \infty$, so $\mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}]$ is defined. Fix $G \in \mathcal{G}$. By the defining property of conditional expectation under $\mathbb{P}$ and the definition of $\mathbb{Q}$,

$$\int_G \mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}] d\mathbb{P} = \int_G \Lambda Y d\mathbb{P} = \int_G Y d\mathbb{Q} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] d\mathbb{Q}.$$

On the other hand, writing $d\mathbb{Q} = \Lambda d\mathbb{P}$ and using that $\mathbb{E}^\mathbb{Q}[Y | \mathcal{G}]$ is $\mathcal{G}$ -measurable, so that it may be taken out of a conditional expectation given $\mathcal{G}$ (Proposition A.1, item 4),

$$\int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] d\mathbb{Q} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] \Lambda d\mathbb{P} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] \mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}] d\mathbb{P}.$$

The two displays agree for every $G \in \mathcal{G}$, and both integrands are $\mathcal{G}$ -measurable, so they coincide almost surely. Dividing by $\mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}]$, which is strictly positive $\mathbb{Q}$ -almost surely, gives the result [50, Lemma 8.9.2]. ■

Suppose then that $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ is such that $\mathcal{E}_\alpha$ is a $\mathbb{P}$ -martingale. Let $\mathbb{Q}$ be defined by $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$, that is, $\mathbb{Q}(A) = \int_A \mathcal{E}_\alpha(T) d\mathbb{P}$ for every $A \in \mathcal{F}$.

Lemma 1.80 (Martingale transfer). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then a $\{\mathcal{F}_t\}$ -adapted process X_t , $t \in [0, T]$ is a $\mathbb{Q}$ -martingale if, and only if, $X(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale.*

Proof: Apply Lemma 1.79 with $\gamma = \mathcal{E}_\alpha(T)$ and $\mathcal{G} = \mathcal{F}_s$. Since $\mathcal{E}_\alpha$ is a $\mathbb{P}$ -martingale, $\mathbb{E}^\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathcal{E}_\alpha(s)$, so for $s \leq t$

$$\mathbb{E}^\mathbb{Q}[X(t) | \mathcal{F}_s] = \frac{\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s]}{\mathcal{E}_\alpha(s)}. \quad (1.74)$$

The numerator simplifies by conditioning on $\mathcal{F}_t$ first and using that $X(t)$ is $\mathcal{F}_t$ -measurable:

$$\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathbb{E}^\mathbb{P}\left[X(t) \mathbb{E}^\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_t] \middle| \mathcal{F}_s\right] = \mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s].$$

Hence $\mathbb{E}^\mathbb{Q}[X(t) | \mathcal{F}_s] = X(s)$ if and only if $\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s] = X(s)\mathcal{E}_\alpha(s)$, which is the assertion [50, Lemma 8.9.3]. ■

Finally, the following key result holds from Lemmas 1.79 and 1.80.

Theorem 1.81 (Girsanov's Theorem). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ such that $\mathcal{E}_\alpha$ is a martingale. Then the process*

$$Z(t) = W(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T] \quad (1.75)$$

is a Brownian motion with respect to the measure $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$.



Proof: The argument is given in Appendix B.6; see also [50, Thm. 8.9.4]. ■

Girsanov's theorem is the formal statement of something we shall use constantly: *changing the measure changes the drift and leaves the volatility alone*. The Brownian motion Z under $\mathbb{Q}$ is the old W with a drift subtracted, and nothing else has moved. The reason is already contained in Theorem 1.70: to verify that Z is a $\mathbb{Q}$ -Brownian motion one need only check that it is a continuous local martingale with $\langle Z \rangle_t = t$, and quadratic variation is a pathwise limit, so it is unaffected by passing to an equivalent measure. Two equivalent measures disagree about how likely a path is, never about which paths exist. This observation is the mathematical content of the statement — which we shall meet again in the next chapter, in Section 2.2 — that an option's price does not depend on the expected return of the underlying, while it depends critically on its volatility.

For the multi-factor models of the next chapter (Section 2.3) we require the vector-valued form.

Theorem 1.82 (Girsanov's Theorem, multidimensional). *Let $\mathbf{W} = (W^1, \dots, W^n)$ be an n -dimensional Brownian motion and let $\alpha = (\alpha^1, \dots, \alpha^n)$ be adapted with $\int_0^T \|\alpha(s)\|^2 ds < \infty$ almost surely. Define*

$$\mathcal{E}_\alpha(t) = \exp \left(\int_0^t \alpha(s)^\top d\mathbf{W}(s) - \frac{1}{2} \int_0^t \|\alpha(s)\|^2 ds \right), \quad (1.76)$$

and suppose Novikov's condition holds in the form $\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \|\alpha(s)\|^2 ds \right) \right] < \infty$. Then $\mathcal{E}_\alpha$ is a martingale, $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$ defines a probability measure equivalent to $\mathbb{P}$, and

$$\mathbf{Z}(t) = \mathbf{W}(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T], \quad (1.77)$$

is an n -dimensional $\mathbb{Q}$ -Brownian motion with independent components. □

The Martingale Representation Theorem

Girsanov's theorem tells us which measures are available. The following tells us which *payoffs* are attainable. Together the two constitute the mathematical content, in the Brownian setting, of the fundamental theorems of asset pricing, proved in Section 2.1 of the next chapter.

Theorem 1.83 (Martingale representation). *Let $\mathbf{W}$ be an n -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $(\mathcal{F}_t^W)_{t \in [0, T]}$ be its augmented natural filtration. Let M be a square-integrable $\mathbb{P}$ -martingale with respect to it. There exists an adapted process $\pi = (\pi^1, \dots, \pi^n) \in L_{\text{ad}}^2([0, T] \times \Omega)$, unique up to $dt \otimes d\mathbb{P}$ -null sets, such that*

$$M_t = M_0 + \int_0^t \pi(s)^\top d\mathbf{W}(s), \quad t \in [0, T], \quad (1.78)$$

almost surely. If M is only a local martingale, the same holds with π in the larger class $\mathcal{L}(\Omega, L^2[0, T])$. □

The theorem should be read as a converse to Theorem 1.52. We showed that every Itô integral is a martingale; here we learn that, provided the filtration is the one generated by $\mathbf{W}$ and nothing more, *every* martingale is an Itô integral: the stochastic integrals do not merely sit inside the space of martingales, they exhaust it.

The proof is an orthogonality argument: one shows that the stochastic integrals form a closed subspace of $L^2(\Omega, \mathcal{F}_T)$ whose orthogonal complement contains only constants, using the fact

that exponential martingales $\mathcal{E}_\alpha(T)$ span a dense set. Existence is therefore obtained without any formula for π , which is worth knowing in advance: the theorem will guarantee that a replicating strategy exists while saying nothing whatever about how to compute it. Recovering π in concrete cases is precisely the delta-hedging computation of Section 2.2, and in models where no closed form is available it is a numerical problem.

Remark 1.84 (Why this matters for pricing). Let G_T be a square-integrable contingent claim and let $\mathbb{Q}$ be an equivalent martingale measure. The process $M_t := \mathbb{E}^{\mathbb{Q}}[G_T/S_T^0 \mid \mathcal{F}_t]$ is by construction a $\mathbb{Q}$ -martingale. If the filtration is generated by the Brownian motion driving the traded assets, Theorem 1.83 applied under $\mathbb{Q}$ produces π with $M_t = M_0 + \int_0^t \pi^\top d\mathbf{Z}$, and π is — after rewriting the integral in terms of the discounted asset prices — a self-financing strategy replicating G_T exactly.

One hypothesis is being used silently. Theorem 1.83 is applied under $\mathbb{Q}$ to the $\mathbb{Q}$ -Brownian motion $\mathbf{Z}$, so what it requires is that the filtration be generated by $\mathbf{Z}$, whereas the filtration we have is the one generated by $\mathbf{W}$. One inclusion is immediate: by Theorem 1.82, $\mathbf{Z}_t = \mathbf{W}_t - \int_0^t \alpha ds$ with α adapted, so $\mathbf{Z}$ is $(\mathcal{F}_t^{\mathbf{W}})_{t \in [0, T]}$ -adapted and $\mathcal{F}_t^{\mathbf{Z}} \subseteq \mathcal{F}_t^{\mathbf{W}}$. The reverse inclusion can fail for a general adapted α , since recovering $\mathbf{W}$ from $\mathbf{Z}$ means solving $\mathbf{W}_t = \mathbf{Z}_t + \int_0^t \alpha ds$ for $\mathbf{W}$; but the two filtrations do coincide whenever α is deterministic, as it is in every application made here — α is constant in Section 2.2 — so that, after the augmentation of Convention 1.7, $(\mathcal{F}_t^{\mathbf{W}})_{t \in [0, T]} = (\mathcal{F}_t^{\mathbf{Z}})_{t \in [0, T]}$. Girsanov changes the measure and leaves the information alone, which is exactly what makes the two theorems composable.

1.2.6 Diffusions and the Kolmogorov Equations

Recall that, for a Markov process, the *transition probability* $P_{s,x}(t, A) = \mathbb{P}(X_t \in A \mid X_s = x)$, for A measurable, $0 \leq s \leq t \leq T$ and $x \in \mathbb{R}^n$. The Chapman–Kolmogorov equation states that

$$P_{s,x}(t, A) = \int_{\mathbb{R}^n} P_{u,z}(t, A) P_{s,x}(u, dz).$$

Definition 1.85 (Diffusion Process). A $\mathbb{R}^n$ -valued process X_t , $t \in [0, T]$ is called a **diffusion process** if its transition probabilities $\{P_{s,x}(t, \cdot)\}$ satisfy the following three conditions for every $t \in [0, T]$, every $x \in \mathbb{R}^n$ and every $\delta > 0$:

- i) $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| > \delta\}} P_{t,x}(t + \varepsilon, dy) = 0$
- ii) $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y - x) P_{t,x}(t + \varepsilon, dy) = \mu(t, x) \in \mathbb{R}^n$ exists,
- iii) $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y - x)(y - x)^\top P_{t,x}(t + \varepsilon, dy) = \Sigma(t, x) \in \mathbb{R}^{n \times n}$ exists,

where the vectors in $\mathbb{R}^n$ are implicitly taken as columns, so that $(y - x)(y - x)^\top$ yields a symmetric positive semi-definite matrix. μ and Σ are called the *drift* and the *diffusion coefficient*, respectively.

Read correctly, the three conditions say that in a short time ε the process moves a distance of order $\sqrt{\varepsilon}$ with overwhelming probability, that the mean of that displacement is of order ε , and



that its covariance is of order ε as well. When conditions ii) and iii) hold, the truncation at level δ turns out not to affect the limits, so we may and shall write them informally as

$$\mu(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[X_{t+\varepsilon} - x \mid X_t = x], \quad \Sigma(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[(X_{t+\varepsilon} - x)(X_{t+\varepsilon} - x)^\top \mid X_t = x].$$

Note how condition i) forbids the process from having jumps: it is precisely the statement that a displacement of size δ in time ε has probability $o(\varepsilon)$. Therefore the Poisson process, for which such a displacement has probability $\lambda\varepsilon + o(\varepsilon)$, is not a diffusion.

Example 1.86. Take an $\mathbb{R}^n$ -valued Brownian motion W_t . Note that $\|W_t\|^2$ is a sum of n independent squares of normal variables, so that $\mathbb{E}\|W_\varepsilon\|^4 = n(n+2)\varepsilon^2$. Hence, for any $c > 0$, Markov's inequality applied to the fourth power gives

$$\mathbb{P}(\|W_{t+\varepsilon} - x\| \geq c \mid W_t = x) = \mathbb{P}(\|W_{t+\varepsilon} - W_t\| \geq c) \leq c^{-4} \mathbb{E}\|W_\varepsilon\|^4 = c^{-4} n(n+2)\varepsilon^2,$$

which is $o(\varepsilon)$; taking $c = \delta$ gives condition i) of Definition 1.85 directly. It is therefore satisfied, and a direct computation of the remaining two limits gives $\mu(t, x) = 0$ and $\Sigma(t, x) = \mathbb{I}_{n \times n}$.

Theorem 1.87. *Let $\sigma(t, x)$ and $\mu(t, x)$ be functions as in Theorem 1.75. Assume they are also continuous on $[0, T] \times \mathbb{R}^n$. The solution to the SDE there defined is a diffusion process where $\mu(t, x)$ is identified and $\Sigma(t, x) = \sigma(t, x) \sigma(t, x)^\top$. $\square$*

The proof of this Theorem is beyond the scope of our purposes and can be found in [50].

We have reached a critical point where diffusions, SDEs and semigroups intersect. These seemingly unrelated mathematical objects give rise to a set of (parabolic) partial differential equations (*PDEs*) that are crucial for computational purposes and pricing financial derivatives. In this context, powerful mathematical tools can be applied.

The first of the approaches, where we consider a semigroup of operators, is included for completeness purposes, showing how its infinitesimal generators may be used to construct diffusion processes. It is also included to illustrate how Functional Analysis arises in this context.

1. Semigroup approach

We restrict to the **stationary case**, in which μ and Σ depend on x alone, and use standard Banach-space language; see [14].

A family $\{P_t\}_{t \geq 0}$ of linear operators on a Banach space $\mathbb{B}$ is a **contraction semigroup** if $P_0 = \mathbf{I}$ and $P_{t+s} = P_t P_s$ (semigroup property), $P_t v \rightarrow v$ as $t \downarrow 0$ for every v (strong continuity), and $\|P_t v\| \leq \|v\|$ (contraction). Its **infinitesimal generator** is

$$\mathcal{A}(v) = \lim_{t \downarrow 0} \frac{1}{t} (P_t(v) - v), \quad (1.79)$$

defined on those v for which the limit exists strongly; it is typically unbounded, as already for the translation semigroup $P_t f(x) = f(x+t)$ on $C_0[0, \infty)$, whose generator is $\mathcal{A}f = f'$. The choice of space is not incidental — strong continuity fails on the larger $C_b[0, \infty)$.

Given a stationary diffusion X_t driven by μ and Σ , one sets $P_t(f)(x) = \int_{\mathbb{R}^n} f(y) P_{0,x}(t, dy)$. Under mild regularity on the transition probabilities — essentially that mass neither escapes to infinity in finite time nor moves a macroscopic distance instantaneously, and that $P_t f$ is

continuous — the family $\{P_t\}$ is a contraction semigroup on $C_0(\mathbb{R}^n)$ and its generator is the second-order elliptic operator

$$\mathcal{A}(f)(x) = \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} + \sum_{i=1}^n \mu_i(x) \frac{\partial f}{\partial x_i}. \quad (1.80)$$

For Brownian motion, where $\Sigma = \mathbb{I}_{n \times n}$ and $\mu = 0$, this is $\mathcal{A}f = \frac{1}{2} \Delta f$.

We shall not develop the semigroup picture further, having obtained from it what we needed: the identification of the generator. It is worth recording, however, what the semigroup property actually encodes, because it is the first of the three approaches to fail in Part II. The identity $P_{t+s} = P_t \circ P_s$ is a restatement of the Markov property: the evolution over $t + s$ can be decomposed into evolution over s followed by evolution over t , because at the intermediate instant the present state is a sufficient statistic for the future. When the driving noise has memory this decomposition is unavailable, the family (P_t) is no longer a semigroup, and the generator ceases to exist as a local operator. Only the pathwise, integral formulation does, which is a large part of why the theory of Part II is built the way it is.

2. Partial Differential Equations Approach

Let $P_{s,x}(t, \cdot)$ be the transition probabilities of a diffusion X_t driven by μ and Σ , and let $F_{s,x}$ be the corresponding distribution function. The argument is a Taylor expansion inside the Chapman–Kolmogorov equation

$$F_{s,x}(t, y) = \int_{\mathbb{R}^n} F_{u,z}(t, y) dF_{s,x}(u, z),$$

and runs as follows. Fix t and y , put $u = s + \varepsilon$, and expand the integrand to second order in z about x . The zeroth-order term gives $F_{s+\varepsilon,x}$; the first-order term contributes $\nabla_x F_{s+\varepsilon,x} \cdot \mu(s, x) \varepsilon$, since by Definition 1.85(ii) the mean displacement over $[s, s+\varepsilon]$ is $\mu(s, x)\varepsilon + o(\varepsilon)$; and the second-order term contributes $\frac{1}{2} \text{Tr}(D_x^2 F_{s+\varepsilon,x} \Sigma(s, x)) \varepsilon$ by Definition 1.85(iii). Rearranging, dividing by ε and letting $\varepsilon \rightarrow 0$ yields

$$-\frac{\partial F_{s,x}}{\partial s} = \sum_{i=1}^n \mu_i \frac{\partial F_{s,x}}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij} \frac{\partial^2 F_{s,x}}{\partial x_i \partial x_j}, \quad (1.81)$$

together with the terminal condition

$$\lim_{s \uparrow t} F_{s,x}(t, y) = \mathbb{1}_{\{x \leq y\}}. \quad (1.82)$$

Equation (1.81) is called the **Kolmogorov backward equation**, because the variable s is the starting time, effectively moving backwards from the terminal time t . It has a converse that we shall not need until the next chapter but may as well announce here: equations of this type are solved by expectations along the diffusion, which is the content of the Feynman–Kac formula (Theorem 2.7 of Chapter 2), and (1.81) is its special case with no discounting.

We shall stay in the one-dimensional case for simplicity now. Assume the transition probabilities of a Markov process X_t possess density functions, that is,

$$P_{s,x}(t, dy) = p_{s,x}(t, y) dy, \quad (1.83)$$

so that the Chapman–Kolmogorov equation takes the form

$$p_{s,x}(t, y) = \int_{\mathbb{R}} p_{u,z}(t, y) p_{s,x}(u, z) dz. \quad (1.84)$$



By using this identity and some auxiliary test functions (integration by parts), it can be proven that

$$\frac{\partial p_{s,x}(t, y)}{\partial t} = -\frac{\partial(\mu(t, y) p_{s,x}(t, y))}{\partial y} + \frac{1}{2} \frac{\partial^2(\Sigma(t, y) p_{s,x}(t, y))}{\partial y^2} \quad (1.85)$$

which, together with the initial condition

$$\lim_{t \downarrow s} p_{s,x}(t, y) = \delta_x(y) \quad (1.86)$$

is known as either the **Kolmogorov forward equation** or the **Fokker–Planck equation**. This equation is much more useful in a financial context because it allows calculations *forward* in time, i.e., evolving from known current market variables.

Theorem 1.88. *Let Σ and μ satisfy the linear growth and Lipschitz conditions in x . Assume, additionally, that there exists $c > 0$ such that $w^\top \Sigma(t, y) w \geq c \|w\|^2$ for all $w \in \mathbb{R}^n$ (uniform ellipticity). Then:*

- a) *The Kolmogorov backward equation has a unique solution $F_{s,x}(t, y)$. Moreover, there exists a continuous Markov process X_t whose transition probabilities are recovered from it, $P_{s,x}(t, dy) = d_y F_{s,x}(t, y)$, the differential being taken in the second spatial argument.*
- b) *Assume additionally that the derivatives of μ and Σ up to second order satisfy the Lipschitz and linear growth conditions. Then the Kolmogorov forward equation has a unique solution.*

□

In terms of the infinitesimal generator, we can summarise the previous discussion in Table 1.3.

Forward Equation	Backward Equation
$\partial p / \partial t = \mathcal{A}^* p$	$-\partial u / \partial s = \mathcal{A} u$

Table 1.3: The two Kolmogorov equations in terms of the generator.

Here $\mathcal{A}^*$ represents the (Hermitic) adjoint of the generator $\mathcal{A}$; see [48] for the functional-analytic details. The backward equation propagates a *payoff* backwards from maturity and is the natural object for pricing a single contract, whereas the forward equation propagates a *density* forwards from today and is the natural object for calibration, since it evolves from known current market data. Dupire’s formula in the next chapter (Subsection 2.3.1) is precisely an application of the second. Figure 1.3 below shows the two evolutions for the geometric Brownian motion.

3. Stochastic Differential Equations Approach

In this case, take μ , Σ and σ such that $\sigma \sigma^\top = \Sigma$. Notice that σU is also a valid solution for any matrix U such that $U U^\top = \mathbb{I}_{n \times n}$ (orthogonal matrix). Therefore, the choice of σ is unique up to multiplication by an orthogonal matrix. Since $UW(t)$ is also a Brownian motion, the transition probabilities of X_t do not depend on the particular choice of σ within its class. We call this class of matrices the *square root* of Σ .

Now, suppose Σ and μ satisfy the linear growth and Lipschitz conditions. It is clear that any square root of Σ satisfies the growth condition. Moreover, if Σ is uniformly elliptic (bounded

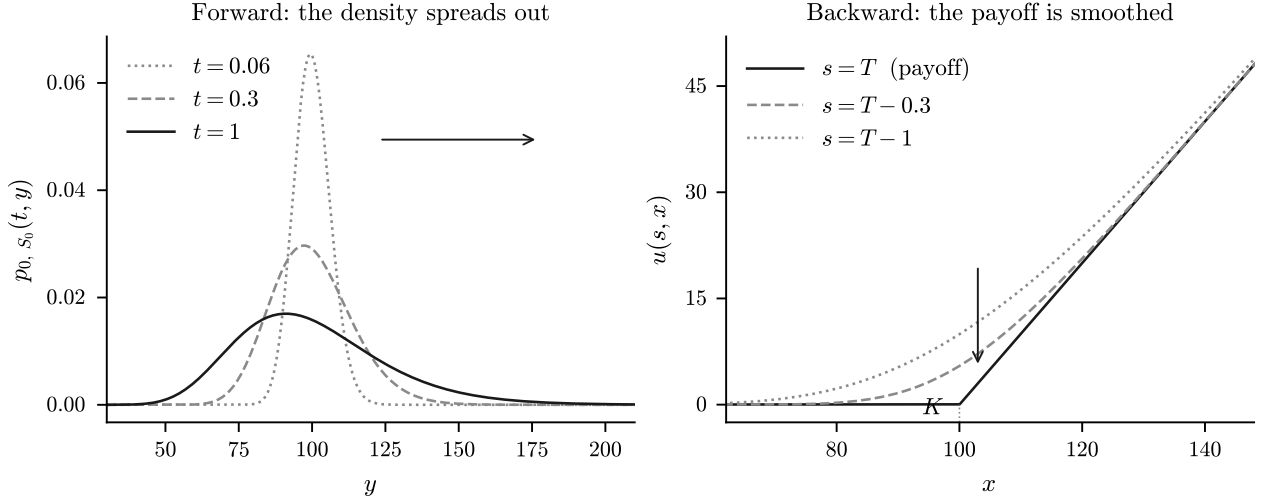


Figure 1.3: The two equations of Table 1.3, for geometric Brownian motion with $\sigma = 0.25$, $r = 0$ and $S_0 = 100$. *Left*: the forward (Fokker–Planck) equation (1.85) transports the density forwards from the point mass at S_0 , which spreads as t increases; this is the direction calibration works in, since S_0 is observed today. *Right*: the backward equation (1.81) transports the call payoff $(x - K)^+$ backwards from maturity, smoothing the kink at the strike as the remaining time $T - s$ grows; this is the direction pricing works in.

away from zero), σ also satisfies the Lipschitz conditions. In the 1D case, this can be seen via the inequality:

$$|\sigma(t, x) - \sigma(t, y)| = \frac{|\Sigma(t, x) - \Sigma(t, y)|}{|\sigma(t, x)| + |\sigma(t, y)|} \leq \frac{|\Sigma(t, x) - \Sigma(t, y)|}{2\sqrt{c}}. \quad (1.87)$$

Having this, we can directly state the main theorem corresponding to the SDE approach.

Theorem 1.89. *The unique solution to*

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ } \mathcal{F}_0\text{-measurable} \quad (1.88)$$

is a diffusion process, whose transition distribution function $F_{s,x}(t, y)$ satisfies the Kolmogorov backward equation. If the additional conditions on part b) of the previous Theorem are satisfied, then X_t has density functions $p_{s,x}(t, y)$, satisfying the corresponding forward equation. $\square$

Example 1.90. We illustrate the previous discussion with the example that will occupy the whole of Chapter 2. Take $n = 1$ and consider geometric Brownian motion with the short rate r as its drift $dX_t = rX_t dt + \sigma X_t dW_t$ on $(0, \infty)$, so that $\mu(x) = rx$ and $\Sigma(x) = \sigma^2 x^2$. By the theorem above its generator is

$$\mathcal{A}f(x) = \frac{1}{2}\sigma^2 x^2 f''(x) + rx f'(x), \quad (1.89)$$

and the backward equation $-\partial u / \partial s = \mathcal{A}u$ is, after adding the discounting effect, exactly the Black–Scholes equation (2.25) of Section 2.2. Its adjoint,

$$\mathcal{A}^*p(x) = \frac{1}{2}\frac{\partial^2}{\partial x^2}(\sigma^2 x^2 p(x)) - \frac{\partial}{\partial x}(rx p(x)), \quad (1.90)$$

governs the forward equation (1.85), and it is the object Dupire inverts in Subsection 2.3.1 to read a local volatility function off market data.



Notes and references

The theoretical framework of this chapter draws primarily from two authoritative sources. The general probability-theoretic material follows the classic exposition of P. Billingsley [8], while the construction of the stochastic integral is based on the comprehensive treatment by H. Kuo [50]. Throughout, the standard reference for the finer points — the Daniell–Kolmogorov and Kolmogorov continuity theorems, the strong Markov property and the reflection principle, and the construction of the essential supremum used in Appendix A.7 — is Karatzas and Shreve [48], which we cite at each of those points and recommend as the natural companion to this chapter and the next. Additionally, references for the underlying functional analysis, implicit in the PDE treatment of diffusions, can be found in the seminal work of H. Brezis [14].

Certain advanced topics, such as the general formulation of stochastic integration with respect to semimartingales or deeper pathwise properties of Brownian motion, have been omitted as they lie beyond the scope of this work. However, the selected contents provide a rigorous and sufficient foundation for the financial theory of Chapter 2 and for the developments of Part II.

Chapter 2

Financial Foundations and Classical Models

Chapter 1 built a machine and did not say much about what it is for. This chapter gives the necessary financial background for using that machine. Everything below rests on the stochastic calculus developed there — Itô’s rule, Girsanov’s theorem and the martingale representation theorem are used constantly, and the square-root process of Example 1.68 reappears as the variance of the Heston model — but the questions being asked are of a different kind. They are questions about *prices*, and the answers are constrained not by what is provable but by what the absence of arbitrage permits.

The chapter is nonetheless mathematical throughout, and deliberately so. The claim we shall keep returning to is that the financial content of this subject is not a layer of interpretation applied on top of the probability: it *is* the probability, read in a particular way. A change of numeraire is a change of measure; a hedging strategy is the integrand supplied by a representation theorem; a risk-neutral density is a second derivative of observed option prices.

2.1 Financial Derivatives

We first introduce the concept of a *derivative* informally and give some examples of the main quoted contracts. The rigorous mathematical treatment of hedging and market completeness is addressed afterwards.

2.1.1 The Concept of a Derivative

In the context of modern mathematical finance, a financial derivative is formally defined as a *contingent claim*. Its value is derived from—or “contingent” upon—the state of some other underlying variables.

Let us fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$, where $\mathbb{P}$ represents the physical (or real-world) probability measure and the filtration $(\mathcal{F}_t)_{t \in [0, T]}$ models the flow of information over time up to a finite horizon T . Mathematically, a financial derivative with maturity T is an $\mathcal{F}_T$ -measurable random variable, denoted by G_T . A few contracts, such as perpetual American options, carry an infinite time horizon ($T = \infty$); these fall outside our finite-horizon framework and form no core part of the derivatives market, so we exclude them.

This measurability condition implies that the value of the derivative at maturity is fully determined by the information available at time T . The function determining this value is often referred to as the *payoff function*, denoted by Φ , so that, for a claim written on the terminal spot, $G_T = \Phi(S_T)$. The key idea behind the subsequent theory is being able to assign *fair prices* to those contingent claims, deduced from quoted (and liquid) market data.

The prices of such claims are not independent of one another. A small set of standardised, highly liquid contracts carries enough information to determine the price of a great many bespoke ones, and the principle enforcing this consistency is the absence of **arbitrage**: informally, the impossibility of a strictly positive risk-free profit from zero net initial investment.

Real markets impose frictions — transaction costs, bid-ask spreads, asymmetric borrowing and lending rates, taxes, and the credit risk borne by whoever must be paid at maturity, which grows with the horizon — and these widen the identity just derived into a band rather than a point. But for the fungible assets and electronic markets we shall be concerned with, the band is tight. Other underlyings (such as commodities) operate with larger spreads, so the models must be adapted to the context.

The Underlying Asset

The process driving the derivative's value is the *underlying asset*, an $(\mathcal{F}_t)$ -adapted process $S = (S_t)_{0 \leq t \leq T}$; a derivative may depend on several. In practice S_t is the price of an equity or index, a bond or interest rate, a commodity such as oil, gold or electricity, or an exchange rate. Derivatives are then categorised by the functional form of their terminal payoff G_T .

1. Forwards These instruments possess a payoff that is a linear function of the underlying asset price at maturity. A prime example is the **forward contract**, an agreement to buy or sell the asset at time T for a pre-specified delivery price K . The payoff at time T for the holder of a long position is:

$$G_T = S_T - K \quad (2.1)$$

Note that this payoff takes values in $[-K, \infty)$ — it is bounded below only because $S_T \geq 0$ — so the holder carries a genuine liability whenever the asset price falls below K .

2. Options Options provide the *right*, but not the *obligation*, to transact. This asymmetry introduces convexity (non-linearity) into the payoff structure, truncating downside risk.

- **European call:** the right to buy S_T at the strike K . The payoff is defined as:

$$G_T = (S_T - K)^+ = \max(S_T - K, 0) \quad (2.2)$$

- **European put:** the right to sell S_T at the strike K . The payoff is defined as:

$$G_T = (K - S_T)^+ = \max(K - S_T, 0) \quad (2.3)$$

- **American options:** as their European counterparts, except that the option may be *exercised* at any time up to maturity T .

Derivatives enable **hedging**: they allow agents to transfer specific risks (e.g., volatility or tail risk) to those willing to bear them. In an idealised complete market, derivatives allow for



the replication of any payoff profile. Some participants take a position for pure speculation, while others genuinely need to hedge, looking for a safer and more efficient way to transfer risk. Collectively, derivatives markets act as an information network: the price of a derivative encodes a *risk-neutral* distribution for the underlying — not the participants' forecast of what will happen, but that forecast reweighted by the price of bearing risk — and Proposition 2.15 below makes that statement exact.

These concepts will be rigorously defined in the next subsection. However, the main question remains: what price should these contracts hold? Before entering the mathematical machinery, we can establish bounds based on physical intuition, much like conservation laws:

- If $T = 0$, meaning we are pricing a claim that expires immediately (and only depends on the underlying at maturity, not on its trajectory), the price must equal the payoff. For a European call with strike K and underlying S_0 , its price must be $(S_0 - K)^+$.
- If an asset gives its buyer strictly more optionality compared to another derivative, its price must be greater than or equal to the latter. For example, the price of an American call must be bounded below by the price of a European call with the same strike and maturity.

2.1.2 Derivative Pricing

We now turn to the rigorous problem of pricing contingent claims. While elementary treatments often assume asset prices follow geometric Brownian motions and interest rates are constant, these assumptions are insufficient for a general mathematical theory.

We work in a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ satisfying the usual conditions (right-continuity and completeness). The time horizon T is finite.

Let the market consist of $n + 1$ assets denoted by the vector process $S = (S^0, S^1, \dots, S^n)$. We assume that each component S^i is an **Itô process**. We distinguish the 0-th asset, S^0 , as the **numeraire**. While the most abstract fundamental theorems of asset pricing rely on the broader class of general semimartingales, Itô processes provide the precise mathematical framework required for the diffusion models prevalent in this dissertation.

Definition 2.1 (Numeraire). A numeraire is a process S^0 that is strictly positive ($S_t^0 > 0$ a.s. for all t) and non-dividend paying. It serves as the unit of account for the market.

We define the **discounted price process** X as the prices of the primary assets expressed in units of the numeraire:

$$X_t = \left(1, \frac{S_t^1}{S_t^0}, \dots, \frac{S_t^n}{S_t^0} \right) \quad (2.4)$$

Because the denominator S_t^0 is strictly positive almost surely, the transformation $f(x, y) = x/y$ is smooth. By a direct application of the multidimensional Itô's formula, the discounted price processes X_t^i are themselves Itô processes.

While S^0 is often chosen to be the bank account $S_t^0 = \exp\left(\int_0^t r_s ds\right)$, the theory holds for *any* valid Itô process numeraire (e.g., a foreign currency or a stock). The bank account is merely a convenient particular case because it explicitly models the time value of money.

This is the point at which to record two objects that will be used constantly. *Capitalising* one unit of currency forward from t to T at the short rate multiplies it by $\exp\left(\int_t^T r_s ds\right)$, which is

S_T^0/S_t^0 ; *discounting* is the inverse operation, and we write

$$D(t, T) := \exp\left(-\int_t^T r_s ds\right) = \frac{S_t^0}{S_T^0} \quad (2.5)$$

for the resulting **discount factor**. The **zero-coupon bond** $B(t, T)$ is a different kind of object: it is the price, agreed and paid at t , of a contract delivering one unit of currency at T . One is a calculation, the other a traded asset, and the two must fit together. Taking S^0 as numeraire and the payoff $G_T \equiv 1$,

$$B(t, T) = S_t^0 \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{S_T^0} \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}[D(t, T) \mid \mathcal{F}_t]. \quad (2.6)$$

The bond is therefore the *expected* discount factor, not the exponential of any single rate. When r is deterministic the expectation does nothing and the two coincide, $B(t, T) = D(t, T)$; that is the case assumed throughout Sections 2.2 and 2.3, and it is why a constant r lets us write $e^{-r(T-t)}$ everywhere without further comment. When r is stochastic the two genuinely differ, discounting no longer factorises out of an expectation, and one switches to the T -forward measure of Definition 2.10 — the measure in which the bond, rather than the bank account, is the unit of account, so that discounting becomes a multiplication by $B(t, T)$ again.

A **trading strategy** is a predictable process $\pi = (\pi^0, \pi^1, \dots, \pi^n)$, where π_t^i represents the number of units of asset i held at time t . The value of the portfolio at time t is the scalar process:

$$V_t(\pi) = \sum_{i=0}^n \pi_t^i S_t^i \quad (2.7)$$

For the strategy to be economically meaningful, it must be **self-financing**. In the language of stochastic integration, this implies that changes in wealth are generated solely by the fluctuations of the asset prices, with no external injection or extraction of capital:

$$dV_t(\pi) = \sum_{i=0}^n \pi_t^i dS_t^i \quad (2.8)$$

When working with discounted prices X , the self-financing condition simplifies. Applying Itô's product rule, it can be shown that a strategy is self-financing if and only if the discounted wealth process V_t/S_t^0 satisfies:

$$\frac{V_t(\pi)}{S_t^0} = \frac{V_0(\pi)}{S_0^0} + \int_0^t \sum_{i=1}^n \pi_u^i dX_u^i = \frac{V_0}{S_0^0} + (\pi \cdot X)_t \quad (2.9)$$

where $(\pi \cdot X)$ denotes the vector stochastic integral.

In continuous time the naive definition of arbitrage is insufficient, because “doubling strategies” — strategies whose intermediate losses are unbounded below — can manufacture a certain profit from zero capital within a finite horizon. To preclude this we restrict the admissible strategies.

Definition 2.2 (Admissible Strategy). A self-financing strategy π is called a -admissible if its discounted value process is uniformly bounded from below by a constant $-a$ almost surely (i.e., $(\pi \cdot X)_t \geq -a$ for all t). A strategy is admissible if it is a -admissible for some $a \geq 0$.

We can now say precisely what is to be excluded. Let $\mathcal{K}$ be the set of random variables representing the terminal wealth of all admissible strategies starting with zero capital:

$$\mathcal{K} = \{(\pi \cdot X)_T \mid \pi \text{ is admissible and } V_0(\pi) = 0\} \quad (2.10)$$



The market admits **no arbitrage** (NA) if $\mathcal{K} \cap L_+^0 = \{0\}$; that is, no admissible strategy starting from nothing finishes non-negative and strictly positive with positive probability. This is the classical condition, and in continuous time it is too weak — not because it tolerates an outright arbitrage, but because the set it constrains is not closed, so that a *limit* of near-arbitrages may escape it. Delbaen and Schachermayer [25] showed that the condition which is genuinely equivalent to the existence of an equivalent martingale measure is *No Free Lunch with Vanishing Risk*. Writing $\mathcal{C}$ for the cone of claims dominated by elements of $\mathcal{K}$, that is $\mathcal{C} = \{g \in L^\infty(\mathcal{F}_T) \mid \exists f \in \mathcal{K}, g \leq f\}$, the market satisfies **NFLVR** if:

$$\bar{\mathcal{C}} \cap L_+^\infty = \{0\} \quad (2.11)$$

where $\bar{\mathcal{C}}$ is the closure of $\mathcal{C}$ in the L^∞ norm. Intuitively, this condition prevents not just exact arbitrage, but also sequences of strategies that approximate an arbitrage opportunity with risk converging to zero.

We are now equipped to state the core result linking market topology to measure theory.

Definition 2.3 (Equivalent Local Martingale Measure). A probability measure $\mathbb{Q}$ is an *equivalent local martingale measure* (ELMM) for the numeraire S^0 if:

1. $\mathbb{Q} \sim \mathbb{P}$ (they share the same null sets).
2. The discounted price process $X_t = S_t/S_t^0$ is a **local martingale** under $\mathbb{Q}$.

Theorem 2.4 (First Fundamental Theorem of Asset Pricing). *Assume the process X is locally bounded. The market satisfies the NFLVR condition if and only if there exists at least one Equivalent Local Martingale Measure $\mathbb{Q}$.* $\square$

The other reasonable question to ask ourselves is whether every contingent claim can be replicated by a self-financing strategy. Before stating the second theorem, we must rigorously define market completeness.

Definition 2.5 (Market Completeness). A market is **complete** if every claim $G_T \in L^\infty(\mathcal{F}_T)$ is *attainable*. That is, for every bounded random variable G_T , there exists an admissible self-financing strategy π and an initial capital $x \in \mathbb{R}$ such that:

$$\frac{G_T}{S_T^0} = x + (\pi \cdot X)_T \quad \mathbb{P}\text{-almost surely.} \quad (2.12)$$

Note that the statement is made under the physical measure $\mathbb{P}$. This is deliberate: at this stage $\mathbb{Q}$ has not been shown to be unique — indeed its uniqueness is exactly what the next theorem characterises — so quantifying over “ $\mathbb{Q}$ -almost surely” would be ambiguous. Since every equivalent local martingale measure shares its null sets with $\mathbb{P}$, the choice is harmless, but it is the honest one.

Completeness is defined above by quantifying over $L^\infty(\mathcal{F}_T)$, which is the Delbaen–Schachermayer convention and the right one for the Fundamental Theorems, boundedness being what makes their separating-hyperplane argument work. The Martingale Representation Theorem 1.83, by contrast, is an L^2 statement, because it is at bottom an assertion about an orthogonal projection in a Hilbert space. The two are reconciled by the inclusion $L^\infty(\mathcal{F}_T) \subset L^2(\mathcal{F}_T)$, valid because $\mathbb{P}$ is a finite measure: every bounded claim is square-integrable, so the representation applies to every claim the definition above quantifies.

Theorem 2.6 (Second Fundamental Theorem of Asset Pricing). *The market is complete if and only if the set of equivalent local martingale measures $\mathcal{M}^e$ is a singleton (i.e., there exists a unique $\mathbb{Q}$).* $\square$

If the market is arbitrage-free and complete (i.e., there exists a unique $\mathbb{Q}$), the fair price at time t of a generic derivative paying G_T at maturity is given by the expectation of the discounted payoff:

$$\frac{V_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}} \left[\frac{G_T}{S_T^0} \middle| \mathcal{F}_t \right] \implies V_t = S_t^0 \mathbb{E}^{\mathbb{Q}} \left[\frac{G_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad (2.13)$$

This formulation reveals that “risk-neutral probabilities” are not absolute; they are tied to the choice of numeraire.

Let $\widetilde{S}^0$ be a different numeraire (strictly positive semimartingale). Let $\mathbb{Q}^{S^0}$ be the measure associated with numeraire S^0 , and $\mathbb{Q}^{\widetilde{S}^0}$ the measure associated with $\widetilde{S}^0$. Prices must be consistent regardless of the unit of account. Therefore, for any asset Z :

$$\frac{Z_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}^{S^0}} \left[\frac{Z_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad \text{and} \quad \frac{Z_t}{\widetilde{S}_t^0} = \mathbb{E}^{\mathbb{Q}^{\widetilde{S}^0}} \left[\frac{Z_T}{\widetilde{S}_T^0} \middle| \mathcal{F}_t \right] \quad (2.14)$$

The change of measure from $\mathbb{Q}^{S^0}$ to $\mathbb{Q}^{\widetilde{S}^0}$ is defined by the Radon–Nikodym derivative, which turns out to be exactly the normalised ratio of the numeraires:

$$\frac{d\mathbb{Q}^{\widetilde{S}^0}}{d\mathbb{Q}^{S^0}} \bigg|_{\mathcal{F}_T} = \frac{\widetilde{S}_T^0 / S_T^0}{\widetilde{S}_0^0 / S_0^0} \quad (2.15)$$

This elegantly shows that changing the numeraire is mathematically equivalent to performing a change of probability measure. This technique is particularly powerful in interest rate theory (e.g., switching to the T -forward measure to efficiently price bond options and swaptions).

Finally, we can now state the Feynman–Kac formula, which provides a powerful link between the PDE approach and the probabilistic representation of derivative prices.

Theorem 2.7 (Feynman–Kac Formula). *Let X be an n -dimensional Itô diffusion process on the interval $[t, T]$ driven, under a measure $\mathbb{Q}$, by the stochastic differential equation*

$$dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dW_s, \quad s \in [t, T], \quad (2.16)$$

with initial condition $X_t = x$, where W is a standard n -dimensional $\mathbb{Q}$ -Brownian motion, and the diffusion matrix is given by $\Sigma(s, x) = \sigma(s, x)\sigma(s, x)^\top$. Let $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous terminal payoff function — continuity is forced by the regularity demanded of u below — and $r : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a bounded continuous function representing the continuously compounded interest rate. Assume $u(t, x)$ is a sufficiently regular function (typically $u \in C^{1,2}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$) that satisfies the following linear parabolic partial differential equation:

$$\frac{\partial u}{\partial t}(t, x) + \sum_{i=1}^n \mu_i(t, x) \frac{\partial u}{\partial x_i}(t, x) + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(t, x) \frac{\partial^2 u}{\partial x_i \partial x_j}(t, x) - r(t, x)u(t, x) = 0, \quad (2.17)$$

for all $(t, x) \in [0, T] \times \mathbb{R}^n$, subject to the terminal condition:

$$u(T, x) = \Phi(x). \quad (2.18)$$



If $u(t, x)$ satisfies a suitable polynomial growth condition preventing the existence of explosive solutions, then $u(t, x)$ admits the stochastic representation:

$$u(t, x) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s, X_s) ds} \Phi(X_T) \middle| X_t = x \right]. \quad (2.19)$$

□

Two further remarks fix the place of the theorem. The first is that it subsumes material already derived: the Kolmogorov backward equation (1.81) of Chapter 1 is exactly (2.17) with $r \equiv 0$ and terminal condition $\Phi = \mathbb{1}_{\{\cdot \leq y\}}$, so the transition distribution function of a diffusion is itself a Feynman–Kac representation — one in which the “payoff” is the indicator of ending below a level and the “price” is the probability of doing so. The second is what makes the theorem a cornerstone of this chapter rather than a technical lemma. Read from (2.17) to (2.19) it turns a pricing equation into a discounted expectation under a martingale measure; read in the opposite direction it turns the expectation (2.13), which is what the absence of arbitrage delivers, into a deterministic PDE that the price must satisfy.

2.2 The Black–Scholes Framework

Having established the general theory of arbitrage-free pricing and martingale measures, we now specialise our market model to the most famous and widely used framework: the Black–Scholes–Merton model, usually called the Black–Scholes model, or simply the BS model.

2.2.1 The Model

We assume a frictionless market: no transaction costs, no bid–ask spread, unrestricted short selling, and a constant risk-free rate r at which one may borrow and lend freely. Under the *physical* measure $\mathbb{P}$ — the measure describing what actually happens — the underlying is modelled by a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (2.20)$$

where $\mu \in \mathbb{R}$ is the expected rate of return, $\sigma > 0$ the volatility, and W a $\mathbb{P}$ -Brownian motion. By Example 1.61 the solution is $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$, which is positive for all t . Inside the model, σ is the diffusion coefficient of the *log*-price: by Example 1.61, $d \log S_t = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW_t$, so that $\mathbb{V}[\log S_t] = \sigma^2 t$. Within this subsection σ is a *parameter* of the model and nothing else; the implied volatility of Definition 2.13 is a separate object, and Subsection 2.2.3 keeps the two apart.

The hedging argument

Consider a derivative whose value is a smooth function $V(t, S)$ of time and the current spot. Form the portfolio

$$\Pi_t = -V(t, S_t) + \Delta_t S_t,$$

consisting of a short position in the derivative and Δ_t units of the underlying. Assuming the position is self-financing in the sense of (2.8), its value evolves as

$$d\Pi_t = -dV_t + \Delta_t dS_t. \quad (2.21)$$

Itô's rule (Theorem 1.59) applied to $V(t, S_t)$, using $(dS_t)^2 = \sigma^2 S_t^2 dt$ from Table 1.1, gives

$$dV_t = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t. \quad (2.22)$$

Substituting (2.20) and (2.22) into (2.21) and collecting terms,

$$d\Pi_t = \underbrace{\left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \mu S_t \left[\Delta_t - \frac{\partial V}{\partial S} \right] \right)}_{\text{drift}} dt + \underbrace{\sigma S_t \left(\Delta_t - \frac{\partial V}{\partial S} \right)}_{\text{diffusion}} dW_t. \quad (2.23)$$

The portfolio is exposed to randomness only through the second term. Choosing

$$\Delta_t = \frac{\partial V}{\partial S}(t, S_t) \quad (2.24)$$

annihilates it. This is the *delta* of the derivative, and (2.24) is the hedge.

Now observe what (2.23) has become. The same choice (2.24) that kills the diffusion term also kills the bracket multiplying μ in the drift. **The expected return of the stock has cancelled**, so a position that is neutral to the random part of S is automatically neutral to its drift as well. What remains,

$$d\Pi_t = \left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt,$$

is riskless. A riskless self-financing portfolio must earn the risk-free rate — otherwise, by borrowing or lending against it, one manufactures an arbitrage in the sense of Subsection 2.1.2 — so $d\Pi_t = r\Pi_t dt$ with $\Pi_t = -V + S_t \partial V / \partial S$. Equating the two expressions and rearranging yields the Black–Scholes equation.

Theorem 2.8 (Black–Scholes–Merton). *Under the dynamics (2.20), the value $V(t, S)$ of a European derivative with payoff $V(T, S) = \Phi(S)$ satisfies the linear parabolic partial differential equation*

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \quad (t, S) \in [0, T) \times (0, \infty), \quad (2.25)$$

together with the terminal condition $V(T, S) = \Phi(S)$.

Proof: The hedging argument above. ■

Equation (2.25) contains r but not μ , which is the claim made above, now proved. It is also exactly the Feynman–Kac equation of Theorem 2.7 for a diffusion with drift rS and diffusion coefficient σS , whence the probabilistic representation

$$V(t, S) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S], \quad dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (2.26)$$

The passage from (2.20) to the dynamics in (2.26) is Girsanov's theorem with $-\alpha = (\mu - r)/\sigma$ the *market price of risk*: the measure change subtracts a drift and leaves σ untouched. Its hypothesis is met here without effort: α is a constant, so $\mathbb{E}\left[\exp\left(\frac{1}{2} \int_0^T \alpha^2 ds\right)\right] = e^{\alpha^2 T/2} < \infty$ and Novikov's condition (Theorem 1.78) holds, whence $\mathcal{E}_\alpha$ is a genuine martingale and $\mathbb{Q}$ is a probability measure equivalent to $\mathbb{P}$. That the hedging argument and the martingale argument give the same answer is not a coincidence but the content of Feynman–Kac.

Finally, the market is complete: by Theorem 1.83 and Remark 1.84, the filtration is generated by the single Brownian motion driving the single traded asset, so every square-integrable claim is attainable and $\mathbb{Q}$ is unique. The delta (2.24) is the integrand π whose abstract existence the representation theorem guarantees; here we have it explicitly.



The closed-form solution

For a European call, $\Phi(S) = (S - K)^+$, evaluating (2.26) against the log-normal density gives

$$C_{\text{BS}}(t, S) = S N(d_+) - K e^{-r(T-t)} N(d_-), \quad d_{\pm} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad (2.27)$$

with $\tau = T - t$ and N the standard normal distribution function. The corresponding put price follows from put–call parity, Proposition 2.12 below.

One might instead assume **normal** (Bachelier) dynamics $dS_t = \mu dt + \sigma dW_t$, as in the original 1900 thesis [3]. This permits negative prices, which is why it was criticised for equities, but it is standard for interest rates and for spread options where negative values are economically meaningful. We do not include the derivation here, but the Bachelier formula for a call is

$$C_{\text{Bach}}(t, S) = (S - K)N(d) + \sigma\sqrt{\tau}N'(d), \quad d = \frac{S - K}{\sigma\sqrt{\tau}}, \quad (2.28)$$

with N' the standard normal density. The Bachelier formula is a special case of the general Feynman–Kac representation (2.19) with $r \equiv 0$ and $\Phi(S) = (S - K)^+$.

The Greeks

The delta (2.24) was singled out above as the hedge, and identified in Remark 1.84 with the integrand π whose existence the Martingale Representation Theorem guarantees. It is not the only sensitivity that matters, and since the whole of Subsection 2.2.4 is an argument about one of the others, we record them here. Write $\tau = T - t$ and let $N'(x) = e^{-x^2/2}/\sqrt{2\pi}$ be the standard normal density.

Proposition 2.9 (Black–Scholes sensitivities). *For the European call (2.27) on a non-dividend-paying underlying,*

$$\Delta := \frac{\partial C}{\partial S} = N(d_+), \quad \Gamma := \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_+)}{S\sigma\sqrt{\tau}}, \quad (2.29)$$

$$\mathcal{V} := \frac{\partial C}{\partial \sigma} = S N'(d_+)\sqrt{\tau}, \quad \theta := \frac{\partial C}{\partial t} = -\frac{S N'(d_+)\sigma}{2\sqrt{\tau}} - r K e^{-r\tau} N(d_-). \quad (2.30)$$

□

The names are delta, gamma, vega and theta respectively. Two identities among these are worth more than the formulae themselves. The first is that Γ and $\mathcal{V}$ are proportional,

$$\mathcal{V} = S^2 \sigma \tau \Gamma, \quad (2.31)$$

as one verifies immediately from (2.29) and (2.30). Convexity in the spot and sensitivity to volatility are therefore the *same* exposure, measured in two units — which is the analytic content of the informal statement that an option buyer is long volatility precisely because the payoff is convex. The second is that substituting (2.29) into the Black–Scholes equation (2.25) rearranges it into

$$\theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r(C - S\Delta), \quad (2.32)$$

which says that for a delta-hedged position (Δ neutralised, so that the right-hand side is just the financing of the residual cash) the time decay θ and the convexity Γ are in exact opposition. This

is the whole economics of running an option book: the gamma that makes the position profit from movement is paid for, day by day, by theta, and (2.32) fixes the exchange rate between them. Equation (2.25) is usually read as a pricing equation; (2.32) is the same equation read as a statement about a hedged portfolio, and it is the form a trader would recognise.

Vega is strictly positive, so a call is worth more when volatility is higher; and it is *not* a sensitivity to a quantity the model admits to be uncertain — in (2.20) the parameter σ is a constant. Differentiating a price with respect to a constant is a symptom that the model is being used outside its own assumptions.

2.2.2 The Forward and Put–Call parity

Suppose first that dividends are paid *continuously* at a constant proportional rate q . The holder of the stock receives $qS_t dt$ over $[t, t + dt]$, so the self-financing condition (2.21) acquires the extra term $\Delta_t q S_t dt$; repeating the argument verbatim replaces the coefficient r of $\partial V / \partial S$ in (2.25) by $r - q$, and r by $r - q$ in $d_{\pm}$. Nothing structural changes and the discounted-spot term acquires its own factor:

$$C(S, \tau) = S e^{-q\tau} N(d_+) - K e^{-r\tau} N(d_-), \quad d_{\pm} = \frac{\log(S/K) + (r - q \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \quad (2.33)$$

Actual equity dividends are *discrete cash and proportional amounts*, announced in advance and paid on known dates. If a cash amount D_i is paid at time t_i together with a proportional dividend d_i , then no-arbitrage forces the spot to drop by that amount across the payment:

$$S_{t_i} = (1 - d_i)S_{t_i^-} - D_i. \quad (2.34)$$

This is genuinely destructive of the framework above. Equation (2.34) says S has a *predictable jump*: the process is not continuous, so Itô's rule as stated does not apply across t_i . Worse, one cannot write $S_t = S_0 \mathcal{E}(\cdot)$ for a positive $\mathcal{E}$ and reproduce (2.34). Also, a fixed cash dividend can in principle exceed the spot price, so a model treating D_i as deterministic does not even guarantee positivity. And since σ multiplies S_t in (2.20), a discontinuous drop in S produces a discontinuous drop in the absolute volatility, which is not what is observed.

The resolution is to stop modelling the spot and model the *forward* instead.

Definition 2.10 (Forward price). The **forward price** $F(t, T)$ is the delivery price agreed at t for a transaction in the underlying at T which has zero value at inception. Equivalently, writing $B(t, T)$ for the price at t of a default-free zero-coupon bond maturing at T , a forward contract struck at K has time- t value

$$V_t(K) = B(t, T) \mathbb{E}^{\mathbb{Q}^T} [S_T - K \mid \mathcal{F}_t], \quad (2.35)$$

where $\mathbb{Q}^T$ is the T -forward measure associated with the numeraire $B(\cdot, T)$. Since $B(t, T) > 0$, the requirement that the contract be costless at inception, $V_t(F(t, T)) = 0$, determines the delivery price uniquely:

$$F(t, T) = \mathbb{E}^{\mathbb{Q}^T} [S_T \mid \mathcal{F}_t]. \quad (2.36)$$

Identity (2.36) says that the forward price is a $\mathbb{Q}^T$ -martingale. It holds by construction, with no assumption whatever on interest rates, on dividends, or on their nature. In particular we



deliberately do *not* write $F(t, T) = S_t e^{(r-q)(T-t)}$, which is valid only for constant r , constant q , and purely proportional dividends. Instead we record the relation abstractly, as an **affine** one:

$$F(t, T) = a_t S_t + b_t, \quad a_t > 0. \quad (2.37)$$

Both coefficients are adapted and model-independent: a_t is the *proportional* part, assembled from the discount curve, the repo or borrow cost and any proportional dividend yield, while b_t is the *cash* part, carrying the announced dividends. Their precise construction is a market-data question, not a modelling one. What matters here is that (2.37) remains valid for time-dependent $r(t)$ and $q(t)$, for discrete cash dividends, for proportional dividends, and for any mixture of them; the purely proportional case of (2.33) is $b_t = 0$. Modelling the forward as a driftless log-normal martingale,

$$dF_t = \sigma F_t dW_t^T, \quad (2.38)$$

therefore has three advantages simultaneously. The dividend discontinuities are absorbed into b_t and never touch the dynamics, which stay diffusive. The drift vanishes, because a forward is already a martingale under its own measure — so there is no drift to specify and none to get wrong. And the resulting formula is calibrated directly against quantities the market quotes.

Thus, we can transform the formula $(S_t - K)^+ = ((F_t - b_t)/a_t - K)^+ = (F_t - (Ka_t + b_t))^+/a_t$ and price the call on the forward, which is continuous and log-normal, rather than on the spot, which is not. The calibration is therefore much simpler, since we only need to work with the forward implied volatility. The result is the celebrated Black-76 formula [10].

Proposition 2.11 (Black-76 [10]). *Under (2.38), the time- t value of a European call with strike K and maturity T is*

$$C(t) = B(t, T) \left[F_t N(d_+) - K N(d_-) \right], \quad d_{\pm} = \frac{\ln(F_t/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad \tau = T - t. \quad (2.39) \quad \square$$

Put–call parity

Before turning to implied volatility we record the one relation in this subject that holds independently of any model, and which we shall invoke repeatedly.

Proposition 2.12 (Put–call parity). *Let $C(t)$ and $P(t)$ be the time- t prices of European call and put options on the same underlying, with the same strike K and maturity T . Then, in any arbitrage-free market,*

$$C(t) - P(t) = B(t, T)(F(t, T) - K). \quad (2.40)$$

Proof: The payoffs satisfy the pointwise identity $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, valid for every value of S_T . Taking $\mathbb{E}^{\mathbb{Q}^T}[\cdot | \mathcal{F}_t]$ and multiplying by $B(t, T)$, the right-hand side becomes $B(t, T)(F(t, T) - K)$ by (2.36). ■

The proof used no dynamics, only the linearity of expectation and an algebraic identity between payoffs. Consequently (2.40) must hold in every model in this dissertation, and it is the standard sanity check on any implementation: a pricer that violates put–call parity is wrong.

2.2.3 Implied Volatility and the Inversion Problem

In professional markets, participants do not “price” options using Black–Scholes to find the dollar value. The market price C^{mkt} is determined by supply and demand. Instead, Black–Scholes is used as a translation mechanism. Since vega is strictly positive (Proposition 2.9),

the pricing function $\sigma \mapsto C_{\text{BS}}(\sigma)$ is strictly *increasing*, and it sweeps out the whole no-arbitrage interval as σ runs over $(0, \infty)$; any quote inside that interval therefore has exactly one preimage.

Definition 2.13 (Implied volatility). Let C^{mkt} be an observed European call price with strike K and maturity T , lying strictly inside the no-arbitrage interval. Its **implied volatility** $\sigma_{\text{imp}}(K, T)$ is the unique $\sigma > 0$ solving

$$C_{\text{BS}}(S, K, T, r, q, \sigma) = C^{\text{mkt}}, \quad (2.41)$$

the dividend yield q entering through (2.33). The map $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$ is the **implied volatility surface**, and it is the object Chapter 4 calibrates against.

Implied volatility, so defined, is the true “price” quoted in the market: a trader who says an option “trades at fifteen” is quoting σ_{imp} , and the dollar amount is recovered from (2.41). Note what the definition does *not* claim. It does not assert that the underlying follows (2.20); it uses C_{BS} only as a strictly monotone change of units, and it would remain well posed even if every modelling assumption of Subsection 2.2.1 were false — as, by the next subsection, they are.

Implied against realised volatility

The word “volatility” is by now doing three quite different jobs, and since the remainder of the dissertation turns on telling them apart, we separate them here once and for all. There is σ , a *parameter* of the model (2.20), constant by assumption and not observable. There is σ_{imp} , a *quote*: read off an observed price through Definition 2.13, and varying with strike and maturity precisely because the model that defines it is false. And there is a *statistic* of the trajectory that actually occurred, which is the object we now make precise.

Definition 2.14 (Realised volatility). Suppose the log-price has instantaneous variance σ_t^2 , so that $d \log S_t = \mu_t dt + \sigma_t dW_t$ for some adapted drift μ_t , which plays no part in what follows. The **realised variance** over $[0, T]$ is the time average

$$\sigma_{\text{real}}^2[0, T] := \frac{1}{T} \int_0^T \sigma_t^2 dt = \frac{1}{T} \langle \log S \rangle_T, \quad (2.42)$$

and σ_{real} is the **realised volatility**. Given prices $S_0, \dots, S_n$ observed at spacing $\Delta = T/n$, it is estimated by

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{i=1}^n \left(\log \frac{S_i}{S_{i-1}} \right)^2. \quad (2.43)$$

Two conventions are worth making explicit. No mean is subtracted, the drift contributing only $O(\Delta)$ to each term; and the normalisation is by T rather than by the number of observations, so that $\hat{\sigma}^2$ estimates a variance *per unit time* and is therefore on the same footing as σ_{imp}^2 .

The second equality in (2.42) is not a definition but a theorem of Chapter 1: by Itô’s rule the quadratic variation of $\log S$ is $\int_0^T \sigma_t^2 dt$, and by Definition 1.31 and Proposition 1.33 that quantity is the limit of the sums of squared increments as the mesh refines. The estimator (2.43) is therefore not a heuristic: it is the pathwise object of Subsection 1.1.4 evaluated on data, and its convergence as $\Delta \rightarrow 0$ is Proposition 1.33 applied to $\log S$.

Realised volatility is a functional of the trajectory that actually occurred: a statistic under $\mathbb{P}$. Implied volatility is extracted from a *price*, and prices are expectations under $\mathbb{Q}$ by (2.13). By Girsanov’s theorem (Theorem 1.81) the two measures agree on the diffusion coefficient of any *given* model but they do not agree on the price of *bearing* volatility, and that price is inside σ_{imp} .



2.2.4 Smile and Skewness

If the assumptions of Black–Scholes were satisfied physically (i.e., if returns were truly log-normal with constant variance), the implied volatility σ_{imp} would be constant across all strike prices K and maturities T . Empirical data reveals this is not the case. When we plot $\sigma_{\text{imp}}(K)$ against K , we observe distinct shapes:

- **The Skew (Smirk):** Typical in equity markets. σ_{imp} decreases as K increases. This reflects “Crash Phobia”—investors bid up the price of low-strike Puts (insurance against market drops), implying higher volatility for downside strikes.
- **The Smile:** Typical in Forex markets. σ_{imp} is lowest at the money ($K \approx F$) and rises for strikes far from the money in *both* directions. This reflects “fat tails” (leptokurtosis) in the return distribution: extreme moves in either direction are more probable than the Gaussian model predicts.

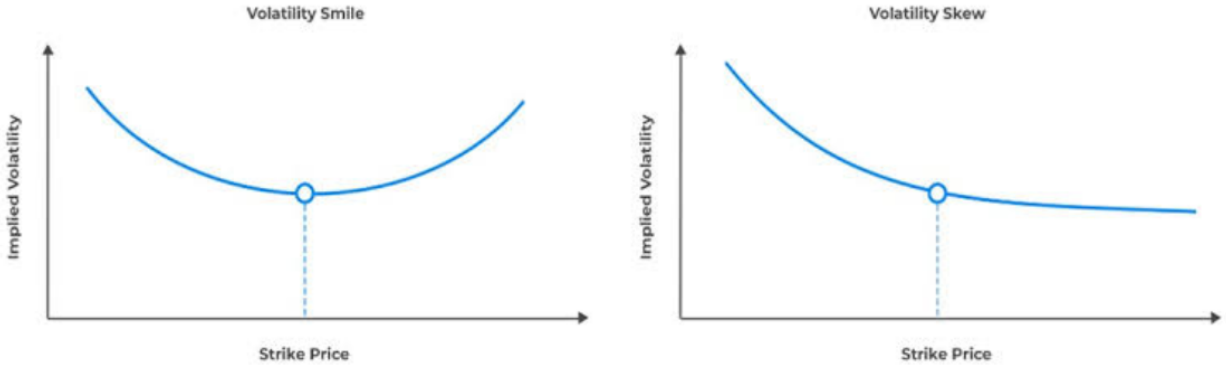


Figure 2.1: The two shapes described above. Left: the usual foreign exchange smile, symmetric about the money and rising into both wings. Right: the typical equity skew, falling monotonically in the strike, so that downside protection carries the highest implied volatility. The curves are illustrative shapes and not drawn from market quotes.

The two shapes, drawn in Figure 2.1, are not merely descriptive. Both say that the market’s risk-neutral distribution is not log-normal, and the following classical result makes the connection exact.

Proposition 2.15 (Breed–Litzenberger [13]). *Let $C(K, T)$ denote the price of a European call of strike K and maturity T , and suppose the law of S_T under the T -forward measure $\mathbb{Q}^T$ admits a density p_T — the risk-neutral density itself whenever rates are deterministic, since $\mathbb{Q}^T = \mathbb{Q}$ in that case. Then*

$$p_T(K) = \frac{1}{B(0, T)} \frac{\partial^2 C}{\partial K^2}(K, T). \quad (2.44)$$

Proof: Write $C(K, T) = B(0, T) \int_K^\infty (x - K) p_T(x) dx$. Differentiating once under the integral sign, the boundary term vanishes because the integrand is zero at $x = K$, thus obtaining $\partial_K C = -B(0, T) \int_K^\infty p_T(x) dx$. Differentiating a second time gives (2.44). ■

The result is remarkable for what it does not assume: no model, no dynamics, only that prices are discounted expectations of the payoff. A continuum of option quotes at a fixed maturity therefore determines the entire marginal distribution of S_T — the market does not merely

express an opinion about volatility, it expresses a complete probability law, and the smile is the visible trace of that law failing to be log-normal.

Two consequences of (2.44) will be used constantly. First, since $p_T \geq 0$, arbitrage-free call prices must be *convex* in strike; a violation is a butterfly arbitrage, executable by buying two wing options and selling two at the middle strike. Second, the formula involves a *second* derivative of market data, which is a numerically delicate problem.

2.3 Other Relevant Models

To account for the volatility smile and skew, we must extend the Black–Scholes framework. There are two primary approaches: making volatility a deterministic function of the state (Local Volatility) or making it a stochastic process (Stochastic Volatility).

2.3.1 Local Volatility Models

The Local Volatility (LV) framework, introduced by Dupire [26], posits that volatility is a deterministic function of the current asset price and time: $\sigma_{\text{loc}} = \sigma_{\text{loc}}(t, S_t)$. The dynamics under the risk-neutral measure are:

$$dS_t = (r - q) S_t dt + \sigma_{\text{loc}}(t, S_t) S_t dW_t \quad (2.45)$$

This model preserves the completeness of the market (there is only one source of randomness W_t) while allowing the model to fit *any* observed arbitrage-free implied volatility surface exactly.

The Dupire formula and Gatheral’s transformation

Dupire’s insight was to link the unknown function $\sigma_{\text{loc}}(K, T)$ directly to observed market option prices. Applying the Fokker–Planck equation (1.85) to the transition density of the diffusion and combining it with Breeden–Litzenberger yields the following.

Theorem 2.16 (Dupire). *Suppose call prices $C(K, T)$ are known for a continuum of strikes and maturities and are consistent with a diffusion of the form $dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t$. Then the local volatility function is uniquely determined by*

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + (r - q) K \frac{\partial C}{\partial K} + q C}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}. \quad (2.46)$$

□

Therefore, there is exactly one local volatility diffusion consistent with a given arbitrage-free implied volatility surface. The calibration problem has a unique solution, not a family of them, and the model can therefore reproduce every European quote simultaneously and exactly. Note also that the denominator is proportional to the risk-neutral density by Proposition 2.15, so (2.46) is well posed precisely when that density is strictly positive.



Static arbitrage and the implied-variance parameterisation

Formula (2.46) is, however, close to unusable as written. It requires one derivative in time and two in strike of raw market data, and market data are quoted on a sparse grid, carry bid-ask spreads, and are contaminated by microstructure noise. Numerical differentiation amplifies all three, and the practical symptom is a negative denominator — that is, a negative local variance, which is meaningless.

The remedy is not better differentiation but a change of variables, together with an interpolation that is guaranteed arbitrage-free by construction. Define the **log-moneyness** and the **total implied variance**

$$k := \log \frac{K}{F(t, T)}, \quad w(k, T) := \sigma_{\text{imp}}^2(k, T) T. \quad (2.47)$$

Total variance is the natural coordinate because it, rather than implied volatility itself, is what accumulates linearly in time under Black–Scholes. We now state the conditions under which a surface w is free of *static* arbitrage — that is, admits no arbitrage exploitable by a buy-and-hold portfolio of European options, with no trading in between.

Proposition 2.17 (Absence of static arbitrage). *A total implied variance surface $w(k, T)$ is free of static arbitrage if the following hold.*

i) (**Calendar spread**) *For each fixed k , the map $T \mapsto w(k, T)$ is non-decreasing:*

$$\frac{\partial w}{\partial T}(k, T) \geq 0. \quad (2.48)$$

ii) (**Butterfly**) *For each fixed T , the function*

$$g(k) := \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_{kk}^2 w}{2} \quad (2.49)$$

satisfies $g(k) \geq 0$ for all k .

iii) (**Wings**) *$w(k, T)$ grows at most linearly in $|k|$ as $|k| \rightarrow \infty$; Lee’s moment formula [51] bounds the asymptotic slopes by 2, this being a necessary condition on an arbitrage-free surface rather than a sufficient one. What is required in addition, in the Gatheral–Jacquier formulation, is the tail condition $d_+(k) \rightarrow -\infty$ as $k \rightarrow +\infty$, which is what forces the call price to vanish in the far wing.*

□

Each condition has a direct financial meaning. Condition i) says a longer-dated option cannot be cheaper than a shorter-dated one of the same moneyness — otherwise one buys the long and sells the short. Condition ii) is convexity of the call price in strike, expressed in the new coordinates, and its violation is a butterfly spread with negative cost and non-negative payoff. Condition iii) constrains the extrapolated wings: by Lee’s formula an asymptotic slope of 2 is the borderline at which moments of S_T cease to exist.

In practice one first fits a parametric form designed to respect i)–iii) — the SVI parameterisation [36] is the standard choice, with SABR common in rates markets — and only then differentiates. Neither guarantees ii) of itself: raw SVI can admit butterfly arbitrage, and it is the constrained variants that exclude it. Since the fitted surface is smooth and arbitrage-free by construction, the differentiation is stable and the local variance is automatically non-negative.

2.3.2 Stochastic Volatility Models

Throughout this subsection, and in Subsection 2.3.3 which follows it, interest rates and dividend yields are taken to be *deterministic*. It is what makes the T -forward measure coincide with the risk-neutral one, $\mathbb{Q}^T = \mathbb{Q}$, so that expectations below may be written under $\mathbb{Q}$ while still exploiting the $\mathbb{Q}^T$ -martingale property of the forward; and it is what allows the discounting term in the American pricing operator of (2.60) to be a deterministic function of time. With stochastic rates both statements require the change of numeraire of Subsection 2.2.2 to be carried through explicitly.

To capture realistic dynamics of the volatility surface itself, we introduce a second source of randomness. In Stochastic Volatility (SV) models, the variance $v_t = \sigma_t^2$ follows its own stochastic differential equation. Under the risk-neutral measure $\mathbb{Q}$, the system is described by:

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t = \mu_v(v_t, t) dt + \sigma_v(v_t, t) dW_t^v. \end{cases} \quad (2.50)$$

A critical parameter is the correlation ρ between the two Brownian motions: $d\langle W^S, W^v \rangle_t = \rho dt$.

- If $\rho < 0$, a drop in price ($S \downarrow$) is associated with an increase in volatility ($v \uparrow$), naturally generating the **skew** observed in equity markets.
- The “vol of vol” parameter (in the diffusion coefficient σ_v) controls the convexity or **curvature** of the smile.

The Heston model

The most prominent SV model is that of Heston [43], in which the variance follows the Cox–Ingersoll–Ross square-root process already analysed in Example 1.68:

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \end{cases} \quad d\langle W^S, W^v \rangle_t = \rho dt, \quad (2.51)$$

where κ is the speed of mean reversion, θ the long-run variance, ξ the volatility of volatility, and ρ the correlation, understood in the sense of Definition 1.71. We recall from Subsection 1.2.3 that (2.51) is well posed by Yamada–Watanabe rather than by the Lipschitz theory, and that strict positivity of v requires the Feller condition $2\kappa\theta \geq \xi^2$ of Proposition 1.69, a condition frequently violated by calibrated parameters. Unlike local volatility, stochastic volatility renders the market **incomplete**. The filtration is generated by two Brownian motions while only one asset is traded, so the hypothesis of Theorem 1.83 fails: a claim whose value depends on W^v cannot be written as a stochastic integral against the traded asset alone. By Theorem 2.6 the equivalent martingale measure is therefore not unique, and pricing requires either specifying a market price of volatility risk or, as is usual in practice, calibrating $(\kappa, \theta, \xi, \rho, v_0)$ directly to liquid quotes.

Theorem 2.18 (Heston’s characteristic function). *Under (2.51), write $\tau = T - t$ and $F_s = S_s e^{(r-q)(T-s)}$ for the forward to T . Then*

$$\varphi(u, \tau) := \mathbb{E}^{\mathbb{Q}} \left[\exp \left(iu \log \frac{F_T}{F_t} \right) \middle| \mathcal{F}_t \right] = \exp \left(C(u, \tau) \theta + D(u, \tau) v_t \right), \quad (2.52)$$



the characteristic function of the centred log-forward $\log(F_T/F_t)$, where C and D solve the system of Riccati ordinary differential equations

$$\frac{\partial D}{\partial \tau} = \frac{\xi^2}{2} D^2 + (\rho \xi i u - \kappa) D - \frac{u^2 + i u}{2}, \quad \frac{\partial C}{\partial \tau} = \kappa D, \quad (2.53)$$

with $C(u, 0) = D(u, 0) = 0$. The system (2.53) admits an explicit solution in terms of elementary functions. $\square$

Given (2.52), European option prices follow by Fourier inversion: the call price is recovered from the characteristic function through an integral of Lewis–Carr–Madan type [53, 15],

$$C(K, T) = B(t, T) \left[F_t - \frac{\sqrt{F_t K}}{\pi} \int_0^\infty \operatorname{Re} \left(e^{-i u k} \varphi \left(u - \frac{i}{2}, \tau \right) \right) \frac{d u}{u^2 + \frac{1}{4}} \right], \quad (2.54)$$

a one-dimensional integral evaluated by quadrature in microseconds, with $k = \log(K/F_t)$ the log-moneyness of Subsection 2.3.1. Calibration, which requires pricing hundreds of options at each step of an optimiser, is thereby feasible, whereas a model requiring Monte Carlo for each European quote would not be.

Since F is a $\mathbb{Q}^T$ -martingale, $\mathbb{E}^\mathbb{Q}[F_T/F_t \mid \mathcal{F}_t] = 1$, so the centred characteristic function (2.52) must satisfy

$$\varphi(-i, \tau) = 1 \quad \text{for every } \tau \geq 0. \quad (2.55)$$

Any implementation of (2.52) should be tested against (2.55), and (2.54) against Black-76 in the limit $\xi \rightarrow 0$, before either is tested against prices: both errors are otherwise silent. In that limit the variance is deterministic, $v_t = \theta + (v_0 - \theta)e^{-\kappa t}$, so the comparison must be made at the implied volatility $\sigma^2 = [\theta\tau + (v_0 - \theta)(1 - e^{-\kappa\tau})/\kappa]/\tau$ and not at an arbitrarily chosen constant.

One implementation caveat deserves a line. The explicit solution of (2.53) involves a complex logarithm, and the naive branch choice makes φ discontinuous in τ once the integrand has wound sufficiently far, corrupting prices at long maturities while leaving short ones correct. We shall write the solution in the form whose principal branch is continuous; the phenomenon is documented in [7].

2.3.3 Path-Dependence and American Options

We conclude this chapter by discussing the challenges posed by path-dependent derivatives, focusing on the most liquid and prominent example: American options.

Because an American option may be exercised at any time up to maturity, its value depends on the law of the underlying at *every* intermediate date, not merely at T . A single implied volatility, which by Definition 2.13 is one number read off one European quote, therefore does not determine it: infinitely many dynamics reproduce that quote while disagreeing about the intermediate marginals, and hence about the exercise boundary. The point matters in practice because options on *single names* (i.e., individual corporate stocks) are almost exclusively American-style. Investors in single equities are reluctant to relinquish the flexibility of early exercise, particularly to capture discrete dividends or navigate corporate actions. In contrast, broad market indices are generally considered more stable and are traded via extremely liquid European options.

Proposition 2.19 (No early exercise for the American call). *Assume $r \geq 0$ and that the underlying pays no dividends before T . Then an American call is never optimally exercised before maturity, and its value coincides with that of the European call with the same strike and maturity.*

Proof: Write $C(t)$ for the European call value at time t . Under $\mathbb{Q}$ the discounted spot is a martingale, so for any $t \leq T$, using the conditional Jensen inequality (Proposition A.1, item 5) applied to the convex function $x \mapsto (x - K)^+$,

$$C(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ | \mathcal{F}_t] \geq e^{-r(T-t)} (\mathbb{E}^{\mathbb{Q}}[S_T | \mathcal{F}_t] - K)^+ = (S_t - Ke^{-r(T-t)})^+,$$

the last step because $\mathbb{E}^{\mathbb{Q}}[S_T | \mathcal{F}_t] = S_t e^{r(T-t)}$ in the absence of dividends. Since $r \geq 0$ gives $Ke^{-r(T-t)} \leq K$, the right-hand side is at least $(S_t - K)^+$, which is the immediate-exercise payoff. Holding therefore dominates exercising at every $t < T$.

It remains to convert this pointwise domination into a statement about stopping times, which does not follow from it directly. The chain above, read at a general t , says precisely that the discounted payoff process $Z_t := e^{-rt}(S_t - K)^+$ satisfies $\mathbb{E}^{\mathbb{Q}}[Z_T | \mathcal{F}_t] \geq Z_t$; that is, Z is a $\mathbb{Q}$ -submartingale — as it must be, being a convex function of a martingale, by Proposition A.1, item 5, together with $r \geq 0$. Doob's optional stopping theorem (Theorem 1.28, case i — every stopping time being bounded by T under Definition 1.9) applied to a submartingale gives, for every $\tau \in \mathcal{T}_{[t, T]}$,

$$\mathbb{E}^{\mathbb{Q}}[Z_T | \mathcal{F}_\tau] \geq Z_\tau.$$

Since $\tau \geq t$ we have $\mathcal{F}_t \subseteq \mathcal{F}_\tau$, so conditioning both sides on $\mathcal{F}_t$ and applying the tower property (Proposition 1.4, item i)) gives

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\tau}(S_\tau - K)^+ | \mathcal{F}_t] \leq \mathbb{E}^{\mathbb{Q}}[e^{-rT}(S_T - K)^+ | \mathcal{F}_t],$$

so the supremum in (2.57) is attained at $\tau = T$. ■

Two things are worth extracting. The inequality is strict whenever $r > 0$ and the option has time value, so the American call is not merely no better than the European one but strictly worse to exercise early: one would be surrendering the interest earned on the strike. For the American *put* the argument fails at the first step regardless of dividends: the same Jensen bound yields only $P(t) \geq (Ke^{-r(T-t)} - S_t)^+$, which lies *below* the immediate-exercise payoff $(K - S_t)^+$, and early exercise is indeed optimal for low enough spot. Note also that put–call parity, Proposition 2.12, has no exact American analogue; only the bounds $S_t - K \leq C_A - P_A \leq S_t - Ke^{-r(T-t)}$ survive.

We aim to show how a local volatility model can be adapted to price American options. However, we must first note a fundamental limitation: Local Volatility models do not perform perfectly for all types of path-dependent derivatives. In particular, because local volatility models make the volatility a deterministic function of the spot, hence perfectly correlated with it, the future smile is determined by where the spot goes, and the model misprices derivatives whose strikes are set at a future date, such as forward-starting options and cliquets.

Nevertheless, to price American options accurately we must model the entire volatility surface prior to maturity, not merely the terminal marginal. The model we use, expressed under the T -forward measure of Subsection 2.2.2, is

$$dF_t = \sigma_{\text{loc}}(t, F_t) F_t dW_t^T. \quad (2.56)$$



Optimal stopping and the Snell envelope

Before writing any partial differential equation, let us state what an American option *is* in probabilistic terms, since the PDE formulation is derived from it and is opaque without it.

The holder may exercise at any stopping time of her choosing, and will choose the one maximising the value of the resulting cash flow. Writing $\mathcal{T}_{[t,T]}$ for the set of stopping times valued in $[t, T]$ and Φ for the exercise payoff, the value at time t is

$$V_t = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^\tau r(s) ds} \Phi(\tau, S_\tau) \mid \mathcal{F}_t \right]. \quad (2.57)$$

The essential supremum, defined in Appendix A.7, is required because the family is uncountable. This is an optimal stopping problem, and its solution is classical. Discounting has to be brought inside before the classical statement applies, since it is the *discounted* value, not V itself, that is a supermartingale. Write

$$Y_t := e^{-\int_0^t r(s) ds} \Phi(t, S_t), \quad U_t := e^{-\int_0^t r(s) ds} V_t, \quad (2.58)$$

so that (2.57) becomes $U_t = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^{\mathbb{Q}}[Y_\tau \mid \mathcal{F}_t]$.

Theorem 2.20 (Snell envelope). *Let Y be a right-continuous adapted process of class D (see Definition A.10). Then the process U of (2.58) is the **Snell envelope** of Y : that is, U is the smallest càdlàg supermartingale dominating Y . It satisfies:*

- i) $U_t \geq Y_t$ for all t (domination);
- ii) U is a $\mathbb{Q}$ -supermartingale;
- iii) U is a $\mathbb{Q}$ -martingale on the continuation region $\mathfrak{C} = \{(t, \omega) : U_t > Y_t\}$;

and, if Y is in addition continuous, the stopping time $\tau^* = \inf\{s \geq t : U_s = Y_s\}$ is optimal in (2.57). A full treatment is given in [64]. $\square$

Continuity is what rules out the reward falling away at the instant it is reached; without it the first contact need not be optimal, and only the envelope itself survives. It holds for the obstacle (2.59) whenever the coefficients a_t and b_t of (2.37) are continuous.

Since $e^{-\int_0^t r(s) ds}$ is strictly positive and deterministic, the continuation region and the optimal stopping rule are the same whether read off U or off V ; the normalisation (2.58) is bookkeeping, not a change of problem.

Property i) says the option is never worth less than exercising it. Property ii) says that holding the option is, on average and after discounting, a losing proposition: waiting has a cost, which is what makes early exercise ever attractive. Property iii) says that this cost is zero exactly where continuation is optimal. The optimal rule is then transparent: continue while $U > Y$, and stop the instant they meet.

Translating Theorem 2.20 into the language of partial differential equations gives a free boundary problem, most rigorously expressed as a linear complementarity problem. Write $\mathfrak{L}$ for the pricing operator associated with (2.56),

$$\mathfrak{L}V := \frac{\partial V}{\partial t} + \frac{1}{2} \sigma_{\text{loc}}^2(t, F) F^2 \frac{\partial^2 V}{\partial F^2} - r(t) V.$$

Exercise delivers the *spot*, not the forward: the holder of an American put receives $K - S_\tau$, and by (2.37) we have $S_t = a_t^{-1}(F_t - b_t)$. Expressed in the forward variable, the obstacle is therefore

$$\Phi(t, F) = \left(K - \frac{F - b_t}{a_t}\right)^+ \quad \text{for a put,} \quad \Phi(t, F) = \left(\frac{F - b_t}{a_t} - K\right)^+ \quad \text{for a call,} \quad (2.59)$$

which is explicitly time-dependent through a_t and b_t .

Theorem 2.21 (American pricing as an LCP). *The value $V(t, F)$ of the American option satisfies, almost everywhere on $[0, T) \times (0, \infty)$, the three simultaneous conditions*

$$\begin{cases} \mathfrak{L}V \leq 0 & (\text{supermartingale property, Theorem 2.20(ii)}) \\ V(t, F) \geq \Phi(t, F) & (\text{domination, Theorem 2.20(i)}) \\ (\mathfrak{L}V) \cdot (V(t, F) - \Phi(t, F)) = 0 & (\text{complementarity, Theorem 2.20(iii)}) \end{cases} \quad (2.60)$$

together with the terminal condition $V(T, F) = \Phi(T, F)$ and Φ as in (2.59). $\square$

The correspondence with Theorem 2.20 is exact. The complementarity condition encodes the dichotomy: at each point either $V > \Phi$, in which case $\mathfrak{L}V = 0$ and the ordinary pricing equation holds, or $V = \Phi$ and the option is exercised. The boundary separating the two regions is not known in advance — it is part of the solution — which is what makes the problem free-boundary rather than merely parabolic, and what rules out any closed-form solution.

Notes and references

The financial framework and pricing methodologies developed in this chapter are built upon cornerstone texts in quantitative finance and stochastic analysis. The rigorous transition from abstract probability to continuous-time asset pricing, including the formulation of equivalent martingale measures and the *No Free Lunch with Vanishing Risk* (NFLVR) condition, relies heavily on the definitive work by Delbaen and Schachermayer [25]. This abstract measure-theoretic approach is bridged with continuous-time applications in the classic textbooks by Shreve [70], Øksendal [63], and Björk [9]. The Feynman–Kac formula of Theorem 2.7, which supplies the bridge between the pricing expectation and the pricing equation and is used in both directions throughout the chapter, is treated in [48, Sec. 4.4 and 5.7].

The explicit formulation of diffusion models is rooted in the foundational breakthroughs of Black and Scholes [11] and Merton [61]. To address the empirical shortcomings of this framework, local volatility was formalised by Dupire [26], with its practical translation into the implied volatility domain extensively covered by Gatheral [35]. For an exhaustive, practitioner-oriented treatment of volatility smile dynamics and stochastic volatility models (such as the Heston model [43]), Bergomi [7] serves as an indispensable reference. Finally, the Fourier pricing route of (2.54), which is what makes calibrating an affine model such as Heston computationally feasible at all, was opened by Carr and Madan [15]; the particular form used here, symmetric in F and K and integrated along the line $\text{Im } u = -\frac{1}{2}$, is due to Lewis [53] rather than to them, and is the variant that is numerically best behaved, and the arbitrage-free interpolation it is calibrated against is Gatheral’s SVI [36], with the wing bounds supplied by Lee’s moment formula [51].



Part II

Modern Approaches

Chapter 3

Rough Paths

3.1 Introduction

Brownian motion has finite p -variation exactly for $p > 2$; Young's integral requires $p < 2$; and the Itô integral of Section 1.2 bridges the gap not by refining an estimate but by *spending probabilistic structure* — adaptedness, an $L^2(\Omega)$ limit in place of a pathwise one, and a definition valid only up to null sets. This allowed us to construct solutions, but we paid a double cost.

The first is that the driving signal may be rougher than Brownian. Itô's construction is not a general theory of integration against irregular paths: it is a theory of integration against *semimartingales*, in the sense of Definition 1.34. The moment the integrator leaves that class, the entire apparatus — the isometry of Theorem 1.51, the martingale property, the localisation of Example 1.11 — becomes unavailable at once.

The second is subtler and, in the end, more consequential. Even where Itô's theory applies, it delivers the solution of a stochastic differential equation as a measurable function on Ω , constructed through an L^2 limit that depends on the filtration and the measure. It does not tell us that the solution depends *continuously* on the driving trajectory, and in fact, as we shall see in Subsection 3.1.3, no such statement is true in the topology one would naïvely choose. A model whose output is a discontinuous function of its input is a model one cannot approximate, cannot discretise with confidence, and cannot calibrate stably. Rough path theory, introduced by Lyons [57], resolves both difficulties simultaneously, and by a single device: it enlarges the notion of “path”. The remainder of this chapter develops that idea. We begin, however, with the process that forces the question upon us.

3.1.1 Fractional Brownian Motion

Brownian motion is characterised, among Gaussian processes with stationary increments, by the independence of those increments. Relaxing exactly that requirement yields the following one-parameter family, introduced by Kolmogorov and given its modern form by Mandelbrot and Van Ness [59].

Definition 3.1 (Fractional Brownian motion). Let $H \in (0, 1)$. A **fractional Brownian motion** with Hurst parameter H is a centred Gaussian process $W^H = \{W_t^H\}_{t \geq 0}$ with $W_0^H = 0$ almost surely and covariance

$$\mathbb{E}[W_s^H W_t^H] = R_H(s, t) := \frac{1}{2} \left(s^{2H} + t^{2H} - |t - s|^{2H} \right), \quad s, t \geq 0. \quad (3.1)$$

That (3.1) is a legitimate covariance — that is, symmetric and positive semi-definite — is a genuine constraint on H , and it is the reason the parameter range is $(0, 1)$ and not larger; a proof is in [60, Ch. 7]. Granting it, the process exists by the standard Gaussian construction, and Kolmogorov's criterion (Theorem 1.37) supplies a continuous modification.

Proposition 3.2 (Elementary properties). *Let W^H be a fractional Brownian motion with Hurst parameter $H \in (0, 1)$. Then:*

a) W^H has stationary increments, with

$$\mathbb{E}[|W_t^H - W_s^H|^2] = |t - s|^{2H}, \quad \text{so } W_t^H - W_s^H \sim \mathcal{N}(0, |t - s|^{2H}). \quad (3.2)$$

b) W^H is self-similar of index H : for every $\lambda > 0$, the processes $\{W_{\lambda t}^H\}_{t \geq 0}$ and $\{\lambda^H W_t^H\}_{t \geq 0}$ have the same law.

c) Almost every sample path is locally γ -Hölder continuous for every $\gamma < H$, and for no $\gamma \geq H$.

d) Almost surely, $V_p(W^H)_t < \infty$ for every $p > 1/H$ and $V_p(W^H)_t = \infty$ for every $p \leq 1/H$.

e) $H = \frac{1}{2}$ recovers Brownian motion. For $H > \frac{1}{2}$ the increments over disjoint intervals are positively correlated, for $H < \frac{1}{2}$ negatively, and in neither case are they independent.

Proof: a) Expanding and using (3.1),

$$\mathbb{E}[(W_t^H - W_s^H)^2] = R_H(t, t) - 2R_H(s, t) + R_H(s, s) = t^{2H} - (s^{2H} + t^{2H} - |t - s|^{2H}) + s^{2H} = |t - s|^{2H}.$$

Since the process is centred and Gaussian, the law of an increment is determined by this variance alone, and depends on s, t only through $t - s$.

b) $R_H(\lambda s, \lambda t) = \lambda^{2H} R_H(s, t)$ is immediate from (3.1). Two centred Gaussian processes with the same covariance have the same law.

c) The Gaussian moments give $\mathbb{E}[|W_t^H - W_s^H|^{2m}] = c_m |t - s|^{2mH}$, so (1.41) holds with $\alpha = 2m$ and $\beta = 2mH - 1$, and Theorem 1.37 yields Hölder exponents $\gamma < H - 1/2m$, which approach H as $m \rightarrow \infty$. The negative half is the exact analogue of Theorem 1.46 i) and is proved by the same Paley–Wiener–Zygmund argument, adapted in [60, Ch. 7].

d) Finiteness for $p > 1/H$ follows from c); the divergence for $p \leq 1/H$ requires the same competition between p -variation and modulus of continuity that produced Proposition 1.44. See [34, Ch. 15] for the details, which we do not repeat.

e) For $H = \frac{1}{2}$, (3.1) collapses to $\min\{s, t\}$, which is Proposition 1.36 a); by Definition 1.35 and the Gaussian characterisation this identifies the process. For the sign of the correlation, take $0 \leq s \leq t \leq u \leq v$ and compute directly from (3.1),

$$2\mathbb{E}[(W_t^H - W_s^H)(W_v^H - W_u^H)] = |v - s|^{2H} + |u - t|^{2H} - |u - s|^{2H} - |v - t|^{2H},$$

which is the second difference of the function $\tau \mapsto \tau^{2H}$, hence positive when that function is convex ($H > \frac{1}{2}$) and negative when it is concave ($H < \frac{1}{2}$). It vanishes identically only at $H = \frac{1}{2}$. ■

Part e) is the modelling content of the definition and deserves its name. For $H > \frac{1}{2}$ the process exhibits *persistence*: an increase tends to be followed by an increase, and the correlations decay



so slowly that they are not summable — the phenomenon known as *long memory*. For $H < \frac{1}{2}$ it exhibits *anti-persistence*, or mean reversion at every scale, and the paths are correspondingly rougher than Brownian. Part d) makes “rougher” precise: the critical exponent $1/H$ exceeds 2 exactly when $H < \frac{1}{2}$.

That last observation is not a curiosity but an obstruction, and we state it as such.

Theorem 3.3 (Rogers). *Fractional Brownian motion is a semimartingale in the sense of Definition 1.34 if and only if $H = \frac{1}{2}$.* $\square$

The proof is in [67], and the mechanism is worth recording even though we do not reproduce the argument. For $H > \frac{1}{2}$ the quadratic variation of Definition 1.31 vanishes, while the total variation is infinite by part d); a continuous semimartingale with zero quadratic variation must be of finite variation, by the uniqueness of the decomposition, and this is a contradiction. For $H < \frac{1}{2}$ the quadratic variation is instead infinite, which is incompatible with the decomposition from the other direction. Only the balance point $H = \frac{1}{2}$ escapes. The consequence is severe: for $H \neq \frac{1}{2}$ none of the semimartingale apparatus of Chapter 1 is available — no Itô integral against W^H , no Itô formula, no martingale representation. Other integration theories for W^H do exist, but they are not the theory built in Chapter 1.

3.1.2 Volatility is Rough

Let σ_t denote realised volatility, estimated from high-frequency returns over a grid of spacing Δ , and consider the q -th absolute moment of the increments of its logarithm,

$$m(q, \Delta) := \frac{1}{N} \sum_{i=1}^N |\log \sigma_{i\Delta} - \log \sigma_{(i-1)\Delta}|^q. \quad (3.3)$$

Gatheral, Jaisson and Rosenbaum [37] observe that, across equity indices, $m(q, \Delta) \propto \Delta^{\zeta_q}$ with $\zeta_q = qH$ — a linear, *monofractal* scaling with no curvature in q — and that the increments are close to Gaussian. Comparison with (3.2) is immediate: these are exactly the scaling relations of a fractional Brownian motion. The estimated exponent is $H \approx 0.1$, and it is remarkably stable across assets and across decades.

This inverts an earlier consensus. The fractional stochastic volatility model of Comte and Renault [22] also used fractional Brownian motion, but with $H > \frac{1}{2}$, in order to reproduce the apparent long memory of realised volatility. The finding above says that the correct exponent is not merely below $\frac{1}{2}$ but very close to 0: volatility is not persistent, it is *rough*. The resulting model replaces the Heston dynamics of Subsection 2.3.2 by

$$\sigma_t = \exp(X_t), \quad dX_t = \nu dW_t^H - a(X_t - m)dt, \quad (3.4)$$

with a so small that, over any horizon of practical interest, X is indistinguishable from a fractional Brownian motion.

A model of the form (3.4) generates an at-the-money skew whose term structure behaves like

$$\psi(\tau) := \left| \frac{\partial \sigma_{\text{imp}}}{\partial k}(\tau, k) \right|_{k=0} \sim \tau^{H-1/2} \quad \text{as } \tau \downarrow 0, \quad (3.5)$$

a power-law explosion at short maturities, where a continuous semimartingale volatility factor gives a skew with a finite limit. This is the one place where H becomes visible in option prices rather than in realised volatility, and it is what §4.2.3 sets out to measure.

3.1.3 What Fails, and Why

Before constructing anything we should be clear that the obstruction is real, and in particular that it is not removed by any sharpening of the classical estimates. There are three layers to this, of increasing depth.

Young integration is out of reach. Recall that $\int g df$ exists pathwise as a limit of Riemann sums whenever f has finite p -variation, g finite q -variation, and $1/p + 1/q > 1$. In Hölder terms, if $f \in \mathcal{C}^\alpha$ and $g \in \mathcal{C}^\beta$ the condition reads $\alpha + \beta > 1$. Now the integrands we care about are of the form $g = F(f)$, which inherits the regularity α of f itself, so the condition becomes $2\alpha > 1$, that is $\alpha > \frac{1}{2}$. By Proposition 3.2 c) this holds for W^H precisely when $H > \frac{1}{2}$. Young's theory therefore covers exactly the persistent case that the data reject, and none of the rough one.

Itô integration is out of reach. By Theorem 3.3, immediately and for all $H \neq \frac{1}{2}$.

The obstruction is not a matter of technique. One might still hope that the map $(f, g) \mapsto \int g df$, manifestly well defined for smooth paths, extends by continuity to the irregular ones — that some cleverer topology on path space would do what the uniform topology fails to do. The following example, due to Lyons, shows that the uniform topology cannot.

Example 3.4 (The oscillation counterexample). For $n \in \mathbb{N}$ define the smooth planar paths

$$x_t^{(n)} := \frac{1}{n} \sin(n^2 t), \quad y_t^{(n)} := \frac{1}{n} \cos(n^2 t), \quad t \in [0, T]. \quad (3.6)$$

Both converge to the zero path uniformly, since $|x^{(n)}|, |y^{(n)}| \leq 1/n$. Yet

$$\int_0^t y_s^{(n)} dx_s^{(n)} = \int_0^t \frac{\cos(n^2 s)}{n} \cdot n \cos(n^2 s) ds = \int_0^t \cos^2(n^2 s) ds = \frac{t}{2} + \frac{\sin(2n^2 t)}{4n^2} \longrightarrow \frac{t}{2},$$

which is not the integral of the zero path against itself. The integral is therefore *not* continuous with respect to uniform convergence.

The picture behind the computation is worth holding on to, because it is the picture behind the whole theory. The paths $(x^{(n)}, y^{(n)})$ trace circles of radius $1/n$ traversed n^2 times per unit time. The circles shrink to a point, which is why the paths converge uniformly to zero; but the *area* swept out is π/n^2 per revolution and there are $n^2/2\pi$ revolutions per unit time, so the total area accumulated per unit time is a constant, independent of n . Uniform convergence sees the shrinking radius. It does not see the accumulating area. And the integral, as the computation shows, depends on the area.

One might now hope for some other norm. There is none.

Proposition 3.5 (Lyons). *Let $\mathcal{W}$ denote the space of continuous paths $[0, T] \rightarrow \mathbb{R}^d$ and let $E^\infty \subseteq \mathcal{W}$ be the smooth paths. There is no norm $\|\cdot\|$ on E^∞ such that, simultaneously, the closure E of E^∞ carries full Wiener measure and the map $(f, g) \mapsto \int g df$ extends continuously from $E^\infty \times E^\infty$ to $E \times E$. $\square$*

A proof is given in [33, Ch. 1]. The statement is a genuine impossibility theorem, not an admission of ignorance, and it forces a change of question. If no topology on *paths* makes the integral continuous, then the integral is not a function of the path alone, and the object we should be topologising is not the path.



Example 3.4 tells us exactly what is missing. What distinguishes $(x^{(n)}, y^{(n)})$ from the zero path is the area it encloses — equivalently, the iterated integral $\iint dx dy$. So we shall carry that quantity along as part of the data, postulate it where it cannot be constructed, and topologise the enlarged object.

3.2 Rough Paths

3.2.1 Tensor Spaces

The iterated integrals we must carry are indexed by pairs of coordinates, so the algebraic home for such objects is the tensor product. We recall only what is needed; the systematic treatment, which requires free Lie algebras and the theory of the free nilpotent group, is developed in [38, Ch. 3] and [34, Ch. 7].

Throughout this chapter $V := \mathbb{R}^d$ with its standard basis $\{e_1, \dots, e_d\}$.

Definition 3.6 (Tensor product). The **tensor product** $V \otimes V$ is the vector space generated by the symbols $v \otimes w$, for $v, w \in V$, subject to bilinearity:

$$(av_1 + v_2) \otimes w = a(v_1 \otimes w) + v_2 \otimes w, \quad v \otimes (aw_1 + w_2) = a(v \otimes w_1) + v \otimes w_2, \quad (3.7)$$

for all $a \in \mathbb{R}$ and $v, v_i, w, w_i \in V$.

Concretely $V \otimes V$ has dimension d^2 with basis $\{e_i \otimes e_j\}_{1 \leq i, j \leq d}$, so that it may be identified with the space of $d \times d$ matrices, an element being written $\xi = \sum_{i,j} \xi^{ij} e_i \otimes e_j$. The point of the notation is that the product is *not* commutative: $e_1 \otimes e_2 \neq e_2 \otimes e_1$. This is essential, since the two iterated integrals $\iint dx^1 dx^2$ and $\iint dx^2 dx^1$ are genuinely different numbers, and it is exactly their difference that Example 3.4 detected.

Definition 3.7 (Truncated tensor algebra). The **truncated tensor algebra of order two** over V is

$$T^{(2)}(V) := \mathbb{R} \oplus V \oplus (V \otimes V), \quad (3.8)$$

whose elements we write $\xi = (\xi_0, \xi_1, \xi_2)$, equipped with the multiplication

$$(\xi \otimes \eta)_n := \sum_{k=0}^n \xi_k \otimes \eta_{n-k}, \quad n = 0, 1, 2, \quad (3.9)$$

all terms of higher order being discarded. We write $T_1^{(2)}(V) := \{\xi \in T^{(2)}(V) : \xi_0 = 1\}$. More generally, the truncation at order N is $T^{(N)}(V) := \bigoplus_{n=0}^N V^{\otimes n}$, with the same multiplication truncated at order N , and the full tensor algebra is $T(V) := \prod_{n \geq 0} V^{\otimes n}$. Only $T^{(2)}$ is used until Section 3.4.

Written out, (3.9) says $\xi \otimes \eta = (\xi_0 \eta_0, \xi_0 \eta_1 + \xi_1 \eta_0, \xi_0 \eta_2 + \xi_1 \otimes \eta_1 + \xi_2 \eta_0)$. The algebra is associative with unit $\mathbf{1} = (1, 0, 0)$, and although a general element has no inverse, every element of $T_1^{(2)}(V)$ does, namely

$$(1, \xi_1, \xi_2)^{-1} = (1, -\xi_1, \xi_1 \otimes \xi_1 - \xi_2), \quad (3.10)$$

as one checks by multiplying out. Thus $T_1^{(2)}(V)$ is a group, and this is the structure in which the whole chapter takes place. The motivation is the following computation, which the reader should regard as the definition of everything that follows.

Example 3.8 (The canonical lift of a smooth path). Let $x : [0, T] \rightarrow V$ be continuously differentiable. For $s \leq t$ set

$$X_{s,t} := x_t - x_s \in V, \quad \mathbb{X}_{s,t} := \int_s^t (x_r - x_s) \otimes dx_r = \int_s^t \int_s^r dx_u \otimes dx_r \in V \otimes V, \quad (3.11)$$

both integrals being ordinary Riemann–Stieltjes integrals, so $\mathbb{X}_{s,t}^{ij} = \int_s^t (x_r^i - x_s^i) \dot{x}_r^j dr$. The pair $\mathbf{X}_{s,t} := (1, X_{s,t}, \mathbb{X}_{s,t})$ takes values in $T_1^{(2)}(V)$.

Two features of (3.11) will survive the passage to irregular paths, and one will not. The two that survive are an algebraic identity, developed in the next subsection, and an analytic bound. The one that does not is the following, which is worth isolating now.

Proposition 3.9 (The symmetric part is determined). *With the notation of Example 3.8,*

$$\text{Sym } \mathbb{X}_{s,t} := \frac{1}{2}(\mathbb{X}_{s,t} + \mathbb{X}_{s,t}^\top) = \frac{1}{2} X_{s,t} \otimes X_{s,t}. \quad (3.12)$$

Proof: In coordinates the claim is $\mathbb{X}_{s,t}^{ij} + \mathbb{X}_{s,t}^{ji} = X_{s,t}^i X_{s,t}^j$. Since x is differentiable, the ordinary product rule gives

$$\frac{d}{dr} \left[(x_r^i - x_s^i)(x_r^j - x_s^j) \right] = (x_r^i - x_s^i) \dot{x}_r^j + (x_r^j - x_s^j) \dot{x}_r^i,$$

and integrating from s to t , where the left-hand side vanishes at $r = s$, yielding the claim. ■

So for a smooth path the symmetric part of the iterated integral carries no information beyond the increment itself: only the *antisymmetric* part, the **Lévy area**

$$A_{s,t} := \frac{1}{2}(\mathbb{X}_{s,t} - \mathbb{X}_{s,t}^\top), \quad (3.13)$$

is new. Geometrically $A_{s,t}^{ij}$ is the signed area enclosed between the chord joining (x_s^i, x_s^j) to (x_t^i, x_t^j) and the arc of the path between them — precisely the quantity that Example 3.4 showed to be invisible to uniform convergence. Whether (3.12) continues to hold once x is no longer differentiable is exactly the distinction between the two classes of rough path introduced in Subsection 3.2.3.

3.2.2 The Multiplicative Property

The algebraic identity announced above is the following, and it is the axiom on which the entire theory rests.

Proposition 3.10 (Chen’s relation for smooth paths). *With the notation of Example 3.8, for all $s \leq u \leq t$,*

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad (3.14)$$

equivalently $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ in the group $T_1^{(2)}(V)$.

Proof: Split the outer integral in (3.11) at u :

$$\mathbb{X}_{s,t} = \int_s^u (x_r - x_s) \otimes dx_r + \int_u^t (x_r - x_s) \otimes dx_r = \mathbb{X}_{s,u} + \int_u^t [(x_r - x_u) + (x_u - x_s)] \otimes dx_r.$$



The first term of the bracket integrates to $\mathbb{X}_{u,t}$. The second is constant in r , so it factors out of the integral, leaving $X_{s,u} \otimes \int_u^t dx_r = X_{s,u} \otimes X_{u,t}$. That the resulting three identities — one at each level — are precisely the statement $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ is immediate from (3.9). ■

At the first level, $X_{s,t} = X_{s,u} + X_{u,t}$ is just the additivity of increments. At the second level it is a genuine constraint, and it says that $\mathbb{X}$ is *not* additive: concatenating two intervals produces a cross-term $X_{s,u} \otimes X_{u,t}$ which is the area contributed by the interaction of the two pieces. The relation is thus the algebraic residue of the additivity of integration, and it is what makes $\mathbb{X}$ a coherent object rather than an arbitrary two-parameter function.

Notation 3.11. Throughout, $\mathbb{X}$ is a function of *two* time variables and is never a path. There is no “ $\mathbb{X}_t$ ”. This is not a notational accident: by (3.14) the increments of $\mathbb{X}$ do not telescope, so $\mathbb{X}$ is not the increment of any $V \otimes V$ -valued path. The first level, by contrast, may be regarded either way, and we write $X_{s,t} = x_t - x_s$ when convenient.

We can now give the central definition. From this point onwards we fix

$$\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right], \quad (3.15)$$

a restriction whose necessity will become visible in Theorem 3.25 and which we discuss in Remark 3.26.

A word on why the exponent is a Hölder one. Lyons’ original theory [57], and the systematic treatment of [34], are formulated in p -variation — the scale of Definition 1.45, in which Proposition 3.2 d) is also stated, so that a reader arriving here expects it. The two settings are essentially interchangeable, the dictionary being $p = 1/\alpha$: a path of finite p -variation is α -Hölder after reparametrisation by its own variation, and every α -Hölder path has finite $1/\alpha$ -variation. We work in Hölder norms because they make the estimates of Subsection 3.3.1 additive in the exponent and therefore easier to read, and because (3.15) then corresponds to $p \in [2, 3)$ — exactly the range in which two levels suffice.

Definition 3.12 (Rough path). An α -**Hölder rough path** over V is a pair $\mathbf{X} = (X, \mathbb{X})$ consisting of $X \in \mathcal{C}^\alpha([0, T], V)$ and a two-parameter function $\mathbb{X} : \{0 \leq s \leq t \leq T\} \rightarrow V \otimes V$ such that:

i) (**Chen’s relation**) for all $s \leq u \leq t$,

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}; \quad (3.16)$$

ii) (**Analytic bound**) the seminorms

$$\|X\|_\alpha := \sup_{s \neq t} \frac{|X_{s,t}|}{|t - s|^\alpha}, \quad \|\mathbb{X}\|_{2\alpha} := \sup_{s \neq t} \frac{|\mathbb{X}_{s,t}|}{|t - s|^{2\alpha}} \quad (3.17)$$

are both finite.

We write $\mathbf{X} \in \mathbf{C}^\alpha([0, T], V)$, abbreviated $\mathbf{C}^\alpha$, and set $\|\mathbf{X}\|_\alpha := \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha}^{1/2}$.

The exponent 2α in (3.17) is not a convention but a necessity: $\mathbb{X}$ is formally a double integral of increments of size $|t - s|^\alpha$, so any object deserving the name must scale that way.

Definition 3.13 (The rough path metric). For $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (\widetilde{X}, \widetilde{\mathbb{X}})$ in $\mathbf{C}^\alpha$ set

$$\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) := \|X - \widetilde{X}\|_\alpha + \|\mathbb{X} - \widetilde{\mathbb{X}}\|_{2\alpha}. \quad (3.18)$$

The second level enters (3.18) to the first power, whereas it entered $\|\mathbf{X}\|_\alpha$ in Definition 3.12 under a square root, so that $\varrho_\alpha(\mathbf{X}, \mathbf{0}) \neq \|\mathbf{X}\|_\alpha$. The metric is *inhomogeneous* and is the one in which the estimates of Subsection 3.3.4 are linear; the norm is *homogeneous*, meaning that it scales correctly under the dilation $(X, \mathbb{X}) \mapsto (\lambda X, \lambda^2 \mathbb{X})$ which the exponent 2α of (3.17) forces. Both are standard, and each is used where its own scaling behaviour is wanted.

Remark 3.14 ($\mathbf{C}^\alpha$ is not a vector space). The set of rough paths is closed under neither addition nor scalar multiplication, since (3.16) is quadratic in X : if $\mathbf{X}$ and $\tilde{\mathbf{X}}$ satisfy it, their sum does not. It is a complete metric space under ϱ_α , and a closed subset of the Banach space $\mathcal{C}^\alpha \oplus \mathcal{C}_2^{2\alpha}$ — the subscript 2 denoting functions of *two* time variables, as $\mathbb{X}$ is — in which it naturally sits, but it is neither a linear space nor separable. The correct picture is a nonlinear “fibre bundle”: over each α -Hölder path X sits the set of admissible second levels, which we now describe. See [33, Ch. 2].

Proposition 3.15 (The lift is not unique). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ and let $f : [0, T] \rightarrow V \otimes V$ belong to $\mathcal{C}^{2\alpha}$. Then $\tilde{\mathbf{X}} := (X, \mathbb{X} + \delta f)$, where $\delta f_{s,t} := f_t - f_s$, is again an element of $\mathbf{C}^\alpha$. Conversely, if $(X, \mathbb{X})$ and $(X, \tilde{\mathbb{X}})$ both lie in $\mathbf{C}^\alpha$, then $\tilde{\mathbb{X}} - \mathbb{X} = \delta f$ for some $f \in \mathcal{C}^{2\alpha}$.*

Proof: For the first claim, note that increments telescope: $\delta f_{s,t} - \delta f_{s,u} - \delta f_{u,t} = 0$, so adding δf to $\mathbb{X}$ leaves (3.16) unchanged, while $\|\delta f\|_{2\alpha} = \|f\|_{2\alpha} < \infty$ preserves the bound. Conversely, set $\Delta_{s,t} := \tilde{\mathbb{X}}_{s,t} - \mathbb{X}_{s,t}$. Subtracting the two instances of (3.16), which share the same right-hand side because the first levels agree, gives $\Delta_{s,t} = \Delta_{s,u} + \Delta_{u,t}$ for all $s \leq u \leq t$. Taking $s = u$ shows $\Delta_{s,s} = 0$, and then $f_t := \Delta_{0,t}$ satisfies $\delta f_{s,t} = \Delta_{0,t} - \Delta_{0,s} = \Delta_{s,t}$, with $\|f\|_{2\alpha} = \|\Delta\|_{2\alpha} < \infty$ by the triangle inequality. ■

This is the single most important structural fact in the subject: **a rough path is not a path**. The second level is genuine additional data, undetermined by the trajectory, and the space of admissible choices over a fixed X is an affine copy of $\mathcal{C}^{2\alpha}$. The theory does not *discover* $\mathbb{X}$; it takes $\mathbb{X}$ as given and proceeds. Different choices are different rough paths, and they yield different integrals and different solutions to differential equations.

3.2.3 Types of Rough Paths

Proposition 3.9 held automatically for smooth paths. Promoting it to a definition isolates the class on which ordinary calculus survives.

Definition 3.16 (Geometric rough paths). A rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ is **weakly geometric** if

$$\text{Sym } \mathbb{X}_{s,t} = \frac{1}{2} X_{s,t} \otimes X_{s,t} \quad \text{for all } s \leq t, \quad (3.19)$$

and **geometric** if it is the limit, in the metric ϱ_α , of canonical lifts of smooth paths in the sense of Example 3.8. We write $\mathbf{C}_{wg}^\alpha$ and $\mathbf{C}_g^\alpha$ respectively.

Proposition 3.17. $\mathbf{C}_g^\alpha \subseteq \mathbf{C}_{wg}^\alpha$.

Proof: Each canonical lift satisfies (3.19) by Proposition 3.9. The condition is a closed one: if $\varrho_\alpha(\mathbf{X}^{(n)}, \mathbf{X}) \rightarrow 0$ then in particular $\mathbb{X}_{s,t}^{(n)} \rightarrow \mathbb{X}_{s,t}$ and $X_{s,t}^{(n)} \rightarrow X_{s,t}$ for each fixed pair s, t , and both sides of (3.19) pass to the limit. ■

The inclusion is strict, but only just, and for our purposes harmlessly: a weakly geometric α -Hölder rough path is a geometric β -Hölder rough path for every $\beta < \alpha$, a result of Friz and



Victoir [34, Ch. 8]. Since our hypotheses always tolerate a slightly smaller exponent, we shall use the two notions interchangeably and say simply *geometric*, flagging the distinction only where it does work.

Example 3.18 (Brownian motion as a rough path). Let W be a d -dimensional Brownian motion. Define, for $s \leq t$,

$$\mathbb{W}_{s,t}^{\text{It}\hat{o}} := \int_s^t (W_r - W_s) \otimes dW_r, \quad (3.20)$$

the integral being the Itô integral of Definition 1.50, and

$$\mathbb{W}_{s,t}^{\text{Strat}} := \mathbb{W}_{s,t}^{\text{It}\hat{o}} + \frac{1}{2}(t-s)\text{Id}, \quad (3.21)$$

where $\text{Id} = \sum_i e_i \otimes e_i$. Then, for $\alpha \in (\frac{1}{3}, \frac{1}{2})$, both $\mathbf{W}^{\text{It}\hat{o}} := (W, \mathbb{W}^{\text{It}\hat{o}})$ and $\mathbf{W}^{\text{Strat}} := (W, \mathbb{W}^{\text{Strat}})$ are almost surely rough paths in the sense of Definition 3.12. The second is geometric; the first is not.

Proof: Chen's relation (3.16) for $\mathbb{W}^{\text{It}\hat{o}}$ is the computation of Proposition 3.10 verbatim, the splitting of the outer integral at u and the extraction of the constant $W_u - W_s$ being valid for the Itô integral by linearity. For (3.21) it holds because the added term $\frac{1}{2}(t-s)\text{Id}$ is an increment and therefore telescopes, exactly as in Proposition 3.15. The analytic bound (3.17) follows from the Itô isometry (Theorem 1.51), which gives $\mathbb{E}[|\mathbb{W}_{s,t}^{\text{It}\hat{o}}|^2] \leq C|t-s|^2$, hence $\mathbb{E}[|\mathbb{W}_{s,t}|^{2m}] \leq C_m|t-s|^{2m}$ by Gaussian hypercontractivity, followed by a two-parameter version of Kolmogorov's criterion; the details are in [33, Ch. 3].

For the geometric assertion, apply the multidimensional Itô rule (Theorem 1.72) to the product $W_{s,t}^i W_{s,t}^j$, using the Itô multiplication table of Chapter 1:

$$W_{s,t}^i W_{s,t}^j = \int_s^t W_{s,r}^i dW_r^j + \int_s^t W_{s,r}^j dW_r^i + \delta_{ij}(t-s),$$

the last term being $\langle W^i, W^j \rangle_{s,t}$ by Proposition 1.33. In tensor notation this reads precisely $2 \text{Sym } \mathbb{W}_{s,t}^{\text{It}\hat{o}} = W_{s,t} \otimes W_{s,t} - (t-s)\text{Id}$, so (3.19) fails for the Itô lift by $-\frac{1}{2}(t-s)\text{Id}$, and holds for the Stratonovich lift, which was defined by adding precisely that discrepancy back. ■

Remark 3.19 (The law of the Lévy area). Since $\mathbb{W}$ has just been constructed, it is worth recording what its distribution is. For $d = 2$ the antisymmetric part (3.13) is the scalar $A_{s,t} = \frac{1}{2}(\mathbb{X}_{s,t}^{12} - \mathbb{X}_{s,t}^{21})$, the signed area swept by the planar Brownian path against its chord, and Lévy's formula gives its conditional characteristic function

$$\mathbb{E}\left[e^{iuA_{s,t}} \mid W_{s,t} = z\right] = \frac{u(t-s)/2}{\sinh(u(t-s)/2)} \exp\left(\frac{|z|^2}{2(t-s)} \left[1 - \frac{u(t-s)}{2} \coth \frac{u(t-s)}{2}\right]\right). \quad (3.22)$$

Unconditionally, $A_{0,t}$ is distributed as tC with C having characteristic function $1/\cosh(u/2)$, so that $\mathbb{V}[A_{0,t}] = t^2/4$. Note that the law is *not* Gaussian, which is the first sign that the second level is a genuinely new object rather than a function of the first; the derivation, via the Cameron–Martin formula for a quadratic functional of Brownian motion, is in [34, Ch. 13]. We do not use (3.22), but it is what one would simulate from if constructing $\mathbb{W}$ directly rather than by discretisation.

The Itô–Stratonovich correction of Chapter 1, which appeared there as a consequence of the left-endpoint convention in the construction of the integral, has reappeared here as the difference between two *elements of the same space* $\mathbf{C}^\alpha$, lying over the same first level and differing by the

increment of $f_t = \frac{1}{2}t \text{Id} \in \mathcal{C}^1 \subset \mathcal{C}^{2\alpha}$. Proposition 3.15 predicted that such pairs exist; Example 3.18 says that the two calculi of Part I are an instance. The choice between them is not forced by the trajectory. It never was.

We turn to the process that motivated the chapter.

Theorem 3.20 (Coutin–Qian). *Let W^H be a d -dimensional fractional Brownian motion with $H \in (\frac{1}{3}, \frac{1}{2}]$ and independent components, and let $W^{H,(n)}$ denote its piecewise-linear interpolation over the dyadic partition of mesh 2^{-n} , canonically lifted as in Example 3.8. Then, for every $\alpha < H$, the lifts converge in probability in the metric ϱ_α to a limit $\mathbf{B}^H = (W^H, \mathbb{X}^H) \in \mathbf{C}_g^\alpha$, which is therefore a geometric rough path.* $\square$

The result is due to Coutin and Qian [23]; a modern proof, via the ϱ -variation of the covariance (3.1), is given in [33, Ch. 10]. The mechanism is instructive: one shows that R_H has finite two-dimensional ϱ -variation with $\varrho = 1/(2H)$, and the construction requires $\varrho < \frac{3}{2}$, which is exactly $H > \frac{1}{3}$.

3.3 Fundamental Results

3.3.1 Integration

We now construct the integral. The strategy has three steps: an abstract lemma which converts local estimates into a global object; a class of admissible integrands; and the theorem that puts them together.

Every construction of an integral in this dissertation has had the same shape — propose a Riemann sum, show it converges. The following lemma, due in this form to Gubinelli, extracts the mechanism.

Lemma 3.21 (Sewing Lemma). *Let $\beta > 1$ and let $\Xi : \{0 \leq s \leq t \leq T\} \rightarrow \mathbb{R}^m$ be continuous with $\Xi_{t,t} = 0$ and*

$$|\delta\Xi_{s,u,t}| \leq C |t - s|^\beta, \quad \text{where } \delta\Xi_{s,u,t} := \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t}, \quad (3.23)$$

for all $s \leq u \leq t$. Then there exists a unique function $\mathcal{I}\Xi$, of the same two variables, which is additive — that is, $(\mathcal{I}\Xi)_{s,t} = (\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ for all $s \leq u \leq t$ — and satisfies

$$|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \leq \frac{C}{1 - 2^{1-\beta}} |t - s|^\beta. \quad (3.24)$$

Moreover $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} \Xi_{u,v}$, the limit being over partitions $\mathcal{P}$ of $[s, t]$ with mesh tending to zero.

Proof (uniqueness): Suppose A and $\tilde{A}$ are both additive and both satisfy (3.24), with constants C_1 and C_2 . Then $D := A - \tilde{A}$ is additive and $|D_{s,t}| \leq (C_1 + C_2)|t - s|^\beta =: K|t - s|^\beta$. Fix $s < t$ and let $\mathcal{P}_n$ be the uniform partition of $[s, t]$ into n pieces. Additivity gives

$$|D_{s,t}| = \left| \sum_{[u,v] \in \mathcal{P}_n} D_{u,v} \right| \leq \sum_{[u,v] \in \mathcal{P}_n} K |v - u|^\beta = n \cdot K \left(\frac{t - s}{n} \right)^\beta = K |t - s|^\beta n^{1-\beta}.$$



Since $\beta > 1$ the right-hand side tends to 0 as $n \rightarrow \infty$, so $D \equiv 0$. The existence half, which proceeds by dyadic refinement, is proved in Appendix B.7. ■

The hypothesis (3.23) is the exact quantitative statement of “ Ξ is almost additive”, and the conclusion is that an almost additive function is uniformly close to a unique genuinely additive one. The strictness of $\beta > 1$ is what the counting argument above consumes: at $\beta = 1$ the factor $n^{1-\beta}$ is 1 and the argument collapses. Everything in this section is an application of the lemma with a specific Ξ , and the whole difficulty is arranging for $\beta > 1$ to hold.

We cannot integrate an arbitrary path against $\mathbf{X}$; the integrand must be compatible with the integrator in a sense that Chapter 1 would have called adaptedness, but which is here entirely deterministic.

Definition 3.22 (Controlled rough path). Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$. A pair (Y, Y') of paths $Y \in \mathcal{C}^\alpha([0, T], W)$ and $Y' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, W))$ is a path **controlled by $\mathbf{X}$** if the remainder defined by

$$R_{s,t}^Y := Y_{s,t} - Y'_s X_{s,t}, \quad Y_{s,t} := Y_t - Y_s, \quad (3.25)$$

satisfies $\|R^Y\|_{2\alpha} < \infty$. We write $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ and call Y' the **Gubinelli derivative** of Y with respect to X .

Equation (3.25) is a first-order Taylor expansion of Y along X : the path Y is required to look, locally, like a linear image of X , with an error of order 2α rather than the α that mere Hölder continuity would give. The pair $(\mathcal{D}_X^{2\alpha}, \|\cdot\|)$ with $\|(Y, Y')\|_{X, 2\alpha} := \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$ is a Banach space once the initial values are fixed [33, Ch. 4].

Remark 3.23 (Y' is data, not a derivative). The notation Y' is suggestive but must not be taken literally: in general Y' is *not* determined by Y and X . If X happens to be smooth, say $X_t = t$, then any Y' whatsoever satisfies (3.25) with a remainder of the required order, since $Y'_s X_{s,t} = O(|t-s|)$ is itself 2α -small when $2\alpha \leq 1$. Uniqueness holds only when X is genuinely irregular — “truly rough” in the terminology of [33, Ch. 6] — which is the case almost surely for Brownian motion and for W^H . Like the second level $\mathbb{X}$ itself, Y' is part of the given data. The theory is consistently one in which one carries more information than the classical objects contain.

Example 3.24. i) The integrator controls itself: $(X, \text{Id}) \in \mathcal{D}_X^{2\alpha}$, with $R^X \equiv 0$.

ii) If $F \in \mathcal{C}_b^2(W, \widetilde{W})$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, then $(F(Y), DF(Y)Y') \in \mathcal{D}_X^{2\alpha}$. Indeed, Taylor’s theorem with integral remainder gives

$$F(Y_t) - F(Y_s) = DF(Y_s)Y_{s,t} + O(|Y_{s,t}|^2) = DF(Y_s)Y'_s X_{s,t} + DF(Y_s)R_{s,t}^Y + O(|t-s|^{2\alpha}),$$

and the last two terms are both of order 2α . This closes the class under the nonlinearities that appear in differential equations.

The naïve Riemann sum $\sum Y_u X_{u,v}$ does not converge: its defect is of order $|t-s|^{2\alpha}$, and $2\alpha \leq 1$, so Lemma 3.21 does not apply. The remedy is to add a second-order term, and the expansion (3.25) tells us precisely which one.

Theorem 3.25 (Gubinelli). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}$. Then the **compensated Riemann sums***

$$\int_s^t Y_r d\mathbf{X}_r := \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} \left(Y_u X_{u,v} + Y'_u \mathbb{X}_{u,v} \right) \quad (3.26)$$

converge, and the limit satisfies

$$\left| \int_s^t Y_r d\mathbf{X}_r - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \leq C \left(\|X\|_\alpha \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'\|_\alpha \right) |t - s|^{3\alpha}, \quad (3.27)$$

for a constant $C = C(\alpha)$. Moreover the indefinite integral is itself controlled by $\mathbf{X}$:

$$\left(\int_0^\cdot Y_r d\mathbf{X}_r, Y \right) \in \mathcal{D}_X^{2\alpha}. \quad (3.28)$$

The proof, an application of Lemma 3.21 to $\Xi_{s,t} := Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}$, is given in Appendix B.8.

Remark 3.26 (Where $\alpha > \frac{1}{3}$ comes from). Computing $\delta\Xi_{s,u,t}$ for the choice above, and using Chen's relation (3.16) to cancel the leading terms — this is the computation in the appendix, and it is the only place where (3.16) is used — one finds

$$-\delta\Xi_{s,u,t} = R_{s,u}^Y X_{u,t} + (Y'_u - Y'_s) \mathbb{X}_{u,t}, \quad (3.29)$$

whose two terms are of order $2\alpha + \alpha$ and $\alpha + 2\alpha$ respectively. So $\beta = 3\alpha$, and Lemma 3.21 demands $\beta > 1$, that is $\alpha > \frac{1}{3}$. Below that threshold the second level no longer suffices and a third iterated integral must be carried; below $\frac{1}{4}$, a fourth; and in general $\lfloor 1/\alpha \rfloor$ levels are required. We have restricted to (3.15) precisely to avoid the algebra that the general case demands, at no cost to the ideas.

Theorem 3.27 (Stability of the integral). *Fix $M < \infty$ and consider $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathbf{C}^\alpha$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\widetilde{Y}, \widetilde{Y}') \in \mathcal{D}_{\widetilde{X}}^{2\alpha}$, all of norm at most M . Then, with $C = C(M, \alpha)$,*

$$\left\| \int_0^\cdot Y d\mathbf{X} - \int_0^\cdot \widetilde{Y} d\widetilde{\mathbf{X}} \right\|_\alpha \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |Y_0 - \widetilde{Y}_0| + |Y'_0 - \widetilde{Y}'_0| + \|Y, Y'; \widetilde{Y}, \widetilde{Y}'\|_{X, \widetilde{X}, 2\alpha} \right), \quad (3.30)$$

where the last term is the distance between two controlled paths lying over different rough paths,

$$\|Y, Y'; \widetilde{Y}, \widetilde{Y}'\|_{X, \widetilde{X}, 2\alpha} := \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha}, \quad (3.31)$$

with $R_{s,t}^Y = Y_{s,t} - Y'_s X_{s,t}$ and $R_{s,t}^{\widetilde{Y}} = \widetilde{Y}_{s,t} - \widetilde{Y}'_s \widetilde{X}_{s,t}$ as in (3.25). $\square$

Note that (Y, Y') and $(\widetilde{Y}, \widetilde{Y}')$ live in the *different* Banach spaces $\mathcal{D}_X^{2\alpha}$ and $\mathcal{D}_{\widetilde{X}}^{2\alpha}$ and cannot simply be subtracted. What is compared instead is the pair of Gubinelli derivatives, which live in a common space, together with the pair of remainders, which also do; the two rough paths enter only through the definitions of R^Y and $R^{\widetilde{Y}}$. This is the standard device, and it is the reason the estimate carries $\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}})$ as a separate term rather than absorbing it.

This is proved in [33, Thm. 4.17] by re-running the proof of Theorem 3.25 on differences. We record it because it, and not Theorem 3.25 itself, is what makes the continuity statement of Subsection 3.3.4 possible.

The construction would be worthless if it disagreed with the theory of Part I where both apply. It does not.

Proposition 3.28 (Consistency). *Let W be a Brownian motion and let $F \in \mathbf{C}_b^3$. Then, almost surely,*

$$\int_0^t F(W_r) d\mathbf{W}_r^{\text{Itô}} = \int_0^t F(W_r) dW_r, \quad \int_0^t F(W_r) d\mathbf{W}_r^{\text{Strat}} = \int_0^t F(W_r) \circ dW_r, \quad (3.32)$$

the right-hand sides being the Itô integral of Definition 1.50 and the Stratonovich integral of Definition 1.56 respectively. $\square$



See [33, Ch. 5]. The first identity is the one to appreciate: the left-hand side is a deterministic limit of Riemann sums, computed path by path with no reference to the filtration, and the right-hand side is an $L^2(\Omega)$ limit that exists only up to null sets. That they agree means the Itô integral, for this class of integrands, admits a pathwise description after all — at the price of carrying $\mathbb{W}^{\text{Itô}}$, an object which is itself defined only through Itô's construction. Rough path theory does not eliminate probability; it isolates it into a single step, performed once, after which everything is deterministic.

We can now make precise the claim of Subsection 3.2.3, that geometric rough paths are those on which ordinary calculus survives. The following quantity measures the failure of (3.19), and turns out to be exactly the correction term in the chain rule.

Definition 3.29 (Bracket of a rough path). For $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ define

$$[\mathbf{X}]_t := X_{0,t} \otimes X_{0,t} - 2 \operatorname{Sym} \mathbb{X}_{0,t} \in V \otimes V. \quad (3.33)$$

Lemma 3.30. *The bracket is a genuine path, in the sense that its increments are given by the corresponding two-parameter expression:*

$$[\mathbf{X}]_t - [\mathbf{X}]_s = X_{s,t} \otimes X_{s,t} - 2 \operatorname{Sym} \mathbb{X}_{s,t}, \quad (3.34)$$

and $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$.

Proof: Expanding $X_{0,t} = X_{0,s} + X_{s,t}$ in the first term of (3.33),

$$X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} = X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s} + X_{s,t} \otimes X_{s,t},$$

while Chen's relation (3.16) gives $\mathbb{X}_{0,t} - \mathbb{X}_{0,s} = \mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t}$, so that

$$2 \operatorname{Sym} \mathbb{X}_{0,t} - 2 \operatorname{Sym} \mathbb{X}_{0,s} = 2 \operatorname{Sym} \mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s}.$$

Subtracting, the two cross terms cancel and (3.34) follows. The regularity is then immediate from Definition 3.12, both terms on the right of (3.34) being of order $|t - s|^{2\alpha}$. $\blacksquare$

By construction $\mathbf{X}$ is weakly geometric precisely when $[\mathbf{X}] \equiv 0$; Definition 3.16 could equally well have been phrased that way.

Proposition 3.31 (Rough chain rule). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ with $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, and let $F \in \mathcal{C}_b^3(V, W)$. Then $(F(X), DF(X)) \in \mathcal{D}_X^{2\alpha}$, the rough integral $\int_0^t DF(X_r) d\mathbf{X}_r$ is defined by Theorem 3.25, and*

$$F(X_t) - F(X_0) = \int_0^t DF(X_r) d\mathbf{X}_r + \frac{1}{2} \int_0^t D^2F(X_r) d[\mathbf{X}]_r, \quad (3.35)$$

the second integral being a Young integral, well defined since $D^2F(X) \in \mathcal{C}^\alpha$, $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$ and $3\alpha > 1$.

In particular, if $\mathbf{X}$ is weakly geometric then $[\mathbf{X}] \equiv 0$ and (3.35) reduces to the classical chain rule

$$F(X_t) - F(X_0) = \int_0^t DF(X_r) d\mathbf{X}_r. \quad (3.36) \quad \square$$

A proof is in [33, Ch. 7]; the mechanism is a second-order Taylor expansion of F along the partition, in which the symmetric part of $\mathbb{X}$ is exactly what the quadratic term of the expansion produces, so that the two agree and cancel precisely when (3.19) holds.

The corollary is the one we have been aiming at since Subsection 3.2.3.

Corollary 3.32 (Itô's formula). *Let W be a d -dimensional Brownian motion and $F \in \mathcal{C}_b^3$. Applying (3.35) to the two lifts of Example 3.18:*

- i) for $\mathbf{W}^{\text{Strat}}$, which is geometric, one obtains $F(W_t) - F(W_0) = \int_0^t DF(W_r) \circ dW_r$, the classical chain rule of Stratonovich calculus;
- ii) for $\mathbf{W}^{\text{Itô}}$ one computes from (3.34) and the identity $2 \text{Sym } \mathbb{W}_{s,t}^{\text{Itô}} = W_{s,t} \otimes W_{s,t} - (t-s)\text{Id}$ established in Example 3.18 that $[\mathbf{W}^{\text{Itô}}]_t = t \text{Id}$, whence

$$F(W_t) - F(W_0) = \int_0^t DF(W_r) dW_r + \frac{1}{2} \int_0^t \Delta F(W_r) dr, \quad (3.37)$$

which is the case $V(t, x) = F(x)$, $X = W$ of Theorem 1.72 of Chapter 1.

Theorem 1.72 covers $\theta(t, X_t)$ for a general n -dimensional Itô process and asks only that $\theta \in \mathcal{C}^{1,2}$; what we have recovered is the time-homogeneous case, driven by Brownian motion, under the stronger hypothesis $F \in \mathcal{C}_b^3$. The rough-path route is therefore not a strengthening of Itô's formula but a *re-derivation of a special case of it on different foundations*. What is bought in exchange is that (3.37) now holds pathwise, for a fixed trajectory and its lift, with both sides continuous in the rough path metric — neither of which the classical statement provides.

In Chapter 1, the second-order term of Itô's formula arose from the non-vanishing quadratic variation of Brownian motion, and appeared as an irreducible feature of stochastic calculus — the price of integrating against a martingale. Here it has arisen from the bracket (3.33) of a particular *lift*, and the Stratonovich lift of the same trajectory has no such term at all. The correction was therefore not forced by Brownian motion alone: the non-vanishing quadratic variation is what makes the two lifts differ at all — for a driver of bounded variation they coincide — but which of them is taken is a choice, made implicitly in Subsection 1.2.1 when the integrand was evaluated at the left endpoint of each subinterval, and made explicitly here by writing down $\mathbb{W}^{\text{Itô}}$ rather than $\mathbb{W}^{\text{Strat}}$.

Let $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (X, \mathbb{X} + \delta f)$ be two lifts of the same path, as in Proposition 3.15. Directly from (3.26),

$$\int_s^t Y d\widetilde{\mathbf{X}} = \int_s^t Y d\mathbf{X} + \int_s^t Y'_r df_r, \quad (3.38)$$

the last integral existing in the Young sense since $Y' \in \mathcal{C}^\alpha$, $f \in \mathcal{C}^{2\alpha}$ and $3\alpha > 1$. Taking $f_t = \frac{1}{2}t \text{Id}$ and $Y' = DF(W)$ recovers, by Example 3.18, the classical relation $\int F(W) \circ dW = \int F(W) dW + \frac{1}{2} \int F'(W) dr$ of Chapter 1. The Itô–Stratonovich correction is thus not a feature of stochastic calculus at all: it is the general formula (3.38) evaluated at a particular f .

3.3.2 Differential Equations

With an integral in hand, differential equations are defined in the only way available: as integral equations. What requires care is that the unknown is now a *pair*.

Definition 3.33 (Rough differential equation). Let $\mathbf{X} \in \mathbf{C}^\alpha$, $\xi \in W$ and $f : W \rightarrow \mathcal{L}(V, W)$. A pair $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ is a **solution** of the rough differential equation

$$dY_t = f(Y_t) d\mathbf{X}_t, \quad Y_0 = \xi, \quad (3.39)$$

if $Y' = f(Y)$ and, for every $t \in [0, T]$,

$$Y_t = \xi + \int_0^t f(Y_r) d\mathbf{X}_r, \quad (3.40)$$



the integral being that of Theorem 3.25, which is well defined because $(f(Y), Df(Y)Y') \in \mathcal{D}_X^{2\alpha}$ by Example 3.24 ii).

Remark 3.34 (Why $Y' = f(Y)$ is imposed). Without it uniqueness fails outright. Consider $X \in \mathcal{C}^1$ lifted canonically, so that $|X_{s,t}| \lesssim |t - s|$. Then for *any* choice of Y' the term $Y'_s X_{s,t}$ is $O(|t - s|)$ and can be absorbed into a 2α -remainder, so every Y' makes (Y, Y') controlled and (3.40) carries no information about it. Requiring $Y' = f(Y)$ pins the derivative to the equation and restores well-posedness. The reader may compare Definition 1.63, where the analogous role is played by the requirement that the solution be adapted.

There is an equivalent formulation, due to Davie, which dispenses with controlled paths entirely and which is the one a numerical analyst would write down.

Proposition 3.35 (Davie's formulation). *A path $Y \in \mathcal{C}^\alpha$ solves (3.39) in the sense of Definition 3.33 if and only if*

$$Y_{s,t} = f(Y_s) X_{s,t} + Df(Y_s) f(Y_s) \mathbb{X}_{s,t} + o(|t - s|) \quad (3.41)$$

uniformly in $s \leq t$. □

The proof is in [33, Prop. 8.8]. Expression (3.41) is worth recognising: truncated after the first term it is the Euler scheme, and as it stands it is the *Milstein scheme*, with $\mathbb{X}_{s,t}$ playing the role of the iterated integral that the Milstein correction requires.

3.3.3 Existence

Theorem 3.36 (Local existence and uniqueness). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} \in \mathbf{C}^\alpha([0, T], V)$ and let $f \in \mathcal{C}_b^3(W, \mathcal{L}(V, W))$. Then for every $\xi \in W$ there exist $T_0 \in (0, T]$ and a unique $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ on $[0, T_0]$ solving (3.39). The time T_0 depends only on α , $\|f\|_{\mathcal{C}_b^3}$ and $\|\mathbf{X}\|_\alpha$, and not on ξ ; consequently the solution extends to a unique global solution on $[0, T]$.*

The proof is given in Appendix B.9. One defines the map

$$\mathcal{M}(Y, Y') := \left(\xi + \int_0^\cdot f(Y_r) d\mathbf{X}_r, f(Y) \right) \quad (3.42)$$

on the Banach space $\mathcal{D}_X^{2\alpha}$, whose fixed points are by Definition 3.33 exactly the solutions. One then shows that $\mathcal{M}$ maps a suitable closed ball into itself and is a contraction on it, provided the time horizon is small; the Banach fixed point theorem does the rest. This is precisely the Picard iteration on which Theorem 1.65 rests in Chapter 1, and the analogy is exact — but the argument is now entirely deterministic, and takes place in the space of controlled paths rather than in $L^2(\Omega)$.

The contraction factor comes from the horizon. Every estimate in Theorem 3.25 carries a factor T_0^α when restricted to $[0, T_0]$, and choosing T_0 small makes that factor as small as desired. The subtle point — and the reason the theorem is stated as it is — is that the admissible T_0 must be *independent of the initial condition*, since otherwise the successive restarts used to reach T might have horizons shrinking to zero and the solution would fail to be global. That independence is secured by the fact, visible in (3.27), that the estimates involve $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$, together with the boundedness of f . With T_0 uniform, one solves on $[0, T_0]$, restarts at Y_{T_0} , and repeats finitely often.

Remark 3.37 (On the regularity of f). Three bounded derivatives are required, one more than the two that Example 3.24 ii) needs to make $f(Y)$ controlled. The extra derivative is spent on the contraction estimate, where one must compare $f(Y)$ with $f(\tilde{Y})$ as *controlled paths*, which involves Df and its Lipschitz constant. This is the analogue of the Lipschitz hypothesis in Definition 1.64; that a rougher driver should demand a smoother coefficient is the recurring trade of this subject. Without boundedness one still obtains a maximal solution, which may explode in finite time exactly as in the ODE case.

3.3.4 Uniqueness and Continuity

We directly state one of the key results of rough path theory.

Theorem 3.38 (Universal Limit Theorem; continuity of the Itô–Lyons map). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $f \in \mathcal{C}_b^3$. For $\mathbf{X} \in \mathbf{C}^\alpha$ and $\xi \in W$ let $\Phi(\xi, \mathbf{X}) := Y$ denote the unique solution of (3.39) given by Theorem 3.36. Then Φ is locally Lipschitz continuous: for every $M < \infty$ there is $C = C(M, \alpha, f)$ such that, whenever $\|\mathbf{X}\|_\alpha, \|\tilde{\mathbf{X}}\|_\alpha \leq M$,*

$$\|\Phi(\xi, \mathbf{X}) - \Phi(\tilde{\xi}, \tilde{\mathbf{X}})\|_\alpha \leq C \left(|\xi - \tilde{\xi}| + \varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) \right). \quad (3.43)$$

The proof is given in Appendix B.10; it combines the stability of rough integration (Theorem 3.27) with the stability of composition, applied to the fixed point characterisation (3.42).

The significance of (3.43) is best measured against what Part I could say. There, the solution of an SDE was produced as an equivalence class of random variables, obtained through an $L^2(\Omega)$ limit; the map from the driving Brownian path to the solution path existed only as a measurable map defined up to null sets, and Example 3.4 showed that it could not be continuous in the uniform topology. Theorem 3.38 says that once the driving signal is *enhanced*, the solution map becomes not merely continuous but locally Lipschitz, and entirely deterministic. All the probability in the theory of stochastic differential equations has been compressed into the single measurable step of constructing $\mathbb{W}$, performed once in Example 3.18.

Three consequences deserve mention. The first is the following theorem:

Corollary 3.39 (Wong–Zakai). *Let W be a Brownian motion and let $W^{(n)}$ its piecewise-linear interpolation over a partition of mesh tending to zero, and let $Y^{(n)}$ solve the ordinary differential equation $dY^{(n)} = f(Y^{(n)}) dW^{(n)}$. Then $Y^{(n)}$ converges in probability, in α -Hölder norm — and almost surely along a nested sequence of partitions, the dyadic one for instance — to the solution of the Stratonovich equation $dY = f(Y) \circ dW$.*

Proof: The canonical lifts of $W^{(n)}$ converge in ϱ_α to $\mathbf{W}^{\text{Strat}}$ — this is the content of [33, Ch. 3], and the Stratonovich rather than the Itô lift appears because each $W^{(n)}$ is smooth, hence its lift is geometric, hence so is the limit by Proposition 3.17; that the limit is $\mathbf{W}^{\text{Strat}}$ specifically, and not some other geometric lift, is part of the cited result. Theorem 3.38 then transfers the convergence to the solutions, and Proposition 3.28 identifies the limit. ■

Second, Theorem 3.38 puts numerical analysis on a pathwise footing. The Milstein scheme (3.41) converges because the map is continuous, and the rate is controlled by (3.43); within Itô theory the corresponding estimates are made in $L^2(\Omega)$, and afresh for each scheme.

Third — and this is the reason the theory is called *universal* — nothing in Theorem 3.38 refers to Brownian motion, to a filtration, or to a probability measure. Any process that can be lifted



to $\mathbf{C}^\alpha$ inherits the entire theory at once: fractional Brownian motion for $H > \frac{1}{3}$ by Theorem 3.20, Gaussian processes with sufficiently regular covariance, suitable Markov processes such as uniformly elliptic symmetric diffusions, and the piecewise-linear paths formed from observed data.

3.4 The Signature Transform

The rough paths of Definition 3.12 carry two levels because two are what an $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ *integrator* requires — the regularity of the integrand is governed instead by the controlled-path condition (3.25). Nothing prevents us from carrying more, and if we carry all of them we obtain an object of independent interest, and the natural direction in which the present work extends.

Theorem 3.40 (Lyons' extension theorem). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ with $\alpha \in (\frac{1}{3}, \frac{1}{2}]$. Then there is a unique extension of $\mathbf{X}$ to a family $(1, X_{s,t}^1, X_{s,t}^2, X_{s,t}^3, \dots)$, with $X^1 = X$, $X^2 = \mathbb{X}$ and $X_{s,t}^n \in V^{\otimes n}$, which satisfies Chen's relation at every level,*

$$X_{s,t}^n = \sum_{k=0}^n X_{s,u}^k \otimes X_{u,t}^{n-k}, \quad s \leq u \leq t, \quad (3.44)$$

and the bounds $|X_{s,t}^n| \leq C_n |t - s|^{n\alpha}$ for every n . □

The theorem is Lyons' [57]; see also [38, Ch. 3]. Its content is that the higher levels are *not* additional data: once two are given, the rest are determined, and determined by a construction which is again an instance of the sewing lemma. This is the precise sense in which Definition 3.12 loses nothing by truncating.

Definition 3.41 (Signature). The **signature** of a rough path $\mathbf{X}$ over $[0, T]$ is the full collection of its iterated integrals over the whole interval,

$$S(\mathbf{X}) := (1, X_{0,T}^1, X_{0,T}^2, X_{0,T}^3, \dots), \quad (3.45)$$

and its **truncation at level N** , written $S^N(\mathbf{X})$, is the finite collection $(1, X_{0,T}^1, \dots, X_{0,T}^N) \in T^{(N)}(V)$. For a path x of bounded variation these are the ordinary iterated integrals

$$X_{0,T}^n = \int_{0 < t_1 < \dots < t_n < T} dx_{t_1} \otimes \dots \otimes dx_{t_n}. \quad (3.46)$$

Example 3.42. For $d = 1$ the tensor products are ordinary products and (3.46) evaluates to $X_{0,T}^n = (x_T - x_0)^n/n!$, independently of the trajectory. The signature of a scalar path is therefore $\exp(x_T - x_0)$ and carries no information beyond the total increment. All the content of the transform lies in the non-commutativity of $\otimes$, which is to say in dimension $d \geq 2$.

Three properties make the signature useful:

Proposition 3.43 (Factorial decay). *For a path x of bounded variation with total variation $L := V_1(x)_T$,*

$$|X_{0,T}^n| \leq \frac{L^n}{n!}. \quad (3.47)$$

Proof: Bounding (3.46) termwise, the integrand has modulus at most $|dx_{t_1}| \cdots |dx_{t_n}|$, and the domain of integration is the simplex $0 < t_1 < \cdots < t_n < T$, which is the $1/n!$ fraction of the cube on which the unrestricted product integrates to L^n . ■

The estimate answers the objection that (3.45) is an infinite object: the terms decay factorially, so a truncation at a modest N retains almost all of the information. This is what makes S^N a usable feature set.

Proposition 3.44 (Shuffle product and linearisation). *Let $\mathbf{X}$ be geometric. Write $X_{0,T}^{n;I}$ for the coordinate of $X_{0,T}^n$ along the basis vector $e_{i_1} \otimes \cdots \otimes e_{i_n}$, indexed by the word $I = (i_1, \dots, i_n)$. Then for any two words $I = (i_1, \dots, i_m)$ and $J = (j_1, \dots, j_n)$ in the letters $\{1, \dots, d\}$,*

$$X_{0,T}^{m;I} \cdot X_{0,T}^{n;J} = \sum_{K \in I \sqcup J} X_{0,T}^{m+n;K}, \quad (3.48)$$

the sum being over all $\binom{m+n}{m}$ interleavings of I and J preserving the internal order of each. □

Relation (3.48) — the *shuffle product formula*, proved in [38, Lem. 1.15] by splitting the product of two simplices into the disjoint union of the simplices indexed by the shuffles — has a consequence out of all proportion to its appearance. It says that the product of two coordinates of the signature is a *linear combination* of other coordinates. Hence every polynomial in the entries of $S(\mathbf{X})$ is again a linear functional of $S(\mathbf{X})$, and any continuous functional of the path can be approximated, by Stone–Weierstrass, by a *linear* functional of its signature. A nonlinear problem on path space becomes a linear problem on signature space. This is the theoretical basis for the use of signatures as features in statistical learning; see [20] and [52].

It remains to be shown that nothing is lost.

Definition 3.45 (Tree-like paths). A *real tree* is a metric space in which any two points are joined by a unique non-self-intersecting path, which is moreover a geodesic. A continuous path $x : [0, T] \rightarrow V$ is **tree-like** if there exist a real tree $\mathcal{T}$ and continuous maps $\phi : [0, T] \rightarrow \mathcal{T}$ and $\psi : \mathcal{T} \rightarrow V$ with $\phi(0) = \phi(T)$ and $x = \psi \circ \phi$. A path is **tree-reduced** if it contains no tree-like piece.

The second notion is the one that makes Theorem 3.46 an honest bijection: the signature is injective not on all paths but on the tree-reduced ones, which are exactly the representatives of the equivalence classes the theorem describes.

Informally, a tree-like path is one that retraces its own steps and thereby cancels itself: it goes out along some route and returns along the same route, enclosing no area. Such excursions contribute nothing to any iterated integral, and the following theorem says that they are the *only* thing the signature forgets.

Theorem 3.46 (Uniqueness of the signature). *Let $\mathbf{X}$ be a weakly geometric rough path. Then $S(\mathbf{X}) = \mathbf{1}$ if and only if $t \mapsto X_{0,t}$ is tree-like. Consequently — applying this to $\mathbf{X} \star \widetilde{\mathbf{X}}^{-1}$, the concatenation of $\mathbf{X}$ with the time-reversal of $\widetilde{\mathbf{X}}$, whose signature is $S(\mathbf{X}) \otimes S(\widetilde{\mathbf{X}})^{-1}$ by the group structure of $T((V))$ — two weakly geometric rough paths have the same signature if and only if they coincide up to tree-like pieces.* □

The bounded-variation case is due to Hambly and Lyons [42]; the extension to rough paths is due to Boedihardjo, Geng, Lyons and Yang [12]. A sketch of the argument, which proceeds by showing that the space of signatures is itself a real tree, is in [38, §4.3].



Back in Chapter 1, discussing the log-normal distribution, we observed that a probability law need not be determined by its moments — Heyde’s construction, recorded in Appendix A.3, exhibits distinct densities sharing every moment with the log-normal — and remarked that “knowing every moment of an object is strictly weaker than knowing the object”. Theorem 3.46 is that statement for *paths*. The iterated integrals are the moments of a path; unlike the moments of a random variable, they determine it, and the only ambiguity is the geometrically trivial one of tree-like excursions, which can be removed outright by the standard device of *time augmentation* — replacing x_t by (t, x_t) , whose first coordinate is strictly increasing and which therefore admits no tree-like piece at all. The same device removes the invariance of (3.46) under reparametrisation, which is otherwise a genuine loss of information whenever the speed at which a path is traversed carries meaning.

Definition 3.47 (Expected signature). Let $\mathbf{X}$ be a random geometric rough path, that is, a random variable with values in $\mathbf{C}_g^\alpha([0, T], V)$. Its **expected signature** is the element of the full tensor algebra $T((V)) = \prod_{n \geq 0} V^{\otimes n}$ of Definition 3.7, obtained by averaging the signature level by level,

$$\mathbb{E}[\mathbf{S}(\mathbf{X})] = \left(1, \mathbb{E}[X_{0,T}^1], \mathbb{E}[X_{0,T}^2], \dots\right), \quad (3.49)$$

whenever each expectation exists.

Theorem 3.48 (Chevyrev–Lyons). *Let $\mathbf{X}$ be a random geometric rough path whose expected signature (3.49) has infinite radius of convergence ($\sum_{n \geq 0} \lambda^n |\mathbb{E}[X_{0,T}^n]| < \infty$ for every $\lambda > 0$). Then $\mathbb{E}[\mathbf{S}(\mathbf{X})]$ determines the law of $\mathbf{X}$, up to the tree-like equivalence of Theorem 3.46.* $\square$

The result is due to Chevyrev and Lyons [21]. It is the precise analogue, for measures on path space, of the elementary fact that a compactly supported law on $\mathbb{R}$ is determined by its moment sequence — and the growth hypothesis plays exactly the role that Carleman’s condition plays in Appendix A.3. At the level of individual objects the correspondence is between the power X^k of a random variable and the iterated integral $X_{0,T}^n$ of a path; there it is Theorem 3.46 that has no counterpart on the left, since the powers of a single real number determine it trivially.

Remark 3.49 (Why this matters for calibration). The three results above are what make the signature usable as a feature map for path-dependent problems. Theorem 3.46 guarantees that no information is discarded by the transform; Proposition 3.43 bounds the tail discarded by truncation, which is cheap once N exceeds the length of the path, and is set against a count of words growing like d^N ; Proposition 3.44 guarantees that any continuous functional of the path is, to arbitrary accuracy, a *linear* functional of the truncated signature, so that a learning problem on path space becomes a regression.

3.5 Rough Models

Subsection 3.1.2 recorded the empirical finding and wrote down the RFSV model (3.4), which is the natural time-series description of a rough log-volatility. For pricing one wants something else: a model specified directly under the pricing measure, rich enough to fit an implied volatility surface, and tractable enough to be calibrated. The model that meets these requirements is the rough counterpart of the Heston model of Subsection 2.3.2, whose tractability rested on its affine structure.

Definition 3.50 (Rough Heston). Let $H \in (0, \frac{1}{2})$ and set $\alpha_H := H + \frac{1}{2} \in (\frac{1}{2}, 1)$. The **rough Heston model** is

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S, \\ v_t = v_0 + \frac{1}{\Gamma(\alpha_H)} \int_0^t (t - s)^{\alpha_H - 1} \left[\kappa(\theta - v_s) ds + \xi \sqrt{v_s} dW_s^v \right], \end{cases} \quad (3.50)$$

with $d\langle W^S, W^v \rangle_t = \rho dt$ in the sense of Definition 1.71.

Setting $\alpha_H = 1$, that is $H = \frac{1}{2}$, collapses the convolution kernel to a constant and recovers the classical Heston system (2.51) exactly. For $\alpha_H < 1$ the kernel $(t - s)^{\alpha_H - 1}$ is singular at $s = t$, and it is this singularity — a fractional integral of order α_H — that roughens the variance path to Hölder regularity H^- , in accordance with Proposition 3.2 c).

Two structural consequences follow immediately, and they pull in opposite directions.

The first is a loss. The convolution in (3.50) makes v_t depend on the entire history of W^v and not merely on v_s , so **the variance process is no longer Markov**, and neither is the pair (S, v) . Every tool of Subsection 1.2.6 therefore becomes unavailable at once: there is no infinitesimal generator, no Kolmogorov equation, no Feynman–Kac representation. The second is a survival, and it is what makes the model usable. The *affine* structure is untouched by the convolution. Affineness was what produced the Riccati system (2.53) in the classical case, and it produces its analogue here.

Theorem 3.51 (El Euch–Rosenbaum). *Let F denote the forward price. Under (3.50) the characteristic function of the centred log-forward, stated at $t = 0$, is*

$$\varphi(u, T) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(iu \log \frac{F_T}{F_0} \right) \right] = \exp \left(\kappa \theta \int_0^T \psi(u, s) ds + v_0 I^{1-\alpha_H} \psi(u, T) \right), \quad (3.51)$$

where $I^{1-\alpha_H}$ denotes the Riemann–Liouville fractional integral of order $1 - \alpha_H$ and $\psi(u, \cdot)$ is the unique continuous solution of the **fractional Riccati equation**

$$D^{\alpha_H} \psi(u, \tau) = \frac{\xi^2}{2} \psi(u, \tau)^2 + (\rho \xi iu - \kappa) \psi(u, \tau) - \frac{u^2 + iu}{2}, \quad I^{1-\alpha_H} \psi(u, 0) = 0, \quad (3.52)$$

D^{α_H} being the Riemann–Liouville fractional derivative of order α_H . $\square$

The statement is made at $t = 0$ deliberately. By the loss recorded above, the time- t conditional law depends on the whole forward variance curve $s \mapsto \mathbb{E}^{\mathbb{Q}}[v_s \mid \mathcal{F}_t]$ and not on v_t alone, so (3.51) does not survive being conditioned on $\mathcal{F}_t$ with v_t in place of v_0 .

The result is due to El Euch and Rosenbaum [29]. Comparison with Chapter 2 is the point of stating it. Equation (3.52) is (2.53) *term for term*: the single difference is that the ordinary derivative ∂_τ has been replaced by a fractional derivative of order $\alpha_H = H + \frac{1}{2}$, and at $H = \frac{1}{2}$ the two coincide. We shortly introduce one of the many notions of fractional derivative.

Definition 3.52 (Riemann–Liouville fractional operators). For $\beta > 0$ and $f \in L^1([0, T])$ the **fractional integral** of order β is

$$(I^\beta f)(t) := \frac{1}{\Gamma(\beta)} \int_0^t (t - s)^{\beta - 1} f(s) ds, \quad I^0 := \text{Id}, \quad (3.53)$$

and for $\beta \in (0, 1)$ the **fractional derivative** of order β is

$$(D^\beta f)(t) := \frac{d}{dt} (I^{1-\beta} f)(t), \quad (3.54)$$

whenever the right-hand side exists.



For integer $\beta = m$ formula (3.53) is Cauchy's repeated-integration formula, so I^m is the m -fold iterated integral and $I^1 = \int_0^\cdot$; the extension to real β is the unique one preserving the semigroup law $I^\beta I^\gamma = I^{\beta+\gamma}$, from which $D^\beta I^\beta = \text{Id}$ and $D^1 = d/dt$ follow. Definition 3.52 is also what appears in (3.50): the variance is $v = v_0 + I^{\alpha_H}[\cdot]$. Note that (3.54) differentiates $I^{1-\beta}f$ and not f , which is why the initial condition in (3.52) is imposed on $I^{1-\alpha_H}\psi$; at $\alpha_H = 1$ it reduces to $\psi(u, 0) = 0$, the initial condition of (2.53).

Remark 3.53 (What is gained and what it costs). Three comments, in decreasing order of comfort.

The model reproduces the skew. By construction the variance has regularity H , so the short-maturity at-the-money skew behaves as $\tau^{H-1/2}$ by (3.5). This is the empirical shape, and no classical stochastic volatility model produces it.

The equation is non-local, hence expensive. Because D^{α_H} is an integral operator, (3.52) cannot be advanced by a local time-stepping scheme: each step requires the entire computed history of ψ , so a discretisation over n steps costs $O(n^2)$ rather than $O(n)$. The standard solver is the Adams predictor–corrector scheme. Since calibration evaluates (3.51) at hundreds of strikes for every trial parameter vector, this cost is the practical bottleneck, and it is the main reason rough Heston is harder to deploy than its classical counterpart.

It is not a theorem of this chapter. Theorem 3.51 is proved by techniques from affine Volterra processes and fractional calculus, and nothing in Sections 3.2 to 3.4 enters its proof. This is a division of labour rather than an inconsistency: at $H \approx 0.1$ the lifting theorems are not available, so a route that does not require them is necessary rather than merely convenient. What rough path theory supplies here is the vocabulary — the regularity exponent, the p -variation scale, the precise sense in which the Markov property has been traded away — rather than the machinery.

3.5.1 Rough Geometric Brownian Motion

Definition 3.54 (Purely rough asset). Fix $H \in (\frac{1}{3}, \frac{1}{2}]$, $\sigma > 0$, $S_0 > 0$, and $r = q = 0$, so that discounting is trivial. Let W^H be a scalar fractional Brownian motion. Also, for $\alpha < H$, let $\mathbf{B}^H = (W^H, \mathbb{X}^H) \in \mathbf{C}_g^\alpha$ be its Coutin–Qian lift (Theorem 3.20), and let $\sigma\mathbf{B}^H := (\sigma W^H, \sigma^2 \mathbb{X}^H)$ be its dilation, which is again a geometric rough path by the scaling recorded after Definition 3.13. The **purely rough asset** is the solution, in the sense of Definition 3.33, of the linear rough differential equation

$$dS_t = \sigma S_t d\mathbf{B}_t^H, \quad S_0 > 0. \quad (3.55)$$

Since the driver is scalar, one ambiguity of the theory is absent here. By (3.19) a geometric lift must satisfy $\text{Sym } \mathbb{X}_{s,t}^H = \frac{1}{2}(W_{s,t}^H)^2$, and in one dimension $\mathbb{X}^H$ is its own symmetric part, so the geometric lift of a scalar path is unique: the choice which Proposition 3.15 identified as the central structural fact of the subject is, for (3.55), no choice at all. Everything below is therefore canonical, and the comparison with the Itô lift made in Proposition 3.56 is a comparison with a *non-geometric* object.

Remark 3.55. The vector field $f(y) = \sigma y$ is not $\mathcal{C}_b^3$, so Theorem 3.36 does not apply as stated; linear equations are covered by a separate and entirely standard argument, in which the solution is constructed on a large ball and the ball is then exhausted. See [38].

Proposition 3.56. *The unique solution of (3.55) is*

$$S_t = S_0 \exp(\sigma W_t^H), \quad t \in [0, T], \quad (3.56)$$

with no second-order correction, and consequently

$$\mathbb{E}[S_t] = S_0 \exp\left(\frac{1}{2} \sigma^2 t^{2H}\right). \quad (3.57)$$

If instead the driver is the Itô lift $\mathbf{W}^{\text{Itô}}$ of Example 3.18 — that is, if $H = \frac{1}{2}$ and the geometric lift is exchanged for the non-geometric one lying over the same trajectory — the solution becomes $S_0 \exp(\sigma W_t - \frac{1}{2} \sigma^2 t)$.

Proof: Put $F(x) := S_0 e^{\sigma x}$, so that $F' = \sigma F$ and $F'' = \sigma^2 F$. The lift is geometric, so its bracket (3.33) vanishes identically and the rough chain rule of Proposition 3.31 takes its classical form (3.36),

$$F(W_t^H) - F(W_0^H) = \int_0^t \sigma F(W_r^H) d\mathbf{B}_r^H,$$

which is exactly the integral equation (3.40) for (3.55); uniqueness is the linear case of Theorem 3.36. For (3.57), Proposition 3.2 a) gives $W_t^H \sim \mathcal{N}(0, t^{2H})$, and $\mathbb{E}[e^{\sigma Z}] = e^{\sigma^2 \mathbb{V}[Z]/2}$ is the Gaussian moment generating function already used for the log-normal law in Subsection 1.1.2.

It is worth seeing the same computation in Davie's formulation (3.41), since there it becomes a Taylor expansion and nothing else. With $f(y) = \sigma y$ one has $f'f = \sigma^2 y$, and the scalar geometric lift is $\mathbb{X}_{s,t}^H = \frac{1}{2}(W_{s,t}^H)^2$ as noted above, so the right-hand side of (3.41) reads

$$\sigma S_s W_{s,t}^H + \frac{1}{2} \sigma^2 S_s (W_{s,t}^H)^2,$$

which is precisely the second-order Taylor expansion of $S_0 e^{\sigma W_t^H}$ about s ; the neglected term is $O(|t-s|^{3\alpha}) = o(|t-s|)$ because $3\alpha > 1$. The Milstein correction of (3.41) is here the second term of an exponential series.

For the last claim, Proposition 3.28 identifies the rough integral against $\mathbf{W}^{\text{Itô}}$ with the Itô integral, so (3.55) reads $dS_t = \sigma S_t dW_t$ and Example 1.61 supplies the solution. The instructive route is the other one: the bracket of the Itô lift is $[\mathbf{W}^{\text{Itô}}]_t = t \text{Id}$ by Corollary 3.32 ii), so (3.35) retains its second-order term and gives $d(S_0 e^{\sigma W_t}) = \sigma(S_0 e^{\sigma W_t}) dW_t + \frac{1}{2} \sigma^2 (S_0 e^{\sigma W_t}) dt$. The exponential therefore acquires a drift it was not asked for, and the factor $e^{-\sigma^2 t/2}$ exists to remove it. ■

The semimartingale property is preserved under an equivalent change of measure, so S fails to be a semimartingale under *every* $\mathbb{Q} \sim \mathbb{P}$ and is in particular a local martingale under none: no equivalent local martingale measure exists. By the First Fundamental Theorem — in the general form [25, Thm. 7.2] rather than the locally bounded Itô statement of Theorem 2.4 — the model violates NFLVR, and Rogers [67] — the same paper that supplies Theorem 3.3 — exhibits the arbitrage explicitly. It is realised by strategies which trade at unboundedly high frequency, and it disappears once a minimal delay between transactions is imposed [19], in the same spirit as the admissibility constraint of Definition 2.2. Therefore, **roughness cannot be placed in the price**. It must then be placed one level down, in the *volatility*, where it alters the law of the price without disturbing its semimartingale property.

The model is simple enough that the transform of Section 3.4 can be computed outright, and doing so exhibits in a few lines each the two facts that section stated abstractly: that a scalar signature is blind, and that the growth hypothesis of Theorem 3.48 is a genuine condition which here happens to hold.



Proposition 3.57 (Expected signature of the driver). *Let $\sigma \mathbf{B}^H$ be as in Definition 3.54 and let $X_{0,T}^n$ denote the level- n term of its signature over $[0, T]$. Then*

$$\mathbb{E}[X_{0,T}^n] = \begin{cases} \frac{\sigma^{2m} T^{2mH}}{2^m m!}, & n = 2m, \\ 0, & n \text{ odd}, \end{cases} \quad (3.58)$$

and, summing the levels,

$$\mathbb{E}[S(\sigma \mathbf{B}^H)_{0,T}] = \exp\left(\frac{1}{2}\sigma^2 T^{2H}\right) = \frac{\mathbb{E}[S_T]}{S_0}. \quad (3.59)$$

Moreover $\sum_{n \geq 0} \lambda^n |\mathbb{E}[X_{0,T}^n]| = \exp\left(\frac{1}{2}\lambda^2 \sigma^2 T^{2H}\right)$ is finite for every $\lambda > 0$, so the hypothesis of Theorem 3.48 is satisfied.

Proof: The path is scalar, so Example 3.42 gives $X_{0,T}^n = \sigma^n (W_T^H)^n / n!$. Since $W_T^H \sim \mathcal{N}(0, T^{2H})$ the odd moments vanish, while $\mathbb{E}[(W_T^H)^{2m}] = (2m-1)!! T^{2mH}$; dividing by $(2m)!$ and using $(2m)! = 2^m m! (2m-1)!!$ yields (3.58). Summing over m produces the exponential series, which (3.57) identifies with $\mathbb{E}[S_T]/S_0$. Replacing σ by $\lambda\sigma$ throughout gives the last claim. ■

What one is entitled to conclude from (3.59) is, however, almost nothing. Example 3.42 showed that the signature of a scalar path depends on its terminal increment alone; by Theorem 3.46 every scalar path is therefore equivalent, up to tree-like pieces, to the straight line joining its endpoints. The law which Theorem 3.48 recovers from (3.59) is thus the law of S_T and nothing more: a single log-normal marginal, carrying no information about the trajectory that produced it. This is precisely the loss that Section 3.4 warned of, and it is repaired there by the same device — time augmentation.

Proposition 3.58 (Expected signature after time augmentation). *Let $\widehat{X}_t := (t, \sigma W_t^H) \in \mathbb{R}^2$ and let $\widehat{\mathbf{X}} = (\widehat{X}, \widehat{\mathbb{X}})$ be its geometric lift, which exists because the second component is lifted by Theorem 3.20 while the remaining entries are Young integrals against a path of bounded variation. Then*

$$\mathbb{E}[\widehat{X}_{0,T}] = (T, 0), \quad (3.60)$$

and, writing $\widehat{\mathbb{X}}^{ij}$ for the component of $\widehat{\mathbb{X}}$ along $e_i \otimes e_j$,

$$\mathbb{E}[\widehat{\mathbb{X}}_{0,T}] = \frac{1}{2} \begin{pmatrix} T^2 & 0 \\ 0 & \sigma^2 T^{2H} \end{pmatrix}. \quad (3.61)$$

Proof: The first level is the increment $\widehat{X}_{0,T} = (T, \sigma W_T^H)$, and $\mathbb{E}[W_T^H] = 0$. For the second level, geometricity (3.19) determines the symmetric part outright and pathwise,

$$\text{Sym } \widehat{\mathbb{X}}_{0,T} = \frac{1}{2} \widehat{X}_{0,T} \otimes \widehat{X}_{0,T} = \frac{1}{2} \begin{pmatrix} T^2 & \sigma T W_T^H \\ \sigma T W_T^H & \sigma^2 (W_T^H)^2 \end{pmatrix},$$

so the two diagonal entries are already identified, and taking expectations with $\mathbb{E}[(W_T^H)^2] = T^{2H}$ gives the diagonal of (3.61). For the off-diagonal, the first coordinate has bounded variation, so $\widehat{\mathbb{X}}_{0,T}^{21} = \sigma \int_0^T W_r^H dr$ is an ordinary Riemann integral; Fubini applies, since $\mathbb{E} \int_0^T |W_r^H| dr = \sqrt{2/\pi} \int_0^T r^H dr < \infty$, and yields $\mathbb{E}[\widehat{\mathbb{X}}_{0,T}^{21}] = \sigma \int_0^T \mathbb{E}[W_r^H] dr = 0$. The remaining entry follows from the symmetric part, since $\widehat{\mathbb{X}}_{0,T}^{12} + \widehat{\mathbb{X}}_{0,T}^{21} = \sigma T W_T^H$ has expectation zero. ■

Compare (3.60) with (3.61). The first level of the expected signature is $(T, 0)$ for *every* value of H : it cannot tell a rough asset from a smooth one. The parameter appears for the first time at level two, in the single entry $\frac{1}{2}\sigma^2 T^{2H}$, and it appears there as an exponent of the time horizon.

At $H = \frac{1}{2}$ the entry reads $\frac{1}{2}\sigma^2 T$, which is the level-two term of Fawcett's formula $\mathbb{E}[S(\mathbf{W}^{\text{Strat}})_{0,T}] = \exp\left(\frac{1}{2}T \text{Id}\right)$ for Brownian motion under the Stratonovich lift [34]. And beyond level two no closed form of Fawcett's type is available to us for $H \neq \frac{1}{2}$: the closed forms of this section stop exactly where the independence of increments stops, which is a fair measure of what the fractional case costs.

3.5.2 The optimal execution problem

The First Fundamental Theorem 2.4 is a statement about *pricing measures*: it says that a market admits no free lunch precisely when an equivalent local martingale measure exists. A rough price has none, so a rough price cannot be the underlying of an arbitrage-free pricing theory. That is the whole of the damage.

It is not, however, the whole of finance. A large part of the subject consists of problems posed under the physical measure $\mathbb{P}$, or under no measure at all, in which no expectation is ever taken: the execution of a large order, the attribution of a hedging profit and loss, the measurement of risk, the estimation of a parameter, the choice of a portfolio. None of these is a pricing problem, none of them requires an equivalent local martingale measure, and none of them is touched by Theorem 3.3. Roughness, meanwhile, is an empirical fact (Subsection 3.1.2). The natural conclusion is that rough path theory is a $\mathbb{P}$ -tool applied to a $\mathbb{Q}$ -innocent class of problems, and this section works one such problem through to a closed-form answer.

Remark 3.59 (Where the lift is, and is not, needed). One preliminary observation removes the restriction $H > \frac{1}{3}$ from everything that is actually *computed* below, even though the definitions that follow continue to state it for the continuum objects. In practice the signature is computed from an *observed*, discretely sampled trajectory, linearly interpolated between ticks. Such a path has bounded variation, its iterated integrals (3.46) are ordinary Riemann–Stieltjes integrals, and no lift is required at all. What rough path theory supplies in that setting is not the construction of the object but the guarantees attached to it: factorial decay (Proposition 3.43) is what licenses truncation, the shuffle relation (Proposition 3.44) is what linearises the problem, Theorem 3.46 is what says nothing has been discarded, and the Universal Limit Theorem 3.38 is what says the answer is stable as the sampling refines rather than an artefact of the grid.

Suppose a trader must liquidate $Q_0 > 0$ shares over a horizon $[0, T]$ — hours, not years. Selling fast moves the price against them; selling slowly exposes them to the market. This is the *optimal execution* problem of Almgren and Chriss [2], and it is the canonical example of a financial problem in which no pricing measure appears: the objective is an expected cost under $\mathbb{P}$, and the trader is not valuing a claim but choosing a schedule.

Definition 3.60 (Execution problem). Let S denote the *unaffected* price, let Q_t be the inventory remaining at time t and let $\pi_t = -\dot{Q}_t$ be the trading rate, so that $Q_t = Q_0 - \int_0^t \pi_r dr$. Selling at rate π_t realises the price $S_t - \eta \pi_t$ with $\eta > 0$ the *temporary impact* coefficient, and any inventory Q_T left at the horizon is liquidated in a block at $S_T - \zeta Q_T$, with $\zeta > 0$. The **implementation shortfall** of the schedule π is

$$\mathcal{C}(\pi) = Q_0 S_0 - \int_0^T \pi_t (S_t - \eta \pi_t) dt - Q_T (S_T - \zeta Q_T), \quad (3.62)$$

the arrival value of the position less what the liquidation actually raises.



Lemma 3.61 (Reduction of the shortfall). *For any C^1 inventory path Q with Q_0 given,*

$$\mathcal{C}(\pi) = - \int_0^T Q_t dS_t + \eta \int_0^T \pi_t^2 dt + \zeta Q_T^2. \quad (3.63)$$

Proof: Since Q is C^1 it has bounded variation, so $\int_0^T Q dS$ is a Young integral and integration by parts is available: $\int_0^T \dot{Q}_t S_t dt = Q_T S_T - Q_0 S_0 - \int_0^T Q_t dS_t$. With $\pi = -\dot{Q}$ this gives $\int_0^T \pi_t S_t dt = Q_0 S_0 - Q_T S_T + \int_0^T Q_t dS_t$. Substituting into (3.62), the two occurrences of $Q_0 S_0$ cancel and so do the two of $Q_T S_T$, leaving (3.63). ■

The three terms of (3.63) are, in order, the market risk carried by the unliquidated inventory, the impact paid for trading, and the cost of the terminal block. Only the first involves the price path, and it involves it through an integral against dS — which is where the regularity of S enters, and where a semimartingale would ordinarily be assumed.

Definition 3.62 (Rough execution model). Fix $H \in (\frac{1}{3}, \frac{1}{2}]$ and $\sigma > 0$, let W^H be a scalar fractional Brownian motion and set

$$S_t = S_0 + \sigma W_t^H, \quad t \in [0, T]. \quad (3.64)$$

The **time-augmented driver** is $\widehat{X}_t = (t, \sigma W_t^H) \in \mathbb{R}^2$, lifted to the geometric rough path $\widehat{\mathbf{X}}$ of Proposition 3.58. We write $\mathbb{S}_t^I$ for the coordinate of $\mathbb{S}(\widehat{\mathbf{X}})_{0,t}$ along the word I in the letters $\{1, 2\}$, indexed as in Proposition 3.44, so that $\mathbb{S}_t^1 = t$ and $\mathbb{S}_t^2 = \sigma W_t^H$.

The price (3.64) is arithmetic rather than exponential, as is standard over an intraday horizon, and it is exactly as illegitimate a *pricing* model as (3.55): it is not a semimartingale, so it admits no equivalent local martingale measure. Nothing in Definition 3.60 asked for one.

3.5.3 Schedules as linear functionals of the signature

We now restrict the schedule to the class in which the problem linearises.

Definition 3.63 (Signature schedule). Let $N \in \mathbb{N}$ and let ℓ be a linear functional on $T^{(N)}(\mathbb{R}^2)$, that is, a finite family of coefficients $(\ell_I)_{|I| \leq N}$ indexed by words. The associated **signature schedule** is

$$\pi_t = \ell(\mathbb{S}^N(\widehat{\mathbf{X}})_{0,t}) = \sum_{|I| \leq N} \ell_I \mathbb{S}_t^I. \quad (3.65)$$

The class is not as special as it looks. By Proposition 3.44 and Stone–Weierstrass, linear functionals of the signature are dense in the continuous functionals of the path; the family (3.65) therefore approximates *any* continuous non-anticipative schedule, and Proposition 3.43 controls the size of the terms discarded at level N . What the restriction buys is that the two operations the objective (3.63) performs on π — integrating it, and squaring it — both stay inside the class. That is the content of the next two observations, and everything afterwards is bookkeeping.

Lemma 3.64 (Integration is concatenation). *For any word I and either letter $j \in \{1, 2\}$, $\mathbb{S}_t^{Ij} = \int_0^t \mathbb{S}_r^I d\widehat{X}_r^j$. In particular, writing ℓj for the functional with coefficients $(\ell j)_{Ij} = \ell_I$ and zero on words not ending in j ,*

$$\int_0^t \pi_r dr = (\ell 1)(\mathbb{S}_t^{N+1}), \quad \int_0^t \pi_r dS_r = (\ell 2)(\mathbb{S}_t^{N+1}). \quad (3.66)$$

Proof: The first assertion is the definition (3.46) of the iterated integral read one level at a time. The two displays follow by linearity, using $d\widehat{X}_r^1 = dr$ and $d\widehat{X}_r^2 = \sigma dW_r^H = dS_r$. ■

Lemma 3.64 is the reason the time coordinate was adjoined in Proposition 3.58. Adjoining t makes *time integration* an operation on words, namely right-concatenation with the letter 1. The inventory, which is the time integral of the schedule, is therefore again a linear functional of the signature — one level higher.

Proposition 3.65 (The objective is a quadratic form in the signature). *Let π be the signature schedule (3.65) and write $\mathbb{S} := \mathbb{S}^{2N+2}(\widehat{\mathbf{X}})_{0,T}$ for the truncated signature of the realised trajectory. Then, on that trajectory,*

$$\mathcal{C}(\pi) = \ell^\top \mathfrak{A}(\mathbb{S}) \ell + \mathfrak{b}(\mathbb{S})^\top \ell + c(\mathbb{S}), \quad (3.67)$$

where, writing e_I for the functional dual to the word I ,

$$\mathfrak{A}_{IJ}(\mathbb{S}) = \eta \left((e_I \sqcup e_J) 1 \right) (\mathbb{S}) + \zeta \left((e_I 1) \sqcup (e_J 1) \right) (\mathbb{S}), \quad (3.68)$$

$$\mathfrak{b}_I(\mathbb{S}) = (e_I 12) (\mathbb{S}) - 2\zeta Q_0 (e_I 1) (\mathbb{S}), \quad (3.69)$$

and $c(\mathbb{S}) = \zeta Q_0^2 - Q_0 \mathbb{S}^2$ does not involve ℓ . Both $\mathfrak{A}$ and $\mathfrak{b}$ are linear in $\mathbb{S}$.

Corollary 3.66. *Writing $\bar{\mathbb{S}} := \mathbb{E}[\mathbb{S}]$ and taking expectations in (3.67),*

$$\mathbb{E}[\mathcal{C}(\pi)] = \ell^\top \mathfrak{A}(\bar{\mathbb{S}}) \ell + \mathfrak{b}(\bar{\mathbb{S}})^\top \ell + \zeta Q_0^2, \quad (3.70)$$

the term $Q_0 \mathbb{E}[\mathbb{S}^2]$ vanishing because $\mathbb{E}[W_T^H] = 0$. The matrix $\mathfrak{A}(\bar{\mathbb{S}})$ is symmetric and positive definite for $\eta, \zeta > 0$, so the minimiser is unique and is

$$\ell^* = -\frac{1}{2} \mathfrak{A}(\bar{\mathbb{S}})^{-1} \mathfrak{b}(\bar{\mathbb{S}}). \quad (3.71)$$

Proof: Take the three terms of (3.63) in turn. By Lemma 3.64, $Q_t = Q_0 - (\ell 1)(\mathbb{S}_t)$, so

$$\int_0^T Q_t dS_t = Q_0 \mathbb{S}_T^2 - (\ell 12)(\mathbb{S}_T),$$

a second application of the lemma with $j = 2$. For the impact term, Proposition 3.44 gives $\pi_t^2 = (\ell \sqcup \ell)(\mathbb{S}_t)$, whence $\int_0^T \pi_t^2 dt = ((\ell \sqcup \ell) 1)(\mathbb{S}_T)$, again by Lemma 3.64. For the terminal term, $Q_T^2 = Q_0^2 - 2Q_0(\ell 1)(\mathbb{S}_T) + ((\ell 1) \sqcup (\ell 1))(\mathbb{S}_T)$. Every resulting expression is a linear functional of $\mathbb{S}_T^{2N+2}$ whose coefficients are quadratic or linear in ℓ ; taking expectations replaces $\mathbb{S}_T$ by $\bar{\mathbb{S}}$ and collects the terms into (3.68)–(3.69), the contribution $Q_0 \mathbb{E}[\mathbb{S}_T^2]$ vanishing because $\mathbb{E}[W_T^H] = 0$. Positive definiteness holds because $\ell^\top \mathfrak{A} \ell = \eta \mathbb{E} \int_0^T \pi_t^2 dt + \zeta \mathbb{E}[(\ell 1)(\mathbb{S}_T)^2] \geq 0$, with equality only if $\pi \equiv 0$ almost surely, which forces $\ell = 0$ because the coordinate functionals $\{\mathbb{S}^I\}_{|I| \leq N}$ are linearly independent in L^2 under the law of $\widehat{X}$. ■

Proposition 3.65 is an *identity*, valid on each trajectory separately: no expectation is taken, no measure is used, and the shuffle relation on which it rests is itself pathwise. The expectation (3.70) is a corollary of it and not the other way round. Everything the method needs is therefore available from realised paths, and the probabilistic model enters only if one chooses to compute $\bar{\mathbb{S}}$ in closed form rather than measure it.

Three further features are worth naming. The problem is finite-dimensional: the unknown is a vector of $\sum_{n \leq N} 2^n$ numbers, and (3.71) is a linear solve. It is *model-free in the coefficients*: $\mathfrak{A}$ and $\mathfrak{b}$ depend on the market only through $\bar{\mathbb{S}}$, so a change of model is a change of one vector of constants and not a change of method. And it requires the signature two levels beyond the schedule, since the objective integrates π once against dt and once against dS .



Remark 3.67 (Calibration is an average). Because (3.67) holds path by path, the coefficients need not be computed from a model at all. Given M historical liquidation windows with realised signatures $\mathbb{S}^{(1)}, \dots, \mathbb{S}^{(M)}$, put $\widehat{\mathbb{S}} = \frac{1}{M} \sum_m \mathbb{S}^{(m)}$ and $\widehat{\ell}^* = -\frac{1}{2} \mathfrak{A}(\widehat{\mathbb{S}})^{-1} \mathfrak{b}(\widehat{\mathbb{S}})$. By the linearity of $\mathfrak{A}$ and $\mathfrak{b}$ this is exactly the empirical risk minimiser over the class (3.65), and it is what [47] implements.

Further, we take $N = 1$ and restrict to the subclass $\ell_1 = 0$, so that the schedule is

$$\pi_t = \ell_0 + \ell_2 \sigma W_t^H, \quad (3.72)$$

a constant rate plus a term proportional to the price move so far: sell faster when the price has risen, or slower, according to the sign of ℓ_2 ; the omitted term $\ell_1 t$ is a deterministic ramp, and we do not claim that it cannot improve the optimum. By Proposition 3.65 the objective needs the expected signature to level $2N + 2 = 4$. The level-one and level-two entries are supplied by Proposition 3.58, and the entries above them by Proposition 3.68 below.

Proposition 3.68 (A level-three entry). *With $\widehat{\mathbf{X}}$ as in Definition 3.62,*

$$\mathbb{E}[\mathbb{S}_T^{212}] = \frac{(2H - 1) \sigma^2 T^{2H+1}}{2(2H + 1)}, \quad (3.73)$$

which vanishes if and only if $H = \frac{1}{2}$, and is strictly negative for $H < \frac{1}{2}$. Moreover

$$\mathbb{E}[(\mathbb{S}_T^{21})^2] = \frac{\sigma^2 T^{2H+2}}{2H + 2}, \quad \mathbb{E}[\mathbb{S}_T^{221}] = \frac{\sigma^2 T^{2H+1}}{2(2H + 1)}. \quad (3.74)$$

Proof: Write $G_t := \int_0^t W_r^H dr$, so that $\mathbb{S}_T^{21} = \sigma G_T$ and, by Lemma 3.64, $\mathbb{S}_T^{212} = \sigma^2 \int_0^T G_t dW_t^H$. Since G is C^1 the integral is a Young integral and integration by parts applies:

$$\int_0^T G_t dW_t^H = G_T W_T^H - \int_0^T W_t^H dG_t = G_T W_T^H - \int_0^T (W_t^H)^2 dt.$$

By Proposition 3.2 a) the covariance is $\mathbb{E}[W_s^H W_t^H] = \frac{1}{2}(s^{2H} + t^{2H} - |t - s|^{2H})$, so Fubini gives

$$\mathbb{E}[G_T W_T^H] = \int_0^T \mathbb{E}[W_s^H W_T^H] ds = \frac{1}{2} \left(\frac{T^{2H+1}}{2H+1} + T^{2H+1} - \frac{T^{2H+1}}{2H+1} \right) = \frac{1}{2} T^{2H+1},$$

while $\mathbb{E} \int_0^T (W_t^H)^2 dt = \int_0^T t^{2H} dt = T^{2H+1}/(2H + 1)$. Subtracting and multiplying by σ^2 yields (3.73). For the first identity in (3.74),

$$\mathbb{E}[G_T^2] = \int_0^T \int_0^T \mathbb{E}[W_s^H W_r^H] ds dr = \frac{T^{2H+2}}{2H + 1} - \frac{T^{2H+2}}{(2H + 1)(2H + 2)} = \frac{T^{2H+2}}{2H + 2},$$

using $\int_0^T \int_0^T |s - r|^{2H} ds dr = 2T^{2H+2}/[(2H + 1)(2H + 2)]$. The last identity is immediate from $\mathbb{S}_t^{22} = \frac{1}{2}(\mathbb{S}_t^2)^2$, which is geometricity (3.19) in one dimension, followed by Lemma 3.64. $\blacksquare$

Remark 3.69 (Two consistency checks). At $H = \frac{1}{2}$ the driver is Brownian, G is adapted and square-integrable, and $\int_0^T G dW$ is a true Itô integral of mean zero; formula (3.73) returns exactly 0. Independently, the shuffle relation (3.48) applied to the words (2) and (2, 1) gives $\mathbb{S}^2 \mathbb{S}^{21} = 2 \mathbb{S}^{221} + \mathbb{S}^{212}$, and the three expectations computed above satisfy it. The second check is a genuine one: it uses a structural identity of the signature to verify a computation performed by Fubini.

The two entries of Proposition 3.68 differ by the factor $2H - 1$ and by nothing else, which has a use of its own.

Corollary 3.70 (A scale-free estimator of the Hurst parameter). *With the notation of Proposition 3.68,*

$$\frac{\mathbb{E}[\mathbb{S}_T^{212}]}{\mathbb{E}[\mathbb{S}_T^{221}]} = 2H - 1, \quad \text{that is} \quad H = \frac{1}{2} \left(1 + \frac{\mathbb{E}[\mathbb{S}_T^{212}]}{\mathbb{E}[\mathbb{S}_T^{221}]} \right). \quad (3.75)$$

Proof: Divide (3.73) by the second identity of (3.74); the denominator is strictly positive. ■

Proposition 3.71 (Optimal signature schedule). *For the model of Definition 3.62 and the schedule class (3.72), the objective (3.70) separates,*

$$\mathbb{E}[\mathcal{C}] = \underbrace{\eta \ell_\emptyset^2 T + \zeta (Q_0 - \ell_\emptyset T)^2}_{\text{schedule}} + \underbrace{\mathbb{E}[\mathbb{S}_T^{212}] \ell_2 + \mathcal{K}_H \ell_2^2}_{\text{signal}}, \quad (3.76)$$

with $\mathcal{K}_H = \sigma^2 T^{2H+1} \left(\frac{\eta}{2H+1} + \frac{\zeta T}{2H+2} \right) > 0$, and the minimisers are

$$\ell_\emptyset^* = \frac{\zeta Q_0}{\eta + \zeta T}, \quad \ell_2^* = \frac{1 - 2H}{4 \eta_H^{\text{eff}}}, \quad \eta_H^{\text{eff}} := \eta + \frac{(2H+1)\zeta T}{2H+2}. \quad (3.77)$$

The gain from the path-dependent term over the best deterministic schedule is

$$\mathbb{E}[\mathcal{C}(\ell^*)] - \mathbb{E}[\mathcal{C}(\ell_\emptyset^*, 0)] = -\frac{(1-2H)^2 \sigma^2 T^{2H+1}}{16(2H+1) \eta_H^{\text{eff}}} \leq 0, \quad (3.78)$$

with equality if and only if $H = \frac{1}{2}$.

Proof: Expand the three terms of (3.63) for the schedule (3.72). The inventory is $Q_t = Q_0 - \ell_\emptyset t - \ell_2 \mathbb{S}_t^{21}$, so $\int_0^T Q_t dS_t = Q_0 \mathbb{S}_T^2 - \ell_\emptyset \mathbb{S}_T^{12} - \ell_2 \mathbb{S}_T^{212}$. Now $\mathbb{E}[\mathbb{S}_T^2] = 0$, and $\mathbb{E}[\mathbb{S}_T^{12}] = \sigma \mathbb{E}[TW_T^H - G_T] = 0$ by parts, so the first term of (3.63) contributes $\mathbb{E}[\mathbb{S}_T^{212}] \ell_2$ and nothing else. The impact term is $\eta \mathbb{E} \int_0^T (\ell_\emptyset + \ell_2 \sigma W_t^H)^2 dt = \eta (\ell_\emptyset^2 T + \ell_2^2 \sigma^2 T^{2H+1} / (2H+1))$, the cross term vanishing with $\mathbb{E}[W_t^H]$. The terminal term is $\zeta \mathbb{E}[(Q_0 - \ell_\emptyset T - \ell_2 \mathbb{S}_T^{21})^2] = \zeta ((Q_0 - \ell_\emptyset T)^2 + \ell_2^2 \sigma^2 T^{2H+2} / (2H+2))$, using $\mathbb{E}[\mathbb{S}_T^{21}] = 0$ and the first identity of (3.74). Collecting gives (3.76), in which $\ell_\emptyset$ and ℓ_2 appear in separate blocks. Each block is a strictly convex scalar quadratic. Setting derivatives to zero, $2\eta T \ell_\emptyset = 2\zeta T (Q_0 - \ell_\emptyset T)$ gives $\ell_\emptyset^*$, and $\ell_2^* = -\mathbb{E}[\mathbb{S}_T^{212}] / (2\mathcal{K}_H)$ gives, after substituting (3.73) and simplifying, the stated form. The gain is $-\mathbb{E}[\mathbb{S}_T^{212}]^2 / (4\mathcal{K}_H)$, which is (3.78). ■

It is tempting to observe that no pricing measure was used anywhere. It is not irrelevant: it is the *source* of the gain, and the accounting is exact. Were S a martingale under $\mathbb{P}$, we would have $\mathbb{E} \int_0^T Q_t dS_t = 0$ for every adapted Q , the first term of (3.63) would vanish identically, and

$$\mathbb{E}[\mathcal{C}] = \eta \mathbb{E} \int_0^T \pi_t^2 dt + \zeta \mathbb{E}[Q_T^2]$$

would not involve the price path at all. The optimal schedule would then be *deterministic*, and no path-dependent rule could beat it in expectation — which is precisely why the classical treatment [2] must adjoin a variance penalty before the problem becomes interesting. What (3.78) measures is the failure of exactly that martingale property, since $\mathbb{E}[\mathbb{S}_T^{212}] \neq 0$ is the statement that $\mathbb{E} \int Q dS \neq 0$ for path-dependent Q .



Remark 3.72 (Is this an arbitrage?). The model admits one, and it should be said plainly. Theorem 3.3 applies to (3.64) verbatim, so there is no equivalent local martingale measure and, by Theorem 2.4, NFLVR fails; Rogers [67] exhibits the arbitrage. Our own computation displays a symptom of it: setting $Q_0 = 0$ and $\eta = \zeta = 0$ in (3.72) leaves the position $Q_t = -\ell_2 \sigma \int_0^t W_s^H ds$, which starts from nothing and has expected profit $-\ell_2 \mathbb{E}[S_T^{212}] > 0$ for every $\ell_2 > 0$ when $H < \frac{1}{2}$, unbounded under scaling of ℓ_2 . A positive expected return is not by itself an arbitrage, but here it is produced by the same anti-persistence that Rogers' construction exploits.

3.5.4 Two assets: the Lévy area

Everything so far has used a scalar asset, and by Example 3.42 a scalar path carries no level-two information of its own: the time coordinate had to be adjoined before the signature said anything at all. With two assets the situation changes qualitatively, and the change is not a matter of degree. This subsection is about the object that appears, which is the one the whole theory is named after.

Let $X = (X^1, X^2)$ be the two-dimensional path of a pair of assets — usually $X_t^i = \log(S_t^i/S_0^i)$, the cumulative log-return — lifted to a geometric rough path $\mathbf{X} = (X, \mathbb{X})$. The time coordinate may be adjoined as in Definition 3.62 whenever it is wanted; it plays no role in what follows and is suppressed.

Proposition 3.73 (The free data at level two). *Let $\mathbf{X}$ be weakly geometric and d -dimensional. Then $\mathbb{X}_{s,t}$ is determined by the increment $X_{s,t}$ together with the $\frac{1}{2}d(d-1)$ entries $A_{s,t}^{ij}$, $i < j$, of the Lévy area (3.13). In particular, for $d = 1$ there are none, and for $d = 2$ there is exactly one.*

Proof: Decompose $\mathbb{X} = \text{Sym } \mathbb{X} + \text{Anti } \mathbb{X}$, the second summand being $A_{s,t}$ by (3.13). Proposition 3.9 identified $\text{Sym } \mathbb{X}_{s,t}$ with $\frac{1}{2}X_{s,t} \otimes X_{s,t}$ for smooth paths, and (3.19) is exactly the requirement that this survive; so the symmetric part is a function of the increment alone. The antisymmetric part is determined by its entries above the diagonal, of which there are $\frac{1}{2}d(d-1)$. ■

Over a single window $\text{Sym } \mathbb{X}_{0,T}^{12} = \frac{1}{2}X_{0,T}^1 X_{0,T}^2$ is the product of the two total returns: it is known to anyone who has looked only at the endpoints, and averaged across windows it is the realised covariance. Everything else level two contains — the whole of what depends on the *trajectory* rather than on where it began and ended — is one number, the scalar

$$A_{s,t} = \frac{1}{2} \left(\mathbb{X}_{s,t}^{12} - \mathbb{X}_{s,t}^{21} \right) = \frac{1}{2} \left(\int_s^t X_{s,r}^1 dX_r^2 - \int_s^t X_{s,r}^2 dX_r^1 \right) \quad (3.79)$$

of Remark 3.19, now carried by a pair of assets rather than by a planar Brownian motion. It vanishes when the two components move proportionally, a straight line enclosing no area.

Remark 3.74 (The area is a profit and loss). The reading that matters here is not geometric. The integral $\int_s^t X_{s,r}^1 dX_r^2$ is the profit of a strategy holding, at each instant, a position in asset 2 equal to the move asset 1 has made since the window opened; and symmetrically for the other term. So $A_{s,t}$ is one half the difference of two crossed-pairs strategies and it is a quantity a desk can actually realise.

Remark 3.75 (Conventions). The area is insensitive to the Itô–Stratonovich choice. Passing from one to the other adds $\frac{1}{2}\langle X^1, X^2 \rangle$ to each of the two integrals in (3.13), and the two additions cancel in the difference; the ambiguity of Chapter 1 lives entirely in $\text{Sym } \mathbb{X}$, which is the quadratic covariation, and not in $\text{Anti } \mathbb{X}$. What the area *is* sensitive to is the choice of lift

itself: by Proposition 3.15 a path does not determine its own second level, and by Proposition 3.73 the whole of that freedom, for a pair, is the single number A . Two markets with identical realised price paths and identical realised covariances but different areas are different rough paths, and by the Universal Limit Theorem 3.38 they produce different solutions to any rough differential equation driven by them — that is, different profit and loss for any path-dependent book. Correlation is not merely an incomplete description of a pair; it is provably the part of level two that carries nothing new.

Returns add across adjacent windows. Areas do not, and the defect is exactly Chen's relation.

Proposition 3.76 (Chen's relation for the area). *For $s \leq u \leq t$,*

$$A_{s,t} = A_{s,u} + A_{u,t} + \frac{1}{2} \det \begin{pmatrix} X_{s,u}^1 & X_{u,t}^1 \\ X_{s,u}^2 & X_{u,t}^2 \end{pmatrix}. \quad (3.80)$$

Proof: Chen's relation (3.16) reads $\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$. Taking the antisymmetric part of the $(1, 2)$ entry, the last term contributes $\frac{1}{2}(X_{s,u}^1 X_{u,t}^2 - X_{s,u}^2 X_{u,t}^1)$, which is the stated determinant. ■

Definition 3.77 (Lead-lag pair). Let W^H and $\widetilde{W}^H$ be independent scalar fractional Brownian motions with the same $H \in (\frac{1}{3}, \frac{1}{2}]$, let $\sigma_1, \sigma_2 > 0$, $\rho \in [-1, 1]$ and $\theta \in \mathbb{R}$, and set

$$X_t^1 = \sigma_1 W_t^H, \quad X_t^2 = \sigma_2 \left(\rho W_t^H + \sqrt{1 - \rho^2} \widetilde{W}_t^H \right) + \theta \int_0^t W_s^H ds. \quad (3.81)$$

The second asset moves with the first contemporaneously, through ρ , and responds to its *past* through θ : asset 1 leads.

Proposition 3.78 (Expected area of a lead-lag pair). *For the pair (3.81),*

$$\mathbb{E}[A_{0,T}] = \frac{\sigma_1 \theta (3 - 2H) T^{2H+1}}{4(2H + 1)}. \quad (3.82)$$

Neither ρ nor σ_2 appears.

Proof: Write $G_t = \int_0^t W_s^H ds$ and expand the two integrals of (3.13). Using $dX^2 = \sigma_2 \rho dW^H + \sigma_2 \sqrt{1 - \rho^2} d\widetilde{W}^H + \theta W_t^H dt$,

$$\int_0^T X^1 dX^2 = \sigma_1 \sigma_2 \rho \int_0^T W^H dW^H + \sigma_1 \sigma_2 \sqrt{1 - \rho^2} \int_0^T W^H d\widetilde{W}^H + \sigma_1 \theta \int_0^T (W_t^H)^2 dt,$$

while $dX^1 = \sigma_1 dW^H$ gives

$$\int_0^T X^2 dX^1 = \sigma_1 \sigma_2 \rho \int_0^T W^H dW^H + \sigma_1 \sigma_2 \sqrt{1 - \rho^2} \int_0^T \widetilde{W}^H dW^H + \sigma_1 \theta \int_0^T G_t dW_t^H.$$

The two ρ terms are *identical* and cancel in the difference (3.13). The two mixed terms have expectation zero: the law of $\widetilde{W}^H$ is invariant under $\widetilde{W}^H \mapsto -\widetilde{W}^H$, which commutes with the piecewise-linear approximations defining the joint geometric lift and reverses the sign of each mixed integral, so each is symmetrically distributed. There remain $\mathbb{E} \int_0^T (W_t^H)^2 dt = T^{2H+1}/(2H + 1)$ and, by Proposition 3.68, $\mathbb{E} \int_0^T G_t dW_t^H = (2H - 1)T^{2H+1}/(2(2H + 1))$. Hence

$$\mathbb{E}[A_{0,T}] = \frac{\sigma_1 \theta}{2} T^{2H+1} \left(\frac{1}{2H + 1} - \frac{2H - 1}{2(2H + 1)} \right) = \frac{\sigma_1 \theta T^{2H+1}}{2(2H + 1)} \cdot \frac{3 - 2H}{2},$$



which is (3.82). ■

At $H = \frac{1}{2}$ the expression reduces to $\sigma_1 \theta T^2/4$, which is checked directly: there $\int G dW$ is an Itô integral of mean zero and only the first term survives. At $\theta = 0$ it vanishes for every ρ , as it must, the Lévy area of a correlated Brownian pair having mean zero.

Remark 3.79 (Correlation is invisible to the area). The cancellation of ρ in the proof is the whole content of this subsection, written as a formula. Proposition 3.73 said that the symmetric part of level two is the endpoints and the antisymmetric part is everything else; Proposition 3.78 exhibits a pair in which the contemporaneous coupling ρ and the lagged coupling θ are separately adjustable, and shows that the area sees the second and is *exactly blind* to the first. A correlation estimate and an area estimate are therefore not two views of the same quantity; they are measurements of orthogonal components of the same tensor.

Inverting (3.82) estimates the lead–lag coefficient,

$$\hat{\theta} = \frac{4(2H+1) \widehat{\mathbb{E}}[A_{0,T}]}{\sigma_1(3-2H) T^{2H+1}},$$

from a single level-two entry. The classical route to the same conclusion is a cross-correlogram scanned over a grid of lags; (3.13) aggregates that scan into one number, at the cost of committing to the shape (3.81) in exchange for not having to choose a maximum lag.

Notes and references

Rough path theory is Lyons’ [57], and the Hölder presentation followed here is that of Friz and Hairer [33], which is the natural companion to this chapter and the source of every proof quoted. The original p -variation formulation, together with the free nilpotent groups that the general case requires and which Definition 3.7 was truncated to avoid, is developed systematically in Friz and Victoir [34]; the lecture notes of Geng [38] are a shorter route to the same material. Three attributions made in passing above deserve their own reference. The multiplicative property (3.16) is Chen’s [18], where the iterated integrals of a smooth path were first organised into a group. The sewing lemma, the controlled paths of Definition 3.22 and the integral of Theorem 3.25 are Gubinelli’s [39], and it is that paper which put the two-level theory into the form used here. The pathwise integral that Subsection 3.1.3 finds out of reach is Young’s [74].

On the fractional side, the process of Definition 3.1 goes back to Kolmogorov and takes its modern form in Mandelbrot and Van Ness [59]; the Gaussian facts collected in Proposition 3.2, including the negative half of the Hölder statement, are in Marcus and Rosen [60, Ch. 7]. Theorem 3.3 is Rogers’ [67], which also exhibits the arbitrage that Subsection 3.5.1 appeals to. The canonical lift of Theorem 3.20 is Coutin and Qian’s [23]. It is worth recording what lies immediately outside it. The piecewise-linear lifts converge only for $H > \frac{1}{4}$, and at no number of levels below that; the Lyons–Victoir extension theorem [58] nevertheless guarantees, by a Hahn–Banach argument, that *every* α -Hölder path admits some lift, at the cost of the axiom of choice and with no canonical choice among them.

The signature of Section 3.4 is Chen’s construction again, read as a transform. Uniqueness for paths of bounded variation is Hambly and Lyons [42], and its extension to rough paths is Boedihardjo, Geng, Lyons and Yang [12]; the expected-signature statement of Theorem 3.48 is Chevyrev and Lyons [21]. Its use as a feature map for learning on path space, which is what

Remark 3.49 is about, is surveyed in [20] and [52]. The execution problem of Subsection 3.5.2 is Almgren and Chriss' [2], of which [16] is the textbook account; its signature formulation, of which Proposition 3.65 and the worked instance (3.72) are the scalar case, is Kalsi, Lyons and Perez Arribas [47]. The same reduction — an objective built from time integrals, integrals against the price and polynomials of the control becomes a quadratic form in one expected signature — underlies the non-parametric pricing and hedging of path-dependent claims in [56], the path-dependent processes of [45], and the signature market models calibrated jointly to SPX and VIX in [24].



Chapter 4

Applications

Everything so far has been either machinery or a model. This chapter is the place where both are made to meet market data. Three studies are carried out. The first two are calibrations to the same option book and are deliberately complementary; the third is not a calibration at all, and measures the underlying series directly.

The first fits an implied volatility surface directly, imposing the static no-arbitrage conditions of Proposition 2.17 as penalties rather than inheriting them from a parametric family, and extrapolating beyond the quoted range by a closed form on which those conditions are *proved* rather than sampled. The second fits the same smiles with the rough Heston model of Definition 3.50, so that a model with dynamics — and with the short-dated skew that Chapter 3 identified as the empirical signature of roughness — is measured against a surface fitted with no dynamics at all. Section 4.1 produces the best possible *description* of today’s option prices; Section 4.2 produces the best possible *explanation* of them within a two-factor rough model.

The third study, Section 4.3, changes both the data and the question. It leaves the option book entirely and works on eleven years of daily closes of two equity indices, computing the level-two signature of the resulting planar path. This is the one place in the thesis where the machinery of Chapter 3 is used as machinery rather than as vocabulary: the objects evaluated are the Lévy area and Chen’s relation themselves, the identities they satisfy are checked as identities, and the estimator built from them in Subsection 3.5.4 is read on real data and then deliberately tested against a case whose answer is known in advance.

Notation 4.1 (Symbols of this chapter). The surface conventions of Chapter 2 are kept without exception: $k = \log(K/F(t, T))$ is log-moneyness, T is the maturity coordinate, $\sigma_{\text{imp}}(k, T)$ is the implied volatility, $w = \sigma_{\text{imp}}^2 T$ the total implied variance and σ_{loc} the local volatility. Two derived quantities recur throughout and are fixed here: the **normalised skew** $\mathfrak{s} := k \sigma_{\text{imp},k} / \sigma_{\text{imp}}$, and the **log-elasticity** in maturity $\lambda_f := \partial \log f / \partial \log T$, used as $\lambda_{\sigma_{\text{imp}}}$ and λ_w .

4.1 Calibration of an Implied Volatility Surface with a PINN

Subsection 2.3.1 established that local volatility is the unique diffusion reproducing a given arbitrage-free implied volatility surface, and the route is Dupire’s formula (2.46):

$$\sigma_{\text{loc}}^2(k, T) = \frac{\partial_T w}{g}, \quad (4.1)$$

whose denominator is exactly the butterfly quantity g of (2.49) and whose numerator is exactly the calendar quantity (2.48). The whole of this section is an attempt to build a surface for which the right-hand side of (4.1) is well defined everywhere, including where nothing is quoted.

The classical response is to fit a parametric slice model — SVI [36] and its descendants — maturity by maturity, and then to repair the slices so that they do not cross. We take a different route. We construct a single globally smooth function of (k, T) , represented by a small neural network in the region where the market quotes and by a closed-form asymptotic family outside it, and we impose the two conditions of Proposition 2.17 as penalties evaluated at collocation points during training. This is the *physics-informed neural network* (PINN) methodology of Raissi, Perdikaris and Karniadakis [66], applied not to a PDE residual but to a pair of differential inequalities.

Neural representations of the implied volatility surface carrying no-arbitrage penalties are not new. Ackerer, Tagasovska and Vatter [1] smooth the surface with a network trained under soft arbitrage constraints, and Chataigner, Crépey and Dixon [17] do the same with the local volatility as the target. What is added here is confined to the extrapolated region: the wings are not learned at all but given in closed form, the no-arbitrage conditions are *proved* on them (Lemma 4.8 and Theorem 4.9) rather than sampled at collocation points, and they are joined to the network with C^3 continuity.

A full account of PINNs, of their capabilities and of their limitations, is outside the scope of this dissertation; the reader interested in these approaches to numerical problems is referred to [66] and to the survey [49]. We record only that they have been successful on direct and inverse problems in disciplines — fluid dynamics above all — where classical approaches are hard, and that for our purposes the decisive feature is a narrow one: automatic differentiation [4] supplies derivatives of the represented function that are exact up to machine precision. A condition such as $g \geq 0$, which is a statement about $\sigma_{\text{imp},k}$ and $\sigma_{\text{imp},kk}$, can therefore be evaluated at a point rather than estimated from a difference quotient.

The distinguishing features of the construction, relative to a naive PINN, are the following.

1. The asymptotic behaviour is *not learned*. Beyond a fixed attachment point on each side the surface is an explicit saturating exponential whose three parameters are read off the network's own local jet and then projected onto a region where the conditions of Proposition 2.17 hold *identically*.
2. The surface is C^3 across every internal frontier by construction, so by (4.1) the local volatility is C^1 and can itself be differentiated.
3. No derivative appearing in the loss is a finite difference in k . All k -derivatives are exact, obtained by forward-mode propagation of a truncated jet through the network.
4. The calendar condition on the extrapolated wings is reduced, by an exact minimisation, to three scalar monotonicity statements in T (Theorem 4.9).

Proposition 2.17 is stated in total-variance coordinates because that is where the conditions are cleanest. The object we shall parameterise, however, is σ_{imp} itself: it is what is quoted, what carries a bid–ask spread, and what has a bounded and well-conditioned range across the whole surface, whereas w varies by two orders of magnitude between one month and five years. The following elementary translation is therefore the bridge between the theory of Chapter 2 and the model of this section, and it is used at every collocation point.



Proposition 4.2 (The butterfly quantity in volatility coordinates). *Let $\sigma_{\text{imp}}(k, T) > 0$ be twice differentiable in k and set $w = \sigma_{\text{imp}}^2 T$. Then the function g of (2.49) satisfies*

$$g = (1 - \mathfrak{s})^2 - \frac{1}{4} T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2 + T \sigma_{\text{imp}} \sigma_{\text{imp},kk}, \quad \mathfrak{s} = \frac{k \sigma_{\text{imp},k}}{\sigma_{\text{imp}}}, \quad (4.2)$$

and the calendar quantity satisfies

$$\partial_T w = \sigma_{\text{imp}}^2 (1 + 2 \lambda_{\sigma_{\text{imp}}}), \quad \lambda_{\sigma_{\text{imp}}} = \frac{T \sigma_{\text{imp},T}}{\sigma_{\text{imp}}}, \quad 1 + 2 \lambda_{\sigma_{\text{imp}}} = \lambda_w. \quad (4.3)$$

The cancellation is what makes the volatility parameterisation workable at all. In (2.49) the slope enters through two terms of opposite sign whose leading parts are both $O(T)$; in (4.2) that part is gone and what survives is a clean decomposition

$$g = g_{\text{slope}} + T \sigma_{\text{imp}} \sigma_{\text{imp},kk}, \quad g_{\text{slope}} := (1 - \mathfrak{s})^2 - \frac{1}{4} T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2, \quad (4.4)$$

in which the curvature appears exactly once, linearly, and with a positive sign. Two consequences follow immediately, and both shape the objective of §4.1.4.

Proposition 4.3 (Qualitative behaviour with short maturities). *Suppose σ_{imp} , $\sigma_{\text{imp},k}$ and $\sigma_{\text{imp},kk}$ remain bounded as $T \downarrow 0$, and work on the branch $\mathfrak{s} < 1$. Both correction terms in (4.2) carry explicit positive powers of T , so $g \rightarrow (1 - \mathfrak{s})^2$ as $T \rightarrow 0^+$, and a margin $g \geq g_\star$ maintained in that limit forces*

$$\mathfrak{s} \leq 1 - \sqrt{g_\star}, \quad (4.5)$$

irrespective of the curvature. This is a statement about the limit, not a constraint imposed exactly at each individual small maturity.

Proof: Immediate from (4.2), the two correction terms being $O(T^2)$ and $O(T)$ with σ_{imp} , $\sigma_{\text{imp},k}$, $\sigma_{\text{imp},kk}$ bounded. At $T = 0$ the requirement reads $(1 - \mathfrak{s})^2 \geq g_\star$, so that on the branch $\mathfrak{s} < 1$ this is (4.5). ■

Remark 4.4 (The curvature enters linearly and benignly). In (4.4) the curvature term $T \sigma_{\text{imp}} \sigma_{\text{imp},kk}$ is linear in $\sigma_{\text{imp},kk}$ with a *positive* coefficient. Consequently:

- i) a given change of slope spread over m times the width costs $1/m$ of the density; in particular, spreading it over three times the width recovers two thirds of the cost;
- ii) positive curvature *raises* g , so curvature is a cost only where it is negative — and that is precisely the case of interest here, since a rising wing approaching a finite asymptote must approach it concavely. At the join this is $\sigma_{\text{imp},kk} = -\mathfrak{s}/\ell < 0$, which is what (4.13) charges for and what forces the floor (4.14) on ℓ .

Inputs

The only inputs the model requires are bid/ask volatility pairs taken from market quotes, together with the strike and maturity they belong to: $(\sigma_{\text{imp}}^{\text{bid}}, \sigma_{\text{imp}}^{\text{ask}}; k, T)$. This implicitly hides one preliminary step, namely the forward curve calibration needed to invert the volatilities out of market prices through Definition 2.13, the forward being the one fixed in Definition 2.10. We assume throughout that this has been done.

The quotes used in this chapter are consensus mid, bid and ask implied volatilities for S&P 500 index options, for the value date 3 March 2025, obtained from the Totem service of S&P Global Market Intelligence [69], a monthly consensus pricing platform for over-the-counter derivatives. Totem quotes are contributed and are therefore European in style and quoted on a fixed moneyness grid rather than on a listed strike ladder, which is why the maturities reach well beyond the horizon of exchange-listed contracts. Access is restricted to contributing institutions, so the data cannot be redistributed with this dissertation; the same source has been used in the academic literature, for instance by Ergun and Uthemann [28]. What follows deals with the volatility surface alone.

The functional form

Fix two attachment points $k_L^* < 0 < k_R^*$ and two collar widths $\Lambda_L, \Lambda_R > 0$, and introduce on each side the *outward coordinate*

$$x_L(k) = k_L^* - k, \quad x_R(k) = k - k_R^*, \quad (4.6)$$

so that $x_s > 0$ is the extrapolation region on that side and $x_s < 0$ points back into the quoted range. The surface is

$$\sigma_{\text{imp}}(k, T) = \mathcal{M}(k, T) + \sum_{s \in \{L, R\}} \left(\mathcal{W}_s(k, T) - \mathcal{M}(k, T) \right) \Psi(z_s(k)) \quad (4.7)$$

where

- $\mathcal{M}$ is a small multilayer perceptron with parameters Θ , positive by construction (§4.1.3);
- $\mathcal{W}_s$ is the analytic wing,

$$\mathcal{W}_s(k, T) = a(T) + b(T) \left(1 - e^{-x_s(k)/\ell(T)} \right), \quad (4.8)$$

with level $a > 0$, saturation length $\ell > 0$ and lift b of either sign, one such triple per side;

- Ψ is a smooth cut-off supported on the collar: with

$$z_s(k) = \min \left\{ 1, \max \left\{ 0, -\frac{x_s(k)}{\Lambda_s} \right\} \right\} \in [0, 1], \quad (4.9)$$

Ψ satisfies $\Psi(0) = 1$ and $\Psi(1) = 0$, and is specified in §4.1.2.

The clamp in (4.9) is what makes (4.7) a single expression rather than a case distinction. For $x_s > 0$ we have $z_s = 0$ and $\Psi = 1$, so $\sigma_{\text{imp}} \equiv \mathcal{W}_s$: the tail is the blend evaluated at its own endpoint, not a separate branch. For $x_s < -\Lambda_s$ we have $z_s = 1$, $\Psi = 0$ and $\sigma_{\text{imp}} \equiv \mathcal{M}$. In between, the two are blended. One formula, three regimes.

Remark 4.5. The two collars never overlap in practice, and the implementation caps $\Lambda_L + \Lambda_R$ at a fixed fraction of $k_R^* - k_L^*$. Where they did overlap, (4.7) would still be well defined but the interpretation as “one wing at a time” would fail.



What is imposed

By Proposition 2.17 and Proposition 4.2, absence of static arbitrage is secured by the two inequalities

$$g \geq 0 \quad \text{and} \quad \partial_T w = \sigma_{\text{imp}}^2 (1 + 2\lambda_{\sigma_{\text{imp}}}) \geq 0, \quad (4.10)$$

with g given by (4.2), the third condition of that proposition — Lee’s wing bound [51] — being automatic for (4.8), as noted in §4.1.1. In practice both are imposed with a strictly positive margin: $g \geq g_*$ and $\lambda_w \geq c_*$ with $g_*, c_* > 0$. The reason is stated once here and used throughout. A surface satisfying $g \geq 0$ with equality attained is one on which the risk-neutral density of Proposition 2.15 vanishes, and by (4.1) the local variance there is infinite; a margin is what keeps (4.1) usable rather than merely non-negative.

4.1.1 The Analytic Wings

Nothing in the architecture prevents a fitted surface from turning over at $|k| = 3$, and Lee’s moment formula [51] is a statement about a limit, not a penalty one can evaluate at a finite collocation point. We therefore replace the network by the three-parameter family (4.8) beyond k^* and *prove* the two conditions (4.10) on that family.

Write the wing in terms of its jet at the join. Dropping the side subscript, with $a = \mathcal{W}(0)$ the level, $s = \sigma_{\text{imp},x}(0)$ the outward slope and ℓ the saturation length,

$$b = s\ell, \quad \sigma_{\text{imp},x}(x) = s e^{-x/\ell}, \quad \sigma_{\text{imp},xx}(x) = -\frac{s}{\ell} e^{-x/\ell}, \quad \mathcal{W}_\infty = a + b. \quad (4.11)$$

It is convenient to use $w_0 = Ta^2$, the total variance at the join, and the dimensionless slope

$$\mathfrak{s}_0 = \frac{|k^*|s}{a}, \quad (4.12)$$

which is the normalised skew $\mathfrak{s}$ of Notation 4.1 evaluated at the join, on either side: at $x = 0$ we have $k = k^*$, $\sigma_{\text{imp}} = a$ and $k\sigma_{\text{imp},k} = |k^*|s$, the signs cancelling on the left wing. Two structural facts are immediate and carry no loss weight:

- $\sigma_{\text{imp},x} = s e^{-x/\ell}$ never vanishes unless $b = 0$, so the surface has *no extremum* beyond k^* ;
- $\sigma_{\text{imp}} \rightarrow \mathcal{W}_\infty$, so $w \sim \mathcal{W}_\infty^2 T$ is bounded in k and condition iii) of Proposition 2.17 holds with room to spare.

The butterfly condition at the join

Substituting (4.11) into $g \geq g_*$ at $x = 0$, using $\sigma_{\text{imp},k}^2 = s^2$ and $\sigma_{\text{imp},kk} = -s/\ell$ and eliminating s through (4.12), gives

$$(1 - \mathfrak{s}_0)^2 - \frac{w_0^2 \mathfrak{s}_0^2}{4k^{*2}} - \frac{w_0 \mathfrak{s}_0}{|k^*| \ell} \geq g_*. \quad (4.13)$$

On a *rising* wing ($\mathfrak{s}_0 > 0$) the last term is a cost, so (4.13) is a *lower* bound on ℓ , namely

$$\ell \geq \ell_{\min} := \frac{w_0 \mathfrak{s}_0 / |k^*|}{(1 - \mathfrak{s}_0)^2 - g_* - (w_0 \mathfrak{s}_0 / 2 |k^*|)^2}. \quad (4.14)$$

Lemma 4.8 below requires $\ell \leq c_\ell |k^*|$ with $c_\ell \leq 1$, so the admissible window in ℓ is non-empty precisely when $\mathfrak{s}_0$ does not exceed a critical value. Imposing (4.13) at the worst admissible length $\ell = c_\ell |k^*|$ and multiplying through by $c_\ell k^{*2}$ turns it into the quadratic

$$(A_\ell - B_\ell) \mathfrak{s}_0^2 - (2A_\ell + w_0) \mathfrak{s}_0 + A_\ell(1 - g_\star) \geq 0, \quad A_\ell = c_\ell k^{*2}, \quad B_\ell = \frac{1}{4} c_\ell w_0^2, \quad (4.15)$$

whose relevant root, rationalised, is

$$\mathfrak{s}_0^\dagger = \frac{2A_\ell(1 - g_\star)}{(2A_\ell + w_0) + \sqrt{4A_\ell^2 g_\star + 4A_\ell w_0 + w_0^2 + 4A_\ell B_\ell(1 - g_\star)}}. \quad (4.16)$$

Three properties of (4.16) deserve comment.

1. It returns the correct root in *both* orientations. For $w_0 < 2|k^*|$ the quadratic (4.15) opens upward and the feasible set is $[0, r_{\text{small}}]$; for $w_0 > 2|k^*|$ it opens downward and the set is $[r_{\text{small}}, r_{\text{large}}]$. In both cases (4.16) returns the binding endpoint.
2. At $T = 0$ it reduces exactly to $1 - \sqrt{g_\star}$ (Proposition 4.3). Indeed $w_0 = Ta^2 \rightarrow 0$ forces $B_\ell \rightarrow 0$, whence the radicand tends to $4A_\ell^2 g_\star$ and $\mathfrak{s}_0^\dagger \rightarrow \frac{2A_\ell(1-g_\star)}{2A_\ell(1+\sqrt{g_\star})} = 1 - \sqrt{g_\star}$.
3. The root is always real. Every one of the four terms of the radicand is non-negative, since $A_\ell, B_\ell, w_0 \geq 0$ and $g_\star \in (0, 1)$.

On a *falling* wing ($\mathfrak{s}_0 < 0$) the last term of (4.13) becomes a credit and $(1 - \mathfrak{s}_0)^2 > 1$, so those two terms both help. The middle term $-w_0^2 \mathfrak{s}_0^2 / 4k^{*2}$ does not, and it grows like w_0^2 : the condition is therefore *not* automatic in ℓ and T . At $k^* = 1$, $a = 1$, $\mathfrak{s}_0 = -0.1$, $\ell = 1$ and $T = 100$ one finds $g(0) = -13.79$, although the asymptote floor below is respected. The maturities carried by this book keep w_0 far from that regime, but (4.13) must be imposed on the falling side as well, not assumed. What binds in addition is the requirement that the asymptote not collapse, $\mathcal{W}_\infty \geq c_\infty a$, which with $b = \mathfrak{s}_0 a / |k^*|$ and $\ell \leq c_\ell |k^*|$ becomes the constant bound

$$\mathfrak{s}_0 \geq -\frac{1 - c_\infty}{c_\ell}. \quad (4.17)$$

Finally, nothing so far prevents the wing from dipping *inward*, into the collar. On a rising wing $\mathcal{W}$ decreases monotonically inward, attaining its minimum at the inner edge $x = -\Lambda$, where $\mathcal{W} = a - b(e^{\Lambda/\ell} - 1)$. Requiring $\mathcal{W} \geq c_{\text{col}} a$ throughout the collar gives a ceiling on $\mathfrak{s}_0$ alone,

$$\mathfrak{s}_0 \leq \frac{(1 - c_{\text{col}}) |k^*|}{\ell_{\min}(e^{\Lambda/\ell_{\min}} - 1)}, \quad (4.18)$$

the worst case being $\ell = \ell_{\min}$ because $\ell \mapsto \ell(e^{\Lambda/\ell} - 1)$ is decreasing. On a falling wing $\mathcal{W}$ increases inward and (4.18) never binds, so it may be applied uniformly without a case split.

Remark 4.6 (Order of operations). The projection forms the network's *request* first and applies the ceilings (4.16), (4.17) and (4.18) *last*. Any clamp applied after a ceiling is a hole in the argument: it can push a parameter back above the bound it was just brought under.

Remark 4.7 (Smooth projections). All projections are performed with a C^∞ soft minimum,

$$\text{softmin}_{c_{\text{soft}}}(u, v) = \min(u, v) - c_{\text{soft}}^{-1} \text{softplus}(-c_{\text{soft}} |u - v|),$$

whose error is at most $c_{\text{soft}}^{-1} \log 2$. A hard min would leave a kink in T wherever the request crosses the ceiling; the drifts of §4.1.1 are estimated by central differences in T , and across a kink such a difference returns the average of two distinct one-sided derivatives — an $O(1)$ error rather than $O(h_T^2)$.



The following lemma is what turns a pointwise condition at the join into a statement about the whole infinite tail. It is the reason the cap $\ell \leq c_\ell |k^*|$ appears at all.

Lemma 4.8 (Monotone density along a rising tail). *Let $\mathcal{W}$ be given by (4.8) with $s > 0$, and suppose the jet at $x = 0$ lies in the admissible window, so that in particular $0 \leq \mathfrak{s}_0 < 1$. If*

$$\ell \leq c_\ell |k^*| \quad \text{with} \quad c_\ell \leq 1, \quad (4.19)$$

then $g(x) \geq g(0)$ for all $x \geq 0$.

The proof is given in Appendix B.11. Under (4.19) the lift obeys $b = \mathfrak{s}_0 a \ell / |k^*| \leq \mathfrak{s}_0 a < a$, and it is exactly this inequality that makes each of the three terms of (4.2) attain its minimum over $[0, \infty)$ at the endpoint $x = 0$.

Consequently $g \geq g_*$ on the entire rising tail as soon as it holds at the join, and the join is where (4.13) was imposed. We do not claim that (4.19) is sharp. The stationarity argument one is tempted to give concerns the *numerator* of $\mathfrak{s}$ alone, while $\mathfrak{s}$ has a varying denominator as well: in fact $\mathfrak{s}'(0) = (\mathfrak{s}_0 / |k^*|) (1 - |k^*| / \ell - \mathfrak{s}_0)$, which at $\ell = |k^*|$ equals $-\mathfrak{s}_0^2 / |k^*| < 0$. What can be said is numerical: over 2×10^5 random admissible boxes, $\min_{x \geq 0} g(x) - g(0)$ was exactly 0 at $c_\ell = 1$ and already -5.4×10^{-4} at $c_\ell = 1.2$. For a falling tail the corresponding sweep gives $\min_x g = 1.000000$ exactly over the boxes sampled.

The calendar condition on the wings

Because k^* does not move with T , the coordinate x of (4.6) is fixed as T varies. Writing $\hat{x} = x / \ell$ for the dimensionless outward coordinate and using (4.3), one obtains $\partial_T w = \sigma_{\text{imp}} \Upsilon$ with $\Upsilon := \sigma_{\text{imp}} + 2T \sigma_{\text{imp}, T}$, and on the wing

$$\Upsilon(\hat{x}) = v_0 + v_1 (1 - e^{-\hat{x}}) + v_2 \hat{x} e^{-\hat{x}}, \quad (4.20)$$

$$v_0 = a(1 + 2\lambda_a), \quad v_1 = b + 2T \dot{b}, \quad v_2 = -2b \lambda_\ell, \quad (4.21)$$

with $\lambda_a = \partial \log a / \partial \log T$ and $\lambda_\ell = \partial \log \ell / \partial \log T$ as in Notation 4.1: $\partial_T w \geq c_\star \sigma_{\text{imp}}^2$ is equivalent to $\Upsilon \geq c_\star \sigma_{\text{imp}}$, and $\Upsilon - c_\star \sigma_{\text{imp}}$ has the same form (4.20) with $v_0 \mapsto v_0 - c_\star a$ and $v_1 \mapsto v_1 - c_\star b$.

Since $\Upsilon'(\hat{x}) = e^{-\hat{x}} [(v_1 + v_2) - v_2 \hat{x}]$ has the single root $\hat{x}^* = (v_1 + v_2) / v_2$ and $\Upsilon''(\hat{x}^*) = -v_2 e^{-\hat{x}^*}$, the point $\hat{x}^*$ is a minimum iff $v_2 < 0$ and a maximum iff $v_2 > 0$; in the latter case, when $v_2 = 0$ and $\hat{x}^*$ is undefined, or when $\hat{x}^* \notin [0, \infty)$, the minimum is attained at an endpoint. Hence

$$\min_{\hat{x} \geq 0} \Upsilon = \min \left(\{ \Upsilon(0), \Upsilon(\infty) \} \cup \{ \Upsilon(\hat{x}^*) : v_2 < 0 \text{ and } \hat{x}^* \geq 0 \} \right) \quad (4.22)$$

— at most three scalars, no grid, no quadrature. The interior candidate has to be discarded when it is undefined or outside the domain: with $v_0 = 0$, $v_1 = 2$, $v_2 = -1$ one has $\hat{x}^* = -1$ and $\Upsilon(\hat{x}^*) = 2 - e < 0$, whereas the true minimum over $[0, \infty)$ is $\Upsilon(0) = 0$. It is exact in k , but it is a *pointwise* statement in T , and that is what the next result removes.

Theorem 4.9 (Wing calendar reduction). *With v_0, v_1, v_2 as in (4.21), write $\Upsilon(0) = v_0$ and $\Upsilon(\infty) = v_0 + v_1$. Then*

$$v_0 \geq 0 \iff a\sqrt{T} \text{ non-decreasing}, \quad v_0 + v_1 \geq 0 \iff \mathcal{W}_\infty \sqrt{T} \text{ non-decreasing}, \quad (4.23)$$

$$v_1 + v_2 = \ell (s + 2T \dot{s}) = \Upsilon'(0), \quad \text{so} \quad v_1 + v_2 \geq 0 \iff s\sqrt{T} \text{ non-decreasing}, \quad (4.24)$$

the last equivalence holding for either sign of s . Moreover, if $\Upsilon'(0) \geq 0$ then

$$\min_{\hat{x} \geq 0} \Upsilon = \min\{v_0, v_0 + v_1\}. \quad (4.25)$$

Consequently, if $a\sqrt{T}$, $\mathcal{W}_\infty\sqrt{T}$ and $s\sqrt{T}$ are each non-decreasing in T , then $\partial_T w \geq 0$ on the entire infinite wing, for every k and every T .

Proof: For any differentiable $X > 0$, $\frac{d}{dT}[X\sqrt{T}] = (X + 2T\dot{X})/(2\sqrt{T})$, so each of the three conditions is of the form $X + 2T\dot{X} \geq 0$. Identity (4.23) for v_0 is immediate from $v_0 = a(1 + 2\lambda_a) = a + 2T\dot{a}$. For $v_0 + v_1$,

$$v_0 + v_1 = (a + 2T\dot{a}) + (b + 2T\dot{b}) = \mathcal{W}_\infty + 2T\dot{\mathcal{W}}_\infty,$$

since $\mathcal{W}_\infty = a + b$. For (4.24), from $b = s\ell$ we have $\dot{b} = \dot{s}\ell + s\dot{\ell}$, so

$$v_1 + v_2 = b + 2T\dot{b} - 2b\frac{T\dot{\ell}}{\ell} = s\ell + 2T\dot{s}\ell + 2Ts\dot{\ell} - 2Ts\dot{\ell} = \ell(s + 2T\dot{s}),$$

and $\Upsilon'(0) = v_1 + v_2$ directly from (4.20); since $\ell > 0$, the sign of $v_1 + v_2$ is that of $s + 2T\dot{s}$, whatever the sign of s itself. For the minimisation (4.25): if $v_2 > 0$ then $\hat{x}^*$ is a maximum and the minimum is an endpoint; if $v_2 = 0$ then Υ is monotone between v_0 and $v_0 + v_1$; and if $v_2 < 0$ with $v_1 + v_2 \geq 0$ then $\hat{x}^* = (v_1 + v_2)/v_2 \leq 0$ and, for all $\hat{x} \geq 0$, $\Upsilon'(\hat{x}) = e^{-\hat{x}}[(v_1 + v_2) - v_2\hat{x}] \geq 0$ because $-v_2\hat{x} \geq 0$, so Υ is non-decreasing and the minimum is $\Upsilon(0) = v_0$. ■

Theorem 4.9 converts a condition that is two-dimensional in (k, T) into three one-dimensional monotonicity statements about scalar functions of T . This is what makes the wings cheap to police.

4.1.2 Smoothness Across the Frontiers

By (4.1) the local volatility involves $\sigma_{\text{imp},k}$, $\sigma_{\text{imp},kk}$ and $\sigma_{\text{imp},T}$. If a downstream calculation differentiates the local volatility itself — a Dupire PDE solved with a high-order scheme, for instance — then σ_{imp} must be C^3 , not merely C^2 . The blend (4.7) has two internal frontiers per side: the join $x = 0$ and the inner collar edge $x = -\Lambda$.

Take $\Psi = 1 - \mathcal{S}_7$ with $\mathcal{S}_7$ a polynomial satisfying $\mathcal{S}_7(0) = 0$, $\mathcal{S}_7(1) = 1$. Writing $\Delta_s = \mathcal{W}_s - \mathcal{M}$ for the gap between wing and core, the blended surface is $\sigma_{\text{imp}} = \mathcal{M} + \Delta_s\Psi(z_s)$ and, since $\partial_k z_s = \mp \mathbb{1}_{\text{collar}}/\Lambda_s$ (the side subscript on $\mathcal{W}_s$ is dropped below wherever only one side is in view),

$$\sigma_{\text{imp},k} = \mathcal{M}_k + (\mathcal{W}_k - \mathcal{M}_k)\Psi + \Delta_s\Psi'\partial_k z_s, \quad (4.26)$$

$$\sigma_{\text{imp},kk} = \mathcal{M}_{kk} + (\mathcal{W}_{kk} - \mathcal{M}_{kk})\Psi + 2(\mathcal{W}_k - \mathcal{M}_k)\Psi'\partial_k z_s + \Delta_s\Psi''(\partial_k z_s)^2. \quad (4.27)$$

Matching the third derivative at $z = 1$ (the inner edge) requires $\Psi(1) = \Psi'(1) = \Psi''(1) = \Psi'''(1) = 0$, while at $z = 0$ (the join) what is required is $\Psi(0) = 1$ together with $\Psi'(0) = \Psi''(0) = \Psi'''(0) = 0$ — the value, not the vanishing, is what differs between the two ends. In terms of $\mathcal{S}_7 = 1 - \Psi$ these are the eight linear conditions $\mathcal{S}_7(0) = 0$, $\mathcal{S}_7(1) = 1$ and $\mathcal{S}'_7 = \mathcal{S}''_7 = \mathcal{S}'''_7 = 0$ at both ends, so there is a unique polynomial of degree 7 satisfying all of them, namely

$$\mathcal{S}_7(z) = 35z^4 - 84z^5 + 70z^6 - 20z^7, \quad \mathcal{S}'_7(z) = 140z^3(1 - z)^3, \quad (4.28)$$

the factorisation making the triple zero at each end immediate.



4.1.3 The Core Network and Exact Derivatives

$\mathcal{M}$ is a multilayer perceptron with the scaled activation $v \mapsto \tanh(mv)/m$, where m is a learnable per-unit gain constrained to $[0.3, 3]$: small m widens the linear region, large m saturates. The output is

$$\mathcal{M} = \text{softplus}(o) + \mathcal{M}_{\min}, \quad (4.29)$$

with $\mathcal{M}_{\min}$ a learnable floor in $[10^{-4}, 0.08]$ and o the last linear pre-activation. Positivity of σ_{imp} — which Definition 2.13 requires, and without which (4.2) is not even well posed — is therefore *structural* and never appears in the loss.

Features

Seven inputs, each with finite derivatives at $T = 0$ so that the origin is an ordinary point of the model:

$$\left[\hat{k}, y, y^2, y^3, T/T_{\text{ref}}, \hat{T}, y\hat{T} \right], \quad (4.30)$$

where $\hat{k}$ is k rescaled to $[-1, 1]$ across $[k_L^*, k_R^*]$, and

$$y = \tanh\left(\frac{k}{k_c \sqrt{T + T_z}}\right), \quad \hat{T} = \frac{\log(1 + T/T_c)}{\log(1 + T_{\text{ref}}/T_c)}. \quad (4.31)$$

The lattice nodes and the quotes have fixed (k, T) , so all seven features of (4.30) and all of their k - and T -derivatives are constants of the calibration. They are built once and indexed thereafter, rather than recomputed — with their square roots, hyperbolic tangents and logarithms — several times per step for sixteen thousand steps.

4.1.4 The Objective

The loss has three blocks: market fit, arbitrage, and shape. Only the arbitrage block rides a schedule; the fit weight is constant, so market fit is the invariant of the optimisation and it is the pressure to be arbitrage-free that is annealed.

Every term is a dimensionless mean per item, normalised by its own natural unit. The convention has a structural payoff: because every term is a mean per item, the number of collocation points changes *where* a constraint is measured without changing the *scale* of any term. The adaptive sampling of §4.1.5 is only legitimate because of this.

Market fit

Let $h = \frac{1}{2}(\sigma_{\text{imp}}^{\text{ask}} - \sigma_{\text{imp}}^{\text{bid}})$ denote the half-spread of a quote and $\sigma_{\text{imp}}^{\text{mid}}$ its mid. The primary residual is ε -insensitive in half-spread units,

$$\varepsilon = \left(\frac{|\sigma_{\text{imp}} - \sigma_{\text{imp}}^{\text{mid}}|}{h} - c_\varepsilon \right)^+, \quad c_\varepsilon = 0.30, \quad (4.32)$$

with a Huber knee beyond. A quote comfortably inside its spread contributes no gradient, so effort is reallocated to quotes that are outside. Underneath sits a weaker absolute-error term in basis points, which keeps “as close to the mid as possible” a live preference and stops the surface parking on the tunnel wall.

Arbitrage

The butterfly and calendar conditions (4.10) each carry two tiers: a soft margin $g_\star$ (resp. $c_\star$) with a quadratic penalty, and a hard floor $g_{\text{hard}} < g_\star$ with a quadratic-plus-quartic barrier under a masked mean.

Two further terms deserve mention, and both are consequences of Proposition 4.2.

- i) *The slope-only floor.* By the decomposition (4.4), the optimiser can always buy g by inflating a positive $\sigma_{\text{imp},kk}$ — convexity that must be paid back with concavity next door, since $\int \sigma_{\text{imp},kk}$ across a slice is fixed by the slopes at its ends. A floor on g_{slope} , which depends on $\mathfrak{s}$ and $\sigma_{\text{imp},k}$ only, closes that loophole.
- ii) *The local variance cap.* By (4.1), a cap $\sigma_{\text{loc}} \leq \sigma_{\text{loc}}^{\text{max}}$ is a butterfly floor $g \geq \partial_T w / (\sigma_{\text{loc}}^{\text{max}})^2$, and the two must be set consistently or one silently overrides the other.

4.1.5 Collocation, Sampling and the Optimiser

The collocation grid is the only place in the design where arbitrage-freedom is *sampled* rather than proved and it is therefore the only place a violation can hide. Two separate structures address this.

- i) *Adaptive collocation in k .* A fine candidate lattice (241 nodes, features precomputed once as in §4.1.3) is divided into strata, one node drawn uniformly from each.
- ii) *The maturity grid contains every quoted maturity exactly,* plus the geometric midpoint between each adjacent pair. The fit pins the surface *at* the quotes and leaves it free *between* them.

Schedule and ratchet

Training uses AdamW [54] with a cosine learning rate and warm restarts [55]. Two schedules run on top: the *envelope* ramps the arbitrage weight from 0.1 to 25 across the main phase, so that the surface first learns the market and is then pushed into the feasible region rather than being confined to it from initialisation.

The *rotary* modulates the arbitrage weight within each period as $(2 + \cos(2\pi n/n_{\text{rot}}))/3$, never reaching zero. The period n_{rot} divides the restart period so that the cycle always ends at maximum enforcement and phase-locks with the learning-rate schedule. Finally, a *ratchet* snapshots the parameters Θ at the end of each rotary period and keeps the best state seen.

4.1.6 Hyperparameters

Table 4.1 collects the settings that carry an argument, now that each has a meaning. The loss weights are not among them, and no single vector would represent them. Training proceeds in stages. Within each stage the learning rate decays, and the weights attached to the arbitrage penalties are rescaled from a localised diagnostic: the domain is partitioned, the violation of



each condition is measured cell by cell, and the weight carried by that condition on that cell is raised in proportion to the measured violation. The weights are therefore a function of position on the grid and of the stage, not constants. What is fixed, and is reported below, are the margins and the caps.

Symbol	Value	Rationale
c_ℓ	1.00	Hypothesis (4.19) of Lemma 4.8, and <i>sharp</i> : the measured excess $\min_{x \geq 0} g(x) - g(0)$ is 0 at $c_\ell = 1$ and -5.4×10^{-4} at $c_\ell = 1.2$.
$c_\dagger$	0.90	Safety factor in $\mathfrak{s}_0 \leq c_\dagger \mathfrak{s}_0^\dagger$; absorbs the softmin error of Remark 4.7 and leaves margin. The Ψ -weighted worst inward collar shortfall is 0.030 at this value.
$g_\star$	0.13	Butterfly margin in (4.13). Set from the measured density attainable by the quotes.
g_{hard}	0.10	Divergent floor. Binds on essentially every book, which is the intended behaviour.
$c_\star$	0.02	Calendar margin. By (4.3), $\lambda_w = 1 + 2\lambda_{\sigma_{\text{imp}}}$, so $c_\star$ demands growth of the total variance like $T^{c_\star}$. At 0.02 this is +1.4% of variance per doubling of maturity and competes directly with the fit on a flat-variance book.
$\ell/ k^\star $	≥ 0.75	By (4.11), $ \sigma_{\text{imp},xx} = s /\ell$, so this floor <i>is</i> a cap on peak wing curvature.
c_∞	0.85	Floor on $\mathcal{W}_\infty/a$ for a falling wing, via (4.17); trades wing freedom against asymptote collapse.
$\Lambda/ k^\star $	≤ 0.25	Collar width. Trades two effects both linear in Λ : injected curvature $\approx 0.361 \Delta_s''' \Lambda$, and quoted range swallowed $= \Lambda$.
c_{soft}	60	Softmin sharpness in Remark 4.7; error $\leq \log 2/60 \approx 0.0116$, an absolute error rather than a relative one, absorbed by $c_\dagger$; the soft minimum lies <i>below</i> the exact one, so it cannot breach a ceiling but can breach a floor.
h_T	0.05	Relative step for the maturity drift, $T e^{\pm h_T}$. The only finite difference in the model.
c_ε	0.30	Fit deadband (4.32) in half-spread units.
Δk_{min}	0.04 / 0.015	Long and short slope-pair spacings.
network	24×3	With the enriched feature basis (4.30) this matches what a 32×4 network delivered at 0.44 of the cost.

Table 4.1: Hyperparameters carrying an argument.

4.1.7 Diagnostics

The instrumentation reports, at each checkpoint:

- **Fit by band and per maturity**, with signed bias. A single aggregate hides a badly fitted smile, because opposite-signed biases cancel.
- **The density budget at the worst node**, split as $g = g_{\text{slope}} + T\sigma_{\text{imp}} \sigma_{\text{imp},kk}$ according to (4.4). If g_{slope} is already below target with non-negative curvature, no amount of weight will reach the target — the constraint is infeasible against the data.
- **Bound activity for the wing jets**, reported against whichever of (4.16), (4.17), (4.18) is active. Reporting only against the ceiling is meaningless when the parameter sits at its floor, which is exactly what a falling call wing does.
- **Collar swallow**: how much quoted range each collar absorbs and how much of the surface at the innermost swallowed quote is wing rather than network.

- **The theorem monitor:** the fraction of maturities at which $v_1 + v_2 < 0$. Zero on both sides means the minimum of Υ is attained at an endpoint and the wing is calendar-free by Theorem 4.9 rather than by sampling.
- **Shape:** the number of trusted quote pairs that move the wrong way. A surface can lie inside every bid/ask and still have the wrong local direction.
- **An independent dense scan** on a mesh finer than, and deliberately not aligned with, the training lattice — uniform in maturity, where the training line is logarithmic — reporting violations separately for the left tail, the core and the right tail.

4.1.8 Results

After calibration to several underlyings for different value dates (that is, for market data from different days), the surface is smooth and fits the market well. No violation of (4.10) was detected on the independent validation grid; on the rising wings the butterfly condition holds analytically under Lemma 4.8, while the calendar guarantee additionally requires the hypotheses of Theorem 4.9 throughout the maturity interval. A finite penalty evaluated at collocation points is not a proof of an inequality on a continuum, and none is claimed here.

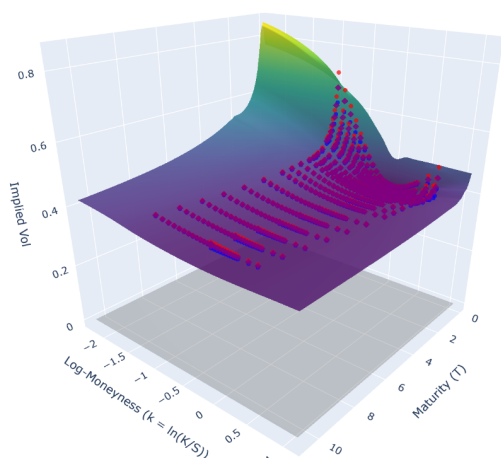


Figure 4.1: Implied volatility surface representation, together with the bid (blue), ask (red) and mid (purple) quotes.

Metric	Count	%
Total Quotes	576	100.0%
Inside Bid/Ask Tunnel	550	95.49%
Outside Tunnel (<i>Below / Above</i>)	26	4.51% (1.9% / 2.6%)
Smiles (T)	22	26–30 quotes/smile

Maturity (T)	Quotes	Inside	Outside	Ratio
0.1971	30	30	0	100.0%
0.27–0.70 (6 smiles)	26 ea.	26 ea.	0	100.0%
0.7693	26	25	1	96.2%
1.0212	26	26	0	100.0%
1.2704	26	15	11	57.7%
1.3662	26	12	14	46.2%
1.52–9.28 (11 smiles)	26 ea.	26 ea.	0	100.0%
Total	576	550	26	95.5%

Table 4.2: Calibration Summary & Smile Breakdown.

The tests performed, mainly for arbitrage purposes, have been passed. In terms of market fit, the main limitation has been noisy quotes as input data. Figure 4.1 above shows the fitted surface against the quotes. It belongs to the SPX index, which is in a sense one of the most complex underlyings to calibrate due to the abundance of market data, tight bid/ask spreads and the presence of a strong skew. The errors measured to the mid are typically below five basis points, and 550 of the 576 quotes fall inside their own bid/ask, which is a good outcome given the complexity of the surface and the number of quotes fitted. The surface is smooth, and the implied volatilities are consistent with market observations.



4.2 Calibration to Rough Heston

Section 4.1 fitted the surface with no dynamics at all: the object produced there is a function of (k, T) constrained to be arbitrage-free, and nothing more. This section fits the same quotes with the rough Heston model of Definition 3.50, in which the surface is not fitted but *implied* by six numbers describing a variance process. The comparison is the point of the chapter, and the question it answers is how much of a real index smile a two-factor rough model can carry.

The book is the one described in §4.1, but the sample and the objective are not those of §4.1.4, and the two studies are therefore not a controlled comparison. Two filters are applied here. Only the sixteen maturities whose T is recoverable exactly are retained, from $T = 0.1040$ to $T = 2.8446$; and within each of them an at-the-money cone $|k| \leq \frac{1}{2}\sqrt{T}$ is imposed, which keeps 2415 of the 2648 quotes. The cone is not a convenience: its width scales as $\sqrt{T}$, so it retains a comparable range of standard deviations at every maturity rather than a comparable range of strikes. The calibration is then performed outright in the model parameters $(v_0, \kappa, \theta, \xi, \rho, H)$, against the half-spread-normalised squared error (4.36) rather than against the composite loss of §4.1.4.

Theorem 3.51 reduces pricing to (3.52), and Remark 3.53 recorded the cost: the fractional derivative is non-local, so a discretisation over n steps carries the whole computed history at each step and costs $O(n^2)$. The solver used here is the fractional Adams predictor–corrector, vectorised across the grid so that one solve prices an entire maturity slice.

The exponent of (3.51) contains $I^{1-\alpha_H}\psi(u, \tau)$, a fractional integral whose kernel $(\tau - s)^{-\alpha_H}$ is singular at the upper endpoint. Quadrature against a singular kernel is delicate and is a standard source of order loss. It is also unnecessary.

Proposition 4.10 (Both terms of the exponent are ordinary integrals). *Let $\psi(u, \cdot)$ solve (3.52) and write $F(u, x) := \frac{\xi^2}{2}x^2 + (\rho\xi iu - \kappa)x - \frac{1}{2}(u^2 + iu)$ for its right-hand side. Then*

$$I^{1-\alpha_H}\psi(u, \tau) = \int_0^\tau F(u, \psi(u, s)) ds, \quad (4.33)$$

so that (3.51) becomes $\varphi(u, T) = \exp\left(\kappa\theta \int_0^T \psi ds + v_0 \int_0^T F(u, \psi) ds\right)$.

Proof: The fractional derivative is $D^{\alpha_H} = \frac{d}{d\tau} \circ I^{1-\alpha_H}$, so (3.52) states that $\frac{d}{d\tau} I^{1-\alpha_H}\psi = F(u, \psi)$. Integrating from 0 to τ and using the initial condition $I^{1-\alpha_H}\psi(u, 0) = 0$ attached to (3.52) gives (4.33). ■

The predictor–corrector already evaluates $F(u, \psi)$ at every node in order to advance, so the right-hand side of (4.33) is available at no cost and is integrated by the trapezoidal rule against a smooth integrand, while the left-hand side would have required a product-integration rule against a singularity. Both terms of the exponent are then ordinary integrals of quantities the solver has already produced.

Prices are recovered by the Lewis form of the Fourier inversion (2.54), evaluated on the line $\text{Im } u = -\frac{1}{2}$ and written for the *undiscounted* call value, so that the bond factor of (2.54) does not appear:

$$\frac{C(k, T)}{F} = 1 - \frac{e^{k/2}}{\pi} \int_0^\infty \text{Re} \left[e^{-iuk} \varphi\left(u - \frac{i}{2}, T\right) \right] \frac{du}{u^2 + \frac{1}{4}}. \quad (4.34)$$

That contour is chosen for a concrete reason: at $u \mapsto u - \frac{i}{2}$ the inhomogeneity of (3.52) becomes $-\frac{1}{2}\left((u - \frac{i}{2})^2 + i(u - \frac{i}{2})\right) = -\frac{1}{2}\left(u^2 + \frac{1}{4}\right)$, which is real and negative for every u . The branch

ambiguity that afflicts the *closed-form* Heston characteristic function does not arise here at all, (3.52) being integrated numerically; what the contour buys is a real, negative inhomogeneity, and the $1/(u^2 + \frac{1}{4})$ weight makes the tail integrable without further damping.

Where the explicit scheme breaks, and why

The predictor–corrector is explicit and therefore has a stability limit, which is reached in practice. Because $\psi(u, 0) = 0$, the first step is approximately $\psi_1 \approx -\frac{1}{2}(u^2 + \frac{1}{4}) \Delta\tau^{\alpha_H} / \Gamma(1 + \alpha_H)$, whereas the true solution settles near the root of the quadratic $F(u, \cdot)$, at the scale $(\frac{1}{2}(u^2 + \frac{1}{4}) / \frac{\xi^2}{2})^{1/2}$. When the first step overshoots that scale, the quadratic term dominates at the second step and the iteration diverges. Demanding no overshoot gives the cut

$$u_{\max} \leq \frac{2\Gamma(1 + \alpha_H)}{\xi \Delta\tau^{\alpha_H}}, \quad \Delta\tau = T/n, \quad (4.35)$$

which is what binds at large vol-of-vol. A second, empirical, cut $u_{\max} \leq c/\sqrt{T}$ with $c = 120$ binds at long maturities. Both sit far outside the region where the integrand of (4.34) is still material — the characteristic function has decayed past 10^{-11} well inside either — so the truncation costs less than 10^{-6} of a vega while removing the divergence entirely. Over a sweep of 1536 parameter and maturity combinations spanning the whole optimisation box, no non-finite price is produced.

4.2.1 The objective

Quotes are weighted by their own half-spread, so that the fit is measured in ticks rather than in volatility points:

$$\Phi(p) = \sum_i \left(\frac{\sigma_{\text{imp}}^p(k_i, T_i) - \sigma_{\text{imp}}^{\text{mid}}(k_i, T_i)}{h_i} \right)^2, \quad h_i = \frac{1}{2}(\sigma_{\text{imp}}^{\text{ask}} - \sigma_{\text{imp}}^{\text{bid}})(k_i, T_i), \quad (4.36)$$

the sum running over the quotes.

The parameter vector is $p = (v_0, \kappa, \theta, \xi, \rho, H)$, all six free, searched by differential evolution over the box of Table 4.3 and then polished by Nelder–Mead at a finer time grid. No parameter is fixed and no penalty or prior is imposed, so the fit reported below is what the surface alone asks for.



Parameter	Box	Reading
v_0	[0.0025, 0.16]	Spot variance; the box is volatility between 5% and 40%.
θ	[0.0025, 0.16]	Long variance, same box.
κ	[0.05, 10]	Mean reversion. In a rough model this is weakly identified: the kernel already supplies the decay of the term structure.
ξ	[0.05, 1.5]	Vol-of-vol; also the parameter that drives the stability cut (4.35).
ρ	$[-0.999, -0.05]$	Spot/vol correlation, constrained negative: the sign of the equity skew is not in question.
H	[0.02, 0.50]	Hurst index. The upper endpoint is the classical Heston case, so the box <i>contains</i> the non-rough model and does not assume the conclusion.

Table 4.3: The optimisation box. Only ρ is sign-constrained; the Hurst index is free over the whole admissible range, including its classical endpoint.

4.2.2 Results

Table 4.4 reports the fit. The global search was a differential evolution over the box of Table 4.3 on a coarse time grid, followed by a bounded local polish on a fine one; the the bound on the polish is not a detail.

Parameter	Value	
v_0	0.02370	spot volatility $\sqrt{v_0} = 15.39\%$
θ	0.06354	long volatility $\sqrt{\theta} = 25.21\%$
κ	2.0033	
ξ	0.8968	
ρ	-0.7346	
H	0.4900	<i>at the smooth end</i> of the box [0.02, 0.50]; $\alpha_H = 0.99$
Φ	76 787	$\Phi/n = 31.80$ on $n = 2415$ quotes
RMSE	44.4 bp	of volatility
inside bid/ask	9.5%	median half-spread 5.5 bp

Table 4.4: Rough Heston calibrated to 2415 SPX quotes, all six parameters free. The Hurst index lands on the boundary of its admissible range.

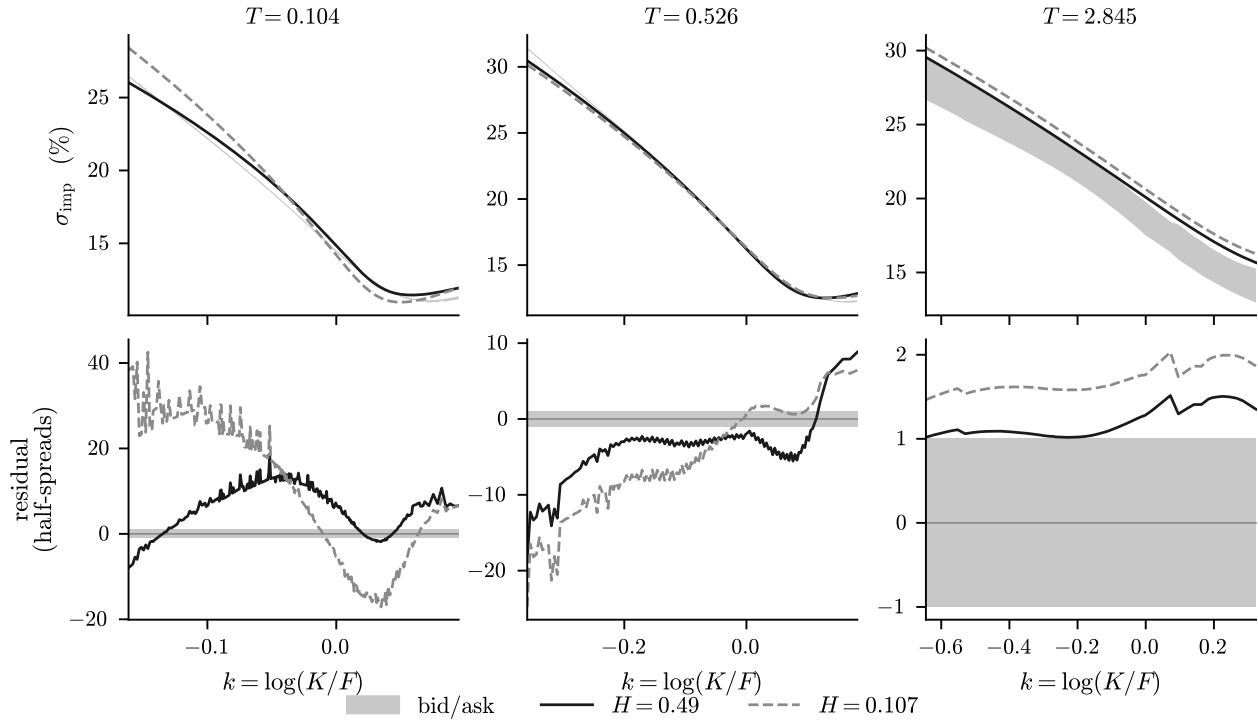


Figure 4.2: The calibrated model against the market at three maturities. Top: implied volatility, with the bid/ask band shaded. Bottom: the residual in half-spread units, the shaded strip being ± 1 half-spread. The two rows disagree about where the model fails, and the disagreement is entirely the term structure of the spread: at $T = 2.845$ the model is more than a volatility point away and still inside two half-spreads, while at $T = 0.104$ it is a quarter of that distance away and tens of half-spreads out.

T	quotes	med. h (bp)	RMSE (bp)	bias (bp)	% of Φ	% of H gap
0.1040	256	5.6	47.0	+36.5	24.3	+57.3
0.1807	266	5.8	29.4	+0.9	9.1	+9.5
0.2574	212	4.8	30.7	-9.6	15.1	+0.5
0.3504	210	4.7	22.1	-8.6	4.8	-1.6
0.4298	118	4.5	28.5	-21.2	5.6	-0.4
0.5257	126	4.2	29.0	-18.4	5.0	+4.0
0.6023	132	4.4	30.0	-17.9	4.0	+5.6
0.6790	134	4.9	23.4	-16.4	3.3	+6.6
0.7748	146	4.9	27.1	-8.8	4.5	+9.4
0.8515	147	5.8	23.5	+8.2	2.8	+4.4
0.9281	147	6.6	29.4	+21.2	4.2	+2.9
1.0240	145	7.5	36.0	+23.1	5.0	+2.7
1.1006	160	8.3	44.4	+43.6	5.9	-0.5
1.3470	99	10.8	78.5	+73.2	5.3	-0.4
1.8480	62	29.1	114.0	+107.4	1.1	+0.2
2.8446	55	109.9	136.5	+135.1	0.1	+0.1

Table 4.5: Fit by maturity at the parameters of Table 4.4. The last two columns are the share of the objective Φ contributed by each maturity, and that maturity's share of the gap $\Phi(H=0.107) - \Phi(H=0.49)$. In volatility the error grows with maturity; weighted by the spread the order reverses, and the shortest maturity carries both a quarter of the objective and well over half of the preference for the smooth kernel.



Three things must be said about this table before anything is concluded from it.

The fit is mediocre in absolute terms. A root mean square error of 44 basis points against a median half-spread of 5.5 is eight half-spreads, and only 9.5% of quotes are matched inside their own bid/ask. Six parameters do not span sixteen maturities of SPX. Figure 4.2 shows the shape of the failure and Table 4.5 localises it.

Where the failure sits depends on the units. Measured in volatility the error grows with maturity: about 25 basis points from three months to one year, rising monotonically to a *bias* of +135 at $T = 2.84$, where bias and RMSE are 135.1 and 136.5 so that essentially the whole error is a parallel offset. Measured in half-spreads — which is the metric (4.36) actually minimises — the ranking reverses completely, because the market quotes the long end 20 times more loosely than the short one: the median half-spread runs from 4.5 basis points around six months to 109.9 at the longest maturity.

The Hurst index goes to the smooth boundary. This is the opposite of what Chapter 3 would lead one to expect, and it is not a numerical accident. Table 4.6 profiles the objective in H , re-optimising the other five parameters at each fixed value.

H	Φ/n	v_0	θ	κ	ξ	ρ
0.030	115.65	0.0223	0.1599*	0.253	0.359	-0.710
0.068	98.17	0.0228	0.1600*	0.260	0.377	-0.715
0.107	82.83	0.0231	0.1600*	0.267	0.392	-0.721
0.145	69.62	0.0233	0.1600*	0.276	0.413	-0.724
0.183	58.65	0.0236	0.1600*	0.286	0.434	-0.726
0.222	50.10	0.0238	0.1600*	0.297	0.458	-0.727
0.260	44.21	0.0240	0.1599*	0.309	0.484	-0.728
0.298	41.00	0.0241	0.1391	0.391	0.521	-0.730
0.337	38.62	0.0240	0.1032	0.637	0.582	-0.731
0.375	36.49	0.0239	0.0857	0.917	0.649	-0.732
0.413	34.61	0.0239	0.0753	1.235	0.724	-0.733
0.452	33.03	0.0238	0.0684	1.596	0.806	-0.734
0.490	31.79	0.0237	0.0635	2.003	0.897	-0.735

Table 4.6: The objective profiled in H . Entries marked * sit on the upper bound of θ . The profile is strictly monotone: every rougher kernel fits strictly worse.

The profile is strictly monotone. At $H = 0.107$, a value typical of the rough volatility literature, the objective is 2.6 times its value at the boundary. Three checks were made before reading anything into this.

- **Is it the box?** For $H \leq 0.26$ the fitted θ sits on its upper bound, so those rows are constrained. Re-optimising with θ allowed up to 0.80 — a long-run volatility of 89%, economically absurd — improves $H = 0.107$ only from $\Phi/n = 82.8$ to 70.3, and does so by driving κ to 0.042. Meanwhile the same widening leaves Φ/n and θ at $H = 0.490$ and $H = 0.375$ unchanged to four decimal places: those optima are interior. Even granted absurd parameters the rough kernel fits more than twice as badly as the smooth one does at its own interior optimum.
- **Is it the solver?** No. Fitting the model's *own* short-dated at-the-money skew to a power of T returns an exponent close to the predicted $H - \frac{1}{2}$ throughout the range: -0.494 against -0.45 at $H = 0.05$, -0.342 against -0.30 at $H = 0.20$, and +0.008

against 0 at the classical endpoint. The roughness is present in the prices the solver produces.

- **Is it the forward convention?** No. If the supplied moneyness were K/S rather than K/F the model would inherit a maturity-proportional strike offset, which is exactly the shape of the long-dated bias in Table 4.5. It is not: the smile minimum sits at 1.7 to 2.1 standard deviations $\sigma_{\text{imp}}\sqrt{T}$ above the money at every one of the sixteen maturities, with no trend. A spot convention would make that ratio grow like $\sqrt{T}$.

4.2.3 What the surface can and cannot identify

It would be easy, and wrong, to report Table 4.6 as evidence that this index is not rough. The correct conclusion is weaker and more interesting: *this data cannot resolve H at all*. What the monotone profile reflects is the short-dated at-the-money term structure, which a one-factor rough Heston with a flat v_0 and θ cannot carry; H is left to absorb that misspecification.

The argument rests on one measurement. Remark 3.53 identified the empirical signature of roughness as the at-the-money skew power law $T^{H-1/2}$, and it is natural to estimate H by regressing the log skew on the log maturity. Doing so on the market quotes gives a slope of -0.454 , that is $H \approx 0.046$: a strongly rough reading, apparently in flat contradiction with Table 4.6. The contradiction dissolves when the same estimator is applied to the fitted models themselves.

Skew regressed over $T \in [0.104, 2.845]$	slope	implied H
market quotes	-0.454	0.046
model at $H = 0.490$	-0.474	0.026
model at $H = 0.107$	-0.507	-0.007

Table 4.7: The skew power law estimated over the available maturity range, applied to the market and to two calibrated models whose true Hurst indices differ by a factor of nearly five. The estimator cannot tell them apart. The legend of Figure 4.3 reports -0.506 for the last of these, the same fit recomputed on the plotted maturity grid; the two agree to the precision the estimate carries.



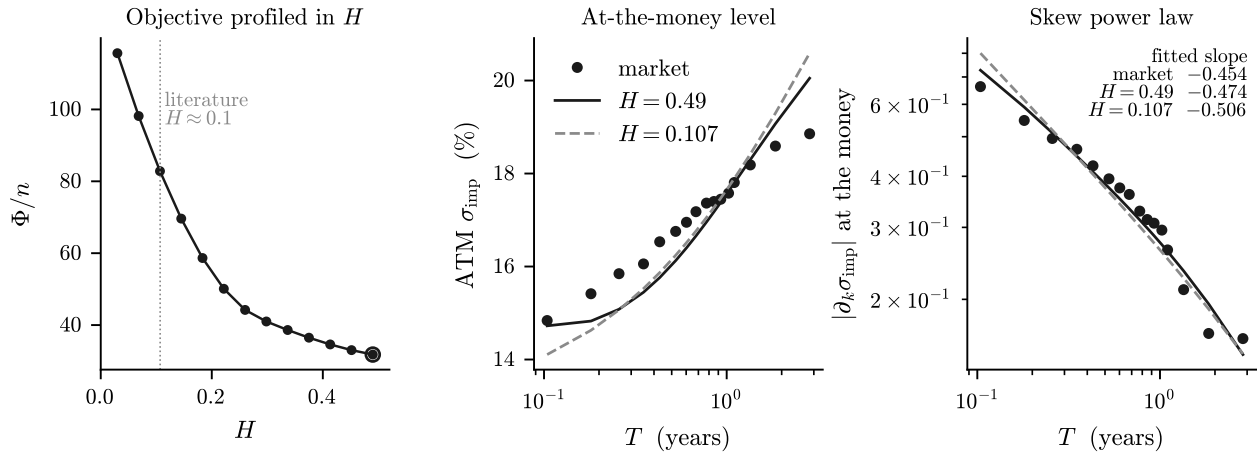


Figure 4.3: Left: the objective profiled in H , the other five parameters re-optimised at each point; it is strictly monotone and its minimum is the boundary $H = 0.49$, marked. Centre: the at-the-money term structure, market against the two calibrated models. Right: the same two models and the market on log axes, with the fitted slopes of $|\partial_k \sigma_{\text{imp}}|$ against T . The three slopes agree to within 0.06 although the two models differ in H by a factor of nearly five, which is Table 4.7 drawn: over this maturity range the skew power law does not separate rough from smooth.

The model with $H = 0.49$ — numerically close to the classical endpoint — reproduces the “rough” exponent -0.474 essentially as well as the genuinely rough model reproduces -0.507 , and both lie on the same side of the market’s -0.454 , within 0.06 of it. The estimator has no power here, and the reason is that $T^{H-1/2}$ is an asymptotic statement as $T \rightarrow 0$. Over a range beginning at 38 days, the mean reversion of *any* Heston-type model decays the skew on its own, and that decay dominates the contribution of the kernel.

One entry of Table 4.7 deserves a line of its own. The implied H read off the rough model is -0.007 , outside the admissible range $(0, \frac{1}{2})$ altogether. That is not a property of the model, whose H is 0.107 by construction, but of the estimator: over this maturity range the regression does not recover the exponent it is meant to measure, and nothing constrains the value it returns to be admissible. An estimate that lands outside the parameter space is the clearest possible symptom of the identification failure this subsection is about.

4.3 The Lévy Area of an Index Pair

Subsection 3.5.4 built a small theory around a single number. It showed that a two-dimensional path carries exactly one free datum at level two (Proposition 3.73), that this datum is a profit and loss (Remark 3.74), that it aggregates across sub-windows by a determinant rather than by addition (Proposition 3.76), and that for the lead-lag pair (3.81) its expectation is (3.82), in which the contemporaneous correlation ρ does not appear at all (Remark 3.79). This section runs all of that on eleven years of market data.

The order of the section is deliberate and is the order in which one should trust a measurement. The identities come first, because they are exact and hold to machine precision or the code is wrong; only then is the estimator read. And because a number that is near zero is easy to produce by accident, the estimator is finally put through a scan in which the correct answer is known in advance.

4.3.1 The data

The pair is the S&P 500 and the EURO STOXX 50, both as *total return* indices, daily closes from 2 January 2015 to 2 September 2026: 3013 observations. Total return series are used rather than price indices so that the two legs are directly comparable; a dividend-stripped index and an accumulating one differ by a slow drift, and a drift is precisely the kind of thing that a level-two statistic will faithfully report.

Within each window the path is

$$X_t^i = \log S_t^i - \log S_{t_0}^i, \quad i = 1 \text{ (S\&P 500)}, 2 \text{ (EURO STOXX 50)},$$

so that $X_{t_0} = 0$ as the signature requires. Windows are consecutive and non-overlapping, of $n \in \{5, 10, 21, 63\}$ observations — a week, a fortnight, a month and a quarter of trading.

One piece of cleaning is necessary and is worth stating because it is not cosmetic. The two markets observe different holiday calendars, and the source carries a shut market forward at its previous close. On such a day one leg has a zero increment while the other does not, which feeds a genuine increment of X^2 against a frozen X^1 into the sum (3.13) — exactly the configuration that manufactures area. The 164 affected rows are removed, leaving 2849 observations.

4.3.2 The three exact identities

Two quadrature rules are computed side by side. Left-point sums, $\mathbb{X}_{s,t}^{12} \approx \sum_k X_{s,t_k}^1 (X_{t_{k+1}}^2 - X_{t_k}^2)$, give the Itô lift; trapezoidal sums give the Stratonovich, or geometric, lift. The two differ in the symmetric part by the realised covariation and agree in the antisymmetric part, which is Remark 3.75. Table 4.8 reports the largest violation observed over every window of every length.

Identity	Source	$\max \text{defect} $
$\text{Sym } \mathbb{X}_{0,T}^{12} = \frac{1}{2} X_{0,T}^1 X_{0,T}^2$	Proposition 3.73	1.7×10^{-18}
$A_{0,T}^{\text{str}} = A_{0,T}^{\text{itô}}$	Remark 3.75	8.7×10^{-19}
$A_{s,t} - A_{s,u} - A_{u,t} - \frac{1}{2} \det(\cdot)$	Proposition 3.76	1.7×10^{-18}

Table 4.8: The identities of Subsection 3.5.4, evaluated on 2849 daily observations of the index pair. All three are satisfied at the level of double-precision rounding.

All three sit at 10^{-18} , which is rounding error and not agreement.

Notice how Chen’s relation is the only one of the three that constrains how information at different time scales fits together, and it is the one with no counterpart in a return-based description. The monthly area is not the sum of the weekly areas; the shortfall is the determinant in (3.80), and it is reproduced exactly.

With the implementation validated, Table 4.9 reads the estimator on the synchronous pairing.

The weekly figure, $t = 1.58$, is not significant. The quarterly figure, $t = 2.00$, is the only one near a conventional threshold; it rests on 45 windows and is one of four horizons inspected, and no correction for that multiplicity leaves it standing.

The standard errors above are $s/\sqrt{M}$ over the M windows, which treats the windows as independent. A circular block bootstrap with block length $\lfloor \sqrt{M} \rfloor$ does not change the reading:



n	windows	$\bar{A}$	s.e.	t	$\bar{A}/\text{scale}$	$ \bar{A} / \text{Sym} $
5	569	$+3.81 \times 10^{-5}$	2.42×10^{-5}	+1.58	+0.057	0.164
10	284	$+1.07 \times 10^{-5}$	5.93×10^{-5}	+0.18	+0.009	0.022
21	135	$+1.52 \times 10^{-5}$	8.79×10^{-5}	+0.17	+0.008	0.022
63	45	$+5.28 \times 10^{-4}$	2.64×10^{-4}	+2.00	+0.120	0.385

Table 4.9: Mean Lévy area over non-overlapping windows of n observations. “scale” is $(\mathbb{E}[(X_{0,T}^1)^2] \mathbb{E}[(X_{0,T}^2)^2])^{1/2}$, which renders the third column dimensionless; the last column compares the area with the symmetric part of the same tensor.

at the monthly horizon the 95% interval for $\bar{A}$ is $[-1.21, 1.45] \times 10^{-4}$ and contains zero, while the same interval for the $\ell = +1$ control of §4.3.3 is $[-13.8, -5.2] \times 10^{-4}$ and does not.

4.3.3 Calibrating the instrument: a lag scan

A null result invites an obvious objection: perhaps the estimator cannot see anything at all. The objection is answerable, because the sign and rough size of the area are known in advance for a deliberately misaligned pair. Pair X_t^1 with $X_{t+\ell}^2$ for $\ell = -3, \dots, 3$ trading days. A positive ℓ pairs today’s S&P with a later STOXX, which imposes an artificial lead of the first index over the second; a negative ℓ imposes the reverse. Proposition 3.78 computes the expected area in the model (3.81), where the coupling enters through a coefficient θ rather than through a time shift; the two are not the same experiment, and we take the proposition as a guide to sign and rough size rather than as a prediction for this scan. What the scan should produce on any reading is a response that reverses with the sign of ℓ and vanishes where the series are truly aligned.

lag ℓ	$\bar{A}$	s.e.	t	$\bar{A}/\text{scale}$
-3	$+8.83 \times 10^{-4}$	2.07×10^{-4}	+4.26	+0.446
-2	$+7.79 \times 10^{-4}$	1.37×10^{-4}	+5.71	+0.403
-1	$+6.80 \times 10^{-4}$	1.06×10^{-4}	+6.41	+0.344
0	$+1.52 \times 10^{-5}$	8.79×10^{-5}	+0.17	+0.008
+1	-8.80×10^{-4}	2.04×10^{-4}	-4.31	-0.453
+2	-1.05×10^{-3}	1.94×10^{-4}	-5.42	-0.557
+3	-9.53×10^{-4}	2.23×10^{-4}	-4.27	-0.511

Table 4.10: Mean Lévy area under an imposed lag of ℓ trading days, monthly windows ($n = 21, 135$ windows). The response reverses in sign with ℓ and changes sign at $\ell = 0$; it is not exactly odd, the magnitudes at $\pm\ell$ differing by some 30% at $|\ell| = 1$.

The scan behaves as expected. Every non-zero lag is detected at $|t| \geq 4.26$; the area reverses with the imposed lag — negative under the convention above, in which a positive ℓ makes the first index lead, and positive under the reverse — so the statistic reports not only that the series are misaligned but *which* of them leads; and the zero crossing sits at $\ell = 0$ and nowhere else.

Three comments, of decreasing generality.

The level-two object is computable and its algebra is exact. Chen’s relation, geometricity and the invariance of the area under the choice of stochastic integral hold identically for any piecewise-linear path, so reproducing them to rounding error on eleven years of real closes tests the implementation rather than the market — which is precisely what one wants of them before

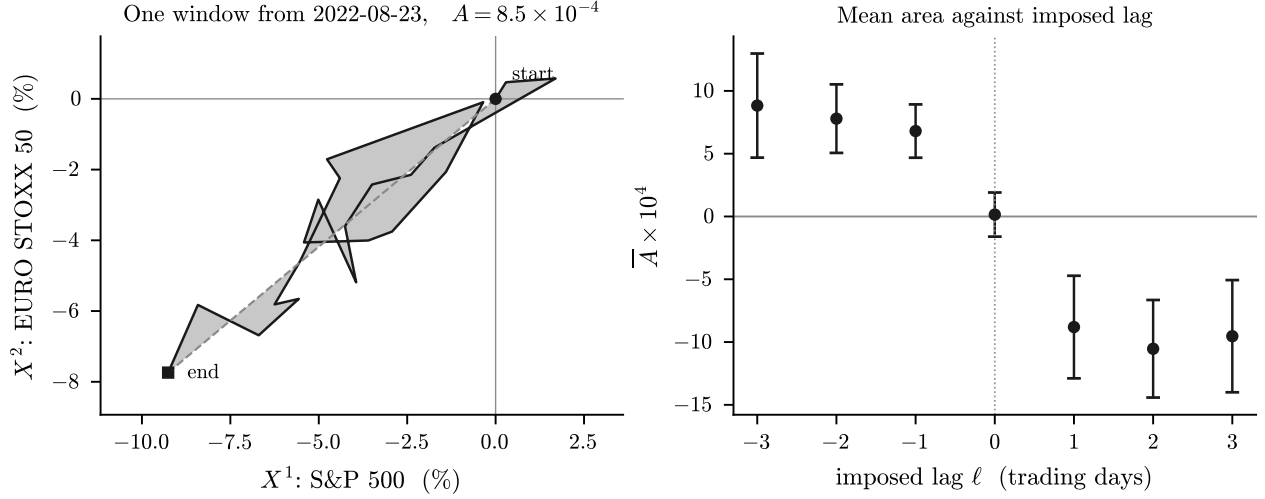


Figure 4.4: Left: one monthly window drawn as a path in the plane of the two cumulative log-returns, with the chord joining its endpoints dashed. The Lévy area $A_{0,T}$ is the shaded signed area between path and chord; the endpoints alone fix $\text{Sym } \mathbb{X}_{0,T}$ and say nothing about it, which is Proposition 3.73. Right: the mean area over 135 such windows as a function of an imposed lag ℓ , with ± 2 standard errors. The response reverses in sign with ℓ though it is not exactly odd in magnitude, is detected at more than four standard errors at every non-zero lag, and crosses zero at $\ell = 0$ alone.

any inference is drawn.

The area is a signed, calibrated detector of lead-lag, and it measures something a correlation cannot. Table 4.10 exhibits an instrument that is null at alignment, grows with misalignment and reports direction through its sign. Remark 3.79 explains why this is not a repackaged correlation: in the model (3.81) the contemporaneous coupling ρ and the lagged coupling θ are separately adjustable, and the area is exactly blind to the first. The symmetric part of the tensor, which on this pair is some fifty times larger, is where the correlation lives. The two statistics are measurements of orthogonal components of one object, and Proposition 3.73 is the statement that between them they exhaust it.

On this pair, at daily frequency, the area does not detect a lead-lag. That is a failure to reject rather than a demonstration of absence, and it should be recorded as such. What the calibration adds is that the instrument is not simply blind: the same statistic that returns $t = 0.17$ here returns $|t| > 4$ against a one-day displacement of the same data. A displacement of a whole day is, however, a very large signal, and the scan says nothing about power against a realistic one. The natural objection — that the S&P and the EURO STOXX close some four and a half hours apart, which is manifestly a lead-lag — is *not* disposed of, and the limitation should be stated plainly. A displacement shorter than the sampling interval does not leave the expected area unchanged. For two-sided Brownian drivers observed at unit spacing with an offset $\delta \in (0, 1)$, the linearly interpolated pair has $\mathbb{E}[A] = \delta/2$ rather than 0, because the overlap of the two increments contributes δ to one cross-covariance and nothing to the other. The closing offset is therefore a genuine confound of the daily measurement rather than something the statistic may ignore, and resolving it requires intraday sampling, at which point the offset moves inside the observable range.



Notes and references

The physics-informed neural network methodology is due to Raissi, Perdikaris and Karniadakis [66]; the survey [49] places it in the wider setting of physics-informed machine learning and is the natural entry point for a reader meeting these methods for the first time. The exactness of the derivatives on which §4.1.3 and §4.1.4 rely is a property of automatic differentiation rather than of neural networks, and the standard reference is the survey of Baydin et al. [4]. The optimiser is AdamW [54] with the warm-restart schedule of [55].

On the financial side nothing in this section is new: the two conditions imposed are those of Proposition 2.17, which is Gatheral’s [35], the wing bound is Lee’s [51], and the local volatility recovered from the fitted surface is Dupire’s [26] in the implied-variance coordinates of [36]. What is new here is the route: the conditions are imposed as penalties on a globally smooth representation rather than inherited from a parametric slice family, and on the extrapolated wings they are established as Lemma 4.8 and Theorem 4.9 rather than sampled.

Section 4.2 rests on the characteristic function of El Euch and Rosenbaum [29], quoted as Theorem 3.51; the wider rough volatility programme in which it sits is surveyed in [6] and [62]. The Fourier route from a characteristic function to a price is [15], evaluated here on the contour $\text{Im } u = -\frac{1}{2}$. The claim that the at-the-money skew behaves as $T^{H-1/2}$, on which §4.2.3 turns, is from [37]. Proposition 4.10 is elementary and is recorded only because it removes the singular quadrature that a direct reading of (3.51) appears to require; the stability cut (4.35) is likewise a practical observation rather than a result. The negative conclusion of §4.2.3 — that a surface beginning at 38 days does not identify H , because the estimator that would do so has no power outside its asymptotic regime — is, as far as the author is aware, not stated in this form in the literature, though the weakness of H under option-based calibration is known folklore.

Section 4.3 draws its theory entirely from Chapter 3: the three identities verified there are Propositions 3.73, 3.76 and Remark 3.75, the model measured is Definition 3.77, and the estimator is Proposition 3.78. The classical route to the same question is a cross-correlogram scanned over a grid of lags, and what (3.82) offers against it is recorded in Remark 3.79: one number in place of a scan, at the cost of committing to the shape of the model. For the use of signatures as market statistics more generally — of which the level-two reading performed here is the smallest instance — see [56], [45] and [24]; the references for the transform itself are collected in the notes to Chapter 3.

Conclusions

The aim of this dissertation was to build a bridge between mathematics and financial markets, from the perspective of an MSc degree in Mathematics. We have presented the key results of the mathematical theory, as well as the necessary financial background, and we have shown how the two interact in practice. We now summarise the main points of the work, and we conclude with some reflections on possible directions for further research.

We began with measure theory and Brownian motion, and built the Itô integral. We also obtained some results on the roughness of Brownian paths. Then we used that machinery to set up the standard pricing framework: arbitrage, the fundamental theorems, Black–Scholes, and the local and stochastic volatility models.

In Part II we crossed the line where the driving signal is rougher than Brownian motion and the semimartingale construction of Part I no longer applies. There we developed rough path theory. This approach is more general, and it links directly with some recent developments in the theory of stochastic processes, such as regularity structures and paracontrolled distributions. The full reach of the theory is beyond our scope, so we restricted ourselves to two levels and to the key results involving the signature. We also looked at Brownian motion as a rough path.

Chapter 4 is where the models met the market, and the three studies there gave three different kinds of answer. The physics-informed surface worked: on an SPX book it placed 95.5% of the quotes inside their own bid/ask, kept the typical error to the mid below five basis points, and showed no violation of the static no-arbitrage conditions on the validation grid, with the extrapolated wings covered analytically. Rough Heston did not fit nearly as well, and it pushed the Hurst parameter to the smooth end of its range. That looks like evidence that this market is not rough, but it is not. As §4.2.3 shows, the estimator that would settle the question has no power over the maturities available: two calibrated models whose H differs by a factor of almost five produce practically the same skew slope. The honest conclusion is that this option book cannot answer that question, which is a more useful thing to know than a fitted number would have been. The third study was the Lévy area for a pair of assets, where the computations of rough path theory were carried out on real data.

The thesis is heterogeneous, in content and in intention, and this is deliberate. It moves from measure theory to neural networks and from theorems to basis points, and we have not sought to hide it: it reflects the author’s professional experience. Several choices in Chapter 4 come from the day-to-day work of a trading floor rather than from a textbook: measuring error in half-spreads instead of volatility points, treating absence of arbitrage as a hard constraint instead of a diagnostic, and asking of a quantity whether a desk could actually realise it. Those are the instincts of a front-office quantitative analyst, and they shaped the applied half of this work.

Three directions remain open, and each is active work today. The signature of Section 3.4 was developed here in full and then used only at its second level; the natural continuation is to let

it drive the model itself, as in the joint calibration to SPX and VIX options of Cuchiero et al. [24] and the path-dependent processes of Abi Jaber et al. [45]. Second, roughness is not the only reading of the evidence in Subsection 3.1.2: Guyon and Lekeufack [40] argue that volatility is mostly *path-dependent* rather than rough, and this dissertation has not tested one account against the other. And on the theoretical side, reaching the exponents the data suggest needs Hairer's regularity structures [41] rather than rough paths, in the form built for rough volatility by Bayer et al. [5]. Any of the three would make a thesis of its own.



Appendix A

Auxiliary Results

This appendix collects results which are used in the main text but whose statements would have interrupted the argument there. Nothing here is original, and proofs are given only where they run to a line or two; references are given in every case. The organising principle is simple: a result belongs in the body if the argument cannot be followed without it, and belongs here if the argument merely *invokes* it. Results which we do prove, but whose proofs are too long to sit comfortably in the body, are deferred instead to Appendix B.

A.1 Complements on conditional expectation

Throughout, $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and all identities are understood to hold almost surely. Proofs may be found in [8] or [27]. Two further properties, the tower property and the L^2 projection characterisation, are used constantly in the main text and are stated there as Proposition 1.4, to which we refer for them rather than repeating them here.

Proposition A.1 (Properties of conditional expectation). *Let $X, Y \in L^1(\Omega)$. Then:*

1. $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$; that is, $\mathbb{E}[X \mid \mathcal{G}]$ and X have the same expectation.
2. If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$.
3. If X and $\mathcal{G}$ are independent — that is, $\{X \in U\}$ and A are independent events for every Borel $U \subseteq \mathbb{R}$ and every $A \in \mathcal{G}$ — then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.
4. (**Taking out what is known**) If Y is $\mathcal{G}$ -measurable and $\mathbb{E}[|XY|] < \infty$, then $\mathbb{E}[XY \mid \mathcal{G}] = Y \mathbb{E}[X \mid \mathcal{G}]$.
5. (**Conditional Jensen's inequality**) If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and both X and $\phi(X)$ are integrable, then $\phi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\phi(X) \mid \mathcal{G}]$.
6. If $X \leq Y$, then $\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$; consequently $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$.
7. For all $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY \mid \mathcal{G}] = a \mathbb{E}[X \mid \mathcal{G}] + b \mathbb{E}[Y \mid \mathcal{G}]$. In particular $\mathbb{E}[\cdot \mid \mathcal{G}]$ is a bounded linear operator from $L^1(\Omega, \mathcal{F})$ into $L^1(\Omega, \mathcal{G})$, a fact used to pass to limits in Theorem 1.52.

□

The convergence theorems also carry over to the conditional setting, in exactly the form one would expect.

Proposition A.2 (Conditional Fatou lemma). *Let $X_n \geq 0$ with $X_n \in L^1(\Omega)$ for every $n \geq 1$, and suppose $\liminf_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then*

$$\mathbb{E} \left[\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G} \right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}] \quad \text{almost surely.}$$

□

Proposition A.3 (Conditional monotone convergence). *Let $0 \leq X_1 \leq X_2 \leq \dots$ and suppose $X = \lim_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then $\mathbb{E}[X \mid \mathcal{G}] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}]$ almost surely.* □

Proposition A.4 (Conditional dominated convergence). *Suppose $X_n \rightarrow X$ almost surely and $|X_n| \leq Y$ for every n , with $Y \in L^1(\Omega)$. Then $\mathbb{E}[X_n \mid \mathcal{G}] \rightarrow \mathbb{E}[X \mid \mathcal{G}]$ almost surely and in $L^1(\Omega)$.* □

A.2 Characteristic functions

Recall that the characteristic function of a real random variable X is $\varphi_X(u) = \mathbb{E}[e^{iuX}]$, $u \in \mathbb{R}$. The following elementary properties are used without comment in the main text.

Proposition A.5. *For any real random variable X :*

- i) $\varphi_X(0) = 1$ and $|\varphi_X(u)| \leq 1$ for all $u \in \mathbb{R}$;
- ii) φ_X is uniformly continuous on $\mathbb{R}$;
- iii) $\varphi_X(-u) = \overline{\varphi_X(u)}$ (Hermitian symmetry);
- iv) $\varphi_{aX+b}(u) = e^{iub} \varphi_X(au)$ for all $a, b \in \mathbb{R}$;
- v) if X and Y are independent, then $\varphi_{X+Y}(u) = \varphi_X(u) \varphi_Y(u)$;
- vi) if $\mathbb{E}[|X|^n] < \infty$, then φ_X is n times differentiable and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k], \quad 1 \leq k \leq n.$$

Conversely, the existence of $2n$ continuous derivatives of φ_X in a neighbourhood of the origin implies $\mathbb{E}[|X|^{2n}] < \infty$.

□

Property ii) deserves a word, since it is the reason the continuity hypothesis in the converse half of Lévy's continuity theorem (Theorem 1.14) is a genuine restriction rather than an automatic one: it constrains the *limit* function, which is not assumed in advance to be a characteristic function.



A.3 Moment indeterminacy of the log-normal law

The following construction, due to Heyde [44], is referred to in Subsection 1.1.2 and shows that a distribution possessing moments of all orders need not be determined by them.

Proposition A.6 (Heyde). *Let f_X denote the density of the standard log-normal law, that is, the density of e^Z with $Z \sim \mathcal{N}(0, 1)$. For $a \in [-1, 1]$ define*

$$f_a(x) = f_X(x) \left(1 + a \sin(2\pi \log x)\right), \quad x > 0. \quad (\text{A.1})$$

Then every f_a is a probability density on $(0, \infty)$, the laws are mutually distinct for distinct a , and all of them share the moment sequence of the log-normal distribution.

Proof: Non-negativity holds because $|a \sin(\cdot)| \leq 1$, and the moment identity reduces to showing that

$$\int_0^\infty x^k f_X(x) \sin(2\pi \log x) dx = 0, \quad k \in \mathbb{N}.$$

Substituting $\log x = t + k$ turns the integrand into an odd function of t multiplied by the Gaussian weight, and the integral vanishes. Taking $k = 0$ shows in particular that each f_a integrates to 1. ■

It is worth being precise about the logical status of this. Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$ is *sufficient* for a law to be determined by its moments, and the log-normal moments, growing like $e^{k^2/2}$, violate it. A violated sufficient condition establishes only that the criterion is silent, never that the conclusion is false: what settles the matter here is Proposition A.6 itself, which exhibits the competing densities explicitly. The growth rate merely explains why no elementary criterion could have helped. See [27] for the general theory.

A.4 Grönwall's inequality

The following is the standard comparison tool for integral inequalities. It is purely deterministic, and in the stochastic setting it is applied pathwise or to expectations.

Lemma A.7 (Grönwall's inequality). *Let $g, u \in L^1[0, T]$ and $\lambda > 0$ satisfy*

$$u(t) \leq g(t) + \lambda \int_0^t u(s) ds, \quad 0 \leq t \leq T.$$

Then

$$u(t) \leq g(t) + \lambda \int_0^t g(s) e^{\lambda(t-s)} ds, \quad 0 \leq t \leq T.$$

In particular, if $g \equiv \kappa$ is constant, then $u(t) \leq \kappa e^{\lambda t}$ for all $0 \leq t \leq T$; and if $\kappa = 0$, then $u \equiv 0$. □

The final sentence is the one actually used. Uniqueness arguments for differential equations almost invariably reduce to showing that a suitable distance between two candidate solutions satisfies a Grönwall inequality with $\kappa = 0$, whence it vanishes identically.

A.5 Radon–Nikodym and the Jordan decomposition

For completeness we record the decomposition underlying Corollary 1.3.

Theorem A.8 (Jordan decomposition). *Every finite signed measure ν on a measurable space $(\Omega, \mathcal{F})$ admits a unique decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. The measure $|\nu| := \nu^+ + \nu^-$ is called the total variation of ν , and $\nu \ll \lambda$ if and only if $|\nu| \ll \lambda$.* $\square$

See [68] for a proof.

A.6 Uniform integrability and the class D

Two integrability notions are invoked in the main text — in the hypotheses of Theorem 1.28 and of Theorem 2.20 respectively — and are recorded here.

Definition A.9 (Uniform integrability). A family $\{X_i\}_{i \in I}$ of integrable random variables is **uniformly integrable** if

$$\lim_{c \rightarrow \infty} \sup_{i \in I} \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > c\}}] = 0. \quad (\text{A.2})$$

The condition says that no member of the family carries a non-negligible amount of its mass arbitrarily far out in the tail, *uniformly in i* . It is strictly stronger than $\sup_i \mathbb{E}|X_i| < \infty$ and strictly weaker than domination by a single integrable variable: if $|X_i| \leq Y$ for all i with $Y \in L^1$, then (A.2) follows from dominated convergence, which is why hypothesis iii) of Theorem 1.28 implies hypothesis ii). Its role is that it is exactly the extra assumption converting convergence in probability into convergence in L^1 , and exactly what is needed for a martingale to be closed by a terminal variable. See [27, Ch. 4].

Definition A.10 (Class D). A process X is of **class D** if the family $\{X_\tau : \tau \text{ a stopping time}\}$, indexed by all stopping times of the filtration, is uniformly integrable.

The class is named for Doob. It is the hypothesis under which the Doob–Meyer decomposition (Theorem 1.32) produces an *integrable* increasing process rather than merely a local one, and under which the Snell envelope of Theorem 2.20 is well defined. Any process dominated by an integrable variable is of class D; in particular, so is any bounded process.

A.7 The essential supremum of a family of random variables

Definition A.11 (Essential supremum). Let $\{X_i\}_{i \in I}$ be a family of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by an arbitrary and possibly uncountable set I . An **essential supremum** of the family, written $\text{ess sup}_{i \in I} X_i$, is a random variable X^* such that

- i) $X_i \leq X^*$ almost surely, for every $i \in I$; and
- ii) if Y satisfies i) in place of X^* , then $X^* \leq Y$ almost surely.

Such an X^* exists and is unique up to null sets.



The construction, for which see [48, Appendix A], produces X^* as the pointwise supremum of a *countable* subfamily, and that is the whole point of the notion. The pointwise supremum $\sup_{i \in I} X_i(\omega)$ over an uncountable I need not be measurable, and even when it is it may exceed every X_i on a set of positive probability in a way that carries no information — an uncountable union of null sets need not be null, which is the same phenomenon that separates modification from indistinguishability in Remark 1.38. The essential supremum repairs this by taking the supremum in the almost-sure order rather than pointwise, and it is the correct object in (2.57), where the family is indexed by the uncountably many stopping times available to the holder of an American option.

Appendix B

Deferred Proofs

This appendix collects those proofs which are given in full but whose length would have interrupted the argument in the main text. The organising principle is the complement of the one governing Appendix A: a result whose proof is short enough to be read in passing is proved where it is stated, a result we merely invoke is recorded without proof in Appendix A, and a result we genuinely use but whose proof runs to more than a few lines is stated in the body and proved here.

Each section below carries a pointer back to the statement it establishes. The arguments of Sections B.1–B.6 follow the treatment of [50]; those of Sections B.7–B.10, which concern Chapter 3, follow [33] and [38].

B.1 Doob's submartingale inequality

We prove Proposition 1.25 of Chapter 1.

Proof: Let Q be an enumeration of the rationals in $[0, T]$. Right-continuity gives $\sup_{0 \leq t \leq T} Z_t = \sup_{r \in Q} Z_r$ almost surely, so it suffices to bound the maximum over a finite set $\{r_1 < \dots < r_k\}$ and let $k \rightarrow \infty$. On such a set, decompose $\{\max_{\nu} Z_{r_{\nu}} \geq \varepsilon\}$ into the disjoint events A_{ν} on which r_{ν} is the *first* index at which the level is reached. Each $A_{\nu} \in \mathcal{F}_{r_{\nu}}$, and on A_{ν} we have $Z_{r_{\nu}} \geq \varepsilon$, so the submartingale property gives

$$\varepsilon \mathbb{P}(A_{\nu}) \leq \mathbb{E}[Z_{r_{\nu}} \mathbf{1}_{A_{\nu}}] \leq \mathbb{E}[\mathbb{E}[Z_T | \mathcal{F}_{r_{\nu}}] \mathbf{1}_{A_{\nu}}] = \mathbb{E}[Z_T \mathbf{1}_{A_{\nu}}] \leq \mathbb{E}[\max\{Z_T, 0\} \mathbf{1}_{A_{\nu}}].$$

Summing over ν and using disjointness bounds $\varepsilon \mathbb{P}(\max_{\nu} Z_{r_{\nu}} \geq \varepsilon)$ by $\mathbb{E}[\max\{Z_T, 0\}]$, uniformly in k . Letting $k \rightarrow \infty$ gives the claim. The martingale case follows by applying this to the submartingale $|X_t|$, which is one by the conditional Jensen inequality. ■

B.2 Density of step processes in $L^2_{\text{ad}}([0, T] \times \Omega)$

We prove Lemma 1.49 of Chapter 1.

Proof: The argument proceeds by successive approximation in three steps, each enlarging the class of integrands. Write $\|f\|^2 := \int_0^T \mathbb{E}[f(t)^2] dt$ for the norm of $L^2_{\text{ad}}([0, T] \times \Omega)$, and recall that it suffices to approximate an arbitrary f within ε by a step process, since the step processes then form a dense subspace.

Step 1: f bounded with continuous sample paths. Let $|f| \leq M$ and let $\mathcal{P}_n = \{0 = t_0^n < \dots < t_{k_n}^n = T\}$ be partitions with $\|\mathcal{P}_n\| \rightarrow 0$. Put

$$f_n(t) := \sum_{i=1}^{k_n} f(t_{i-1}^n) \mathbf{1}_{[t_{i-1}^n, t_i^n)}(t).$$

Each f_n is a step process, and it is *adapted* because the coefficient $f(t_{i-1}^n)$ is $\mathcal{F}_{t_{i-1}^n}$ -measurable and multiplies the indicator of an interval beginning at t_{i-1}^n — this is the left-endpoint choice, and it is the only reason adaptedness survives. For almost every ω the path $t \mapsto f(t, \omega)$ is continuous on the compact $[0, T]$, hence uniformly continuous, so $f_n(t, \omega) \rightarrow f(t, \omega)$ for every t . Since $|f_n - f|^2 \leq 4M^2$, which is integrable on $[0, T] \times \Omega$ (a finite measure space), dominated convergence gives $\|f_n - f\| \rightarrow 0$.

Step 2: f bounded, sample paths not assumed continuous. Let $|f| \leq M$. Extend f by 0 outside $[0, T]$ and mollify in time: for $n \in \mathbb{N}$ set

$$g_n(t) := n \int_{t-1/n}^t f(s) ds = \int_0^1 f\left(t - \frac{u}{n}\right) du.$$

Each g_n has continuous sample paths, since it is an indefinite integral, and $|g_n| \leq M$. Adaptedness is preserved because $g_n(t)$ averages f over $[t - 1/n, t]$ only, using no information after t ; this is where averaging *backwards* rather than symmetrically matters. By the Lebesgue differentiation theorem, $g_n(t, \omega) \rightarrow f(t, \omega)$ for almost every t , for almost every ω , and dominated convergence again gives $\|g_n - f\| \rightarrow 0$. Step 1 now approximates each g_n by step processes, and a diagonal argument produces step processes converging to f .

Step 3: general $f \in L_{\text{ad}}^2([0, T] \times \Omega)$. Truncate at level n by setting $f^{(n)} := (f \wedge n) \vee (-n)$. Each $f^{(n)}$ is bounded and remains adapted, being a composition of f with a continuous deterministic function. Moreover $|f^{(n)} - f|^2 \leq f^2$ pointwise and $f^{(n)} \rightarrow f$ pointwise, so dominated convergence — with dominating function f^2 , integrable precisely because $f \in L_{\text{ad}}^2$ — yields $\|f^{(n)} - f\| \rightarrow 0$. Combining with Step 2 completes the proof.

Adaptedness is the only delicate point throughout, and it is worth noting that each of the three steps preserves it for a different reason: the left-endpoint evaluation in Step 1, the backward averaging window in Step 2, and the deterministic truncation in Step 3. ■

B.3 Continuity of the sample paths of an Itô integral

We prove Theorem 1.54 of Chapter 1.

Proof: Let $\{f_n\}$ be step processes converging to f in $L_{\text{ad}}^2([0, T] \times \Omega)$; the associated processes $X_t^n = \int_0^t f_n dW$ have continuous paths by construction, being finite sums of continuous terms. By Theorem 1.52 the difference $X^n - X^m$ is a martingale, so Doob's L^2 maximal inequality (Proposition 1.26) together with Itô's isometry gives

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^n - X_t^m|^2 \right] \leq 4 \mathbb{E} [|X_T^n - X_T^m|^2] = 4 \int_0^T \mathbb{E} [(f_n - f_m)^2] dt \longrightarrow 0.$$

Passing to a subsequence for which the right-hand side is summable, the Borel–Cantelli lemma yields $\sup_{t \leq T} |X_t^{n_k} - X_t| \rightarrow 0$ almost surely. A uniform limit of continuous functions is continuous, so almost all paths of X are continuous on $[0, T]$. ■



B.4 The Itô integral as a limit of Riemann sums

We prove Theorem 1.55 of Chapter 1.

Proof: Let $f_{\mathcal{P}}(t) = \sum_i f(t_{i-1}) \mathbf{1}_{[t_{i-1}, t_i)}(t)$ be the left-endpoint step approximation, so that the Riemann sum is exactly $I(f_{\mathcal{P}})$. By Itô's isometry it suffices to show that $\int_0^T \mathbb{E}[(f - f_{\mathcal{P}})^2] dt \rightarrow 0$. Expanding the square, this integral is a sum of terms of the form $\mathbb{E}[f(t)^2] - 2\mathbb{E}[f(t)f(t_{i-1})] + \mathbb{E}[f(t_{i-1})^2]$, each built from the function $(s, t) \mapsto \mathbb{E}[f(t)f(s)]$ evaluated at points within $\|\mathcal{P}\|$ of the diagonal. That function is continuous by hypothesis, hence uniformly continuous on the compact $[0, T]^2$, so the whole expression tends to 0 uniformly as $\|\mathcal{P}\| \rightarrow 0$. ■

B.5 Lévy's characterisation theorem

We prove Theorem 1.70 of Chapter 1.

Proof: The one-dimensional case is proved by applying Itô's rule to $t \mapsto e^{iuX_t}$ for fixed $u \in \mathbb{R}$. Since X is a continuous local martingale with $\langle X \rangle_t = t$, the second-order term contributes $-\frac{1}{2}u^2 e^{iuX_t} dt$, and after localisation and taking conditional expectations one obtains, for $s < t$,

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = 1 - \frac{1}{2}u^2 \int_s^t \mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s] dr.$$

This is *not* yet a deterministic equation: for each r the quantity $\mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s]$ is an $\mathcal{F}_s$ -measurable random variable, so the unknown is a random function and no uniqueness theorem for integral equations applies to it as it stands. The standard device removes the randomness by testing. Fix $A \in \mathcal{F}_s$, multiply through by $\mathbf{1}_A$, take expectations, and interchange expectation and the dr integral, which is legitimate by Fubini since the integrand has modulus one. Writing

$$h_A(t) := \mathbb{E}[\mathbf{1}_A e^{iu(X_t - X_s)}],$$

we obtain the genuinely deterministic linear integral equation

$$h_A(t) = \mathbb{P}(A) - \frac{1}{2}u^2 \int_s^t h_A(r) dr, \quad t \geq s,$$

whose unique solution is $h_A(t) = \mathbb{P}(A) e^{-u^2(t-s)/2}$. Since this holds for every $A \in \mathcal{F}_s$ and the right-hand side is the integral over A of the constant $e^{-u^2(t-s)/2}$, we conclude

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = e^{-u^2(t-s)/2} \quad \text{almost surely.}$$

The conditional characteristic function of $X_t - X_s$ given $\mathcal{F}_s$ is therefore deterministic and equal to that of $\mathcal{N}(0, t - s)$, which is simultaneously independence of the increment from $\mathcal{F}_s$ and the correct Gaussian law; Proposition 1.40 then supplies independence of increments. The multidimensional case is identical, applied to $e^{iu^\top X_t}$ with the cross-variations $\langle X^i, X^j \rangle_t = \delta_{ij}t$ making the exponent $-\frac{1}{2}\|u\|^2(t - s)$. ■

B.6 Girsanov's theorem

We prove Theorem 1.81 of Chapter 1.

Proof: By Theorem 1.70 it suffices to show that, under $\mathbb{Q}$, the process Z is a continuous local martingale null at the origin with $\langle Z \rangle_t = t$. Continuity and $Z_0 = 0$ are clear.

For the martingale property we use Lemma 1.80, which reduces the claim to showing that $Z(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale. Itô's product formula gives

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t) dZ_t + Z_t d\mathcal{E}_\alpha(t) + dZ_t d\mathcal{E}_\alpha(t).$$

Now $dZ_t = dW_t - \alpha(t)dt$ and $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW_t$, so by the Itô table $dZ_t d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dt$. Substituting, the two dt terms cancel exactly:

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t)(dW_t - \alpha(t)dt) + Z_t \alpha(t)\mathcal{E}_\alpha(t)dW_t + \alpha(t)\mathcal{E}_\alpha(t)dt = (1 + Z_t \alpha(t))\mathcal{E}_\alpha(t)dW_t.$$

The right-hand side has no drift, so $Z\mathcal{E}_\alpha$ is a $\mathbb{P}$ -local martingale, and under the stated integrability it is a martingale; hence Z is a $\mathbb{Q}$ -martingale.

Finally, $\langle Z \rangle_t = \langle W \rangle_t = t$, because subtracting the finite-variation term $\int_0^t \alpha(s)ds$ does not change the quadratic variation, and because quadratic variation is a pathwise limit and is therefore unaffected by the passage to the equivalent measure $\mathbb{Q}$. The same argument applied to $Z_t^2 - t$ confirms the compensator. Lévy's theorem now identifies Z as a $\mathbb{Q}$ -Brownian motion. ■

B.7 The sewing lemma

We prove the existence half of Lemma 3.21 of Chapter 3; uniqueness was established where the lemma was stated.

Proof: Fix $s < t$ and let $\mathcal{P}^k$ denote the dyadic partition of $[s, t]$ into 2^k intervals of equal length $2^{-k}(t - s)$. Write

$$I_{s,t}^k := \sum_{[u,v] \in \mathcal{P}^k} \Xi_{u,v}$$

for the associated approximation, so that $I_{s,t}^0 = \Xi_{s,t}$. The whole proof consists in showing that $(I_{s,t}^k)_k$ is Cauchy, and the mechanism is that passing from $\mathcal{P}^k$ to $\mathcal{P}^{k+1}$ splits each interval exactly once, so that the difference of consecutive approximations is a sum of defects of the form (3.23).

Step 1: consecutive approximations. Each interval $[u, v]$ of $\mathcal{P}^k$ is divided by $\mathcal{P}^{k+1}$ at its midpoint $w = (u + v)/2$, and the corresponding contribution to $I_{s,t}^{k+1}$ is $\Xi_{u,w} + \Xi_{w,v}$ in place of $\Xi_{u,v}$. Hence

$$I_{s,t}^{k+1} - I_{s,t}^k = - \sum_{[u,v] \in \mathcal{P}^k} \delta \Xi_{u,w,v},$$

and by the hypothesis (3.23), since each of the 2^k intervals has length $2^{-k}(t - s)$,

$$|I_{s,t}^{k+1} - I_{s,t}^k| \leq 2^k \cdot C \left(2^{-k}(t - s) \right)^\beta = C |t - s|^\beta 2^{-k(\beta-1)}. \quad (\text{B.1})$$

Step 2: convergence. Because $\beta > 1$ the exponent $-(\beta - 1)$ is negative, so the bounds (B.1) form a convergent geometric series. Summing them,

$$|I_{s,t}^m - I_{s,t}^k| \leq C |t - s|^\beta \sum_{j \geq k} 2^{-j(\beta-1)} = C |t - s|^\beta \frac{2^{-k(\beta-1)}}{1 - 2^{-(\beta-1)}}, \quad m > k,$$



so $(I_{s,t}^k)_k$ is Cauchy and converges; call the limit $(\mathcal{I}\Xi)_{s,t}$. Taking $k = 0$ in the display and recalling $I_{s,t}^0 = \Xi_{s,t}$ gives

$$\left| (\mathcal{I}\Xi)_{s,t} - \Xi_{s,t} \right| \leq \frac{C |t-s|^\beta}{1-2^{-(\beta-1)}} = \frac{C}{1-2^{1-\beta}} |t-s|^\beta,$$

which is exactly (3.24). Note that this is the step in which $\beta > 1$ is most visibly consumed: at $\beta = 1$ the series in Step 2 diverges, and correspondingly the constant $(1-2^{1-\beta})^{-1}$ blows up.

Step 3: a uniform bound over arbitrary partitions. The dyadic construction produces a limit along dyadic partitions; to know that every partition sequence produces the *same* limit — which is what additivity requires — we need an estimate valid for all partitions at once.

Let $\mathcal{P} = \{s = t_0 < t_1 < \dots < t_n = t\}$ be arbitrary, with $n \geq 2$. Since

$$\sum_{j=1}^{n-1} (t_{j+1} - t_{j-1}) = (t_n + t_{n-1}) - (t_0 + t_1) \leq 2(t-s),$$

some interior point t_j satisfies $t_{j+1} - t_{j-1} \leq 2(t-s)/(n-1)$. Deleting it changes the sum by exactly one defect,

$$\left| \sum_{\mathcal{P}} \Xi - \sum_{\mathcal{P} \setminus \{t_j\}} \Xi \right| = \left| \delta \Xi_{t_{j-1}, t_j, t_{j+1}} \right| \leq C \left(\frac{2(t-s)}{n-1} \right)^\beta.$$

Repeating until only $\{s, t\}$ remains and summing the resulting bounds,

$$\left| \sum_{\mathcal{P}} \Xi - \Xi_{s,t} \right| \leq C (2|t-s|)^\beta \sum_{m \geq 1} m^{-\beta} = 2^\beta \zeta(\beta) C |t-s|^\beta, \quad (\text{B.2})$$

the series converging precisely because $\beta > 1$. The bound (B.2) holds for every partition of every subinterval, with no reference to the dyadic scheme.

Step 4: partition-independence, additivity, and the Riemann-sum limit. Let $\mathcal{P}$ be any partition of $[s, t]$ and $\mathcal{Q}$ any refinement of it. Applying (B.2) on each interval $[u, v] \in \mathcal{P}$ to the partition induced there by $\mathcal{Q}$,

$$\left| \sum_{\mathcal{Q}} \Xi - \sum_{\mathcal{P}} \Xi \right| \leq 2^\beta \zeta(\beta) C \sum_{[u,v] \in \mathcal{P}} |v-u|^\beta \leq 2^\beta \zeta(\beta) C |\mathcal{P}|^{\beta-1} |t-s|.$$

Given two partitions, comparing each with their common refinement shows that the sums $\sum_{\mathcal{P}} \Xi$ form a Cauchy family as $|\mathcal{P}| \rightarrow 0$. The limit therefore exists, does not depend on the sequence of partitions chosen, and in particular coincides with the dyadic limit $(\mathcal{I}\Xi)_{s,t}$ constructed in Step 2 — so the estimate (3.24), with its better constant, is retained. This also proves the final assertion of the lemma, that $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}| \rightarrow 0} \sum_{\mathcal{P}} \Xi$.

Additivity is now immediate: for $s \leq u \leq t$, concatenating partitions of $[s, u]$ and of $[u, t]$ yields partitions of $[s, t]$ whose mesh tends to zero, and whose sums converge to $(\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ by construction and to $(\mathcal{I}\Xi)_{s,t}$ by the preceding paragraph. ■

B.8 Gubinelli's integration theorem

We prove Theorem 3.25 of Chapter 3.

Proof: The entire proof consists in applying Lemma 3.21 to the two-parameter function

$$\Xi_{s,t} := Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}, \quad (\text{B.3})$$

which is exactly the summand appearing in the compensated Riemann sums (3.26). We must verify the hypothesis (3.23) with $\beta = 3\alpha > 1$.

Step 1: computing the defect. Fix $s \leq u \leq t$. By definition,

$$\delta\Xi_{s,u,t} = \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} - Y_s X_{s,u} - Y'_s \mathbb{X}_{s,u} - Y_u X_{u,t} - Y'_u \mathbb{X}_{u,t}.$$

Now use the two structural facts at our disposal. Additivity of the first level, $X_{s,t} = X_{s,u} + X_{u,t}$, collapses the terms in Y_s to $Y_s X_{u,t}$; and Chen's relation (3.16), in the form $\mathbb{X}_{s,t} - \mathbb{X}_{s,u} = \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$, collapses the terms in Y'_s to $Y'_s \mathbb{X}_{u,t} + Y'_s X_{s,u} X_{u,t}$. Therefore

$$\delta\Xi_{s,u,t} = (Y_s - Y_u) X_{u,t} + Y'_s X_{s,u} X_{u,t} - (Y'_u - Y'_s) \mathbb{X}_{u,t} = -(Y_{s,u} - Y'_s X_{s,u}) X_{u,t} - (Y'_u - Y'_s) \mathbb{X}_{u,t},$$

and recognising the first bracket as the remainder $R_{s,u}^Y$ of Definition 3.22 gives precisely (3.29):

$$-\delta\Xi_{s,u,t} = R_{s,u}^Y X_{u,t} + (Y'_u - Y'_s) \mathbb{X}_{u,t}.$$

It is worth pausing on this line, since it is the whole reason the theory is built as it is. The second-order term $Y'_s \mathbb{X}_{s,t}$ was inserted into (B.3) for no other purpose than to produce, via Chen's relation, the cross-term $Y'_s X_{s,u} X_{u,t}$ which cancels the leading part of $(Y_s - Y_u) X_{u,t}$. Without it the defect would be of order $\alpha + \alpha = 2\alpha \leq 1$ and the sewing lemma would not apply — which is exactly the failure of the naïve Riemann sums.

Step 2: the estimate. Bounding the two terms separately, using Definitions 3.12 and 3.22 and $u - s, t - u \leq t - s$,

$$|\delta\Xi_{s,u,t}| \leq \|R^Y\|_{2\alpha} |u - s|^{2\alpha} \|X\|_{\alpha} |t - u|^{\alpha} + \|Y'\|_{\alpha} |u - s|^{\alpha} \|\mathbb{X}\|_{2\alpha} |t - u|^{2\alpha} \leq C_{\Xi} |t - s|^{3\alpha},$$

with $C_{\Xi} := \|X\|_{\alpha} \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'\|_{\alpha}$. Since $\alpha > \frac{1}{3}$ we have $3\alpha > 1$, so Lemma 3.21 applies and yields a unique additive $\mathcal{I}\Xi$ with $|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \leq C C_{\Xi} |t - s|^{3\alpha}$. By the last assertion of that lemma, $(\mathcal{I}\Xi)_{s,t}$ is the limit of $\sum_{\mathcal{P}} \Xi_{u,v}$, which is (3.26); and the displayed bound is (3.27).

Step 3: the integral is controlled. Write $Z_t := (\mathcal{I}\Xi)_{0,t}$, which is well defined by additivity. From the estimate just obtained,

$$Z_{s,t} - Y_s X_{s,t} = Y'_s \mathbb{X}_{s,t} + O(|t - s|^{3\alpha}),$$

so $R_{s,t}^Z := Z_{s,t} - Y_s X_{s,t}$ obeys $|R_{s,t}^Z| \leq \|Y'\|_{\infty} \|\mathbb{X}\|_{2\alpha} |t - s|^{2\alpha} + C C_{\Xi} |t - s|^{3\alpha}$, which is $O(|t - s|^{2\alpha})$ since $3\alpha > 2\alpha$. Hence $\|R^Z\|_{2\alpha} < \infty$ and $(Z, Y) \in \mathcal{D}_X^{2\alpha}$, which is (3.28). That Z is itself α -Hölder follows from $|Z_{s,t}| \leq |Y_s| |X_{s,t}| + |R_{s,t}^Z|$. $\blacksquare$



B.9 Existence & uniqueness (rough differential equations)

We prove Theorem 3.36 of Chapter 3.

Proof: Throughout, C denotes a constant depending only on α and $\|f\|_{\mathcal{C}_b^3}$, allowed to change from line to line, and we abbreviate $M := \|\mathbf{X}\|_\alpha$. Recall from Definition 3.33 that solutions on $[0, \tau]$ are exactly the fixed points of

$$\mathcal{M}(Y, Y') = \left(\xi + \int_0^\cdot f(Y_r) d\mathbf{X}_r, f(Y) \right),$$

which is well defined on $\mathcal{D}_X^{2\alpha}([0, \tau])$ by Example 3.24 ii) together with Theorem 3.25.

Step 1: the ball. Since the space $\mathcal{D}_X^{2\alpha}$ is affine rather than linear once the initial data are fixed, we centre the ball at the canonical first-order approximation

$$\mathcal{W}_t := \xi + f(\xi)X_{0,t}, \quad \mathcal{W}'_t := f(\xi), \quad t \in [0, \tau],$$

which lies in $\mathcal{D}_X^{2\alpha}$ with $R^\mathcal{W} \equiv 0$, and set

$$B_\tau(R) := \left\{ (Y, Y') \in \mathcal{D}_X^{2\alpha}([0, \tau]) : Y_0 = \xi, Y'_0 = f(\xi), \|(Y, Y') - (\mathcal{W}, \mathcal{W}')\|_{X, 2\alpha} \leq R \right\},$$

a closed subset of a Banach space, hence a complete metric space.

Step 2: invariance. Let $(Y, Y') \in B_\tau(R)$. Applying (3.27) to the integrand $(f(Y), Df(Y)Y')$, whose controlled norm is bounded by $C(1 + R)$ by Example 3.24 ii), one obtains

$$\|\mathcal{M}(Y, Y') - (\mathcal{W}, \mathcal{W}')\|_{X, 2\alpha} \leq C|f(\xi)|M + C(1 + R)M\tau^\alpha.$$

The first summand does not involve τ and is bounded by $C\|f\|_\infty M$; choose $R := 2C\|f\|_\infty M$, so that this summand is at most $R/2$. Then choose τ small enough that $C(1 + R)M\tau^\alpha \leq R/2$. With these choices $\mathcal{M}(B_\tau(R)) \subseteq B_\tau(R)$.

Two features of this step are what make the theorem global, and they should be noted explicitly. First, the estimate (3.27) involves $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$; and f is bounded, so $|f(\xi)| \leq \|f\|_\infty$ irrespective of ξ . Hence neither R nor τ depends on the initial condition. Second, the factor τ^α is produced by restricting the time horizon, and it is the only source of smallness available — which is why the argument is local in time and not in space.

Step 3: contraction. For $(Y, Y'), (\tilde{Y}, \tilde{Y}') \in B_\tau(R)$, the same estimate applied to the difference of the integrands, together with the local Lipschitz property of $Y \mapsto f(Y)$ as a map of controlled paths — this is where the third derivative of f is used, and it is proved in [33, Lem. 7.3] — gives

$$\|\mathcal{M}(Y, Y') - \mathcal{M}(\tilde{Y}, \tilde{Y}')\|_{X, 2\alpha} \leq C(1 + R)M\tau^\alpha \|(Y, Y') - (\tilde{Y}, \tilde{Y}')\|_{X, 2\alpha}.$$

Reducing τ further so that $C(1 + R)M\tau^\alpha \leq \frac{1}{2}$ makes $\mathcal{M}$ a contraction of modulus $\frac{1}{2}$ on $B_\tau(R)$. The Banach fixed point theorem supplies a unique fixed point, that is, a unique solution on $[0, \tau]$, and $\tau =: T_0$ depends only on α , $\|f\|_{\mathcal{C}_b^3}$ and M , as claimed.

Step 4: patching. Because T_0 is independent of the initial condition, the solution on $[0, T_0]$ may be restarted at Y_{T_0} to produce a solution on $[T_0, 2T_0]$, and so on; after $\lceil T/T_0 \rceil$ steps the interval $[0, T]$ is exhausted. That the concatenation is again a controlled path, and that it solves (3.40) across the junctions, follows from the additivity of the rough integral.

For global uniqueness, suppose $(\tilde{Y}, \tilde{Y}')$ is another solution on $[0, T]$ and let $\sigma := \sup\{t : \tilde{Y} = Y \text{ on } [0, t]\}$. The two agree at σ by continuity, so applying Step 3 on $[\sigma, \sigma + \tau]$ — legitimate since τ does not depend on the initial value Y_σ — forces them to agree there also, contradicting the maximality of σ unless $\sigma = T$. ■

B.10 Continuity of the Itô–Lyons map

We prove Theorem 3.38 of Chapter 3.

Proof: Let $\mathbf{X}, \tilde{\mathbf{X}} \in \mathbf{C}^\alpha$ with $\|\mathbf{X}\|_\alpha, \|\tilde{\mathbf{X}}\|_\alpha \leq M$, and let $(Y, f(Y))$ and $(\tilde{Y}, f(\tilde{Y}))$ be the corresponding solutions started at ξ and $\tilde{\xi}$. The difficulty of the statement is that the two solutions live in *different* Banach spaces, $\mathcal{D}_X^{2\alpha}$ and $\mathcal{D}_{\tilde{X}}^{2\alpha}$, so no norm compares them directly; this is why Theorem 3.27 was stated in terms of the quantity

$$d_{X, \tilde{X}, 2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}') := \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha},$$

which is not a metric but does compare the two, and it is this quantity that we estimate.

Step 1: transferring the estimate through the fixed point. By Theorem 3.36 the solutions are the fixed points of the respective maps (3.42), so that $(Y, f(Y)) = \mathcal{M}(Y, f(Y))$ and likewise with tildes. Applying the stability of rough integration, Theorem 3.27, to the two integrals, and writing $\Xi := f(Y)$, $\tilde{\Xi} := f(\tilde{Y})$ for the integrands,

$$d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y})) \leq C\left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| + \tau^\alpha d_{X, \tilde{X}, 2\alpha}(\Xi, \Xi'; \tilde{\Xi}, \tilde{\Xi}')\right). \quad (\text{B.4})$$

The last term is the price of the nonlinearity, and it is handled by the local Lipschitz estimate for composition with $f \in \mathcal{C}_b^3$ [33, Thm. 7.6], which bounds it by

$$d_{X, \tilde{X}, 2\alpha}(\Xi, \Xi'; \tilde{\Xi}, \tilde{\Xi}') \leq C\left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| + d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y}))\right).$$

Step 2: absorbing. Substituting the second display into (B.4) gives, with a new constant,

$$d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y})) \leq C\left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}|\right) + C\tau^\alpha d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y})).$$

Choosing τ small enough that $C\tau^\alpha \leq \frac{1}{2}$ — possible with τ depending only on M , α and f , exactly as in Theorem 3.36 — the final term may be absorbed into the left-hand side, leaving

$$d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y})) \leq 2C\left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}|\right)$$

on $[0, \tau]$. This is the required bound in the comparison quantity.

Step 3: from the comparison quantity to the Hölder norm, and iteration. Writing $Y_{s,t} - \tilde{Y}_{s,t} = f(Y_s)X_{s,t} - f(\tilde{Y}_s)\tilde{X}_{s,t} + R_{s,t}^Y - R_{s,t}^{\tilde{Y}}$ and using the elementary inequality $|ab - \tilde{a}\tilde{b}| \leq |a||b - \tilde{b}| + |a - \tilde{a}||\tilde{b}|$ on the first pair of terms, the bound of Step 2 converts into

$$\|Y - \tilde{Y}\|_\alpha \leq C\left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}|\right)$$

on $[0, \tau]$, which is (3.43) on that interval. Since τ depends only on M and not on the initial conditions, the argument may be restarted at τ with initial values Y_τ and $\tilde{Y}_\tau$, whose difference is already controlled by the right-hand side; after finitely many such steps the estimate covers $[0, T]$, with a constant that has been multiplied by a fixed factor at each step. ■



B.11 Monotone density along a rising wing

We prove Lemma 4.8 of Chapter 4.

Proof: The butterfly quantity (4.2) is invariant under the reflection $k \mapsto -k$, under which $\sigma_{\text{imp},k} \mapsto -\sigma_{\text{imp},k}$ and $\sigma_{\text{imp},kk} \mapsto \sigma_{\text{imp},kk}$: the product $k \sigma_{\text{imp},k}$ is unchanged, and so are the remaining two terms. It therefore suffices to treat the right wing, where $k^* > 0$ and $k = k^* + x$ with $x \geq 0$, so that $\sigma_{\text{imp},k} = \sigma_{\text{imp},x}$ and $\sigma_{\text{imp},kk} = \sigma_{\text{imp},xx}$.

Introduce the substitution

$$u = e^{-x/\ell} \in (0, 1],$$

which is a strictly decreasing bijection from $[0, \infty)$ onto $(0, 1]$, with $u = 1$ at $x = 0$. By (4.8) and (4.11),

$$\sigma_{\text{imp}} = \mathcal{W}_{\infty} - bu, \quad \sigma_{\text{imp},k} = su, \quad \sigma_{\text{imp},kk} = -\frac{s}{\ell}u, \quad k = k^* - \ell \log u, \quad (\text{B.5})$$

where $\mathcal{W}_{\infty} = a + b$ and $s > 0$, $b = s\ell > 0$ by hypothesis. We show that each of the three terms of (4.2) attains its minimum over $x \geq 0$ at the endpoint $x = 0$.

Step 1: the hypothesis (4.19) bounds the lift. From (4.12), $s = \mathfrak{s}_0 a / k^*$, so

$$b = s\ell = \frac{\mathfrak{s}_0 a \ell}{k^*} \leq \mathfrak{s}_0 a c_{\ell} \leq \mathfrak{s}_0 a < a, \quad (\text{B.6})$$

using $\ell \leq c_{\ell} k^*$, $c_{\ell} \leq 1$ and $\mathfrak{s}_0 < 1$. The wing may not lift the volatility by more than its own level at the join. This single inequality drives Steps 2 and 3.

Step 2: the second and third terms. Both are governed by the same function. Set

$$\mathcal{J}(u) := \sigma_{\text{imp}} \sigma_{\text{imp},k} / s = (\mathcal{W}_{\infty} - bu)u = \mathcal{W}_{\infty}u - bu^2, \quad u \in (0, 1].$$

Then $\mathcal{J} > 0$ on $(0, 1]$, since $\mathcal{W}_{\infty} - bu \geq \mathcal{W}_{\infty} - b = a > 0$, and $\mathcal{J}'(u) = \mathcal{W}_{\infty} - 2bu$. By (B.6) we have $b < a$, hence $\mathcal{W}_{\infty} = a + b > 2b \geq 2bu$ for every $u \leq 1$, so $\mathcal{J}' > 0$ on $(0, 1]$: the function $\mathcal{J}$ is strictly increasing there and attains its maximum at $u = 1$, where $\mathcal{J}(1) = \mathcal{W}_{\infty} - b = a$. Therefore

$$0 < \mathcal{J}(u) \leq \mathcal{J}(1) = a \quad \text{for all } u \in (0, 1]. \quad (\text{B.7})$$

Now the second term of (4.2) is $-\frac{1}{4}T^2\sigma_{\text{imp}}^2\sigma_{\text{imp},k}^2 = -\frac{1}{4}T^2s^2\mathcal{J}(u)^2$, which by (B.7) is minimised exactly where $\mathcal{J}$ is maximal, that is at $u = 1$; and the third term is $T\sigma_{\text{imp}}\sigma_{\text{imp},kk} = -\frac{Ts}{\ell}\mathcal{J}(u)$, which is likewise minimised at $u = 1$. Since $u = 1$ corresponds to $x = 0$, both terms are bounded below by their values at the join.

Step 3: the first term. By (B.5) the normalised skew along the wing is

$$\mathfrak{s}(x) = \frac{k \sigma_{\text{imp},k}}{\sigma_{\text{imp}}} = \frac{s(k^* + x)e^{-x/\ell}}{\sigma_{\text{imp}}(x)}.$$

The numerator is $s\mathcal{R}(x)$ with $\mathcal{R}(x) = (k^* + x)e^{-x/\ell}$, and

$$\mathcal{R}'(x) = e^{-x/\ell} \left(1 - \frac{k^* + x}{\ell}\right) \leq e^{-x/\ell} \left(1 - \frac{k^*}{\ell}\right) \leq 0 \quad \text{for } x \geq 0,$$

the last inequality being exactly the hypothesis $\ell \leq c_{\ell} k^* \leq k^*$. The denominator $\sigma_{\text{imp}}(x) = \mathcal{W}_{\infty} - be^{-x/\ell}$ is strictly increasing in x , since $b > 0$. A non-increasing positive numerator over

an increasing positive denominator is non-increasing, so $\mathfrak{s}$ is non-increasing on $[0, \infty)$, with $\mathfrak{s}(0) = \mathfrak{s}_0$ and $\mathfrak{s} \geq 0$ throughout. Hence $0 \leq \mathfrak{s}(x) \leq \mathfrak{s}_0 < 1$, and since $\mathfrak{s} \mapsto (1 - \mathfrak{s})^2$ is decreasing on $[0, 1]$,

$$\left(1 - \mathfrak{s}(x)\right)^2 \geq \left(1 - \mathfrak{s}_0\right)^2 \quad \text{for all } x \geq 0.$$

Conclusion. Each of the three terms of (4.2) is bounded below by its value at $x = 0$, whence $g(x) \geq g(0)$ for all $x \geq 0$.

The hypothesis $c_\ell \leq 1$ is sharp in the following sense. At $c_\ell = 1$, that is $\ell = k^*$, one has $\mathcal{R}'(0) = 0$: the first term is stationary at the join, and the argument of Step 3 has no slack left. For $c_\ell > 1$ we get $\mathcal{R}'(0) = 1 - k^*/\ell > 0$, so $\mathfrak{s}$ increases initially, $(1 - \mathfrak{s})^2$ decreases, and $g(x) < g(0)$ for small $x > 0$ unless the gain from Step 2 happens to compensate. This is what the numerical sweep reported after Lemma 4.8 measures. ■



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